HPE Networking ClearPass 6.14 Installation Guide

Part Number:
Published: May 2026
Edition:

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Abstract

The HPE Networking ClearPass 6.14 Installation Guide provides detailed information for installing or upgrading ClearPass on a virtual appliance, hardware appliance, or cloud instance.

Part Number:
Published: May 2026
Edition:

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About the ClearPass 6.14 Installation Guide

The ClearPass 6.11 Installation Guide provides detailed information for installing or upgrading ClearPass on a virtual appliance, hardware appliance, or cloud instance.

For the latest information about ClearPass 6.11 features and issues, please refer to the ClearPass 6.x.x Release Notes.

This guide contains the following sections:

- System Requirements for ClearPass 6.14 — Provides important system requirements information for this release.
- Upgrading to ClearPass 6.14 — Provides considerations and instructions for version upgrades.
- Updating Within the Same Major Version — Provides considerations and instructions for patch updates.
- Virtual appliance installation sections — Provide instructions for installing, upgrading, and morphing on virtual appliances, including Hyper-V, KVM, and VMware vSphere (ESXi).
- Hardware Appliance Installation — Provides instructions for installing and configuring ClearPass on a hardware appliance and performing other administrative tasks.
- Cloud deployment sections — Provide instructions for deploying ClearPass as a VA on hosted cloud services and external context servers, including Amazon Web Services (AWS), Google Administration Console MDM, and Microsoft Azure cloud service.



NOTE

Periodic revisions to ClearPass documentation are sometimes posted; however, some browsers might display a cached previous version. To ensure that you are viewing the latest version of the documentation, you may want to clear your browser's download history and cached images and files.

Subtopics

Revision History
Long Support or Short Support Release?
Contacting Support
Legal Statements

Intended Audience
Conventions

Revision History

The following table provides the revision history of this document.

Table 1. Revision History

Revision	Change Description
November 4, 2026	Added HPE-VME Installation section.
October 2, 2026	Added system requirements topic for HPE-VME hypervisor.
October 1, 2026	Updated support lists for the **6.14.1** patch release. Updated "Sample Times Required for Upgrade" topic.
May 27, 2026	Initial release, ClearPass 6.14 Installation Guide.

Long Support or Short Support Release?

ClearPass 6.14.0 is a Long Support Release (LSR).

- In a Long Support Release, HPE Aruba Networking introduces new features and new hardware, and "parks" hardware (last major release supported) as needed. A ClearPass LSR version includes certifications and active support for the first three years of release and will then provide security updates only for the final two years of release prior to reaching the EOM and EOST milestones. A maximum of two ClearPass releases are supported at any point in time: one Long Support Release (LSR) and one Short Support Release (SSR). For more information, see the HPE Aruba Networking **End of Life Policy** page.
- In a Short Support Release, HPE Aruba Networking introduces new features and new hardware, but does not "park" any hardware. A ClearPass SSR version reaches End of Support on the date when the next SSR or LSR version is released. A maximum of

two ClearPass releases are supported at any point in time: one Long Support Release (LSR) and one Short Support Release (SSR). For more information, see the HPE Aruba Networking **End of Life Policy** page.

Contacting Support

Main Site	**hpe.com/us/en/networking/hpe - aruba - networking.html**
Support Site	**HPE Networking Support Portal**
Airheads Social Forums and Knowledge Base	**community.arubanetworks.com**
North American Telephone	1 - 800 - 943 - 4526 (US and Canada Toll - Free Number)
International Telephones	**arubanetworking.hpe.com/contact - support/**
Software Licensing Site	**licensemanagement.hpe.com/**
End - of - Life Information	**https://www.hpe.com/psnow/doc/a00143052enw**
Security Incident Response Team	Site: **support.hpe.com/hpesc/public/km/Security - Bulletin - Library** Email: **networking - sirt@hpe.com**

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Subtopics

Use of Cookies
Open Source Code

Use of Cookies

Cookies are small text files that are placed on a user's computer by web sites the user visits. They are widely used in order to make web sites work, or work more efficiently, as well as to provide information to the owners of a site. Session cookies are temporary cookies that last only for the duration of one user session.

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San Jose, CA 95002

USA

Intended Audience

The intended audience for this installation guide includes customers, partners, system administrators, and Aruba/HPE field System Engineers. Please note that this document is not a training guide, and it is assumed that the reader has, at a minimum, foundational training in ClearPass essentials.

The user of this guide should have a working knowledge of the following:

- AAA technologies (RADIUS, TACACS+, 802.1X, MAC authentication, and Web authentication)
- Layer-2 and Layer-3 networking
- User Identity stores, such as Active Directory



NOTE

Providing information about network device configurations and capabilities is outside the scope of this guide. For information on these topics, refer to the documentation provided by the vendor of your network equipment.

Conventions

The following conventions are used throughout this guide to emphasize important concepts:

Table 1. Typographical Conventions

Type Style	Description
Italics	This style is used to emphasize important terms and to mark the titles of books.

Type Style	Description
`System items`	This fixed - width font depicts the following: • Sample screen output • System prompts • Filenames, software devices, and specific commands when mentioned in the text
`Commands`	In the command examples, this bold font depicts text that you must type exactly as shown.
<Arguments>	In the command examples, italicized text within angle brackets represents items that you should replace with information appropriate to your specific situation. For example: `# send <text message>` In this example, you would type “send” at the system prompt exactly as shown, followed by the text of the message you wish to send. Do not type the angle brackets.
`[Optional]`	Command examples enclosed in brackets are optional. Do not type the brackets.
`{Item A | Item B}`	In the command examples, items within curled braces and separated by a vertical bar represent the available choices. Enter only one choice. Do not type the braces or bars.

The following informational icons are used throughout this guide:

NOTE
Indicates helpful suggestions, pertinent information, and important things to remember.

CAUTION
Indicates a risk of damage to your hardware or loss of data.

WARNING
Indicates a risk of personal injury or death.

System Requirements for ClearPass 6.14

This section provides important system requirements information specific to this release. It should be read carefully before Installing ClearPass 6.11.0.

Subtopics

End of Support
Virtual Appliance Requirements
Cloud Instance Deployment Requirements
Supported Browsers
ClearPass OnGuard Agent and Agentless OnGuard Requirements
ClearPass Onboard Requirements
New Installations: Caveats

End of Support

This section describes ClearPass and third-party systems, software, and features that are no longer supported or that are approaching their end-of-support date.

Subtopics

ClearPass 6.14 Milestones
ClearPass 6.14 Deprecated Features
ClearPass 6.14 Deprecation Notices
Third-Party Vendor Operating System End-of-Support

ClearPass 6.14 Milestones

Adjustments may be made to the dates shown below. For the most recent information and for details on the End of Life policy, refer to the **End of Life Policy** page.

The ClearPass 6.14 release is a Long Support Release (LSR).

- Initial Release Date (T0): May 27, 2026
- End of Support (EoST): Next LSR release date (minimum 3 years)

  Releases are now either Long Support Release (LSR) or Short Support Release (SSR). Long Support Releases are supported through all subsequent Short Support Releases for customers who desire a release without the need to update to new features. Short Support Releases are released more regularly to include the addition of new features and functionality.

ClearPass 6.14 Deprecated Features

The following features are no longer supported in ClearPass 6.11:

- Support for DH Groups 14, 19, PRF-HMAC-SHA256, HMAC-SHA256 has been deprecated in ClearPass 6.14.0 and later versions in IPsec Tunnel configurations when CC mode is enabled.
- The PingID SSO social login provider is deprecated in ClearPass 6.14.0 and later versions, and is now removed from the UI's provisioning, web login, and self-registration pages. Instead, the same functionality is now included in the new Generic Open ID Connect provider introduced in this release. (CP-49316)

  ClearPass 6.14 does not support the Aruba Auto Sign-On service template. (CP-50784, CP-50964)

  ClearPass 6.14 does not support 32-bit versions of Linux and Windows operating systems for the OnGuard Agents and Agentless OnGuard. (CP-51628, CP-54227)
- Deprecation of the following attributes from OnGuard Agent Global Settings was announced in the ClearPass 6.7.0 Release Notes. In the 6.14.0 Release, these attributes are not supported and they are removed from the user interface of 6.14.0 and later versions. If these attributes were configured in an existing ClearPass deployment, then after upgrading to 6.14.0 or a later version they will be removed from the configuration: (CP-52315)

- [Deprecated] Allowed Subnets for Wired access
- [Deprecated] Allowed Subnets for Wireless access

- ClearPass 6.14 does not support the following cryptographic algorithms and ciphers. When a system is upgraded to 6.14, if its configuration database includes certificates with the SHA-1 signature algorithm, the upgrade is blocked and a warning message is displayed. Deprecated certificates that are not enabled are deleted after user verification. For more information, see CP-55730. (CP-55730, CP-56172, CP-56030)
  - SHA-1 signature algorithm in digital signatures and certificates
  - 3DES and RC4 encryption algorithms
  - RSA and Diffie-Hellman parameters less than 2048 bits
  - DSA signature algorithm
  - CBC mode ciphers in SSH
  - IKEv1 IPsec protocol
- ClearPass 6.14 does not support the DES_CBC privacy protocol in any operating mode. Customers who use this protocol need to migrate to other available privacy protocol options. As part of this change, configuration options for DES_CBC privacy protocol have been removed from the following pages in 6.14 and later releases: (CP-55862, CP-55882, CP-55883)
  - **Administration > External Accounts > Add**
  - **Administration > External Servers > SNMP Trap Receivers > Add**
  - **Administration > Server Manager > Server Configuration > Service Parameters > ClearPass Network Services**
  - **Administration > Server Manager > Server Configuration > System Monitoring**
  - **Configuration > Network > Devices > Add Device > SNMP Read Settings**
  - **Configuration > Network > Devices > Add Device > SNMP Write Settings**
- ClearPass 6.14 includes cryptography libraries that are FIPS 140-3 certified. The FIPS 140-3 standard mandates use of the TLS Extended Master Secret (EMS) extension in order to be FIPS-compliant. Starting with the 6.14 release, when ClearPass is in FIPS mode clients that connect by TLS must use cryptographic libraries that support the EMS extension or the connection will be rejected. This impacts RADIUS, Captive Portal authentications, RadSec connections, and HTTPS connections for UI logins from older clients. Affected clients must be upgraded to recent versions that support the EMS extension for TLS connections. (CP-56860, CP-57116, CP-57117, CP-58043)

- ClearPass 6.14 does not support the MDM vendor JAMF through the native ClearPass MDM functionality. Instead, JAMF functionality is available through ClearPass Extensions, which support the latest JAMF APIs and provide enhanced capabilities. (CP-56875, CP-56868)
- ClearPass 6.14 does not support SSH keys of type Ed25519. Support for ED25519 cryptographic keys is not available, and any configurations or integrations relying on ED25519-based authentication are not supported. As a result, systems or clients using ED25519 keys may experience authentication failures or connection issues. (CP-57133, CP-57259, CP-57260)

  The CLI command **system install-image** is not supported in 6.14. (CP-57174)

  ClearPass 6.14 does not support the BIOS firmware boot option during ClearPass upgrades and new installations. Starting with the ClearPass 6.14 release, only the UEFI firmware boot option is supported. Before beginning the upgrade to 6.14, ClearPass deployments that were installed on 6.11 or 6.12 using BIOS firmware must be reconfigured to use EFI firmware instead. (CP-57257, CP-57272)

  The C2000 and C3000 hardware appliances have reached end of support (EOS/EOSL). These appliances cannot be upgraded to ClearPass 6.14.0. (CP-56976, CP-59168, CP-59193)
- The Remote Assistance feature is deprecated in ClearPass 6.11, 6.12, and later versions. Although the **Administration > Support > Remote Assistance** page still appears in the user interface in 6.11 and 6.12, the functionality is removed and it cannot be used. The page will be removed in the 6.14 release. (CP-53753)
- ClearPass 6.11 was the last major version that supported OnGuard on the Windows 32-bit operating system.
- ClearPass 6.11 was the last major version that supported OnGuard on Ubuntu 18.04.
- Facebook's parent company Meta terminated its cloud identity program Facebook Wi-Fi on June 12, 2023. As of that date Facebook Wi-Fi stopped working as a cloud identity authentication provider in any ClearPass version. In the ClearPass 6.12.0 release, Facebook Wi-Fi is removed from the list of cloud identity authentication providers in ClearPass Guest's Web Login, Self-Registration, and Onboard Provisioning Settings login forms.
- Beginning in ClearPass 6.11 the use of m4 Amazon Web Services is no longer supported. Amazon indicates that all customers must move to m5 instances to be able to support operating ClearPass.
- ClearPass 6.10 was the last major version that supported VMware ESXi 6.5 and 6.7. ClearPass now requires ESXi 7.0 U3 or later.

- ClearPass 6.10 was the last major version that supported Windows Server 2016 with Hyper-V. ClearPass now requires Windows server 2019 with Hyper-V or later.
- ClearPass 6.10 was the last major version that supported installing the ClearPass OnGuard Agent on an Ubuntu 16.04 system.
- ClearPass 6.10 was the last major version that supported the WorldPay Hotspot transaction processor.
- ClearPass 6.9 was the last major version that supported the following appliance hardware models that were sold as CP-HW-5K or CP-HW-25K. These models are now prevented from upgrading to later major versions:
  - C3000 (R620)
  - C3000 (R630)
- ClearPass Policy Manager 6.8 was the last major version that supported the following appliance hardware models that were originally sold as CP-HW-5K or CP-HW-25K. These models will be prevented from upgrading to version 6.9 or 6.10:
  - C2000 (R210)
  - C2000 (R220)
  - C3000 (R610)
- The following TipsAPI (XML), Guest SOAP APIs, and Guest XML-RPC APIs are no longer supported, and are replaced by the indicated RESTful APIs:
  - **GuestUser TipsAPI** is replaced by **GuestManager** RESTful APIs
  - **OnboardDevice** TipsAPI is replaced by **Onboard** RESTful API
  - **Guest SOAP** APIs are replaced by the **GuestManager**, **Onboard**, **OperatorLogins**, and **SmsServices** RESTful APIs
  - **Guest XML-RPC** APIs are replaced by the **GuestManager**, **Onboard**, **OperatorLogins**, and **SmsServices** RESTful APIs
- The legacy Okta authentication source is now deprecated, and the ability to create new Okta authentication sources has been removed. Okta is fully supported as an identity provider using modern protocols instead. For more information, see the **ClearPass Configuration Guide: Onboard + Cloud Identity Providers** article in the AirHeads Community. (CP-34682).
- The External Backup Puller tool is deprecated and not supported after ClearPass 6.8. To push backups for Linux and Windows systems, customers should use the SFTP, SCP, or NFS methods instead. For information about using SFTP or SCP instead to

push the data, see the “File Backup Servers” section of the *Policy Manager User Guide*.

ClearPass 6.14 Deprecation Notices

The following ClearPass features will not be supported after ClearPass 6.11:

- Microsoft might remove their NPLogonNotify and NPPasswordChangeNotify APIs in a future Windows release. When that happens, and starting with that Windows version, in the ClearPass OnGuard Agent for Windows the **Enable to use Windows Single Sign-On** option will no longer work. For more information, refer to Microsoft's **Deprecated features for Windows client** page.
- ClearPass continually builds on the unified REST API framework to support a wide variety of use cases. All future development will focus on this framework. Although the legacy APIs will still be available and you can continue to use them, no new functionality will be developed for them. We encourage customers to migrate any planned or existing applications to interface with the new REST API framework instead. If you still use the legacy TipsAPI (XML), Guest SOAP APIs, or Guest XML-RPC APIs, we recommend that you migrate to the appropriate RESTful API as soon as possible.

Third-Party Vendor Operating System End-of-Support

Third-party operating systems are supported only during the time between their official release and the end of support from the vendor. While some systems may continue to operate and some functionality will continue to be exposed to work with these, they are officially unsupported. Our TAC organization will not be able to provide service to support requests related to clients running OS versions not officially supported by their vendor. This list includes the following:

- Android 10.x and earlier
- Apple macOS 13.x (Ventura)
- Apple macOS 12.x (Monterey)
- Apple macOS 11.x (Big Sur)
- Apple macOS 10.x (Catalina)
- Apple iOS & iPadOS 14.x and earlier

- Linux Ubuntu 32-bit versions
- Linux CentOS 7
- Linux CentOS Stream 8
- Linux RHEL 7
- Linux RHEL 8
- Linux Ubuntu 21.04 and earlier
- Microsoft Windows 11 versions no longer supported based on the Microsoft® Windows lifecycle fact sheet.
- Microsoft Windows XP
- Microsoft Windows Vista
- Microsoft Windows 11 21H2
- Microsoft Windows 11 22H2
- Microsoft Microsoft Windows 10
- Microsoft Microsoft Windows 8.1 and earlier
- Microsoft Windows Server 2012 R2 and earlier
- Microsoft Windows Server 2012
- Microsoft Windows Server 2008
- Microsoft Windows Server 2003

Virtual Appliance Requirements

Please carefully review all virtual appliance (VA) requirements, including functional IOP ratings, and verify that your system meets these requirements. These requirements supersede earlier requirements that were published for ClearPass 6.x installations. For more information on selecting the best appliance for your deployment, see the **ClearPass Scaling & Ordering Guide**.

Virtual appliance requirements are adjusted to align with the shipping ClearPass hardware appliance specifications. If you do not have the VA resources to support a full workload, then you should consider ordering a ClearPass hardware appliance.

To ensure scalability, dedicate or reserve the processing and memory to the ClearPass virtual appliance instance. You must also ensure that the disk subsystem can maintain

the IOP's throughput as detailed in the hypervisor requirements sections below. Most virtualized environments use a shared disk subsystem assuming that each application will have bursts of I/O without a sustained high I/O throughput. ClearPass requires a continuous sustained high data I/O rate. We recommend that virtual appliance deployments use thick provisioning for drives for optimal performance.

The CLABV appliance image is for training, configuration testing, and demonstrations. The C1000V, C2000V, and C3000V appliance images should be used for performance and scale testing or to mimic a production environment.

For complete information on installing, configuring, or morphing an ESXi™, Hyper-V®, or KVM hypervisor, see the *Virtual Appliance Installation* sections in this Installation Guide.



NOTE

ClearPass does not support nested or multi-level virtualization deployments — for example, Nutanix → Hyper-V → ClearPass, or ESXi → Ubuntu/KVM → CPPM. ClearPass is also not supported on ESXi, Hyper-V, or KVM hypervisors hosted within public cloud environments such as AWS, Azure, or GCP. Only directly hosted, qualified hypervisor deployments listed as supported in the System Requirements chapter are supported.



NOTE

Virtual appliances might display different input/output per second (IOPS) values during boot time depending on the hypervisor in use. This issue might result in the displayed IOPS values having differences between boot and current times. To address this, the warnings during boot for disk performance have been suppressed. (CP-36221)



NOTE

In ClearPass 6.11.0 and later versions, the maximum single cluster size is limited to 32 servers. This includes the publisher, standby publisher, subscribers, dedicated Insight server, and standby Insight servers.

Subtopics

Supported Hypervisors
HPE Morpheus VM Essentials (HPE-VME) Requirements
Hyper-V Requirements
KVM Requirements
Nutanix Acropolis Hypervisor (AHV) Requirements
VMware vSphere Hypervisor (ESXi) Requirements

Supported Hypervisors

The following hypervisors are supported, as well as all updates for the versions shown here. Hypervisors that run on a client computer such as VMware Player are not supported.

- HPE Morpheus VM Essentials (HPE-VME) 8.1.2 on Ubuntu 24.04.x and later and HVM OS.
- Hyper-V: Windows Server 2025 with Hyper-V, Windows Server 2022 with Hyper-V, and Windows Server 2019 with Hyper-V.
- KVM on CentOS Stream 10, CentOS Stream 9, Ubuntu 24.04 LTS, and Ubuntu 26.04 LTS.
- Nutanix AHV: Prism Central pc.7.5.0.5

- VMware vSphere Hypervisor (ESXi) 9.0, 8.0 U3, and ESXi9

NOTE

HPE Morpheus VM Essentials is supported starting from 6.14.1. For hardware compatibility information, refer to the **HPE Aruba Networking compatibility matrix**.

NOTE

Nutanix is supported starting from 6.14.0. Upgrade from 6.11.X or 6.12.X to 6.14.0 is not applicable for this platform.

CAUTION

Installing or upgrading to ClearPass 6.12.0 on an ESXi 7.0.0 system fails, and the system becomes unreachable and cannot be recovered. If you are on ESXi 7.0 you must apply the latest ESXi patch updates and ensure it is updated to ESXi 7.0 Update 3c or later before upgrading or installing ClearPass. You should also make a backup of your ClearPass 6.11.x system before installing or upgrading ClearPass. (CP-49208)

CAUTION

ClearPass does not support nested deployments for hypervisors. ClearPass ESXi, Hyper-V, or KVM deployments are supported only for on-premises solutions.

NOTE

ClearPass does not support nested or multi-level virtualization deployments — for example, Nutanix → Hyper-V → ClearPass, or ESXi → Ubuntu/KVM → CPPM. ClearPass is also not supported on ESXi, Hyper-V, or KVM hypervisors hosted within public cloud environments such as AWS, Azure, or GCP. Only directly hosted, qualified hypervisor deployments listed as supported in the System Requirements chapter are supported.

Some third-party vendor operating systems might have reached their vendor's End of Life (EOL) date since ClearPass 6.11 was released. Aruba will attempt to preserve compatibility with those legacy operating systems until the next major ClearPass release following the vendor's EOL date.

HPE Morpheus VM Essentials (HPE-VME) Requirements

After you have verified these requirements, for complete information on installing, configuring, or morphing an HPE-VME hypervisor, see the "Virtual Appliance Installation: HPE-VME" section in this Installation Guide.

CLABV (Evaluation QCOW2 disk image)

- 4 reserved virtual CPUs
- 6 GB RAM
- 400 GB disk space required (reserved disk with storage container)

  2 Gigabit virtual switched ports

C1000V (Virtual Appliance QCOW2 Disk Image)

- 8 reserved virtual CPUs
- 8 GB RAM
- Disk space: 1000 GB disk space required (reserved disk with storage container)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C2000V (Virtual Appliance QCOW2 Disk Image)

- 8 reserved virtual CPUs
- 16 GB RAM
- Disk space: 1000 GB disk required (reserved disk with storage container)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C3000V (Virtual Appliance QCOW2 Disk Image)

- 24 reserved virtual CPUs
- 64 GB RAM
- Disk space: 1800 GB disk required (reserved disk with storage container)

- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

Hyper-V Requirements

After you have verified these requirements, for complete information on installing, configuring, or morphing a Hyper-V® hypervisor, see the Virtual Appliance Installation: Microsoft Hyper-V section in this Installation Guide.

NOTE

When ClearPass 6.14.0 is freshly installed on a VMware or Hyper-V instance, in a C1000V or C2000V VA it occupies 16 GB RAM by default, matching the N1000 appliance. Users should be aware that the system requirements have not changed and remain the same at 8 GB RAM. The size of the instance can be changed to 8 GB RAM to form a C1000V appliance type.

NOTE

If you are upgrading an existing deployment from an older system with smaller disk space, you might want to work with Aruba's Technical Assistance Center (TAC) to reprovision the system in order to avoid potentially running out of disk space. ClearPass Policy Manager 6.9 was the last major version that was able to run on a system with smaller disk space than the current system requirements.

CLABV (Evaluation VHDX)

- 4 reserved virtual CPUs
- 6 GB RAM
- 400 GB disk space required (thick provisioned)

C1000V (Virtual Appliance VHDX)

- 8 reserved virtual CPUs
  - Underlying CPU is recommended to have a **PassMark** ® of 3000 or higher
- 8 GB RAM

- Disk space: 1000 GB disk space required (thick provisioned)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C2000V (Virtual Appliance VHDX)

- 8 reserved virtual CPUs
  - Underlying is recommended to have a **PassMark** ® of 9600 or higher
- 16 GB RAM
- Disk space: 1000 GB disk required (thick provisioned)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C3000V (Virtual Appliance VHDX)

- 24 reserved virtual CPUs
  - Underlying CPUs are recommended to have a **PassMark** ® of 16,000 or higher
- 64 GB RAM
- Disk space: 1800 GB disk required (thick provisioned)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

KVM Requirements

After you have verified these requirements, for complete information on installing, configuring, or morphing a KVM hypervisor, see the Virtual Appliance Installation: KVM section in this Installation Guide.

> **NOTE**
> When ClearPass 6.14.0 is freshly installed on a VMware or Hyper-V instance, in a C1000V or C2000V VA it occupies 16 GB RAM by default, matching the N1000 appliance. Users should be aware that the system requirements have not changed and remain the same at 8 GB RAM. The size of the instance can be changed to 8 GB RAM to form a C1000V appliance type.

> **NOTE**
> Users should be aware that the **Disk bus** type must be set to **IDE** for new ClearPass Policy Manager installations and upgrades on KVM hypervisors. Do not use SCSI/VirtIO as the bus type.

CLABV (Evaluation RAW Disk Image)

- 4 reserved virtual CPUs
- 6 GB RAM
- Disk space: 400 GB disk space required (thick provisioned)
- 2 Gigabit virtual switched ports

C1000V (Virtual Appliance RAW Disk Image)

- 8 reserved virtual CPUs
  - Underlying CPU is recommended to have a **PassMark** ® of 3000 or higher
- 8 GB RAM
- Disk space: 1000 GB disk space required (thick provisioned)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C2000V (Virtual Appliance RAW Disk Image)

- 8 reserved virtual CPUs
  - Underlying is recommended to have a **PassMark** ® of 9600 or higher

- 16 GB RAM
- Disk space: 1000 GB disk space required (thick provisioned)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C3000V (Virtual Appliance RAW Disk Image)

- 24 reserved virtual CPUs
  - Underlying CPUs are recommended to have a **PassMark** ® of 16,000 or higher
- 64 GB RAM
- Disk space: 1800 GB disk space required (thick provisioned)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

Nutanix Acropolis Hypervisor (AHV) Requirements

After you have verified these requirements, for complete information on installing, configuring, or morphing a Nutanix Acropolis Hypervisor (AHV) ®, see the Nutanix AHV Installation Process Overview section in this Installation Guide.

CLABV (Evaluation RAW disk image)

- 4 reserved virtual CPUs
- 6 GB RAM
- 400 GB disk space required (reserved disk with storage container)

  2 Gigabit virtual switched ports

C1000V (Virtual Appliance RAW Disk Image)

- 8 reserved virtual CPUs
- 8 GB RAM
- Disk space: 1000 GB disk space required (reserved disk with storage container)

- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C2000V (Virtual Appliance RAW Disk Image)

- 8 reserved virtual CPUs
- 16 GB RAM
- Disk space: 1000 GB disk required (reserved disk with storage container)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C3000V (Virtual Appliance RAW Disk Image)

- 24 reserved virtual CPUs
- 64 GB RAM
- Disk space: 1800 GB disk required (reserved disk with storage container)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

VMware vSphere Hypervisor (ESXi) Requirements

After you have verified these requirements, for complete information on installing, configuring, or morphing an ESXi™ hypervisor, see the Virtual Appliance Installation: VMware vSphere (ESXi) section in this Installation Guide.

NOTE

When ClearPass 6.14.0 is freshly installed on a VMware or Hyper-V instance, in a C1000V or C2000V VA it occupies 16 GB RAM by default, matching the N1000 appliance. Users should be aware that the system requirements have not changed and remain the same at 8 GB RAM. The size of the instance can be changed to 8 GB RAM to form a C1000V appliance type.

NOTE

If you are upgrading an existing deployment from an older system with smaller disk space, you might want to work with Aruba's Technical Assistance Center (TAC) to reprovision the system in order to avoid potentially running out of disk space. ClearPass Policy Manager 6.9 was the last major version that was able to run on a system with smaller disk space than the current system requirements.

CLABV (Evaluation OVF)

- 4 reserved virtual CPUs
- 6 GB RAM
- 400 GB disk space required (thick provisioned)

C1000V (Virtual Appliance OVF)

- 8 reserved virtual CPUs
  - Underlying CPU is recommended to have a **PassMark** ® of 3000 or higher
- 8 GB RAM
- Disk space: 1000 GB disk space required (thick provisioned)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C2000V (Virtual Appliance OVF)

- 8 reserved virtual CPUs

- Underlying CPU is recommended to have a **PassMark** ® of 9600 or higher

- 16 GB RAM
- Disk space: 1000 GB disk space required (thick provisioned)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

C3000V (Virtual Appliance OVF)

- 24 reserved virtual CPUs
  - Underlying CPUs are recommended to have a **PassMark** ® of 16,000 or higher
- 64 GB RAM
- Disk space: 1800 GB disk space required (thick provisioned)
- 2 Gigabit virtual switched ports
- The input/output operations per second (IOPS) values should be reasonable for the database server

Cloud Instance Deployment Requirements

Please carefully review all cloud instance requirements and verify that your deployment meets these specifications. These requirements supersede earlier requirements that were published for ClearPass Policy Manager 6.11.x instance deployments. For more information on selecting the best specifications for your instance deployment, see the **ClearPass Scaling & Ordering Guide**.

Cloud instance requirements are adjusted to align with the shipping ClearPass Policy Manager hardware appliance specifications. For complete information on deploying a cloud instance, see the specific cloud instance sections in this Installation Guide.

NOTE

In ClearPass 6.11.0 and later versions, the maximum single cluster size is limited to 32 servers. This includes the publisher, standby publisher, subscribers, dedicated Insight server, and standby Insight servers.

NOTE

ClearPass does not support nested or multi-level virtualization deployments — for example, Nutanix → Hyper-V → ClearPass, or ESXi → Ubuntu/KVM → CPPM. ClearPass is also not supported on ESXi, Hyper-V, or KVM hypervisors hosted within public cloud environments such as AWS, Azure, or GCP. Only directly hosted, qualified hypervisor deployments listed as supported in the System Requirements chapter are supported.

Subtopics

Amazon Web Services (AWS) Requirements
Microsoft Azure Requirements

Amazon Web Services (AWS) Requirements

After you have verified these requirements, for complete information on deploying, configuring, or morphing an Amazon Web Services (AWS) instance, see the Cloud Deployments: Amazon Web Services (AWS) section in this Installation Guide.

C1000V: m5.large, m7a.large, and m7i.large Instance Types

The m5.large, m7a.large, and m7i.large AWS instance types are the equivalent of the C1000V appliance in ClearPass 6.10.0 and later.

- 2 reserved virtual CPUs
- 8 GB RAM
- 1000 GB disk space required

C2000V: m5.xlarge, m7a.xlarge, and m7i.xlarge Instance Types

The m5.xlarge, m7a.xlarge, and m7i.xlarge AWS instance types are the equivalent of the C2000V appliance in ClearPass 6.10.0 and later.

- 4 reserved virtual CPUs
- 16 GB RAM
- 1000 GB disk space required

C3000V: m5.4xlarge, m7a.4xlarge, and m7i.4xlarge Instance Types

The m5.4xlarge, m7a.4xlarge, and m7i.4xlarge AWS instance types are the equivalent of the C3000V appliance in ClearPass 6.10.0 and later.

- 16 reserved virtual CPUs
- 64 GB RAM
- 1800 GB disk space required

Microsoft Azure Requirements

After you have verified these requirements, for complete information on deploying, configuring, or morphing a Microsoft Azure instance, see the Cloud Deployments: Microsoft Azure Cloud Service section in this Installation Guide.

C1000V: Standard_D2s_v3 Instance Type

The Standard_D2s_v3 Azure instance type is the equivalent of the C1000V appliance.

- 2 reserved virtual CPUs
- 8 GB RAM
- 1000 GB disk space required

C2000V: Standard_D4s_v3 Instance Type

The Standard_D4s_v3 Azure instance type is the equivalent of the C2000V or C2010V appliance.

- 4 reserved virtual CPUs
- 16 GB RAM
- 1000 GB disk space required

C3000V: Standard_D16s_v3 Instance Type

The Standard_D16s_v3 Azure instance type is the equivalent of the C3000V or C3010V appliance.

- 16 reserved virtual CPUs
- 64 GB RAM
- 1800 GB disk space required

C1000V: Standard_D2ds_v5 Instance Type

The Standard_D2ds_v5 Azure instance type is the equivalent of the C1000V appliance.

- 2 reserved virtual CPUs
- 8 GB RAM
- 1000 GB disk space required

C2000V: Standard_D4ds_v5 Instance Type

The Standard_D4ds_v5 Azure instance type is the equivalent of the C2000V or C2010V appliance.

- 4 reserved virtual CPUs
- 16 GB RAM
- 1000 GB disk space required

C3000V: Standard_D16ds_v5 Instance Type

The Standard_D16ds_v5 Azure instance type is the equivalent of the C3000V or C3010V appliance.

- 16 reserved virtual CPUs
- 64 GB RAM
- 1800 GB disk space required

Supported Browsers

For the best user experience, we recommend you update your browser to the latest version available. Supported browsers for ClearPass are:

- Apple Safari 26.x and later on macOS.
- Google Chrome for macOS and Windows.

- Microsoft Edge on Windows 11 and macOS.
- Mozilla Firefox on Windows 11 and macOS 14.x (Sonoma) and later.

Browsers in Compatibility or Emulation Mode are Unsupported

NOTE

ClearPass does not support running a browser in compatibility or emulation mode. Users should be aware that when an otherwise supported browser version emulates a lower, unsupported version, it is actually using the functionality of the unsupported version. For example, when an Internet Explorer (IE) 11 browser is running in compatibility mode emulating IE 7, it is essentially running IE 7 and is therefore unsupported.

ClearPass OnGuard Agent and Agentless OnGuard Requirements

This section describes system requirements for the OnGuard Agent and Agentless OnGuard, supported third-party products, and Dissolvable Agent requirements.

Subtopics

OnGuard Agent and Agentless OnGuard System Requirements
OnGuard Supported Third-Party Products
OnGuard Dissolvable Agent Version Requirements

OnGuard Agent and Agentless OnGuard System Requirements

Be sure that your client system meets the following requirements before installing the ClearPass OnGuard Agent or using Agentless OnGuard:

- 1 GB RAM recommended, 512 MB RAM minimum
- 300 MB disk space
- CentOS Stream 9 and CentOS Stream 10
- Fedora 44

- macOS 14.x (Sonoma) to 27.x (Golden Gate)
- RHEL 9.x, 10.x
- SUSE Linux Enterprise 15
- Ubuntu 22.04 LTS, 24.04 LTS, or 26.04 LTS
- Windows 11 (23H2 and later), Windows Server 2025, Windows Server 2022, Windows Server 2019, and Windows Server 2016 are all supported with no service pack requirements.
- ARM processor architectures are supported for the OnGuard Agents for Windows (Persistent, Dissolvable, and Agentless). They are **not** supported on any version of Linux.

  NOTE

  For the required additional libraries for Ubuntu 26.04, refer to the readme.txt file included in the OnGuard installer package for details.

  Some third-party vendor operating systems might have reached their vendor's End of Life (EOL) date since ClearPass 6.11 was released. Aruba will attempt to preserve compatibility with those legacy operating systems until the next major ClearPass release following the vendor's EOL date.



NOTE

Windows Core versions are not supported in ClearPass 6.14.x.

Subtopics

OnGuard Installations: Caveats

OnGuard Installations: Caveats

This section provides information you should be aware of before you install ClearPass OnGuard.

CAUTION

Installing the ClearPass OnGuard Unified Agent on a Linux operating system will remove an existing VIA installation. To continue using VPN functionality on Linux, log in to ClearPass Policy Manager as the administrator, go to **Administration > Agents and Software Updates > OnGuard Settings**, and select **Install and enable Aruba VPN component** from the **Linux Installer Mode** drop-down list. This caveat does not apply to other operating systems.

NOTE

A graphical user interface (UI) is required for the OnGuard Persistent and Native Dissolvable Agent. The OnGuard Persistent and Native Dissolvable Agents are not supported on operating systems without a UI.

NOTE

ClearPass 6.8 was the last major version that supported the Windows System Health Validator and Windows Security Health Validator plugins. These are not supported for Windows 11 and higher versions.

NOTE

The ClearPass OnGuard Native Dissolvable Agent is not supported on Linux CentOS, RHEL, or Ubuntu systems in a KDE desktop environment.

NOTE

Users should be aware that Agentless OnGuard is not supported in FIPS or CC mode.

OnGuard Supported Third-Party Products

This section describes third-party processes that should be allowlisted for the OnGuard Persistent Agent and for Agentless OnGuard on each OS type, as well as a summary of some of the third-party antivirus and malware products used in testing.

Processes to Allowlist for the Windows OnGuard Persistent Agent

NOTE

For the Windows OnGuard Persistent Agent to work properly, please allowlist the following executable files and installation folders in your antivirus products:

- ClearPassAgentController.exe
- ClearPassAgentHelper.exe
- ClearPassInstallerHelper.exe
- ClearPassOnGuard.exe
- ClearPassOnGuardAgentService.exe
- ClearPassUSHARemediate.exe
- C:\Program Files (x86)\Aruba Networks\ClearPassOnGuard\
- C:\Program Files\Aruba Networks\ClearPassOnGuard\
- wa_3rd_party_host_32.exe
- wa_3rd_party_host_64.exe

Processes to Allowlist for Windows Agentless OnGuard

NOTE

For Windows Agentless OnGuard to work properly, please allowlist the following executable files and installation folders in your antivirus products:

- ArubaOnGuard.exe
- ArubaOnGuardHelper.exe
- C:\ProgramData\Aruba Networks\AgentlessOnGuard\
- wa_3rd_party_host_32.exe
- wa_3rd_party_host_64.exe

- <Working Directory>\AgentlessLauncher.exe

> **NOTE**
>
> *<Working Directory>* is the working directory path configured in **Agentless OnGuard Settings** on the ClearPass server.

Processes to Allowlist for the macOS OnGuard Persistent Agent

> **NOTE**
>
> For the macOS OnGuard Persistent Agent to work properly, please allowlist the following executable files and installation folders in your antivirus products:

- "/Applications/Aruba Networks/ClearPass OnGuard.app"
- "/Applications/Aruba Networks/ClearPass OnGuard.app/Contents/MacOS/ClearPass OnGuard"
- "/Applications/Aruba Networks/ClearPass OnGuard.app/Contents/PlugIns/OnGuard.bundle/Contents/MacOS/OnGuard"
- "/Applications/Aruba Networks/ClearPass OnGuard.app/Contents/PlugIns/OnGuard.bundle/Contents/MacOS/OnGuardService"
- "/Applications/Aruba Networks/ClearPass OnGuard.app/Contents/PlugIns/OnGuard.bundle/Contents/MacOS/ClearPassAgentHelper"
- "/Applications/Aruba Networks/ClearPass OnGuard.app/Contents/PlugIns/OnGuard.bundle/Contents/MacOS/InstallerHelper.app"
- "/Applications/Aruba Networks/ClearPass OnGuard.app/Contents/PlugIns/OnGuard.bundle/Contents/MacOS/OnGuardService"
- "/Applications/Aruba Networks/ClearPass OnGuard.app/Contents/PlugIns/OnGuard.bundle/Contents/MacOS/ServiceDaemon"

  Note for macOS: Some paths include spaces. The customer should put such paths within double quotes (" ") when allowing these paths in 3rd party applications.

Processes to Allowlist for macOS Agentless OnGuard

NOTE

For macOS Agentless OnGuard to work properly, please allowlist the following executable files and installation folders in your antivirus products:

- "/Library/Application Support/AgentlessOnGuard/AgentlessOnGuard"
- "/Library/Application Support/AgentlessOnGuard/ClearPassAgentHelper"
- "/Library/Application Support/AgentlessOnGuard/AgentlessOnGuardRemover"

Note for macOS: Some paths include spaces. The customer should put such paths within double quotes ("") when allowing these paths in 3rd party applications.

Processes to Allowlist for the Linux OnGuard Persistent Agent

NOTE

For the Linux OnGuard Persistent Agent to work properly, please allowlist the following executable files and installation folders in your antivirus products:

- /usr/share/clearpass-onguard/clearpass-agentcontroller-service
- /usr/share/clearpass-onguard/clearpass-onguard
- /usr/share/clearpass-onguard/clearpass-onguard-agent-service
- /usr/share/clearpass-onguard/clearpass-onguard-autostart
- /usr/share/clearpass-onguard/nm-viavpn-service
- /usr/share/clearpass-onguard/via-cli
- /usr/share/clearpass-onguard/via-login-handler
- /usr/share/clearpass-onguard/via-logout-handler
- /usr/share/clearpass-onguard/via-reset-ui-handler
- /usr/share/clearpass-onguard/via-vpn-srv
- /usr/bin/clearpass-agentcontroller-service
- /usr/bin/clearpass-onguard-agent-service
- /usr/bin/via-vpn-srv

Processes to Allowlist for Linux Agentless OnGuard

NOTE

For Linux Agentless OnGuard to work properly, please allowlist the following executable files and installation folders in your antivirus products:

- /usr/bin/clearpass-agentless-onguard-service
- /usr/share/clearpass-agentless-onguard/clearpass-agentless-onguard-helper
- /usr/share/clearpass-agentless-onguard/clearpass-agentless-onguard-remover

Third-Party Software Validation Summary

In current laboratory tests for ClearPass Policy Manager 6.11.16, we use the following third-party software for our validations. Due to the large number of products available, this list may change at any time:

Table 1. Third - Party Software Summary

Product Type	Product Name
Antivirus	Avast Pro Antivirus (Windows)
	Avira Mac Security (macOS)
	ESET Cyber Security Pro (macOS)
	F - Secure Anti - Virus for Mac (macOS)
	Kaspersky Internet Security (macOS)
	Kaspersky Total Security (Windows)
	McAfee Endpoint Security Threat Prevention (Windows)
	Sophos Anti - Virus (Windows)
	Symantec Endpoint Protection (Windows)
	Windows Defender (Windows)
Antispyware	McAfee Host Intrusion Prevention (Windows)
	McAfee VirusScan Enterprise (Windows)

Product Type	Product Name
Firewall	Mac OS X Built - In Firewall (macOS)
	McAfee Endpoint Protection for Mac (macOS)
	Microsoft Windows Firewall (Windows)
Disk Encryption	BitLocker Drive Encryption (Windows)
	FileVault (macOS)
Patch Management	McAfee ePolicy Orchestrator Agent (Windows)
	Microsoft Windows Update Agent (Windows)
	Software Update (macOS)
	System Center Configuration Manager (SCCM) (Windows)
Virtual Machine	Oracle VM VirtualBox (Windows)
	VirtualBox (macOS)
	VMware Fusion (macOS)



NOTE

Some third-party anti-malware products are not supported by ClearPass OnGuard. For complete lists of third-party products supported by OnGuard, go to **Policy Manager > Administration > Support > Documentation**. For products supported by the OESIS V4 SDK, click the **OnGuard Agent Support Charts for Plugin Version 2.0** link. Next, click the link for the appropriate product type and operating system.

OnGuard Dissolvable Agent Version Requirements

This section provides version information for the dissolvable agents. For more information on the dissolvable agents, refer to the ClearPass Policy Manager online help.

> **NOTE**
> A graphical user interface (UI) is required for the OnGuard Persistent and Native Dissolvable Agent. The OnGuard Persistent and Native Dissolvable Agents are not supported on operating systems without a UI.

Subtopics

OnGuard Native Dissolvable Agent Browser Version Information

OnGuard Native Dissolvable Agent Browser Version Information

In current laboratory tests for ClearPass 6.11.16, the browser versions shown in Table 1 were verified for the ClearPass OnGuard native dissolvable agent. There are considerations to be aware of with some browser versions. For more information, click the issue ID number next to the browser's name.

Some third-party vendor operating systems might have reached their vendor's End of Life (EOL) date since ClearPass 6.11 was released. HPE Aruba Networking will attempt to preserve compatibility with those legacy operating systems until the next major ClearPass release following the vendor's EOL date.

> **NOTE**
> The native dissolvable agent is not currently supported with the Firefox browser.

> **NOTE**
> The following list might include some legacy operating systems that are no longer supported by their vendors and are therefore unsupported in ClearPass. Users should be aware that testing for these legacy systems is on a best-effort basis.

Table 1. Native Dissolvable Agent Latest Supported Browser Versions for This Release

Operating System	Browser
Linux - CentOS Stream 10 64 - bit	Chrome 154.x

Operating System	Browser
Linux - CentOS Stream 9 64 - bit	Chrome 154.x
Linux - Fedora 44 64 - bit	Chrome 154.x
Linux - Redhat (RHEL) 10 64 - bit	Chrome 154.x
Linux - Redhat (RHEL) 9.6 64 - bit	Chrome 154.x
Linux - SUSE Enterprise Desktop 15 SP7 64 - bit	Chrome 154.x
Linux - Ubuntu 26.04 64 - bit LTS	Chrome 154.x
Linux - Ubuntu 24.04 64 - bit LTS	Chrome 154.x
Linux - Ubuntu 22.04 64 - bit LTS	Chrome 154.x
macOS 27.x (Golden Gate)	Safari 27.x (CP-23775)
	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge Chromium 153.x
macOS 26.x (Tahoe)	Safari 27.x (CP-23775)
	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge Chromium 153.x
macOS 15.x (Sequoia)	Safari 27.x (CP-23775)
	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge Chromium 153.x
macOS 14.x (Sonoma)	Safari 27.x (CP-23775)
	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge Chromium 153.x
Windows 11 26H2	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge 153.x
Windows 11 26H1	Chrome 154.x (CP-28513, CP-28615)

Operating System	Browser
	Microsoft Edge 153.x
Windows 11 25H2	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge 153.x
Windows 11 24H2	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge 153.x
Windows 11 23H2	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge 153.x
Windows 11 22H2	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge 153.x
Windows 11 21H2	Chrome 154.x (CP-28513, CP-28615)
	Microsoft Edge 153.x
Windows Server 2025 Datacenter Edition	Chrome 154.x (CP-28615)
Windows Server 2025 Standard Edition	Chrome 154.x (CP-28615)
Windows Server 2022 Datacenter Edition	Chrome 154.x (CP-28615)
Windows Server 2022 Standard Edition	Chrome 154.x (CP-28615)
Windows Server 2019 Datacenter Edition	Chrome 154.x (CP-28615)
Windows Server 2019 Standard Edition	Chrome 154.x (CP-28615)

ClearPass Onboard Requirements

Onboard's QuickConnect wizard does not support over-the-air provisioning for Windows RT, Windows Phone, or Windows 10 S. Certificate provisioning and enrollment for these devices can be configured through Onboard's **Device Provisioning** pages.

NOTE

ClearPass Onboard is not for use with managed systems such as domain joined computers where the information is available for distribution with a Group Policy Object (GPO).

NOTE

Starting with the ClearPass 6.12.6 release, QuickConnect for Windows is supported on WiFi 7 network adapters.

Subtopics

Onboard Supported Platforms

Onboard Supported Platforms

The platforms supported by ClearPass Onboard and the version requirements for each platform are summarized in the following table.

Some third-party vendor operating systems might have reached their vendor's End of Life (EOL) date since ClearPass 6.11 was released. HPE Aruba Networking will attempt to preserve compatibility with those legacy operating systems until the next major ClearPass release following the vendor's EOL date.

Table 1. Platforms Supported by ClearPass Onboard

Platform	Example Devices	Version Required for Onboard Support	Notes
Android	Samsung Galaxy	Android 17	2
	Motorola Droid	Android 16	
	Google Pixel	Android 15.x	
		Android 14.x	
		Android 13.x	
		Android 12.x	
Apple iOS	iPhone	iOS 15.x and later	1, 3
	iPad	iPadOS 15.x and later	

Platform	Example Devices	Version Required for Onboard Support	Notes
	iPod Touch		
Apple macOS	MacBook Pro MacBook Air	macOS 14.x to 27.x (Sonoma to Golden Gate)	1
		macOS 14.x to 27.x (Sonoma to Golden Gate)	2
Chrome OS	Laptop	Chrome OS 152 Chrome OS 147 Chrome OS 143 Chrome OS 140 Chrome OS 139	
Microsoft Windows	Laptop Desktop Tablet	Windows 11 (26H2) Windows 11 (26H1) Windows 11 (25H2) Windows 11 (24H2) Windows 11 (23H2) Windows Server 2025 Windows Server 2022 Windows Server 2019	2, 4
Ubuntu	Desktop Server	26.04 24.04 22.04 20.04	

[1] **Note 1:** Uses the "Over-the-air provisioning" method.
[2] **Note 2:** Uses the "Onboard provisioning" method.
[3] **Note 3:** Onboard may also be used to provision VPN settings, Exchange ActiveSync settings, and passcode policy on these devices.
[4] **Note 4:** Windows 10 Mobile is not currently supported in Onboard.

New Installations: Caveats

This section provides information you should be aware of before you perform a new ClearPass installation.

- Please carefully review all sections of the Upgrading to ClearPass 6.14 chapter before proceeding with the 6.11 backup, fresh installation, or configuration data restoration.
- When creating a database backup of a lower version of ClearPass to be restored on ClearPass 6.11, best practice is to create separate backups for the configuration db (tipsdb) and Insight (insightdb) databases. Backup files can be very large, and creating separate backups ensures that they will be restored correctly without overloading the system and blocking the restoration operation.
- If you are performing a new ClearPass 6.11.0 installation on a VMware ESXi server, **and only** if the MAC address of **Network adapter1** is higher than that of **Network adapter2**, additional steps are required after the installation. Before you begin the installation, please review After You Install on ESXi Servers: Establishing NW Connectivity in this Installation Guide.
- In ClearPass 6.11.0 and later versions, the maximum single cluster size is limited to 32 servers. This includes the publisher, standby publisher, subscribers, dedicated Insight server, and standby Insight servers.

Upgrading to ClearPass 6.14

This section provides instructions for upgrading your ClearPass appliance. An upgrade is the process of moving from one major release version to another—for example, from 6.12.x to 6.11.0. This section describes accessing upgrade images, considerations to be aware of, and instructions for restoring the log database after the upgrade (optional).

- Upgrade images are available within ClearPass Policy Manager from the **Software Updates** portal at **Administration > Agents and Software Updates > Software Updates**. HTTPS access must be allowed to clearpass.arubanetworks.com.
- Upgrade images and preparation patches are also available for download on the **HPE Networking Support Portal**.
- To upgrade a cluster to 6.11.0, we recommend using the **Cluster Upgrade** page in the Policy Manager user interface. For more information, see the "Cluster Update and Upgrade" section in the ClearPass Policy Manager User Guide. For information about known issues with cluster upgrades, please refer to the "Cluster Upgrade and Update" sections in the Release Notes.

Subtopics

Upgrade Paths and Version Considerations
Before You Upgrade
After You Upgrade: Restoring Log DB and Access Tracker Records
After You Install or Upgrade on ESXi Servers: Establishing NW Connectivity
After You Upgrade: Restoring Custom Service Parameter Configurations
After You Upgrade: Rollback Limitations

Upgrade Paths and Version Considerations

Direct upgrades to 6.11.0 are only supported from 6.12.x or 6.11.x or any hotfix applied to them, with some caveats as described in this section.

For all other versions, direct upgrades are not supported. The specific patch update and upgrade paths described below must be followed instead.

Before you proceed with any upgrade, you should always apply the latest available patch updates for your current release. For information on the patch update procedure, see Updating Within the Same Major Version .



NOTE

Before you upgrade, there are several limitations to be aware of regarding rollback behaviors after the upgrade. For details, see After You Upgrade: Rollback Limitations.

From 6.12.x and 6.11.x

Through the Software Updates Portal or through the Cluster Upgrade Portal — Direct upgrades to 6.11.0 are supported for all 6.12.x and 6.11.x versions when the upgrade is done through the Software Updates portal's **Import Upgrade Image** option, or through the Cluster Upgrade portal, or downloaded or installed through the web service, or via the CLI using the **system upgrade** command.

From 6.10.x

Direct upgrades from 6.10.x to 6.11.0 are not supported. You must first take a backup of the 6.10.x ClearPass instance and restore it on a fresh 6.11.x instance. After that is done, you can upgrade the instance from 6.11.x to 6.11.0.

From 6.9.x

Direct upgrades from 6.9.x to 6.11.0 are not supported. You must first take a backup of the 6.9.x ClearPass instance and restore it on a fresh 6.11.x instance. After that is done, you can upgrade the instance from 6.11.x to 6.11.0.

Before You Upgrade

Before you begin the upgrade process, please review the following important items:

- The stand-by-publisher value should be set to false during the upgrade process to avoid a false failover.
- Plan downtime accordingly. Upgrades can take longer (several hours) depending on the size of your configuration database. A large number of audit records (hundreds of thousands) due to Mobile Device Management (MDM) integration can significantly increase upgrade times. Refer to the sample times shown in Table 1 in Sample Times Required for Upgrade .
- Review the hypervisor disk requirements. These are described in Virtual Appliance Requirements of the "System Requirements for ClearPass " section.
- Any log settings that were modified prior to the upgrade are not retained and are reset to the default. The administrator should configure any custom log settings again after the upgrade.



NOTE

Gen9 C2000 and C3000 hardware appliances cannot be upgraded to ClearPass 6.14.0, as these appliances have already reached end of support (EOS/EOSL). If an upgrade is attempted on these appliances, the error message "ClearPass upgrade is not supported on Gen9 hardware (C2000/C3000)" is displayed.



NOTE

Log Database and Access Tracker records are not restored as part of the upgrade. If required, you can manually restore them after the upgrade. For more information, please review After You Upgrade: Restoring Log DB and Access Tracker Records .



NOTE

Before you upgrade, there are several limitations to be aware of regarding rollback behaviors after the upgrade. For details, see After You Upgrade: Rollback Limitations.

- Before initiating the upgrade process in ClearPass, we recommend you set the **Auto backup configuration options** to **Off** (if it was set to other values such as Config or Config|Session). The reason for disabling this setting is to avoid interference between the auto backup process and the migration process. To change this setting, go to **Administration > Server Manager > Server Configuration > Cluster Wide Parameters > Database > Auto backup configuration options** and set it to **Off**.
- If expired licenses are present prior to the upgrade, heavy CPU usage due to unstable Database Change Notice (DBCN) events might occur. To avoid this, perform the following steps before you upgrade:
  - Delete any expired evaluation or subscription licenses of any type (Entry, Access Upgrade, Access, Onboard, OnGuard).
  - Check for DBCN unstable errors in the **Event Viewer** before you perform the upgrade. If you see any such errors, contact TAC to resolve them before the upgrade.
- Users should be aware that custom skins that were installed offline are not migrated during an upgrade. If you have offline custom skins and are planning an upgrade, contact TAC before the upgrade. They can provide access to the offline custom skin files, which must be manually reinstalled after the upgrade.
- If you have a custom authentication source configured to use the session log database, additional steps are required after upgrade. You have such an authentication source configured if you have a source of type **Generic SQL DB** in **ClearPass Policy Manager > Configuration > Authentication > Sources** with server name **localhost** or **127.0.0.1** and with the database name **tipsLogDb**. In such cases, manually restoring the session log database is required after the upgrade completes (see After You Upgrade: Restoring Log DB and Access Tracker Records). Please contact Customer Support for configuration recommendations to move away from using the session log database as an authentication source.
- During an upgrade to ClearPass 6.14.0, any attributes in a custom change of authorization (CoA) dictionary will be removed, which can result in failure to export RADIUS services or Access Tracker events if they reference the custom dictionary. To retain attributes in custom CoA dictionaries after the 6.14.0 upgrade, first export the custom CoA dictionary before the upgrade and then re-import it after the upgrade.
- In ClearPass 6.11.0 and later versions, the maximum single cluster size is limited to 32 servers. This includes the publisher, standby publisher, subscribers, dedicated Insight server, and standby Insight servers.

Subtopics

Sample Times Required for Upgrade

Sample Times Required for Upgrade

To help you estimate how much time the upgrade might take, the tables in this section show representative numbers for upgrade times under test conditions. Remember that the figures here are only examples. The actual time required for your upgrade depends on several factors:

- Your hardware or virtual appliance model. In the case of virtual appliance (VA) installations, upgrade times vary significantly based on the IOPS performance of your VA infrastructure.
- The size of the configuration database to be migrated.
- For Insight nodes, the size of the Insight database.
- For subscriber nodes, the bandwidth and latency of the network link between the subscriber and the publisher.

The following table shows examples of upgrade times for standalone appliances. Test results are shown for hardware appliances and for virtual appliances.

Table 1. Standalone Appliances — Sample Total Times Required for Upgrade, Hardware and Virtual Appliances

Appliance Type	Total Backup Size	Config DB Size	Insight DB Size	Log DB Size	Total Upgrade Time
C1000 (HW)	221 MB	509 MB	176 MB	834 MB	150 minutes
	950 MB	466 MB	8276 MB	16 GB	212 minutes
	1.4 GB	455 MB	19 GB	13 GB	374 minutes
C1000V (VA)	221 MB	522 MB	175 MB	836 MB	19 minutes
	950 MB	466 MB	8277 MB	16 GB	24 minutes
	1.4 GB	463 MB	19 GB	13 GB	25 minutes
N1000	221 MB	522 MB	174 MB	836 MB	80 minutes

Appliance Type	Total Backup Size	Config DB Size	Insight DB Size	Log DB Size	Total Upgrade Time
(HW)	950 MB	466 MB	8274 MB	16 GB	87 minutes
	1.4 GB	453 MB	19 GB	13 GB	94 minutes
C2000V (VA)	221 MB	522 MB	175 MB	835 MB	26 minutes
	950 MB	466 MB	9353 MB	16 GB	44 minutes
	1.4 GB	463 MB	19 GB	13 GB	66 minutes
C2010 (HW)	221 MB	522 MB	176 MB	839 MB	38 minutes
	950 MB	466 MB	8279 MB	16 GB	40 minutes
	1.4 GB	197 MB	19 GB	13 GB	53 minutes
C2020 (HW)	221 MB	522 MB	176 MB	840 MB	51 minutes
	950 MB	466 MB	8275 MB	16 GB	59 minutes
	1.4 GB	197 MB	19 GB	13 GB	73 minutes
C3000V (VA)	221 MB	522 MB	176 MB	836 MB	45 minutes
	950 MB	459 MB	8.5 GB	16 GB	49 minutes
	1.4 GB	461 MB	19 GB	13 GB	73 minutes
N3000 (HW)	221 MB	522 MB	175 MB	842 MB	26 minutes
	950 MB	466 MB	8276 MB	16 GB	22 minutes
	1.4 GB	463 MB	19 GB	13 GB	27 minutes
C3010 (HW)	221 MB	522 MB	175 MB	840 MB	35 minutes
	950 MB	466 MB	9794 MB	16 GB	43 minutes
	1.4 GB	463 MB	19971 MB	13 GB	52 minutes

The following table shows examples of upgrade times for appliances in a cluster. Test results are shown for publishers and subscribers, and for various combinations of hardware appliances and virtual appliances.

Table 2. Cluster Appliances — Sample Total Times Required for Upgrade, Hardware and Virtual Appliances

Example #	Publisher or Subscriber?	Config DB Size	Insight DB Size	Appliance Type	Total Upgrade Time
1	Publisher (WAN)	374 MB	—	C2000V (VA)	22 mins
	Subscriber	374 MB	19 GB	C1000V (VA)	21 mins
	Subscriber	374 MB	19 GB	C2010 (HW)	40 mins
2	Publisher	527 MB	20 GB	N1000 (HW)	93 mins
	Subscriber	527 MB	20 GB	C2010 (HW)	43 mins
	Subscriber (WAN)	527 MB	—	C2000V (VA)	18 mins
3	Publisher	735 MB	56 GB	C2020 (HW)	123 min
	Subscriber 1	735 MB	56 GB	C3000 V(VA)	76 min
	Subscriber 2 (WAN)	735 MB	—	C2000V (VA)	77 min
4	Publisher	1141 MB	62 GB	N3001 (HW)	42 mins
	Subscriber 1	1141 MB	62 GB	C2000V (VA)	111 mins
	Subscriber 2 (WAN)	1141 MB	—	C3010 (HW)	23 mins

Azure VM Generation 1 to Generation 2 Conversion

This chapter describes the procedure for converting the VM boot mode from Legacy BIOS to UEFI by upgrading the Azure VM from Generation 1 (Gen1) to Generation 2 (Gen2) with the Trusted Launch security type.

> **NOTE**
> ClearPass Policy Manager 6.14.0 requires the VM to boot in UEFI mode. VMs currently running on Microsoft Azure were deployed as Gen1 VMs and therefore boot in BIOS mode.

> **NOTE**
> ClearPass Policy Manager can be upgraded to version 6.14.0 only after the VM is upgraded to Gen2 with Trusted Launch, including Secure Boot, vTPM, and boot integrity monitoring.

Subtopics

Compatibility
Prerequisites Before Starting the Conversion
Converting the VM Through the Azure Portal
Converting the VM Through the Azure CLI
Troubleshooting After the Conversion

Compatibility

The ClearPass appliance image for versions 6.11.x, 6.12.x, and 6.14.0 support both BIOS + UEFI boot modes. The OS disk already includes a GPT partition table, an EFI System Partition mounted at /boot/efi, and the Secure Boot components (`shim` and `grub2-efi-x64`) pre-installed. The appliance currently boots in Legacy BIOS mode because it was originally deployed as a Gen1 VM. Converting it to Gen2 with Trusted Launch enables it to boot into UEFI mode with Secure Boot enabled.

Gen1 Linux VMs created from supported Azure Marketplace distributions, such as RHEL, Canonical, and SUSE, already have the GPT disk layout and EFI System Partition required for Trusted Launch. Therefore, no guest OS volume changes are required before the upgrade.

Prerequisites Before Starting the Conversion

About this task

Before converting the Azure VM from Gen1 to Gen2, make sure that the following prerequisites are met.

- The VM size and OS version must support Trusted Launch, as well as custom images.
- If Azure Backup is enabled, the VM must use the Enhanced backup policy since the standard backup policy is not supported.
- Azure Site Recovery (ASR) must be disables for the conversion. It can be re-enable after conversion is completed.
- Verify that the OS disk is UEFI-ready. Connect to the Gen1 Linux VM through SSH and run the following commands:

Table 1.

Validation Step	Command	Result
Connect to the Gen1 Linux VM	`ssh <support-user>@<vm>`	Shell access
Identify boot device	`bootDevice=$(echo "/dev/$(sudo lsblk -no pkname $(sudo df /boot | awk 'NR==2 {print $1}'))")`	`/dev/sdb`
Check disk (partition) type	`sudo blkid $bootDevice -o value -s PTTYPE`	`gpt`
EFI System Partition present	`sudo fdisk -l $bootDevice | grep EFI | awk '{print $1}'`	`/dev/sdb1`
`/boot/efi` configured in fstab	`sudo grep -qs '/boot/efi' /etc/fstab && echo present || echo missing`	present

NOTE

Create a restore point or snapshot of the VM before the conversion to provide a rollback option.

Converting the VM Through the Azure Portal

About this task

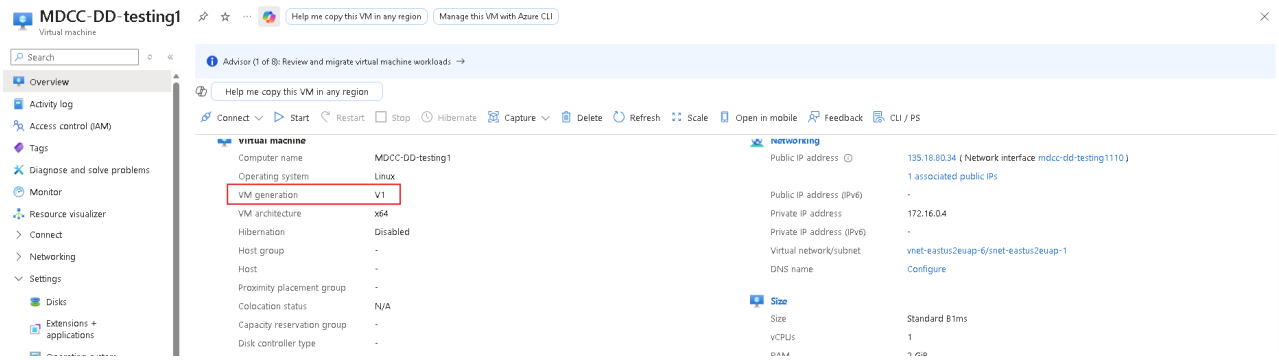
> **WARNING**
> Converting a Gen1 VM to Gen2 with Trusted Launch cannot be reversed to the original Gen1 configuration. Trusted Launch can be disabled and it converts the VM to Gen2 with Standard security, however it is a one-way operation and Trusted Launch cannot be re-enabled on the same VM.

Use the Azure portal to convert the VM from Gen1 to Gen2 and configure Trusted Launch with Secure Boot and vTPM.

Procedure

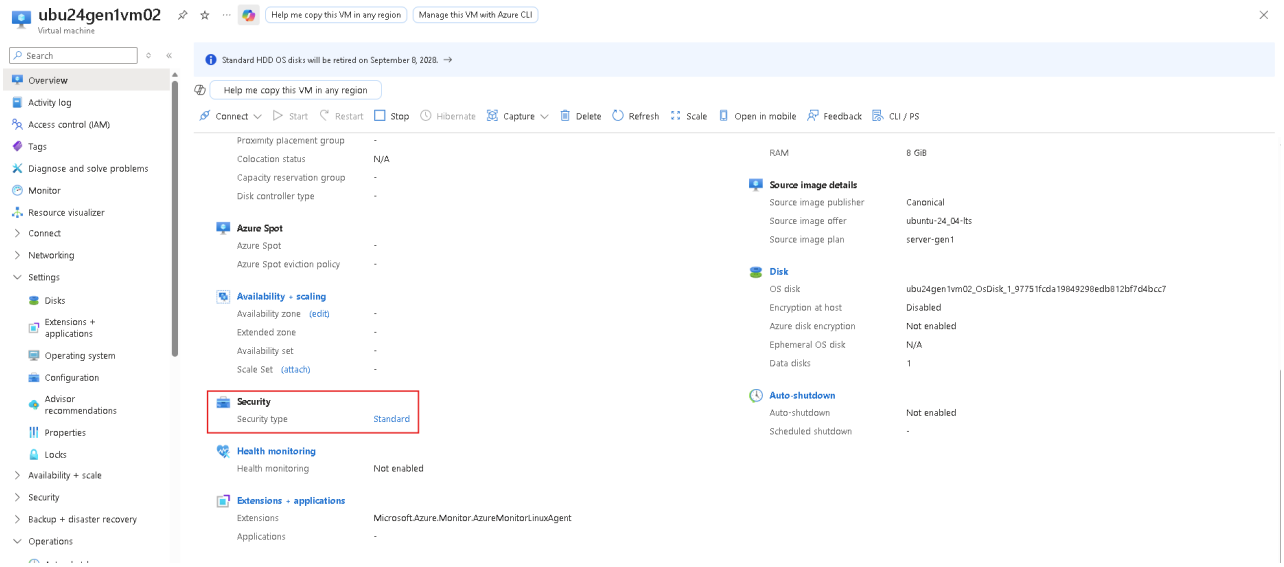
1. Complete the Guest OS volume validation mentioned in the prerequisites, then sign in to the Azure portal.
2. Open the VM and confirm the VM generation is V1 and the VM is Running.

Figure 1. Verifying the VM Generation



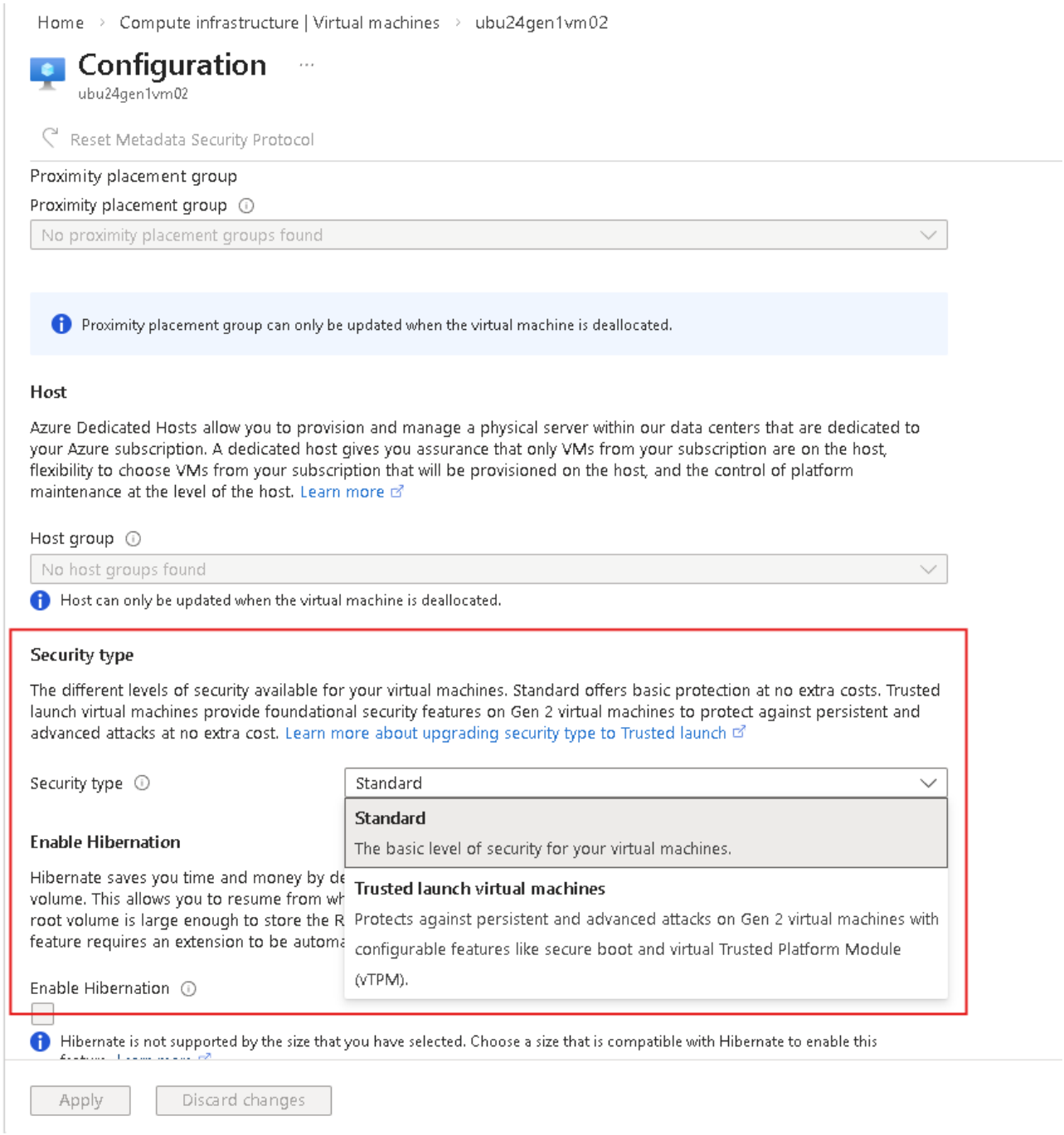
3. On the **Overview** page, under **Security type**, select **Standard**. The VM Configuration page opens.

Figure 2. Selecting the VM Security Type



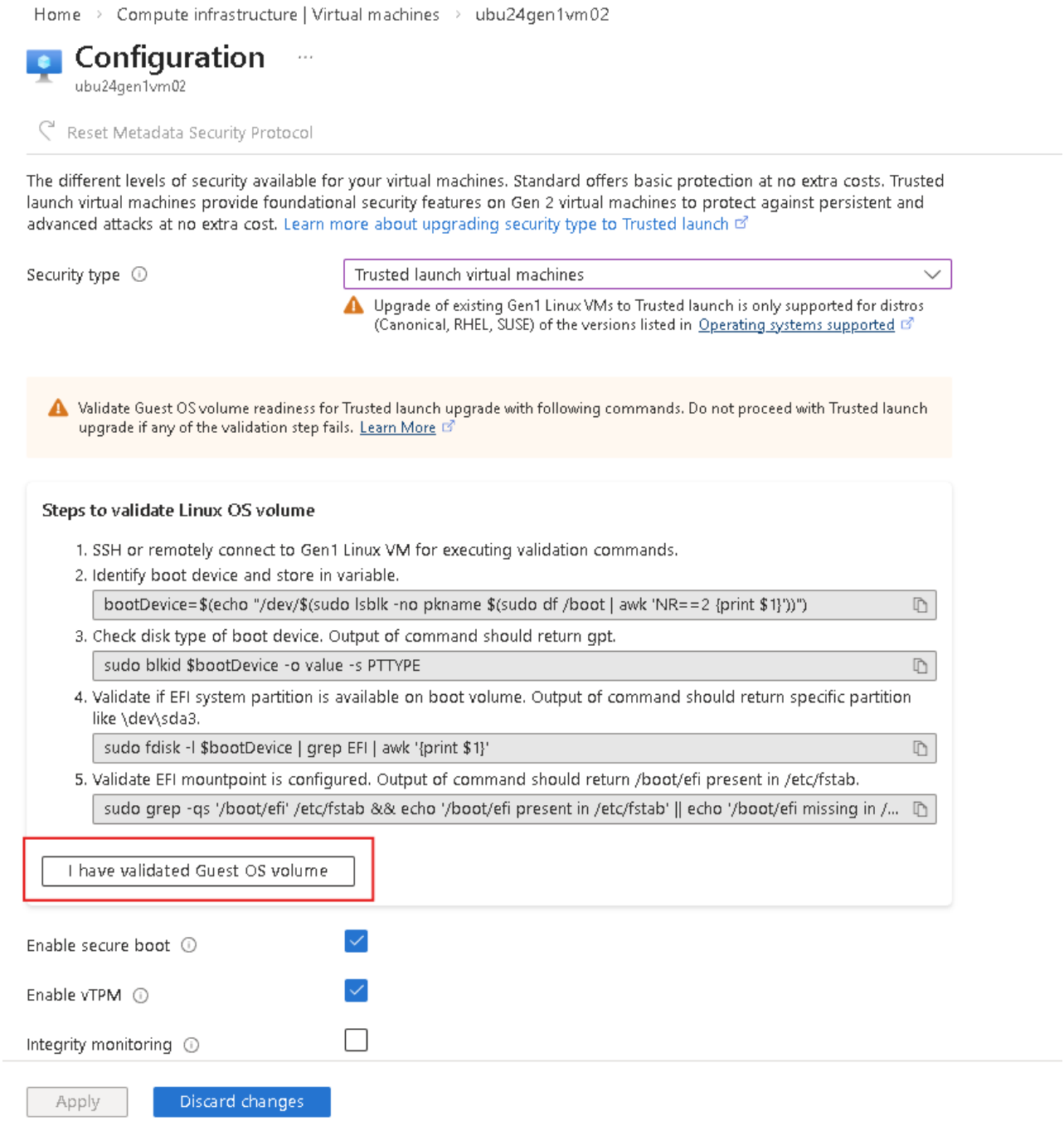
4. On the **Configuration** page, under **Security type**, open the **Security type** drop-down menu.

Figure 3. Opening the Security Type Menu



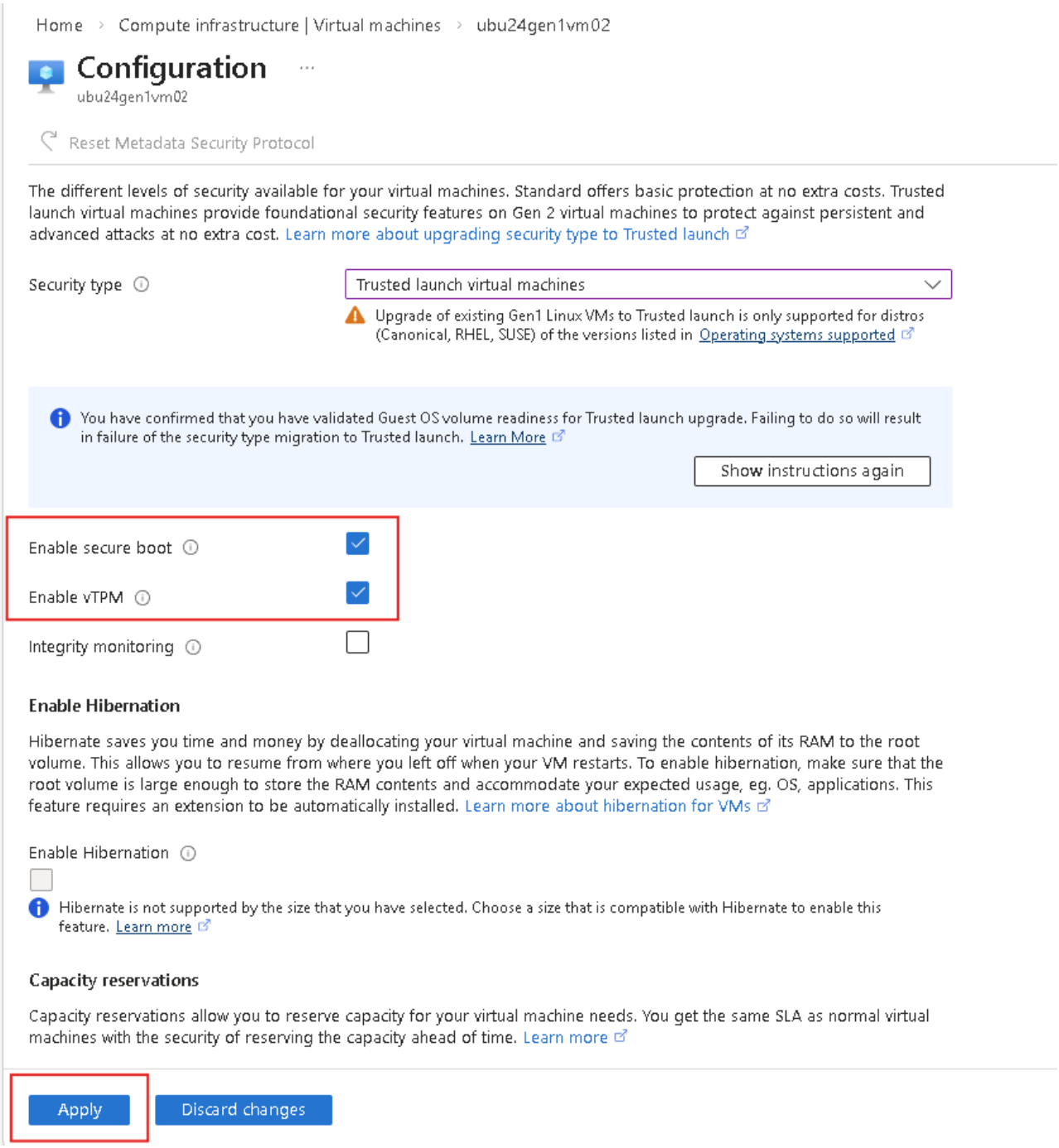
5. Select **Trusted launch virtual machines**, then acknowledge that the Guest OS volume was updated and validated.

Figure 4. Validating the Guest OS Volume



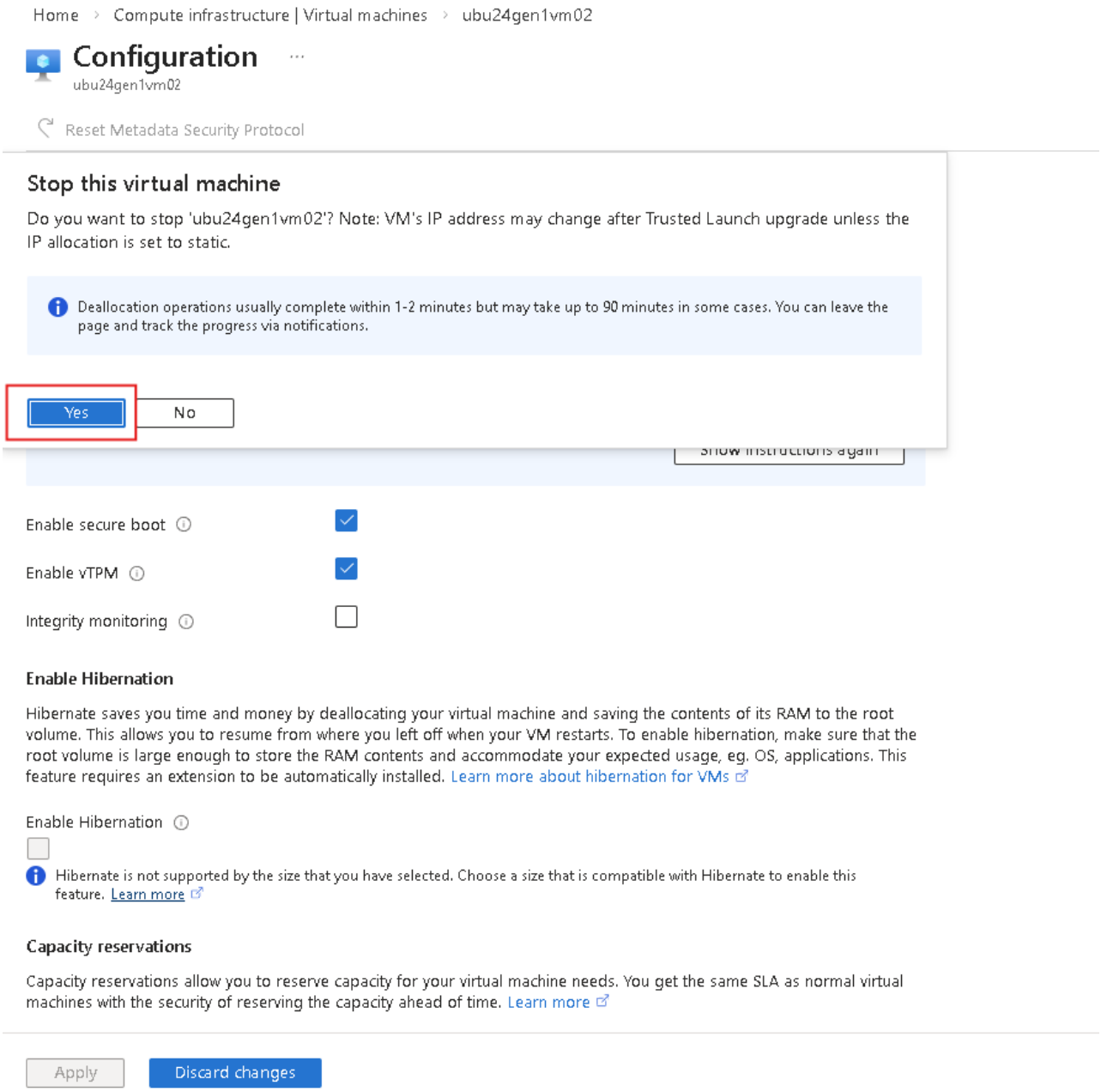
6. Enable Secure Boot and vTPM, then select **Save**.

Figure 5. Enabling Secure Boot and vTPM



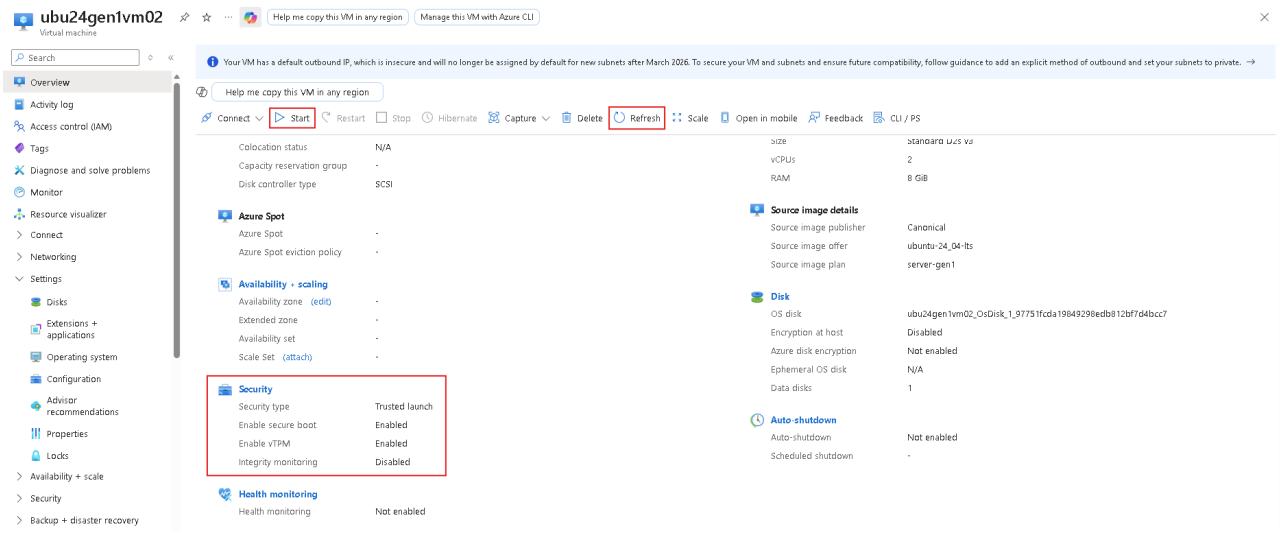
7. The VM must be deallocated to complete the upgrade. Confirm the prompt by selecting **Yes**.

Figure 6. Stopping the VM



8. After the update is complete, close the **Configuration** page and confirm the **Security type** on the **Overview** page.

Figure 7. Confirming the Trusted Launch configuration



9. Start the upgraded Trusted launch VM and verify SSH sign-in (Linux).what

Results

The following example shows the expected values for a successful conversion. The example includes ClearPass 6.11.1 and 6.12.0 nodes from the same hybrid BIOS and UEFI appliance image family as ClearPass 6.14.0.

Table 1.

VM (Role)	Generation	Security Type	Secure Boot	vTPM	In-OS Firmware
ClearPass 6.11.1 — node 1	V2	TrustedLaunch	Enabled	Enabled	UEFI
ClearPass 6.11.1 — node 2	V2	TrustedLaunch	Enabled	Enabled	UEFI
ClearPass 6.11.1 — node 3	V2	TrustedLaunch	Enabled	Enabled	UEFI
ClearPass 6.12.0 — node 1	V2	TrustedLaunch	Enabled	Enabled	UEFI
ClearPass 6.12.0 — node 2	V2	TrustedLaunch	Enabled	Enabled	UEFI
ClearPass 6.12.0 — node 3	V2	TrustedLaunch	Enabled	Enabled	UEFI

Converting the VM Through the Azure CLI

About this task

> **WARNING**
> Converting a Gen1 VM to Gen2 with Trusted Launch cannot be reversed to the original Gen1 configuration. Trusted Launch can be disabled and it converts the VM to Gen2 with Standard security, however it is a one-way operation and Trusted Launch cannot be re-enabled on the same VM.

Use the Azure CLI to convert the VM from Gen1 to Gen2 and configure Trusted Launch with Secure Boot and vTPM.

> **NOTE**
> This procedure is specially useful when performing the conversion in multiple nodes.

Procedure

1. Sign in and select the subscription.

```
az login
az account set --subscription <subscription-id>
```

2. Deallocate/power off the VM.

```
az vm deallocate --resource-group <rg> --name <vm>
```

3. Enable Trusted Launch (Gen2 + Secure Boot + vTPM).

```
az vm update --resource-group <rg> --name <vm> \
    --security-type TrustedLaunch \
    --enable-secure-boot true --enable-vtpm true

"securityProfile": {
    "securityType": "TrustedLaunch",
    "uefiSettings": { "secureBootEnabled": true, "vTpmEnabled": true }
}
```

4. Start the VM.

```
az vm start --resource-group <rg> --name <vm>
```

Results

The following example shows the expected values for a successful conversion when validating from inside a node.

```
firmware   : UEFI            (/sys/firmware/efi present)
Secure Boot: enabled         (mokutil --sb-state)
```

Troubleshooting After the Conversion

About this task

If a subscriber does not return within the expected time after the reboot started in CUT tool, verify that it has stalled during early boot.

Procedure

1. In the Azure portal, open the VM and go to **Serial** console and confirm that boot diagnostics are enabled, then review the live boot output.
2. If the output is frozen or stops at an early-kernel stage with repeated timestamps and a halt at `Bringing up secondary CPUs ..`, that means that the node has stalled during first boot.
3. Restart the VM from the Azure portal, go to **Overview > Restart**. On the next boot the node comes up cleanly, rejoins the cluster, and synchronizes automatically.

Example

The following is a sample serial-console boot log that shows a stalled node:

```
[    0.030573] Kernel command line:
BOOT_IMAGE=/vmlinuz-5.14.0-570.23.1.el9_6.x86_64 root=UUID=1a25e030-
e55a-47f1-93c6-031673659497 ro panic=30 console=ttyS0,9600n8 console=tty0
fips=0 boot=UUID=52a7b05f-cee3-4ae6-bc9f-2e7b238c7e49 clocksource=acpi_pm
clocksource_failover=acpi_pm pcie_aspm=off selinux=0 biosdevname=0
net.ifnames=1 crashkernel=auto rhgb LANG=en_US.UTF-8 edd=off
transparent_hugepage=never audit=1 pti=on audit_backlog_limit=8192
page_poison=1 vsyscall=none slub_debug=P
[    0.030654] fips mode: disabled
[    0.030713] PCIe ASPM is disabled
[    0.030844] audit: enabled (after initialization)
[    0.030871] audit: audit_backlog_limit: 8192
[    0.030945] Unknown kernel command line parameters
"rhgb BOOT_IMAGE=/vmlinuz-5.14.0-570.23.1.el9_6.x86_64 boot=UUID=52a7b05f-
cee3-4ae6-bc9f-2e7b238c7e49 clocksource_failover=acpi_pm biosdevname=0
LANG=en_US.UTF-8 edd=off", will be passed to user space.
[    0.030968] random: crng init done
[    0.032237] Dentry cache hash table entries: 1048576 (order: 11, 8388608
```

```
bytes, linear)
[    0.032874] Inode-cache hash table entries: 524288 (order: 10, 4194304
bytes, linear)
[    0.032962] Fallback order for Node 0: 0
[    0.032966] Built 1 zonelists, mobility grouping on.  Total pages:
2048728
[    0.032967] Policy zone: Normal
[    0.033339] mem auto-init: stack:off, heap alloc:off, heap free:off
[    0.033345] software IO TLB: area num 2.
[    0.270947] Memory: 985712K/8331928K available (16384K kernel code,
5766K rwdata, 13628K rodata, 4048K init, 7384K bss, 337548K reserved, 0K
cma-reserved)
[    0.270953] ******************************************************
[    0.270954] **   NOTICE NOTICE NOTICE NOTICE NOTICE NOTICE NOTICE   **
[    0.270955] **                                                      **
[    0.270955] ** This system shows unhashed kernel memory addresses   **
[    0.270956] ** via the console, logs, and other interfaces. This    **
[    0.270956] ** might reduce the security of your system.            **
[    0.270956] **                                                      **
[    0.270957] ** If you see this message and you are not debugging    **
[    0.270957] ** the kernel, report this immediately to your system   **
[    0.270958] ** administrator!                                       **
[    0.270958] **                                                      **
[    0.270959] **   NOTICE NOTICE NOTICE NOTICE NOTICE NOTICE NOTICE   **
[    0.270959] ******************************************************
[    0.271163] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=2, Nodes=1
[    0.271177] Kernel/User page tables isolation: enabled
[    0.271222] ftrace: allocating 49058 entries in 192 pages
[    0.282265] ftrace: allocated 192 pages with 2 groups
[    0.283253] Dynamic Preempt: voluntary
[    0.283311] rcu: Preemptible hierarchical RCU implementation.
[    0.283312] rcu:     RCU event tracing is enabled.
[    0.283313] rcu:     RCU restricting CPUs from NR_CPUS=8192 to
nr_cpu_ids=2.
[    0.283315]  Trampoline variant of Tasks RCU enabled.
[    0.283315]  Rude variant of Tasks RCU enabled.
[    0.283316]  Tracing variant of Tasks RCU enabled.
[    0.283317] rcu: RCU calculated value of scheduler-enlistment delay is
100 jiffies.
[    0.283318] rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=2
[    0.283328] RCU Tasks: Setting shift to 1 and lim to 1
rcu_task_cb_adjust=1.
[    0.283330] RCU Tasks Rude: Setting shift to 1 and lim to 1
```

```
rcu_task_cb_adjust=1.
[    0.283331] RCU Tasks Trace: Setting shift to 1 and lim to 1
rcu_task_cb_adjust=1.
[    0.289061] Using NULL legacy PIC
[    0.289062] NR_IRQS: 524544, nr_irqs: 440, preallocated irqs: 0
[    0.289117] rcu: srcu_init: Setting srcu_struct sizes based on
contention.
[    0.289660] kfence: initialized - using 2097152 bytes for 255 objects at
0xffff8ff3f6e00000-0xffff8ff3f7000000
[    0.289702] Console: colour dummy device 80x25
[    0.289705] printk: legacy console [tty0] enabled
[    0.290125] printk: console [ttyS0] enabled
[    1.042363] ACPI: Core revision 20230331
[    1.044676] Failed to register legacy timer interrupt
[    1.047541] APIC: Switch to symmetric I/O mode setup
[    1.054942] x2apic enabled
[    1.061466] APIC: Switched APIC routing to: physical x2apic
[    1.064607] Hyper-V: Host Build 10.0.26102.1438-1-0
[    1.067554] Hyper-V: enabling crash_kexec_post_notifiers
[    1.070985] Hyper-V: Using IPI hypercalls
[    1.073087] APIC: send_IPI() replaced with hv_send_ipi()
[    1.075980] APIC: send_IPI_mask() replaced with hv_send_ipi_mask()
[    1.079306] APIC: send_IPI_mask_allbutself() replaced with
hv_send_ipi_mask_allbutself()
[    1.083857] APIC: send_IPI_allbutself() replaced with
hv_send_ipi_allbutself()
[    1.087820] APIC: send_IPI_all() replaced with hv_send_ipi_all()
[    1.090775] APIC: send_IPI_self() replaced with hv_send_ipi_self()
[    1.093822] clocksource: tsc-early: mask: 0xffffffffffffffff max_cycles:
0x1e3383a4a97, max_idle_ns: 440795274565 ns
[    1.099380] Calibrating delay loop (skipped), value calculated using
timer frequency.. 4190.42 BogoMIPS (lpj=2095210)
[    1.100375] Last level iTLB entries: 4KB 64, 2MB 8, 4MB 8
[    1.100375] Last level dTLB entries: 4KB 64, 2MB 0, 4MB 0, 1GB 4
[    1.100375] Spectre V1 : Mitigation: usercopy/swapgs barriers and __user
pointer sanitization
[    1.100375] Spectre V2 : Mitigation: Retpolines
[    1.100375] Spectre V2 : Spectre v2 / SpectreRSB mitigation: Filling RSB
on context switch
[    1.100375] Spectre V2 : Spectre v2 / SpectreRSB : Filling RSB on VMEXIT
[    1.100375] RETBleed: WARNING: Spectre v2 mitigation leaves CPU
vulnerable to RETBleed attacks, data leaks possible!
[    1.100375] RETBleed: Vulnerable
```

```
[    1.100375] Speculative Store Bypass: Vulnerable
[    1.100375] MDS: Vulnerable: Clear CPU buffers attempted, no microcode
[    1.100375] TAA: Vulnerable: Clear CPU buffers attempted, no microcode
[    1.100375] MMIO Stale Data: Vulnerable: Clear CPU buffers attempted, no
microcode
[    1.100375] x86/fpu: Supporting XSAVE feature 0x001: 'x87 floating point
registers'
[    1.100375] x86/fpu: Supporting XSAVE feature 0x002: 'SSE registers'
[    1.100375] x86/fpu: Supporting XSAVE feature 0x004: 'AVX registers'
[    1.100375] x86/fpu: Supporting XSAVE feature 0x020: 'AVX-512 opmask'
[    1.100375] x86/fpu: Supporting XSAVE feature 0x040: 'AVX-512 Hi256'
[    1.100375] x86/fpu: Supporting XSAVE feature 0x080: 'AVX-512 ZMM_Hi256'
[    1.100375] x86/fpu: xstate_offset[2]:  576, xstate_sizes[2]:  256
[    1.100375] x86/fpu: xstate_offset[5]:  832, xstate_sizes[5]:   64
[    1.100375] x86/fpu: xstate_offset[6]:  896, xstate_sizes[6]:  512
[    1.100375] x86/fpu: xstate_offset[7]: 1408, xstate_sizes[7]: 1024
[    1.100375] x86/fpu: Enabled xstate features 0xe7, context size is 2432
bytes, using 'compacted' format.
[    1.100375] Freeing SMP alternatives memory: 40K
[    1.100375] pid_max: default: 32768 minimum: 301
[    1.100375] LSM: initializing
lsm=lockdown,capability,landlock,yama,integrity,bpf
[    1.100375] landlock: Up and running.
[    1.100375] Yama: becoming mindful.
[    1.100375] LSM support for eBPF active
[    1.100375] Mount-cache hash table entries: 16384 (order: 5, 131072
bytes, linear)
[    1.100375] Mountpoint-cache hash table entries: 16384 (order: 5, 131072
bytes, linear)
[    1.100375] smpboot: CPU0: Intel(R) Xeon(R) Platinum 8171M CPU @ 2.60GHz
(family: 0x6, model: 0x55, stepping: 0x4)
[    1.100675] Performance Events: unsupported p6 CPU model 85 no PMU
driver, software events only.
[    1.101418] signal: max sigframe size: 3632
[    1.102480] rcu: Hierarchical SRCU implementation.
[    1.102483] rcu:     Max phase no-delay instances is 400.
[    1.104849] NMI watchdog: Perf NMI watchdog permanently disabled
[    1.105487] smp: Bringing up secondary CPUs ..
```

After You Upgrade: Restoring Log DB and Access Tracker Records

Users should be aware that it is recommended to only back up the ClearPass Policy Manager configuration data that will then be used to restore configurations in 6.11.0. If required, you can manually restore Log Database records after the upgrade through either the application or the CLI.

Restoring the Log DB Through the User Interface

About this task

To restore the Log DB after an upgrade through the UI, restore from the auto-generated **upgrade-backup.tar.gz** file (available at **Administration > Server Manager > Local Shared Folders**).

The restoration process could take several hours, depending on the size of your session log database. All services are accessible and will handle requests during the restoration, but there will be a performance impact while the restoration is in progress. We recommend that you perform this operation during a planned change window.

The restoration process will continue in the background even if the UI is closed or the session times out. A “Restore complete” event is logged in the Event Viewer when the restoration is complete.

This process needs to be repeated on each server in the cluster that should retain the session log database.

Procedure

1. Go to **Administration > Server Manager > Server Configuration** and click **Restore** for the server.
2. In the **Restore Policy Manager Database** window, select the **File is on server** option, and select the **upgrade-backup.tar.gz** file.
3. Also select the following options:
   - **Restore CPPM session log data (if it exists on the backup)**
   - **Ignore version mismatch and attempt data migration**

- **Do not back up the existing databases before this operation**

> **NOTE**
>
> Do not use the **Password** field in this form unless your backup file was password-encrypted. This field is only to be used for that file-encryption password, if it exists. Otherwise, leave the Password field empty. If you enter an admin or cluster password in this form, the restoration will fail.

4. Uncheck the **Restore CPPM configuration data** option.
5. Click **Start**.

Restoring the Log DB Through the CLI

To restore the Log Database after the upgrade process is complete, use the **restore** command. Go to **Administration > Server Manager > Local Shared Folders** and download the **upgrade-backup.tar.gz** file. Host the file at an **scp** or `http` location accessible from the ClearPass appliance and execute the command restore *<location/upgrade-backup.tar.gz>* -l –i –b.

The restoration process could take several hours depending on the size of your session log database. All services are accessible and handling requests during the restoration, but there will be a performance impact while the restoration is in progress. We recommend that you perform this operation during a planned change window.

> **NOTE**
>
> The restoration process will abort if the CLI session is closed or times out. We recommend that you initiate the restoration from the user interface, especially if you have a large number of Access Tracker and Accounting records.

This process needs to be repeated on each server in the cluster that should retain the session log database.

The `restore` command syntax is as follows:

```
Usage:
   restore user@hostname:/<backup-filename> [-l] [-i] [-b] [-c] [-r] [-n|-
N] [-s]
   restore http://hostname/<backup-filename>[-l] [-i] [-b] [-c] [-e] [-n|-
N] [-s]
   restore <backup-filename> [-l] [-i] [-b] [-c] [-r] [-n|-N] [-s]
   -b -- do not backup current config before restore
   -c -- restore CPPM configuration data
```

```
   -l -- restore CPPM session log data as well if it exists in the backup
   -r -- restore Insight data as well if it exists in the backup
   -i -- ignore version mismatch and attempt data migration
   -n -- retain local node config like certificates etc. after restore
(default)
   -N -- do not retain local node config after restore
   -s -- restore cluster server/node entries from backup.
    The node entries will be in disabled state on restore
```

After You Install or Upgrade on ESXi Servers: Establishing NW Connectivity

About this task

For new ClearPass installations on a VMware ESXi server, or if you are upgrading ClearPass from 6.12.x or 6.11.x to 6.11.0 on a VMware ESXi server, **and only** if the MAC address of **Network adapter1** is higher than that of **Network adapter2**, additional steps are required after the upgrade.

After installing or upgrading, follow the steps below in order for ClearPass to have network connectivity:

Procedure

1. After you install or upgrade to ClearPass 6.11.0, log in to the console as appadmin and use the CLI command **<system shutdown>** to shut down the ClearPass server. This step must be done only through the console.
2. After the command is executed, wait for the virtual appliance to shut down completely.
3. Edit the ClearPass virtual appliance settings in the vSphere client and remove the two Ethernet adapters that are named **Network adapter1** and **Network adapter2**.
4. Add two new network adapters with the names **Network adapter1** and **Network adapter2** and of type **Ethernet Adapter**:
   - **Network adapter1** should be the management port connected to SwitchManagement. This network adapter should have the lowest MAC address.

- **Network adapter2** should be the data port connected to SwitchData. This network adapter should have a higher MAC address than **Network adapter1**.

> **NOTE**
>
> After you add the two network adapters, verify that **Network adapter 1** has a lower MAC address than **Network adapter 2**. Interfaces map based on the lowest MAC address, such that the lowest MAC address becomes the management port and the second lowest MAC address becomes the data port.

5. Save the new settings and start the ClearPass virtual appliance.
6. Log in to the ClearPass console using the appadmin account, and then run the following CLI command to refresh the network settings:

```
system refresh-network
```

7. After the refresh command is executed, reboot the ClearPass virtual appliance to establish network connectivity.

> **NOTE**
>
> When creating a database backup of a lower version of ClearPass to be restored on ClearPass 6.11, best practice is to create separate backups for the configuration (tipsdb) and Insight (insightdb) databases. Backup files can be very large, and creating separate backups ensures that they will be restored correctly without overloading the system and blocking the restore operation.

After You Upgrade: Restoring Custom Service Parameter Configurations

Due to significant changes that occur during a major release, some service parameters are not migrated during upgrade. If the default values for any of these parameters were replaced with custom values in the earlier ClearPass version, then when you upgrade to 6.11.0 those custom values will be overwritten by the default values. After the upgrade, you must manually recreate the custom service parameter configurations.

The service parameters shown below are not migrated:

- For the **System Monitor Service**:
  - **1 Min CPU load average Threshold**
  - **5 Min CPU load average Threshold**
  - **15 Min CPU load average Threshold**
- For the **Policy Server Service**:
  - **Number of request processing threads**
  - **HTTP Thread Pool Size**
  - **Authentication Thread Pool Size**
- For the **RADIUS Server Service**:
  - **TLS Session Cache Limit**
  - **Maximum Number of Threads**
  - **Number of Initial Threads**
- For the **ClearPass System Services**:
  - **Maximum Requests**
  - **Maximum Clients**

After You Upgrade: Rollback Limitations

Before upgrading to ClearPass 6.11.0, be aware of the following limitations:

- After upgrading to 6.11.0 from 6.11.14, only reverting to 6.11.14 is supported. Rolling back to earlier versions of 6.11.x (lower than 6.11.14) is not supported.
- After upgrading to 6.11.0 from 6.11.14, if the system is reverted to 6.11.14 or is rolled back to a future 6.11.x version later than 6.11.14 (for example, 6.11.15 or 6.11.16), then moving forward to 6.11.0 again would require upgrading to the image again.
- After upgrading to 6.11.0 from any 6.12.x release, if the system is rolled back or updated to any 6.12.x version, then moving forward to 6.11.0 again would require upgrading to the 6.11.0 image again.
- After upgrading to 6.11.0 from 6.11.x or 6.12.x, switching between images will only work if the respective image being moved to is made active through the system

boot-image -a <partition number> CLI command. Secure boot would fail and the image would fail to boot if the image is not made active through the CLI command.

Updating Within the Same Major Version

This section provides instructions for updating your ClearPass appliance. An update is the process of applying a minor patch release within the same major version—for example, from 6.11.15 to 6.11.16.

- Updates are available from the **Software Updates** portal in ClearPass Policy Manager. This section describes how to install a patch update either through the **Software Updates** portal, as an offline update, or through the **Cluster Update** page.
- Patch updates and preparation patches are also available for download on the **HPE Networking Support Portal**.
- To update a cluster to 6.11.16, we recommend using the **Cluster Update** page in the Policy Manager user interface. For more information, see the "Cluster Update and Upgrade" section in the ClearPass Policy Manager User Guide. For information about known issues with cluster updates, please refer to the “Installation, Upgrade, and Update” sections in the Release Notes.
- During an update, the log database is retained. No extra steps are needed to retain the session log history during an update.

> **NOTE**
>
> The stand-by-publisher value should be set to false during the patch update process to avoid a false failover.

Subtopics

Installation Instructions Through the Software Updates Portal
Installation Instructions for an Offline Update
Installation Instructions Through the Cluster Update Page
After You Update: Performing a Patch Rollback

Installation Instructions Through the Software Updates Portal

About this task

> **NOTE**
> This method may still be used to manually update appliances in a cluster, beginning with the publisher and then each subscriber; however, we recommend using the **Cluster Update** page to automate the process.

If HTTPS access is allowed to clearpass.arubanetworks.com, ClearPass appliances will show the latest patches on the **Software Updates** portal:

Procedure

1. In ClearPass Policy Manager, go to **Administration > Agents and Software Updates > Software Updates**.
2. In the **Firmware and Patch Updates** area, find the latest patch and click the **Download** button in its row.
3. After the patch is downloaded, click **Install**.
4. When the installation is complete, if the status on the **Software Updates** portal is shown as **Needs Restart**, click the **Reboot** button to restart ClearPass. After the restart, the status for the patch is shown as **Installed**.

Installation Instructions for an Offline Update

About this task

If you do not have HTTPS access to clearpass.arubanetworks.com and you need to do an offline update, you may download the signed patch from the HPE Networking Support Portal, upload it to the ClearPass appliance, and then install it through the user interface:

Procedure

1. Download the appropriate patch update from the HPE Networking Support Portal (**networkingsupport.hpe.com/**).
2. Open ClearPass Policy Manager and go to **Administration > Agents and Software Updates > Software Updates**.
3. At the bottom of the **Firmware and Patch Updates** area, click **Import Updates**.
4. Browse to the downloaded patch file and then click **Import**.

5. When the import is complete, click **Install**.
6. When the installation is complete, if the status on the **Software Updates** portal is shown as **Needs Restart**, click the **Reboot** button to restart ClearPass. After the restart, the status for the patch is shown as **Installed**.

Installation Instructions Through the Cluster Update Page

About this task

The **Cluster Update** page automates the process of updating a cluster. The publisher is automatically updated first before any selected subscribers. In large cluster deployments (greater than 6) we recommend updating the subscribers in batches of no more than five at a time.

To update the cluster:

Procedure

1. In ClearPass Policy Manager, go to **Administration > Support > Agents and Software Updates**.
2. Download or import the patch you wish to deploy, and then click the **Cluster Update** link.
3. In the **Update Info** area, select the desired patch from the **Update Image Name** drop-down list.
4. Click the **Start Update** link. The **Start Cluster Update** window opens.
5. Select the cluster subscribers to be updated, and then click **Update**.

   For more information about the **Cluster Update** page, see the "Cluster Update and Upgrade" section in the ClearPass Policy Manager User Guide. For information about known issues with cluster updates, please refer to the “Cluster Upgrade and Update” sections in the Release Notes or contact TAC for technical assistance.

After You Update: Performing a Patch Rollback

After a cumulative patch is applied, the CLI command **system patch-rollback** allows administrators to revert to the last version that was installed on the system. For example:

- If a system was at 6.11.6 and cumulative update 6.11.7 is applied, it can be reverted to 6.11.6 through the **system patch-rollback** command.
- If a system was at 6.11.4 and cumulative update 6.11.7 is applied, it can be reverted to 6.11.4 because that was the last installed version. It cannot be reverted to other versions such as 6.11.5 or 6.11.3.

  This command can be used if a problem is found after the patch update, such as an issue identified in production that was not identified during testing, resulting in a degradation of capabilities.

  Before using this command to revert to the last installed 6.11.x version, you must first download the **ClearPass 6.14.0 Rollback RPM Package** from the **Software Updates** portal and install it.

  As part of the patch rollback feature, users should be aware of the following:

- As best practice, users should always take a data backup before an update.
- This command cannot be used after an upgrade to revert to an earlier major version (for example, 6.12.x).
- Rolling back hotfix patches is not supported. ClearPass only supports rollbacks of regular cumulative patch versions.
- In a cluster where the / partition size exceeds 50% usage, perform a cleanup before the rollback operation.
- When this command is used in a cluster, it must be done to all the appliances in the cluster. The publisher must be rolled back first, followed by each subscriber individually. The rollback must be completed on all appliances in the cluster within 24 hours after the publisher rollback is initiated in order to maintain the cluster status.
- Any custom skins that are installed in the current version are retained after the rollback to the earlier version.
- System rollback events are logged in the **Event Viewer**.

Virtual Appliance Installation: HPE-VME

This section describes procedures for installing and upgrading ClearPass on an HPE Morpheus VM Essentials (HPE-VME) virtual appliance.

Subtopics

HPE-VME Hypervisor Installation Process Overview
Meeting HPE-VME Server Specifications
Installing ClearPass on an HPE-VME Virtual Appliance

HPE-VME Hypervisor Installation Process Overview

About this task

The process of installing ClearPass on an HPE Morpheus VM Essentials (HPE-VME) virtual appliance (VA) is done in four stages:

Procedure

1. Create a Service Plan in the HPE-VME console.
2. Upload the QCOW2 disk software image to VME as a Virtual Image.
3. Create the VM and provision the ClearPass instance.
4. Configure the network adapter.

   Instructions for these procedures are provided in Installing Policy Manager on an ESXi Virtual Appliance .

> **NOTE**
>
> Review the Release Notes for the current release before you install ClearPass.

> **NOTE**
>
> Cloning a virtual appliance to facilitate a ClearPass deployment is not recommended or supported.

Meeting HPE-VME Server Specifications

Before you perform any installations, upgrades, or updates, ensure that your system meets all virtual appliance requirements. Please refer to System Requirements for ClearPass 6.14 and "HPE Morpheus VM Essentials (HPE-VME) Requirements". Carefully

review all virtual appliance (VA) requirements, including functional IOP ratings, and verify that your system meets these requirements.

Prerequisites:

- VME must be installed and its management UI must be up and reachable with a valid VME configuration before you begin deploying ClearPass.
- The VME infrastructure must be configured correctly. Ensure that the Group, Cloud, Cluster, and Network configurations are all in place before you start deploying ClearPass.

An HPE-VME version can be morphed to a larger version by using the morph-vm command. For more information, see Morphing a VME Version in this Installation Guide, and the Command Line Interface > System Commands > system morph-vm section in the ClearPass Policy Manager User Guide.

Installing ClearPass on an HPE-VME Virtual Appliance

After you have verified the requirements for deploying HPE Networking ClearPass on an HPE-VME virtual appliance, this section provides complete information on installing, configuring, and morphing the VME instance.

Subtopics

Service Plan Creation
Image Upload
VM Creation
Verifying the Network Adapter MAC Address Order
Morphing a VME Version

Service Plan Creation

About this task

About this task

A Service Plan defines the compute resources (CPU, memory, and storage) that are assigned to a ClearPass virtual appliance when it is deployed on VME. Create a Service Plan before provisioning the ClearPass instance.

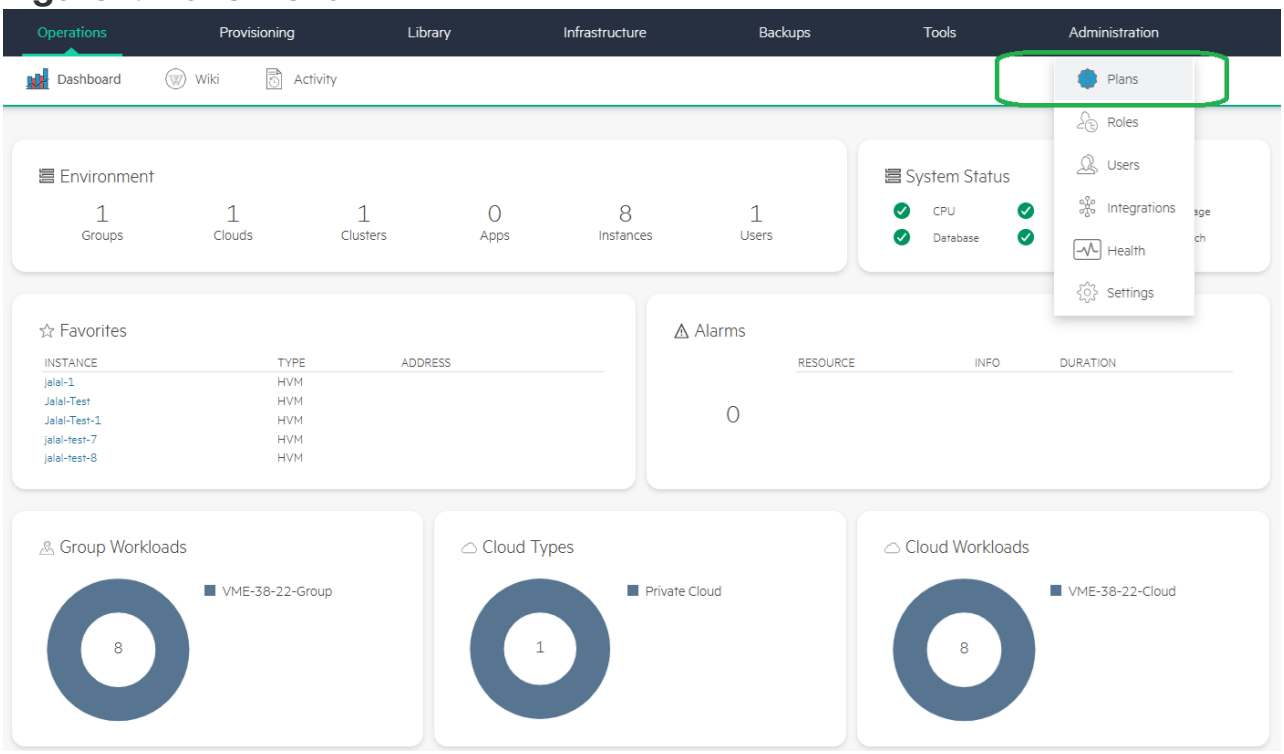
> **NOTE**
> The Custom options (Custom Memory and Custom Cores) must be enabled. These options are mandatory to allow the appliance to be morphed to a higher model later. If they are not enabled, morphing to a larger virtual appliance model will not be possible.

Procedure:

Procedure

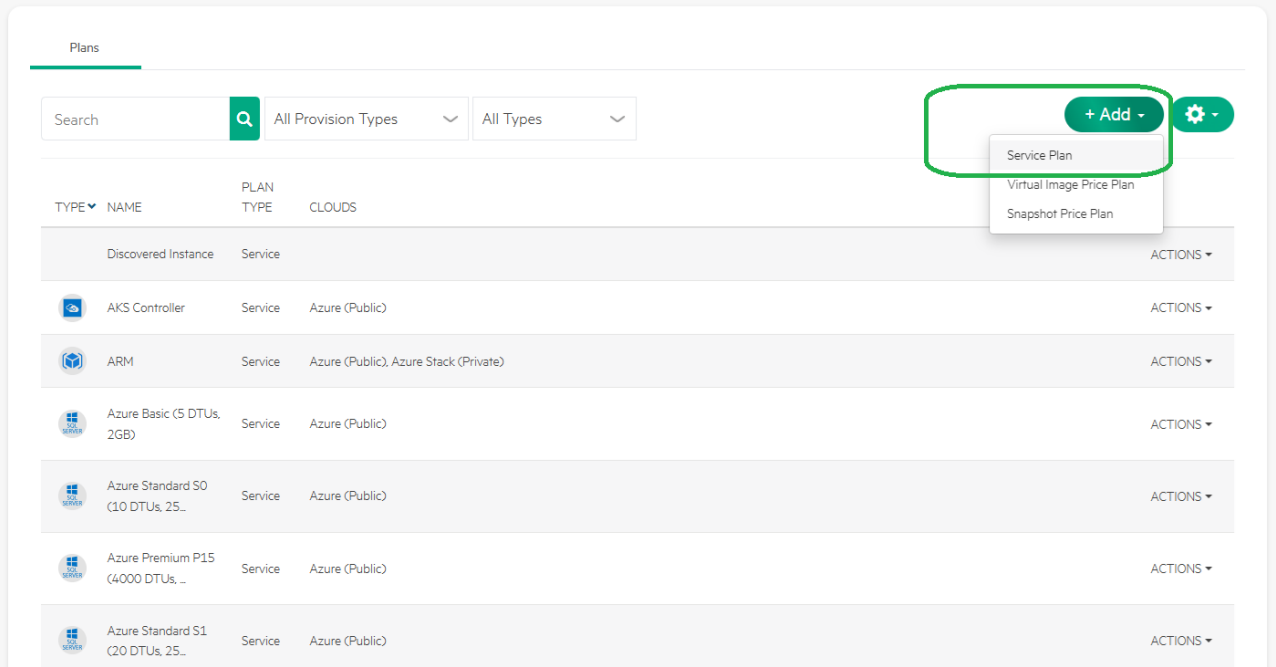
1. Log in to the VME (HPE Morpheus) management console.
2. Navigate to Administration > Plans.

Figure 1. Plans Menu



3. Click + Add and select Service Plan. The New Service Plan form is displayed.

Figure 2. Adding a Service Plan



4. Configure the Service Plan fields as listed in the following table.

Table 1.

Field	Value
Name	CPPM
Active	Enabled (checked)
Code	1k-Plan (used as a unique identifier user's choice
Display Order	0
Provision Type	KVM
Server Type	(leave blank)
Region Code	(leave blank)
Storage	0 GB
Customize Root Volume	Enabled (checked)
Customize Extra Volumes	Enabled (checked)
Add Volumes	Enabled (checked)
Max Disks Allowed	0 (0 = unlimited)
Memory	8 GB

Field	Value
Custom Memory	Enabled (checked) -- mandatory for morphing
Core Count	8
Custom Cores	Enabled (checked) -- mandatory for morphing
Cores Per Socket	1
GPU Count	0

Figure 3. Service Plan Configuration

NAME CPPM

ACTIVE

CODE CPPM

Used as a unique identifier in the API and CLI

DISPLAY ORDER 0

PROVISION TYPE KVM

SERVER TYPE

REGION CODE

STORAGE 0 GB

CUSTOMIZE ROOT VOLUME

CUSTOMIZE EXTRA VOLUMES

ADD VOLUMES

MAX DISKS ALLOWED 0

Sets the maximum amount of disks allowed, (0 = unlimited)

MEMORY 8 GB

CUSTOM MEMORY

CORE COUNT 8 #

CUSTOM CORES

CORES PER SOCKET 1 #

GPU COUNT 0 #

▶ Custom Ranges

Key points:

- Set Provision Type to KVM.
- Set Storage to 0 and enable all three storage options (Customize Root Volume, Customize Extra Volumes, and Add Volumes). This lets you assign the reserved (thick) disk during VM creation.
- Enable Custom Memory and Custom Cores. These custom options are required so the appliance can be morphed to a higher model.

5. Click Save. The new CPPM service plan is created and appears in the Plans list.

Results

The CPPM Service Plan is now available and can be selected when provisioning a ClearPass instance (see VM creation). The following figure shows the completed CPPM service plan listed in the Plans list.

Figure 4. List of CPPM Service Plans

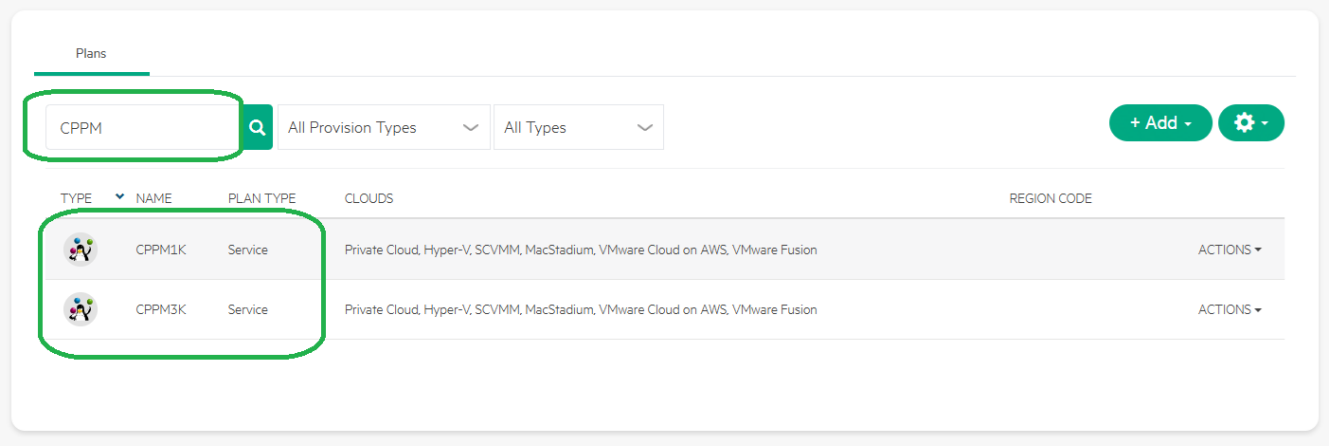


Image Upload

About this task

About this task

The ClearPass software image (QCOW2 disk image) must be uploaded to VME as a Virtual Image before an instance can be provisioned.

The ClearPass RAW image is distributed as a compressed archive (ZIP). Before uploading, extract the archive and use the extracted (.raw) file for the upload — do not upload the ZIP file itself.

> **NOTE**
> The ClearPass image is provided in RAW disk-image format only. Before uploading, convert the RAW image to QCOW2 using the following command, then upload the resulting QCOW2 image as described below.

```
qemu-img convert -f raw -O qcow2 -o preallocation=full <KVM.raw> <KVM.qcow2>
```

The image can be added in one of two ways:

- File: download the ClearPass image ZIP, extract the. qcow2 file locally, and upload it from your local network.
- URL/Path: provide a direct URL to the extracted. qcow2 image.

Important:

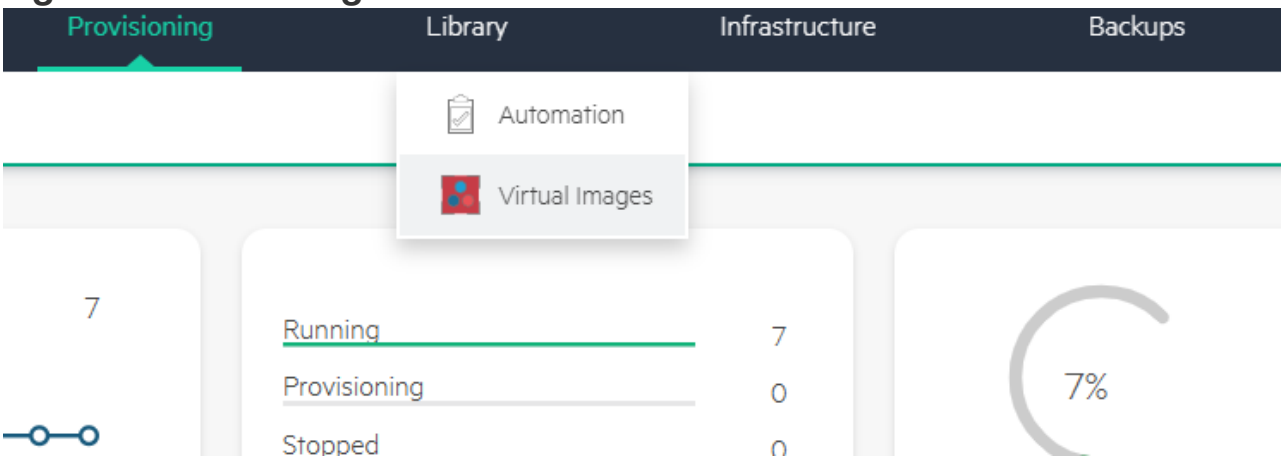
Image upload can take some time to complete. After clicking Add File, do not close or navigate away from the page. Keep the browser window / UI active and wait until the upload has completed; otherwise the upload may fail.

Procedure:

Procedure

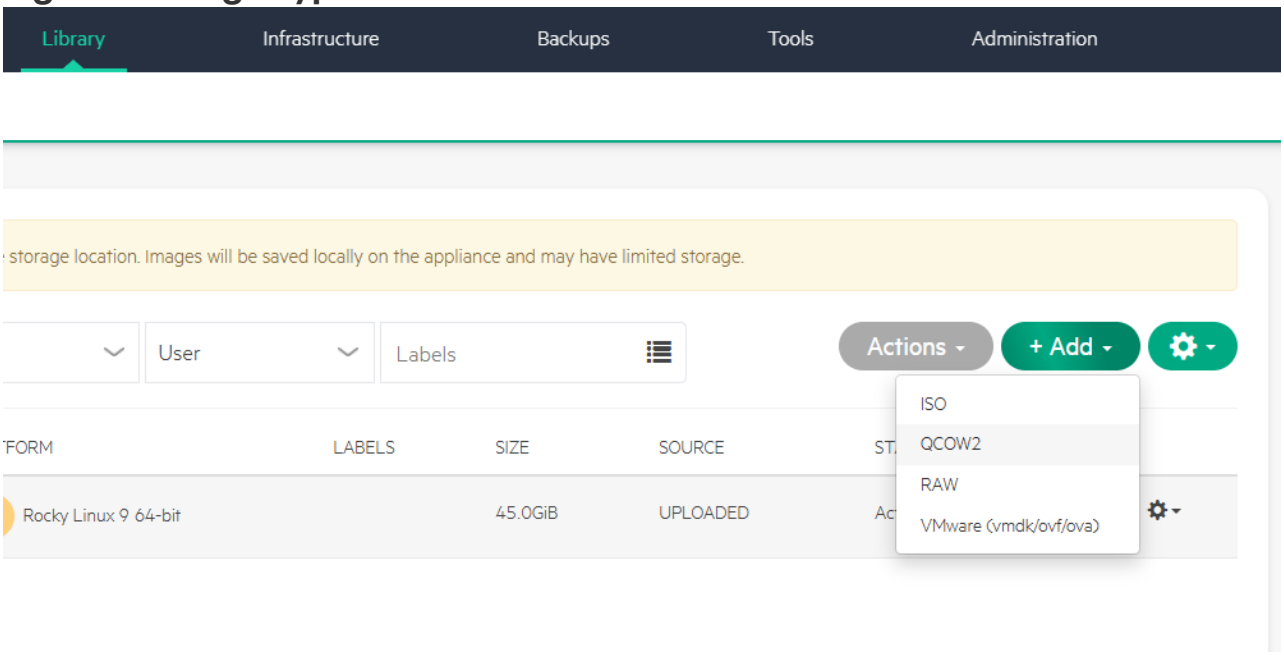
1. Log in to the VME (HPE Morpheus) management console.
2. Navigate to Library > Virtual Images.

Figure 1. Provisioning



3. Click + Add and select QCOW2. (The image-type menu also lists ISO, RAW, and VMware options.)

Figure 2. Image Type Selection



4. In the Upload Virtual Image dialog, configure the fields as listed in the following table.

Table 1.

Field	Value
Name	<image name, e.g. CPPM-6.14>
Labels	(optional)
Operating System	Rocky Linux 9 64-bit
Minimum Memory	0 MB (default)
Is Cloud Init Enabled?	Enabled (default) — leave as-is
Cloud Guest Customization?	Disabled (default)
Sysprepped / Generalized Image?	Disabled (default)
Install Agent?	Enabled (default) — leave as-is

Figure 3. Virtual Image Configuration

NAME

LABELS

A comma separated list of labels that can be used to group/organize items.

OPERATING SYSTEM Select

MINIMUM MEMORY 0 MB

IS CLOUD INIT ENABLED?

If this template uses cloud-init (or cloudbase-init for Windows), this should be enabled. This will allow Morpheus to pass in user data and metadata to the instance as well as configure network and users.

CLOUD GUEST CUSTOMIZATION?

Utilizes agent tools to kick off guest customization scripts for vmware or kvm windows instances. This is useful when not using pre-sysprepped windows images.

SYSPREPPED / GENERALIZED IMAGE?

Required if the template is already sysprepped and shutdown.

INSTALL AGENT?

Determines whether or not the Morpheus advanced automation agent will be installed on the workload during provisioning. This has additional requirements. Please see documentation for specifics.

USERNAME

5. Under Create Image By, select the image source and configure it as listed in the following table.

Figure 4. Image Source Configuration

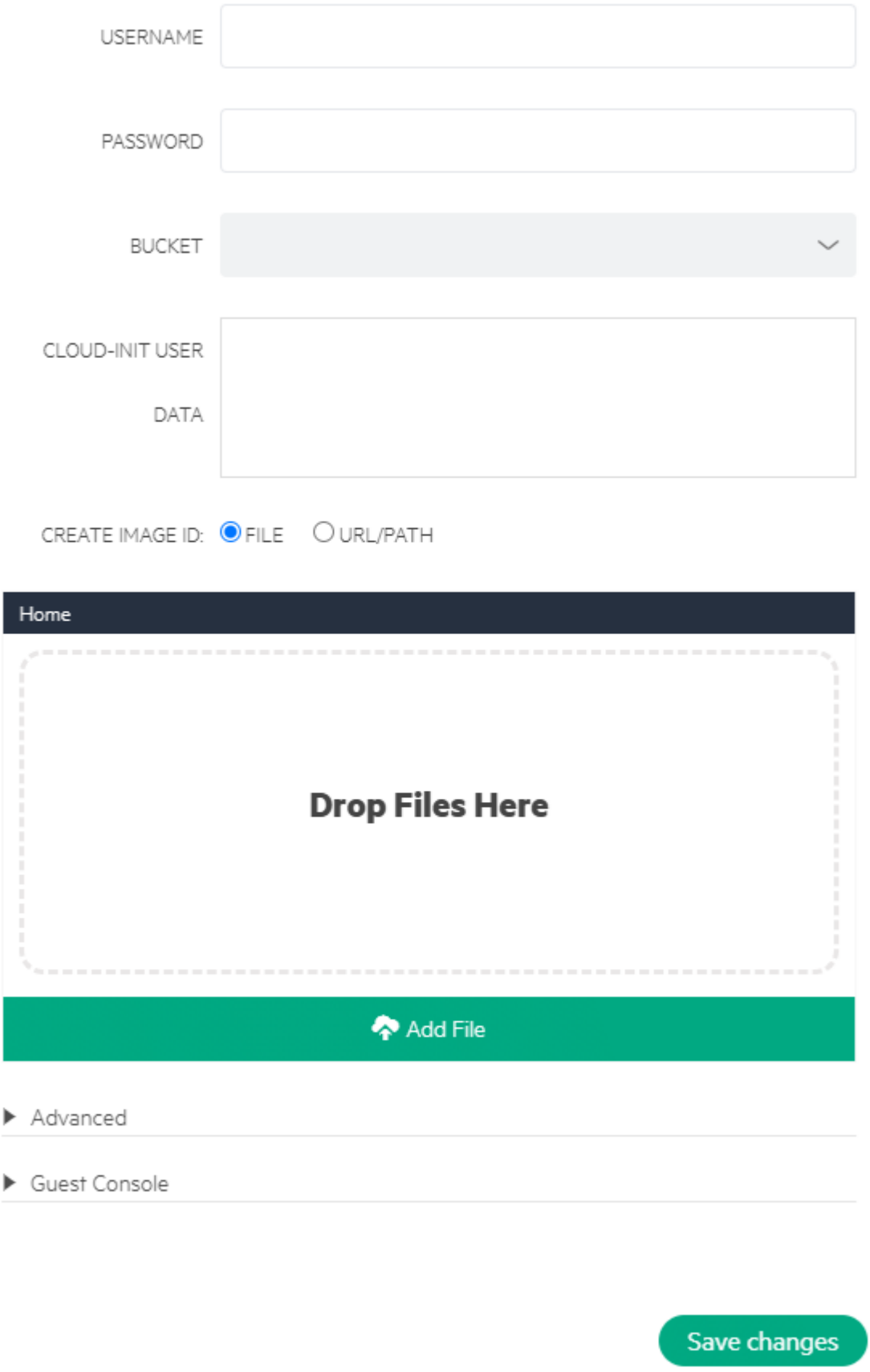

USERNAME
PASSWORD
BUCKET
CLOUD-INIT USER DATA
CREATE IMAGE ID:
FILE
URL/PATH
Home
Drop Files Here
Add File
Advanced
Guest Console
Save changes

Table 2.

Option	Action	Configuration
File (upload from local machine/network)	Select File, then click Add File	Click Add File (or drag the QCOW2 file into the Drop Files Here area) and choose the locally downloaded ClearPass QCOW2 image.
URL/Path (direct URL)	Select URL/Path, then enter the URL	Enter the direct URL to the ClearPass QCOW2 image.

Important:

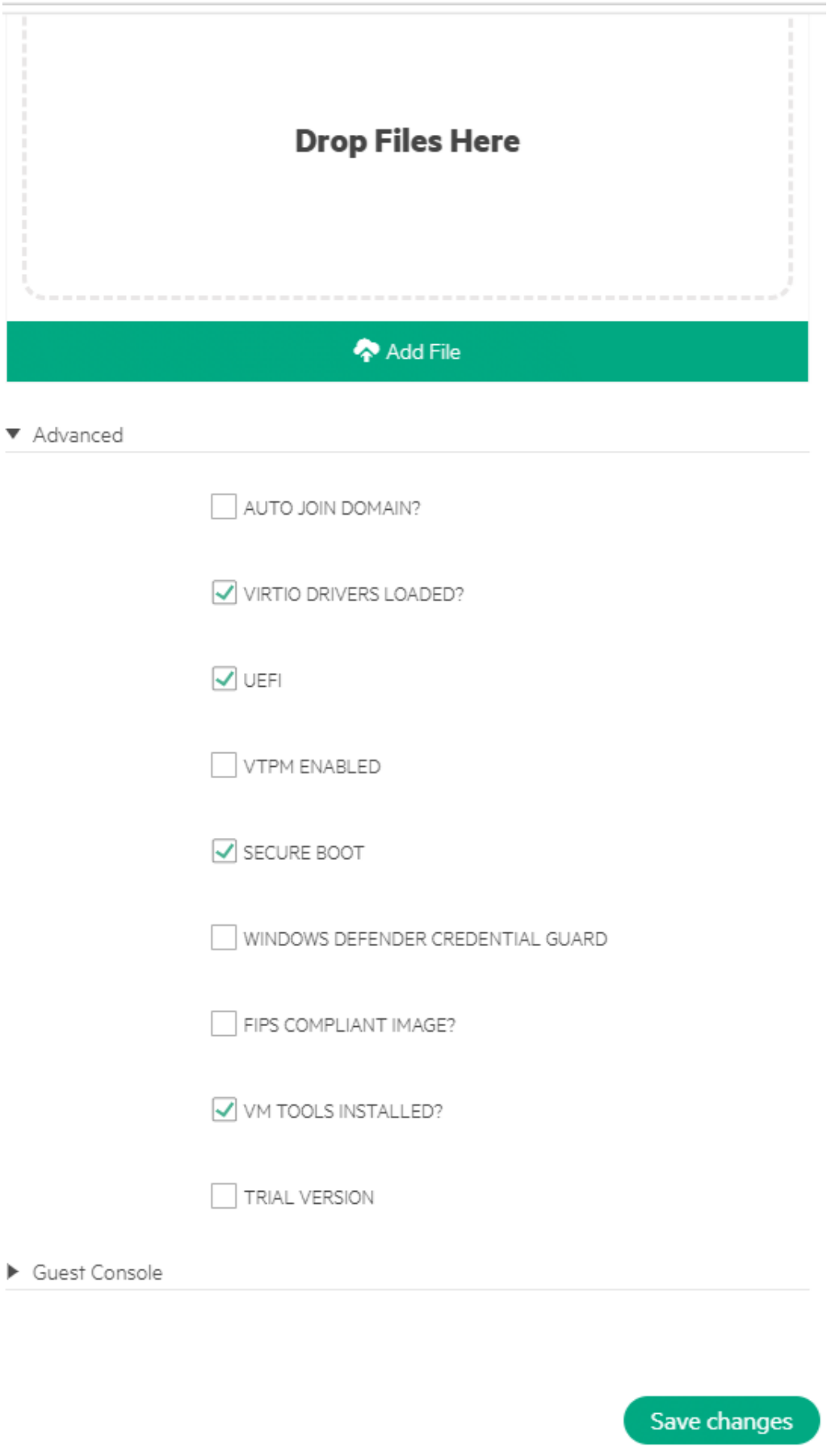
For either option, wait until the upload/transfer has completed, and keep the UI active (do not close the browser tab) until it finishes.

6. Expand Advanced and configure the boot options as listed in the following table:

Table 3.

Field	Value
UEFI	Enabled (checked) — must be enabled
Secure Boot	Enabled (checked) — must be enabled
VirtIO Drivers Loaded	Enabled (default)
VM Tools Installed	Enabled (default)

Figure 5. Advanced Configuration


Drop Files Here
Add File
Advanced
AUTO JOIN DOMAIN?
VIRTIO DRIVERS LOADED?
UEFI
VTPM ENABLED
SECURE BOOT
WINDOWS DEFENDER CREDENTIAL GUARD
FIPS COMPLIANT IMAGE?
VM TOOLS INSTALLED?
TRIAL VERSION
Guest Console
Save changes

7. Click Save Changes to start the upload.

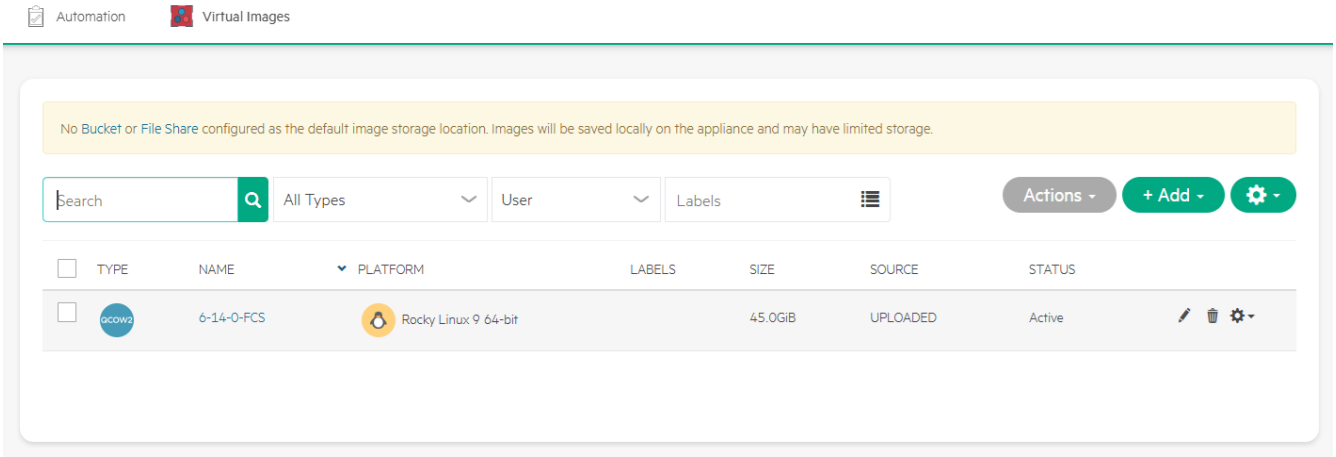
   Two actions are available in this dialog:

   - Add File — selects/attaches the QCOW2 image (Option A).
   - Save Changes — creates the virtual image and begins the upload.

Results

The ClearPass QCOW2 image is uploaded to VME and appears in the Virtual Images list once the upload completes (Status: Active). It can then be used to provision a ClearPass instance.

Figure 6. ClearPass Active Image



VM Creation

About this task

About this task

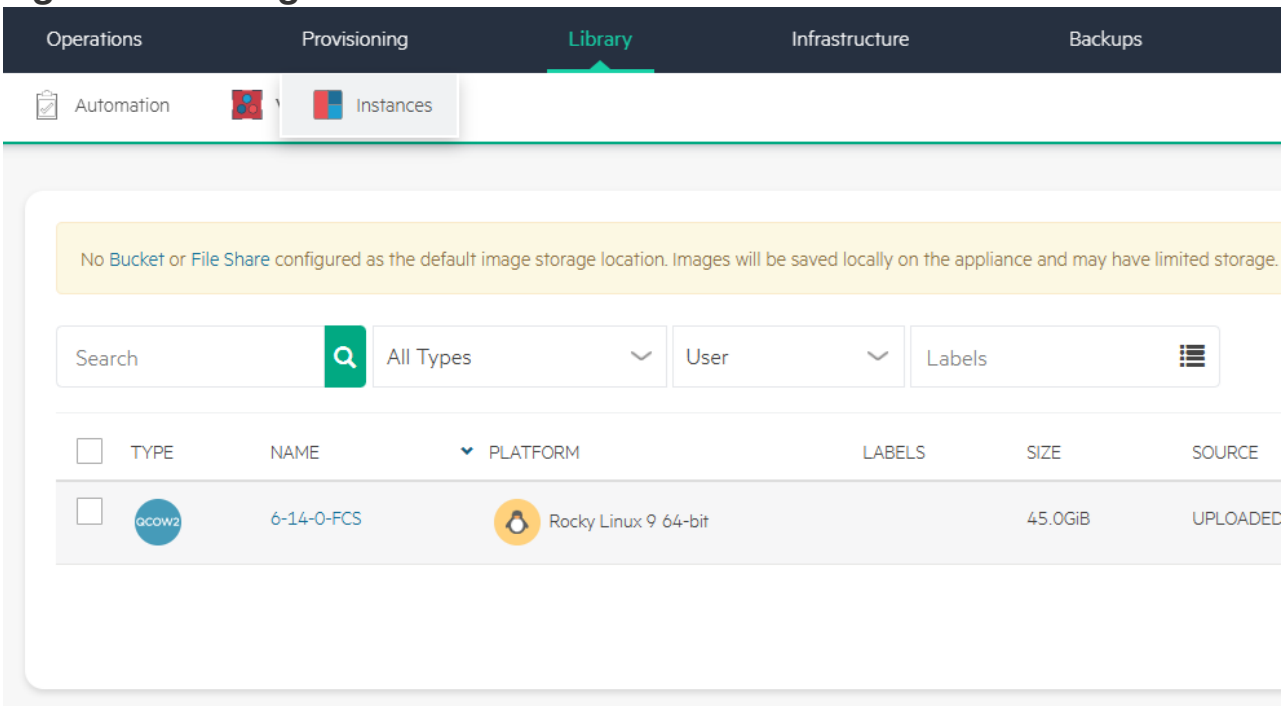
After the ClearPass QCOW2 image has been uploaded to the Virtual Images library, provision a new ClearPass instance (virtual machine) from Provisioning > Instances. The Create Instance wizard walks through five stages: Type, Group, Configure, Automation, and Review.

Procedure:

Procedure

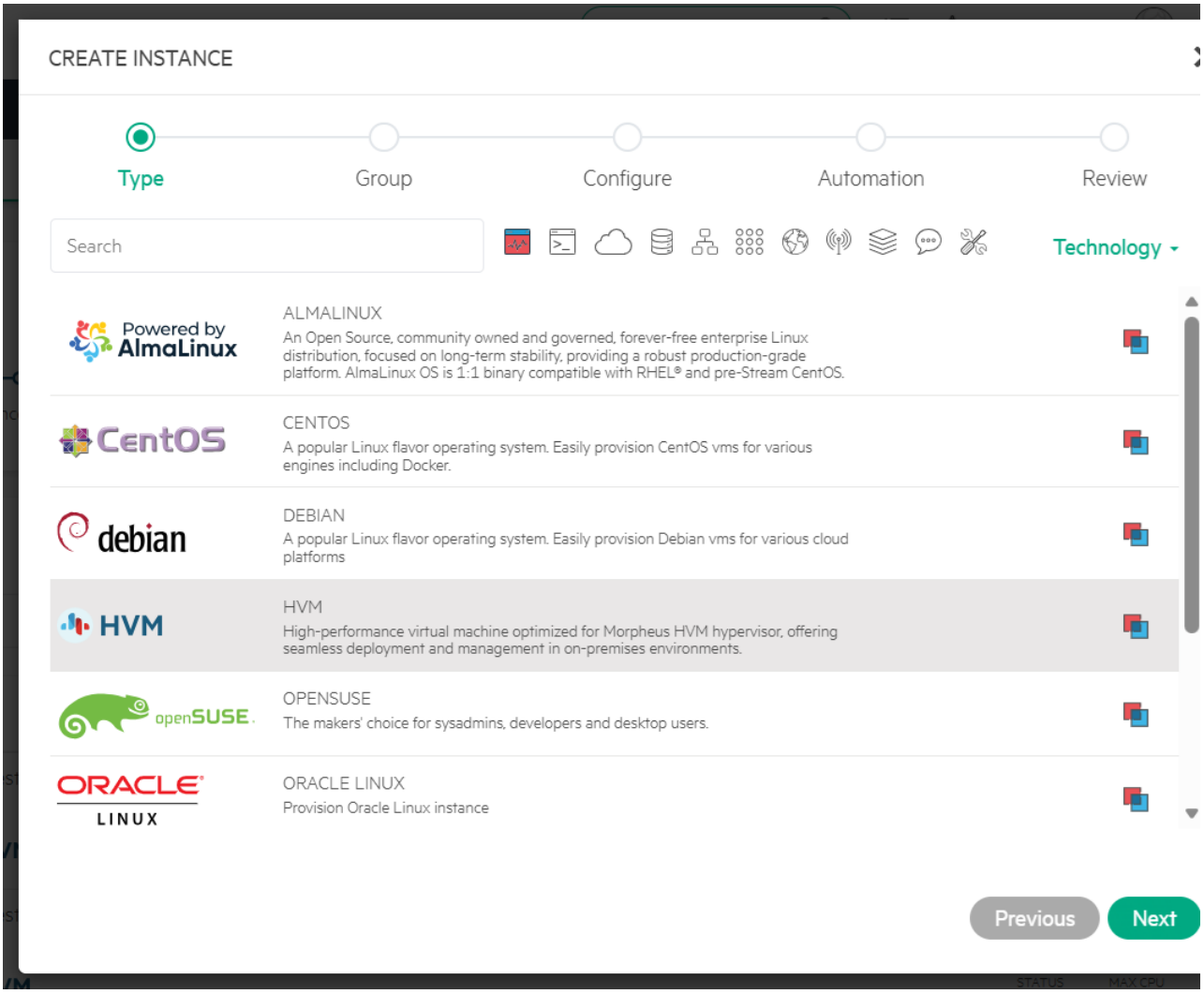
1. From the top menu, navigate to Provisioning > Instances and then click + Add to open the Create Instance wizard.

Figure 1. Creating an Instance



2. Type stage — select HVM as the instance type, then click Next.

Figure 2. HVM Selection

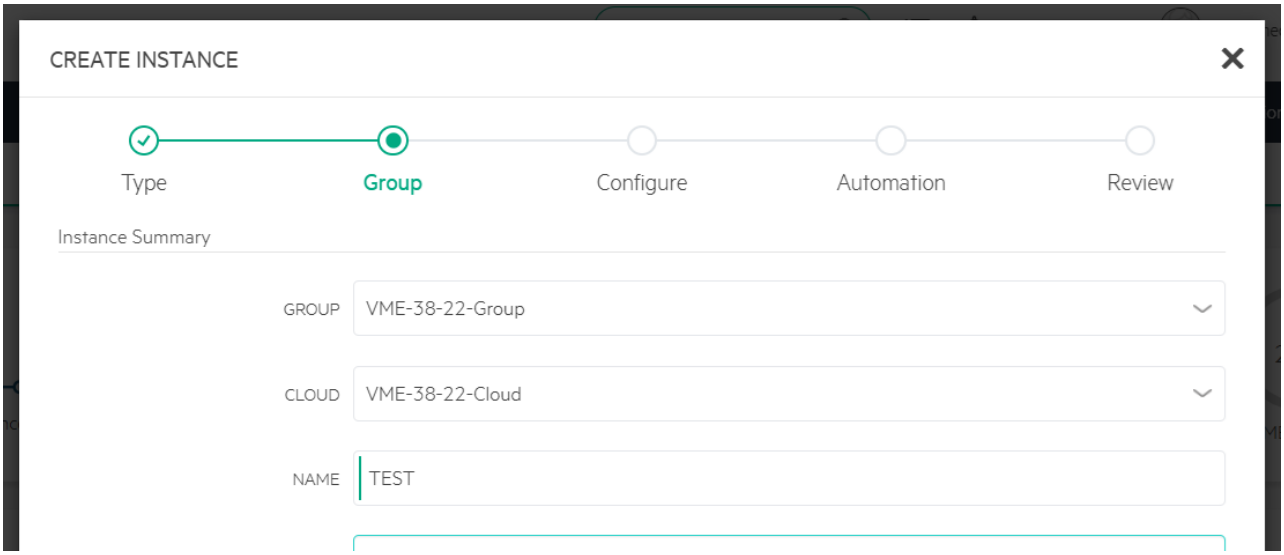


3. Group stage — configure the fields as listed in the following table, then click Next.

Table 1.

Group	VME-38-22-Group (select your group)
Cloud	VME-38-22-Cloud (select your cloud)
Name	<instance name, e.g. CPPM-6.14>
Environment	(optional)
Labels	(optional)

Figure 3. Instance Summary



4. Configure stage — configure the fields as listed in the following table.

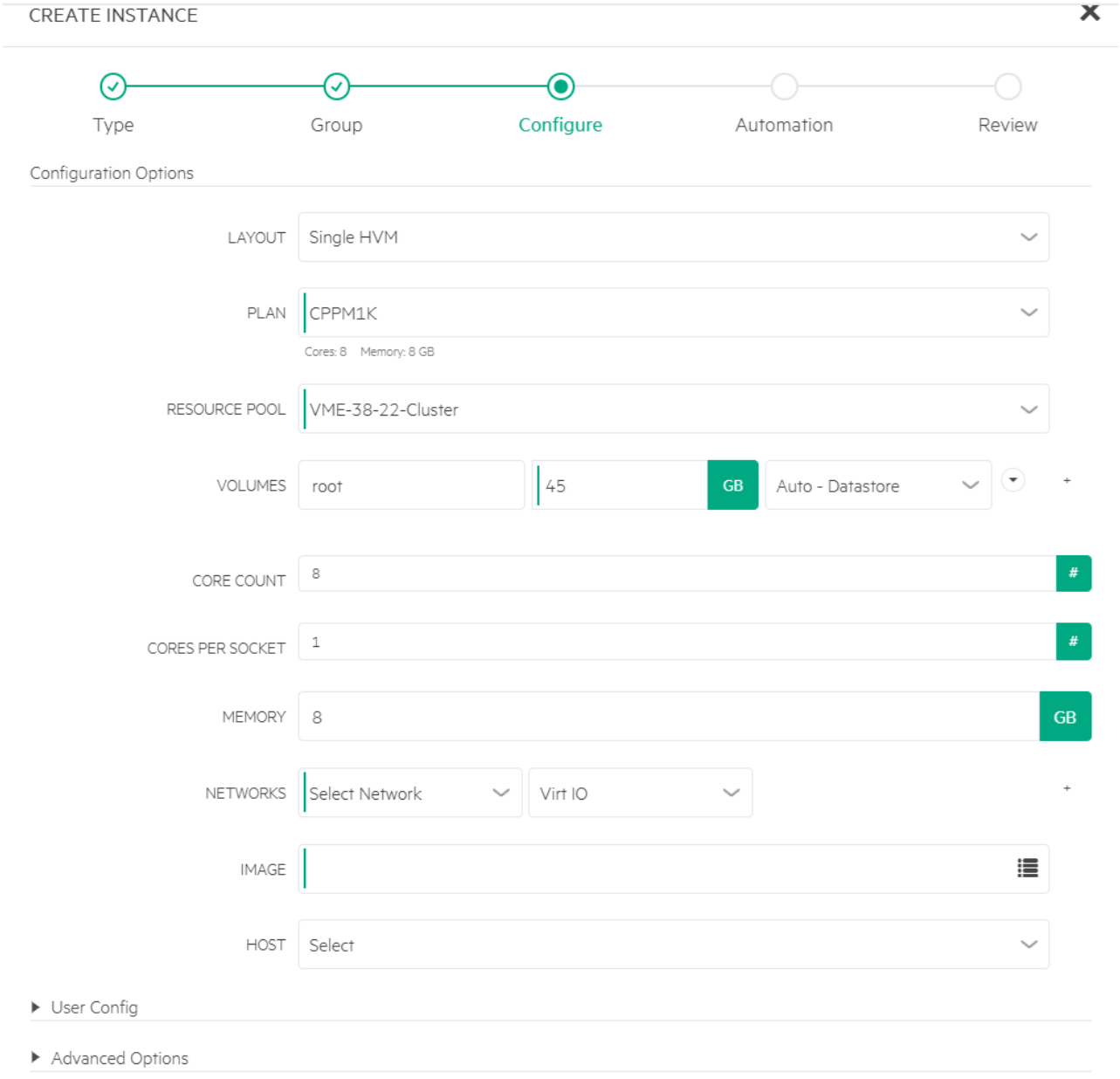
> **NOTE**
>
> In the Plan field, select the ClearPass service plan you created earlier under Administration (for example, CPPM-16C). Once the plan is selected, the Core Count and Memory values are populated automatically from the plan definition.

Table 2.

Layout	Single HVM
Plan	Select the ClearPass service plan created earlier (e.g. CPPM-16C). CPU and memory auto-populate from the plan.
Resource Pool	VME-38-22-Cluster (select your cluster)
Volumes	root — 45 GB (Auto - Datastore). Add data disks as required (see step 4a–4b).
Core Count	Auto-populated from the selected plan (e.g. 8).
Cores Per Socket	1
Memory	Auto-populated from the selected plan (e.g. 8 GB).

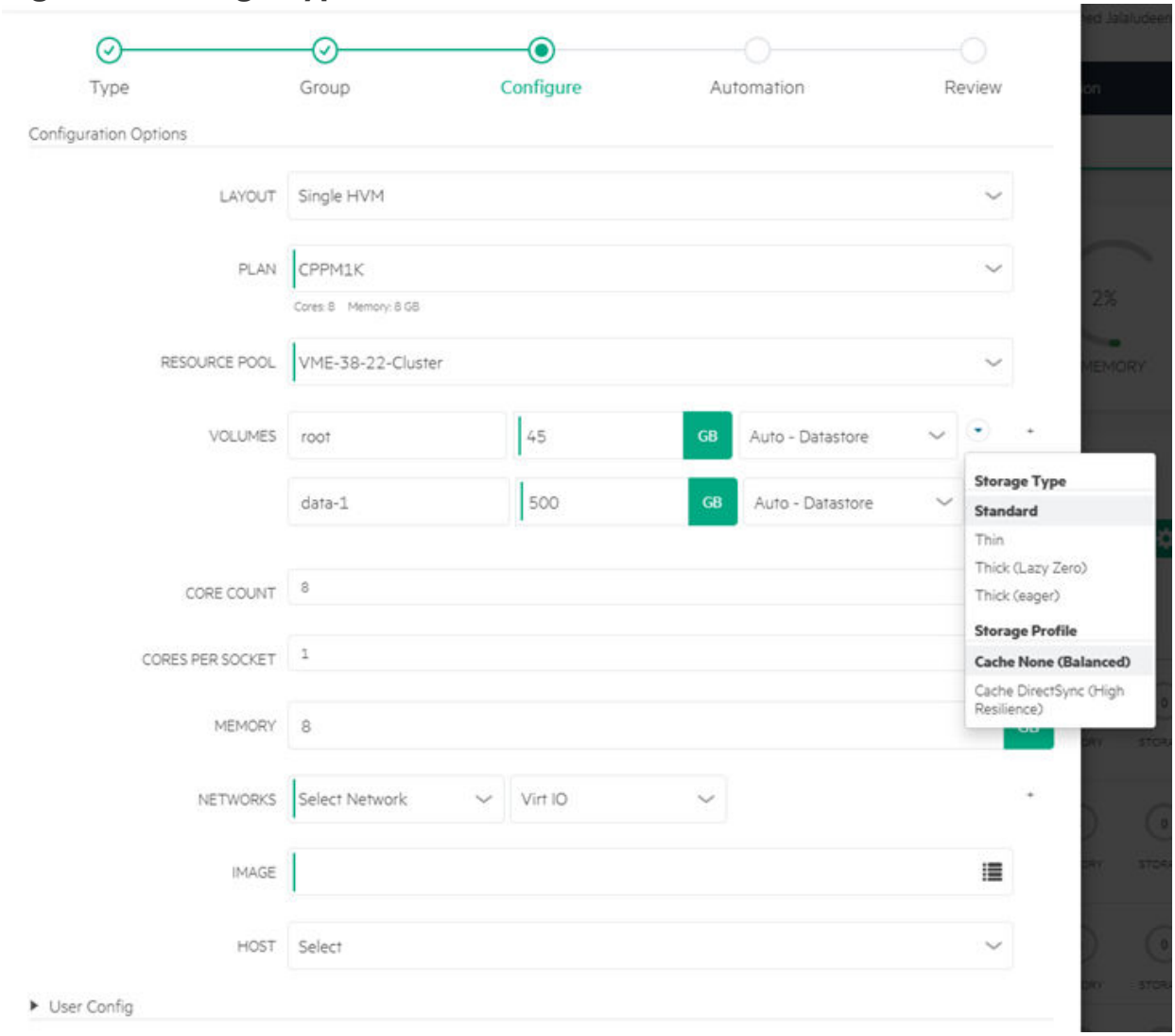
Networks	Management (Virt IO) and Compute (Virt IO). Add adapters as required (see step 4c).
Image	Select the uploaded ClearPass QCOW2 image (see step 4d).
Host	Select the host based on the configuration (see step 4e).

Figure 4. CPU and memory Configuration



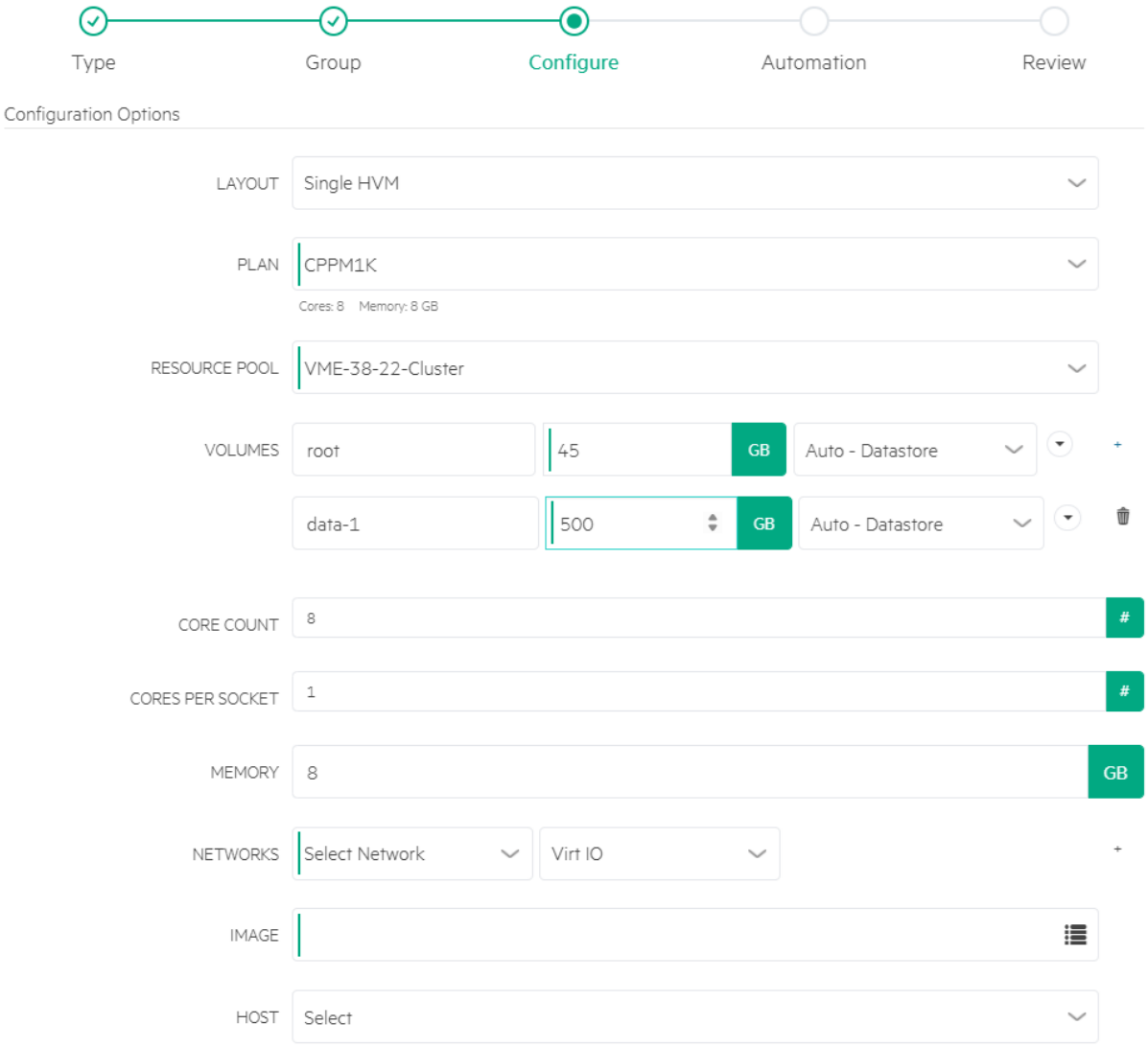
a. Set the disk provisioning type. Click the down-arrow next to the respective Volumes row and, under Storage Type, select Thick (Lazy Zero).

Figure 5. Storage Type Selection



b. To add an additional disk, click the + option next to the Volumes section and specify the size as per platform model (for example, C2000 — 1000 GB). Set its Storage Type to Thick (Lazy Zero) as in step 4a.

Figure 6. Storage Volume Selection



c. Under Networks, select the required network from the drop-down (for example, Management). To add another interface, click the + option and select the additional network (for example, Compute). Configure one adapter for the MGMT interface and one for the data interface.

Figure 7. Network Selection

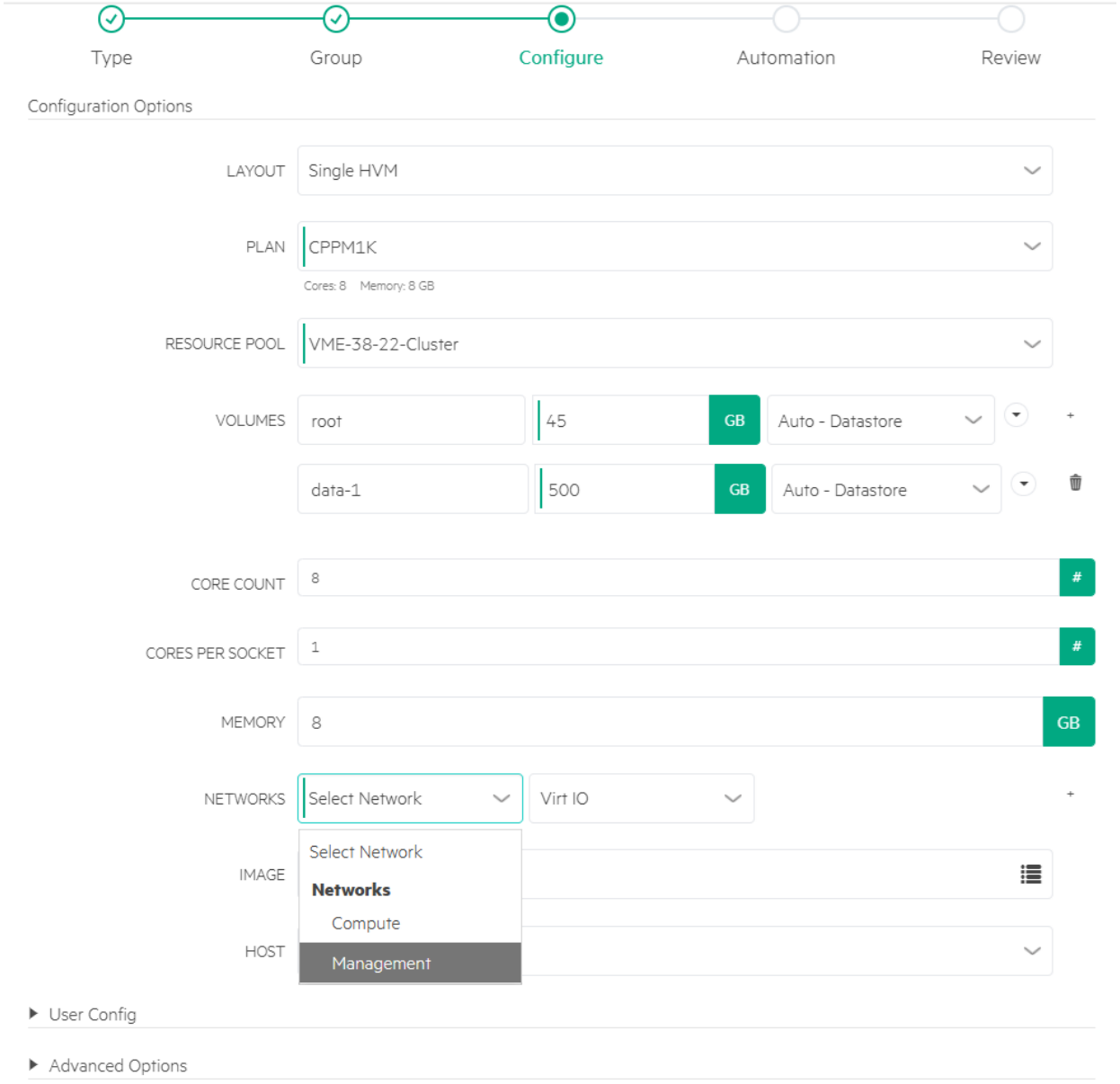
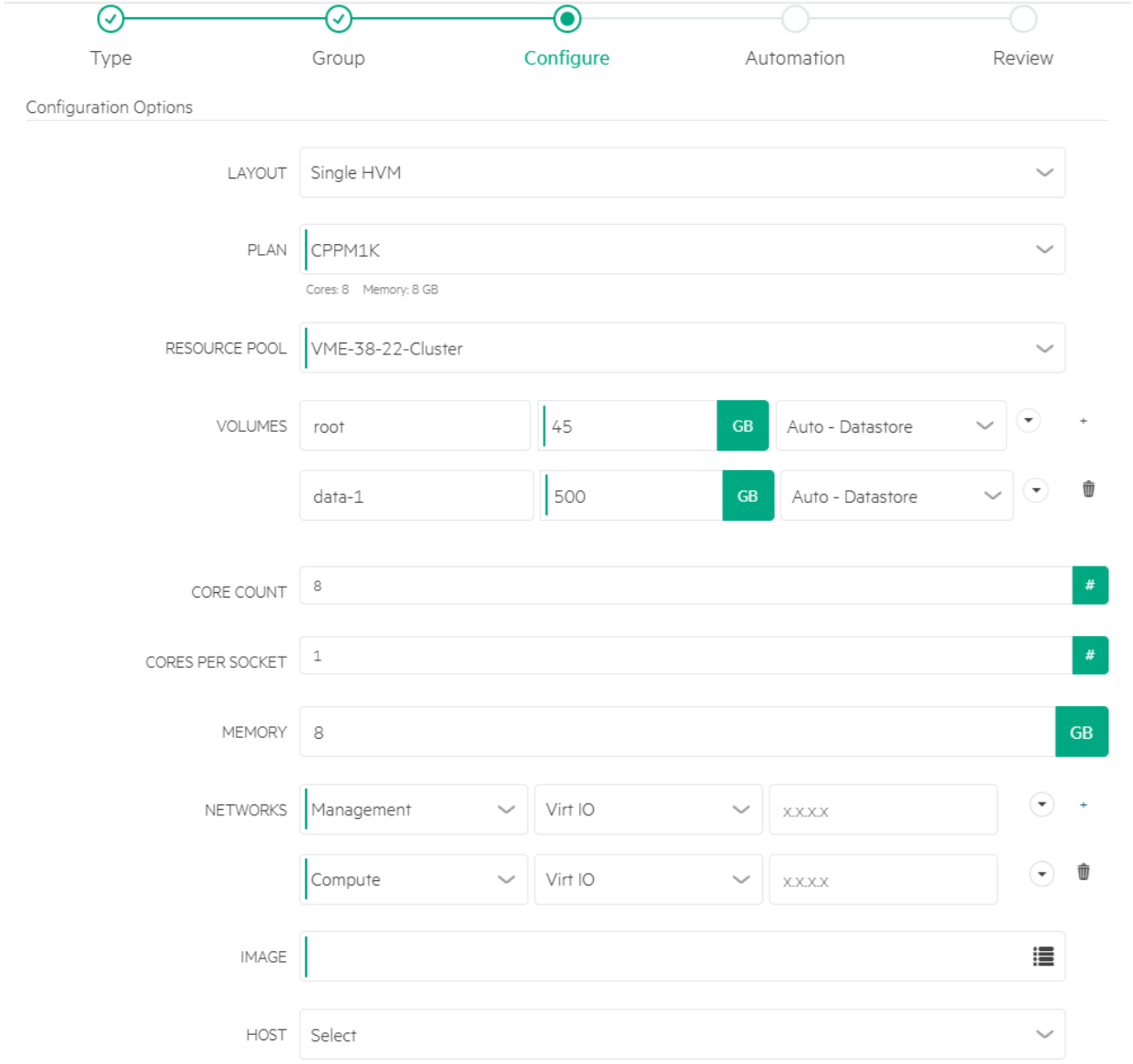
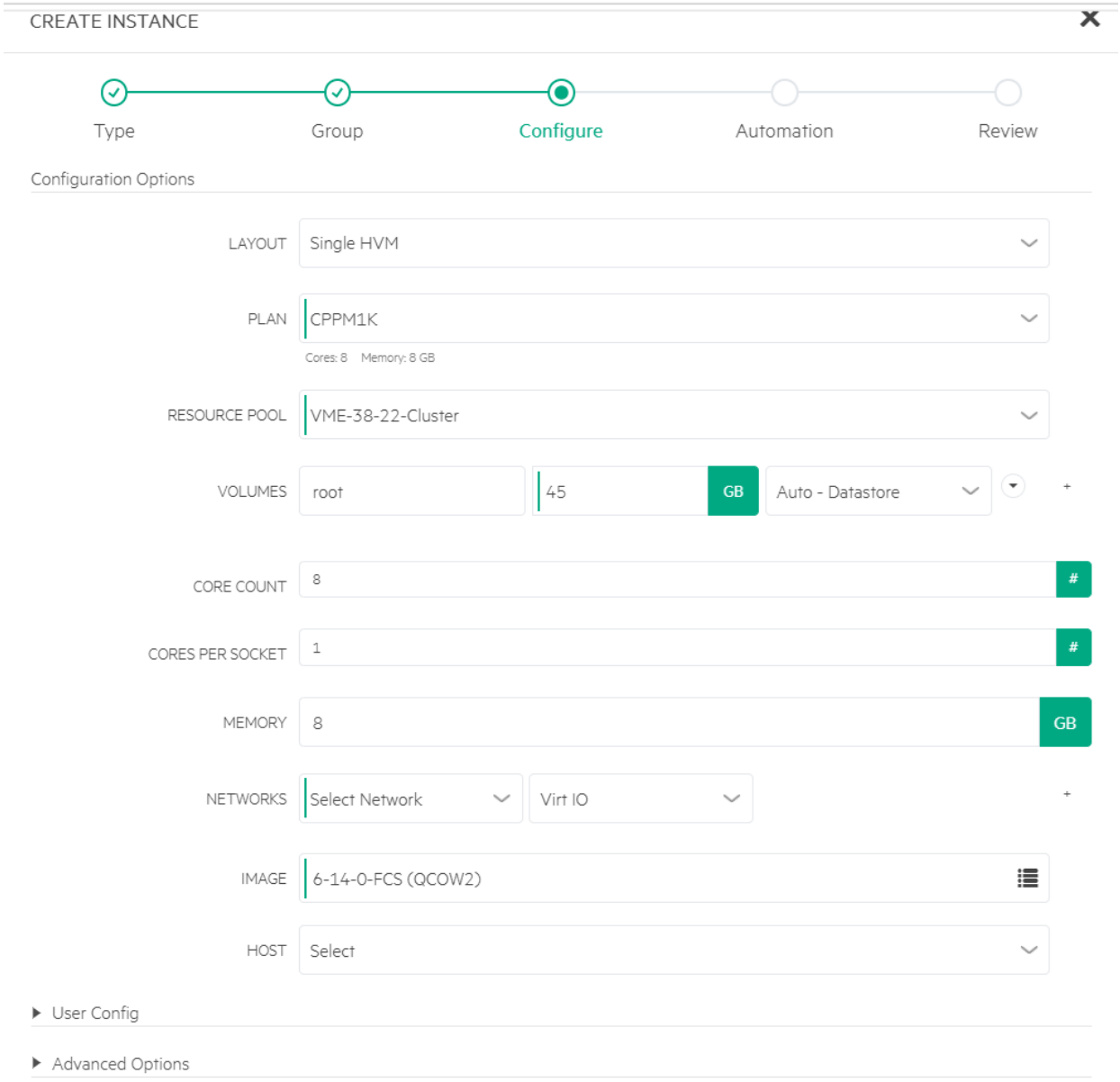


Figure 8. Network Adapter Selection



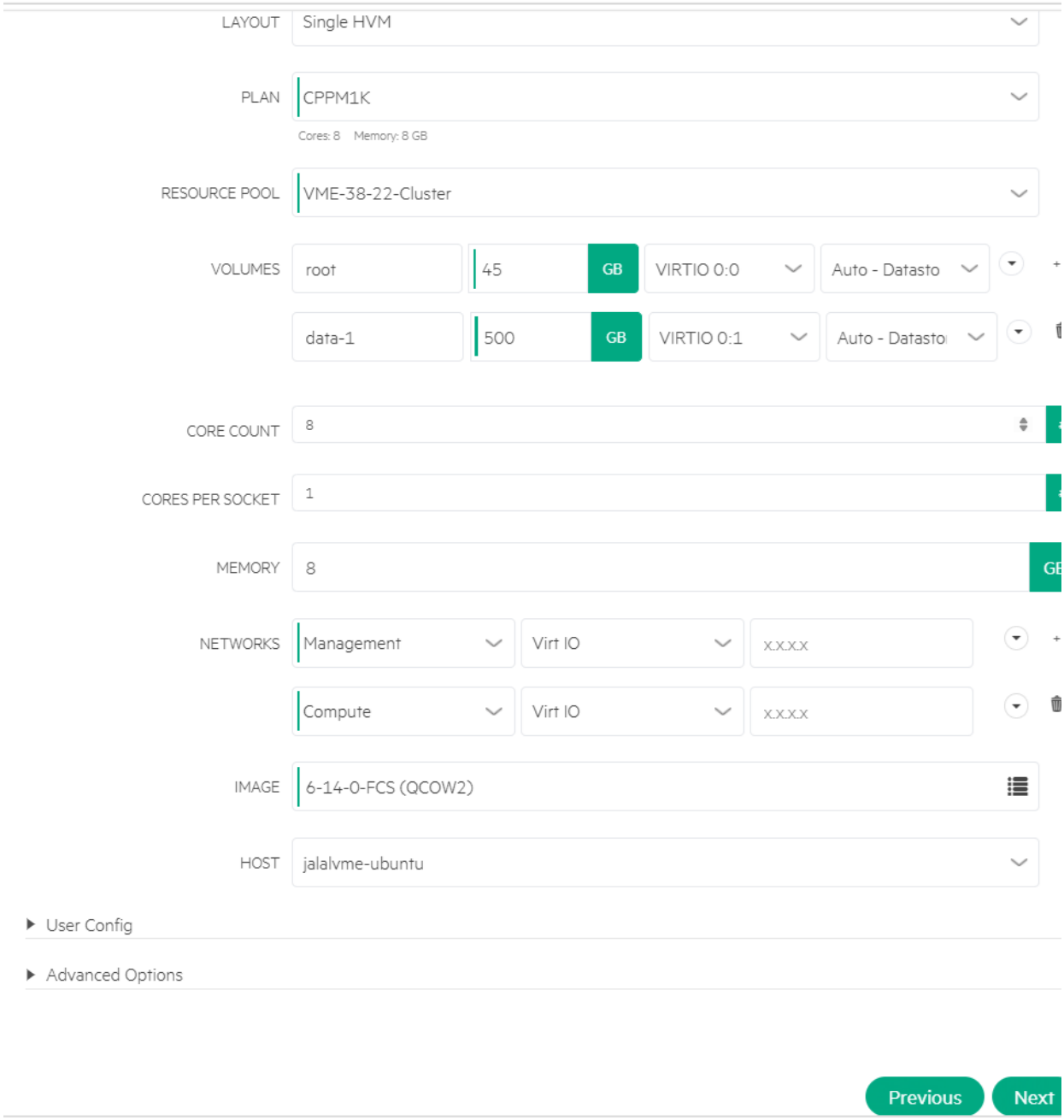
d. In the Image field, select the ClearPass QCOW2 image that was uploaded earlier (for example, 6-14-0-FCS (QCOW2)).

Figure 9. Image Selection



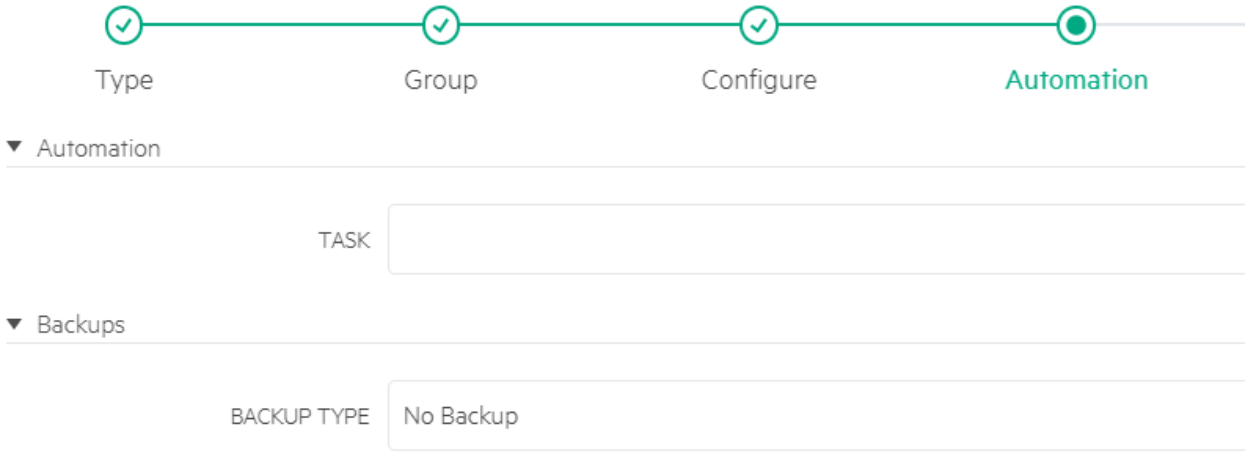
e. In the Host field, select the target host based on the configuration made. When all fields are complete, click Next.

Figure 10. Host Selection



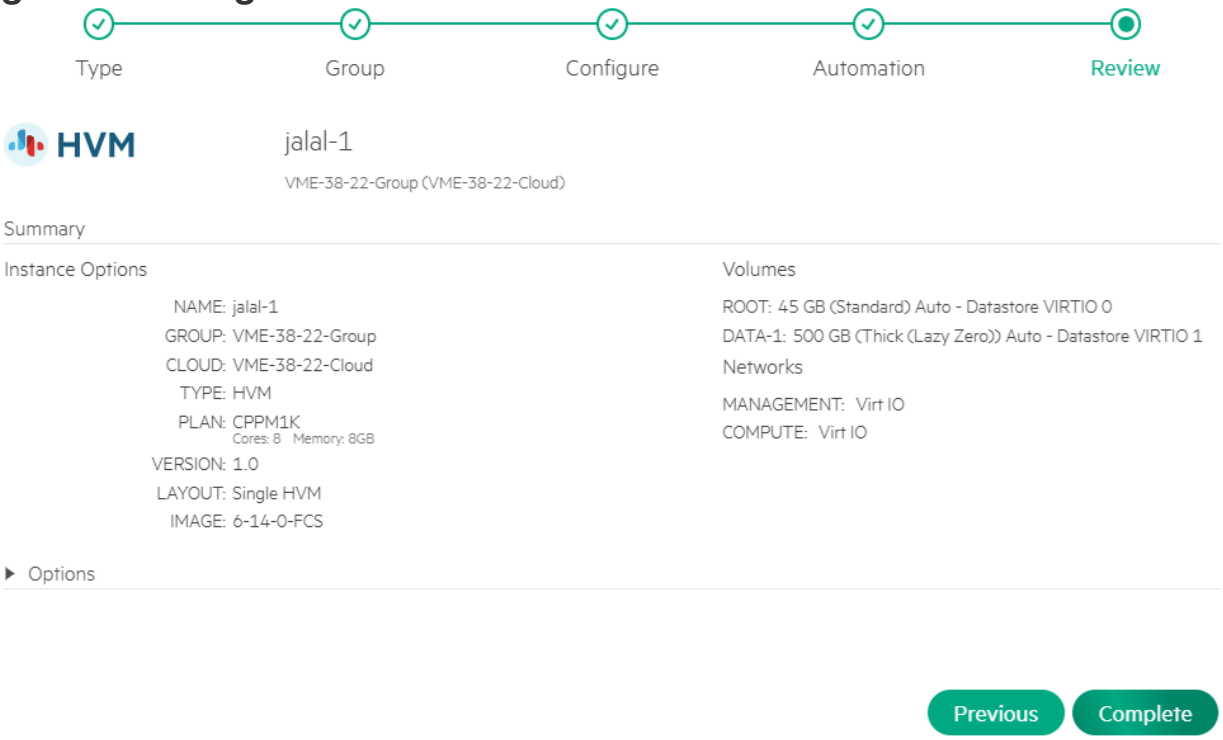
5. Automation stage — leave Task empty and set Backup Type to Default option then click Next.

Figure 11. Automation Configuration



6. Review stage — verify the instance summary (Instance Options, Volumes, and Networks), then click Complete to provision the instance.

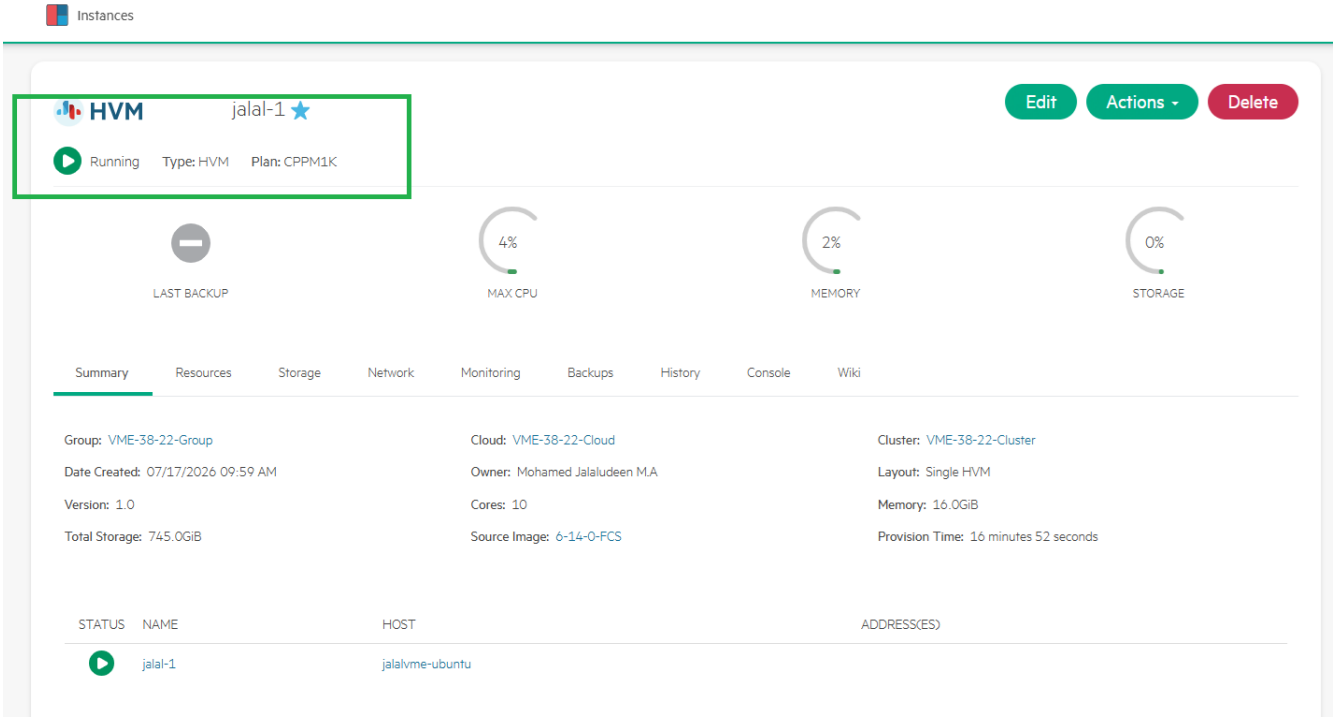
Figure 12. Configuration Review



Results

VME begins provisioning the ClearPass instance. Once provisioning finishes, the instance appears in the Instances list and can be powered on and accessed. The following figure shows the provisioned ClearPass instance (Running) with its plan, cluster, CPU, memory, and storage summary.

Figure 13. Running ClearPass Instance



Verifying the Network Adapter MAC Address Order

About this task

About this task

The ClearPass appliance requires the Management (MGMT) interface to have the lower MAC address of the two network adapters. After the instance is provisioned, verify the MAC address order and, if required, reconfigure the adapters until the MGMT interface holds the lower MAC address.

Important:

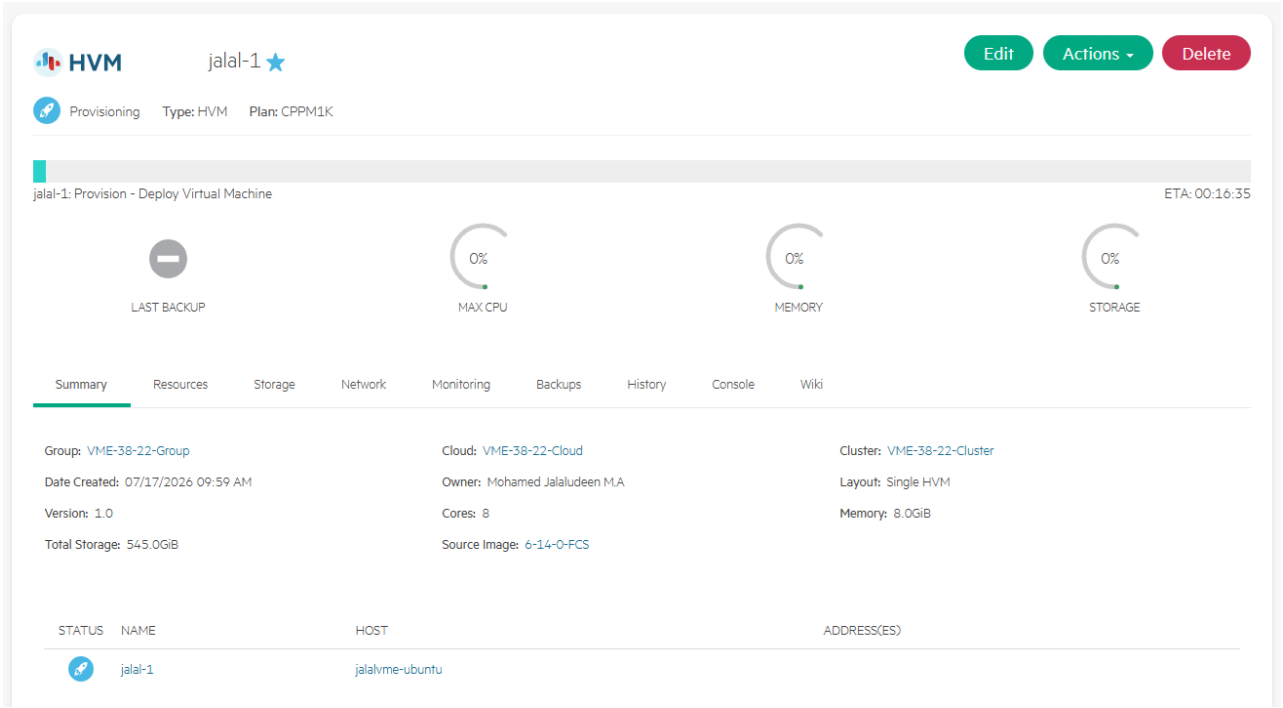
Provisioning the instance can take some time to complete. Wait for the instance status to show that provisioning has finished before continuing.

Procedure:

Procedure

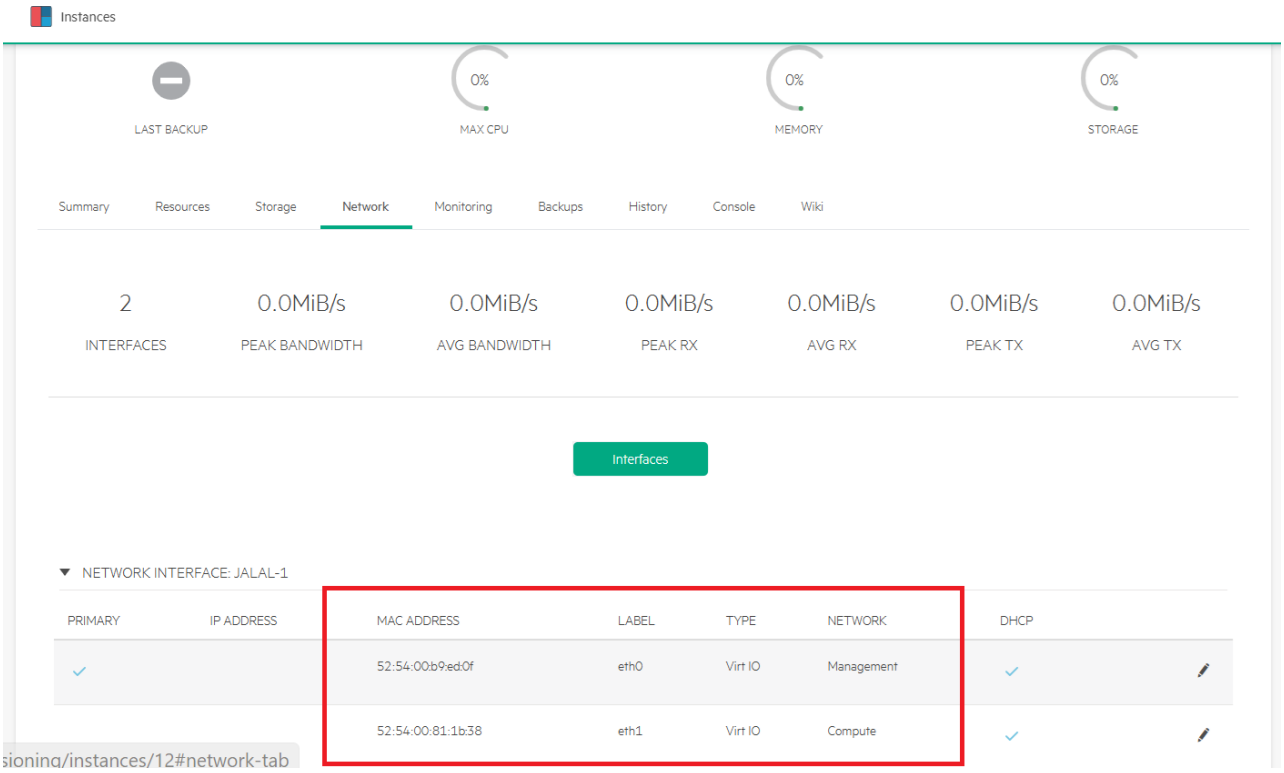
1. After clicking Complete on the Review page, wait for the VM creation to complete. The instance page shows the provisioning progress; wait until it has finished.

Figure 1. Instance Provisioning Completion



2. Once provisioning completes, open the instance and click the Network tab at the bottom of the page. Check the MAC address of each interface and confirm that the Management (MGMT) interface has the lower MAC address.

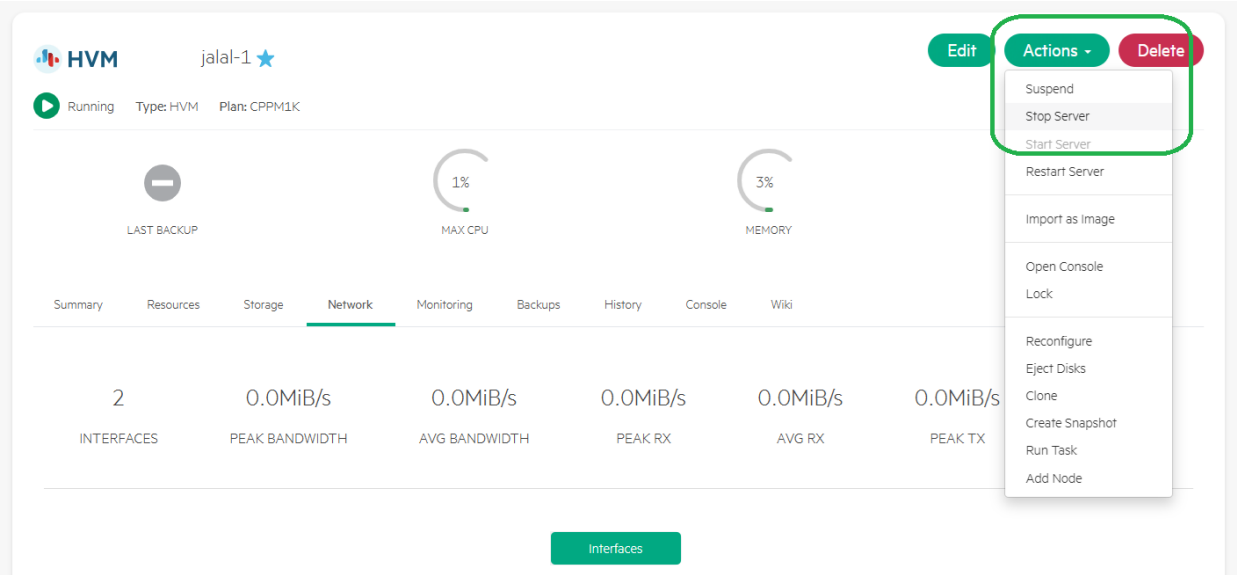
Figure 2. Verifying MAC Addresses



3. If the MGMT interface does NOT have the lower MAC address, reconfigure the adapters as follows:

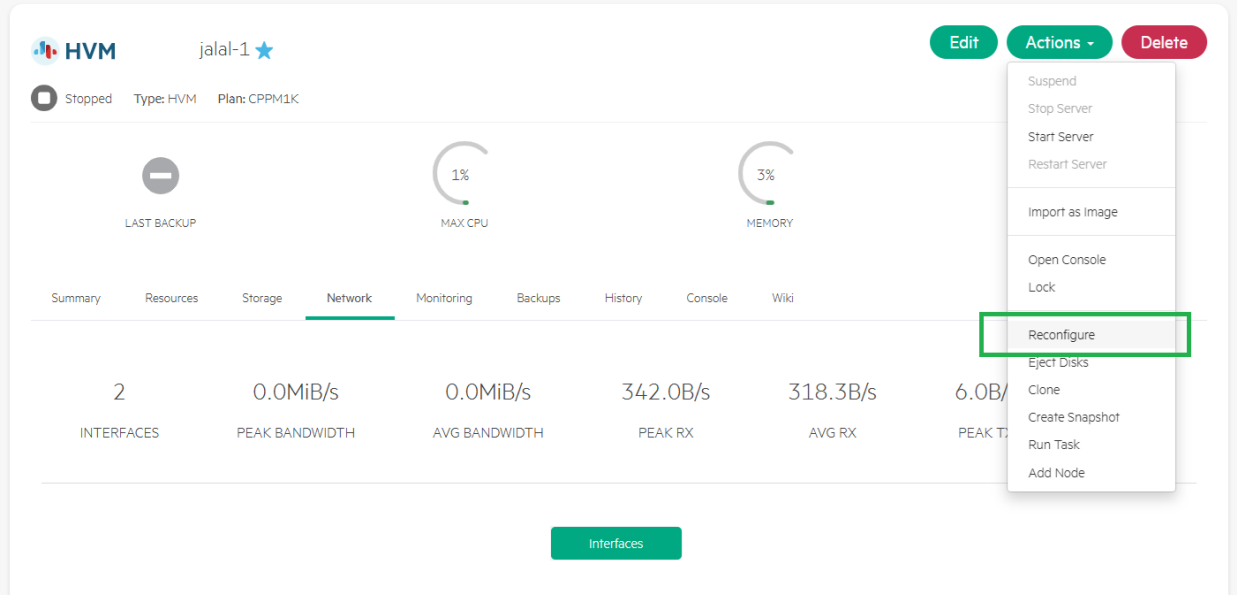
   a. Click Actions > Stop Server to stop the instance.

Figure 3. Stopping the server



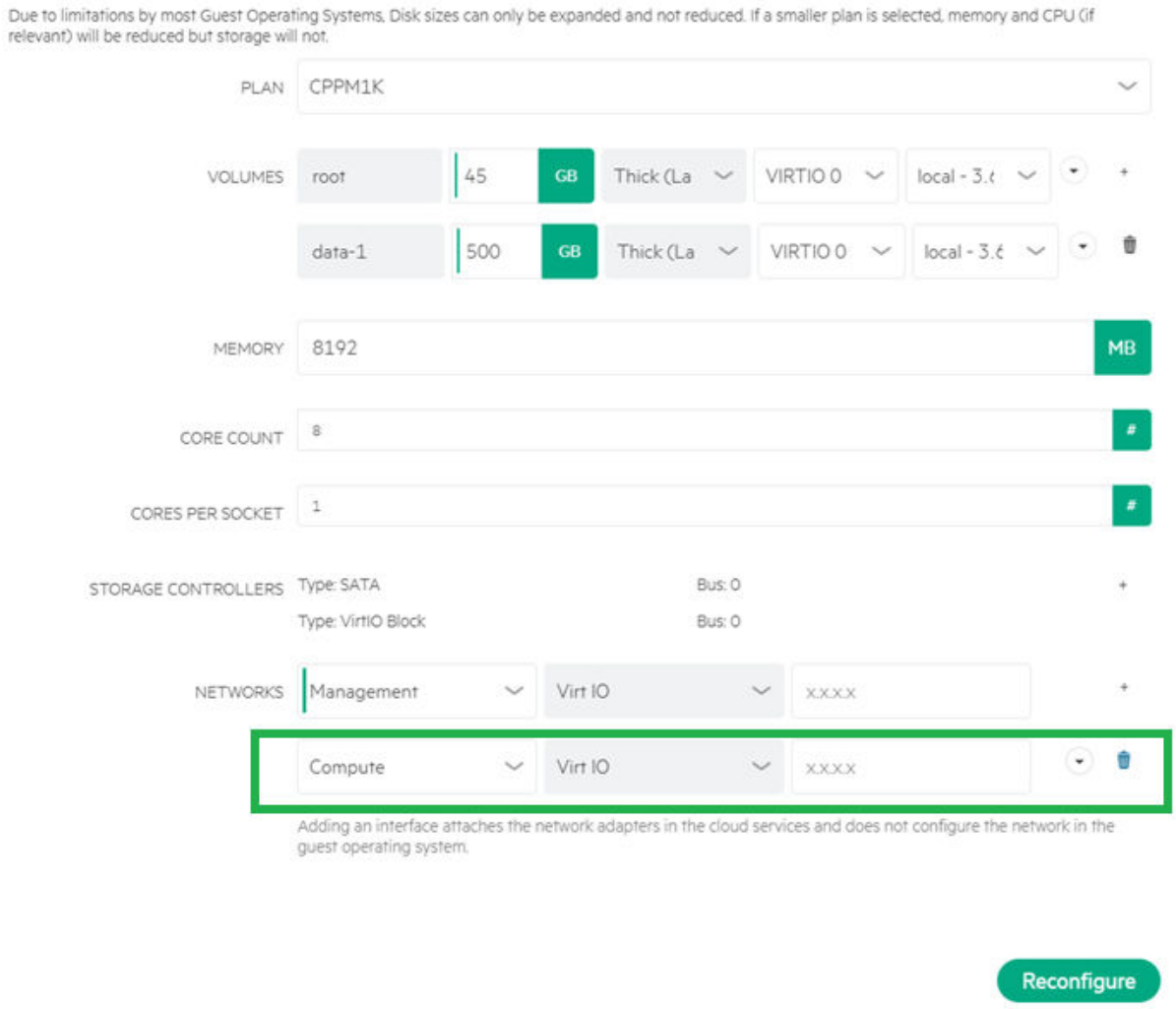
   b. Click Actions > Reconfigure.

Figure 4. Reconfigure Menu



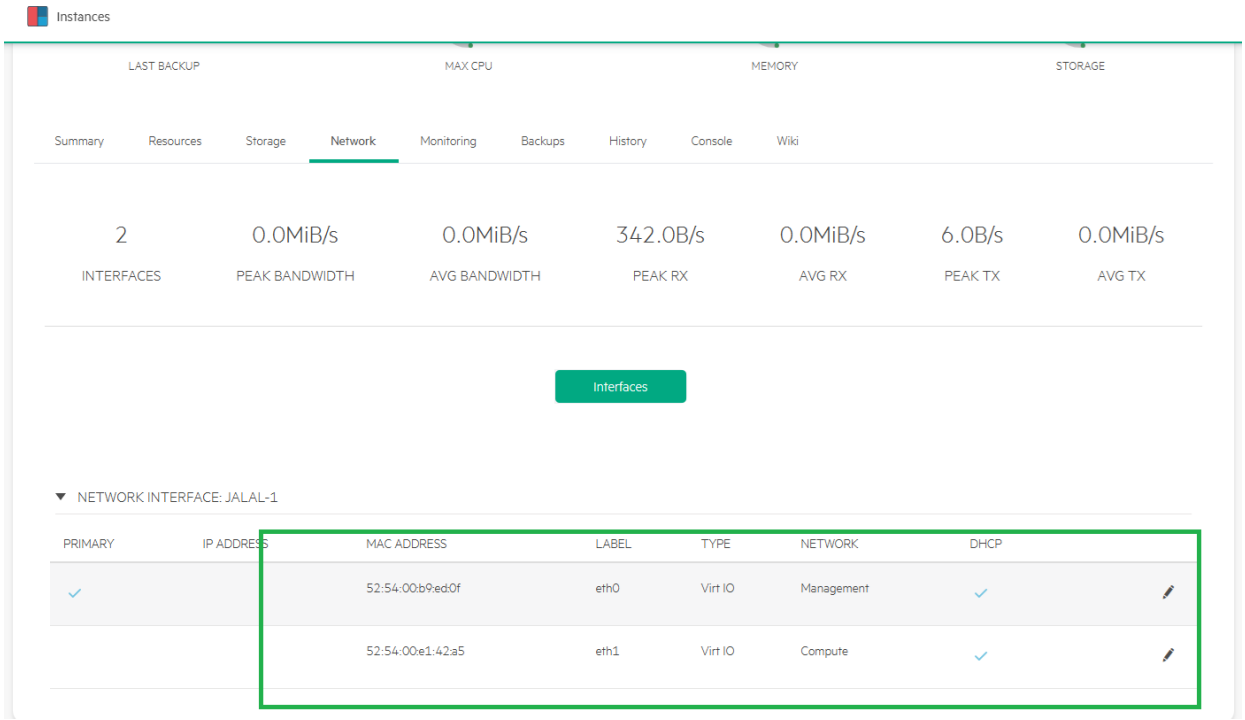
   c. In the Reconfigure Instance dialog, delete the Compute interface (click the delete icon next to it), then add the Compute interface again, and click Reconfigure.

Figure 5. Adding the Compute Interface



d. Return to the Network tab and re-check the MAC addresses.

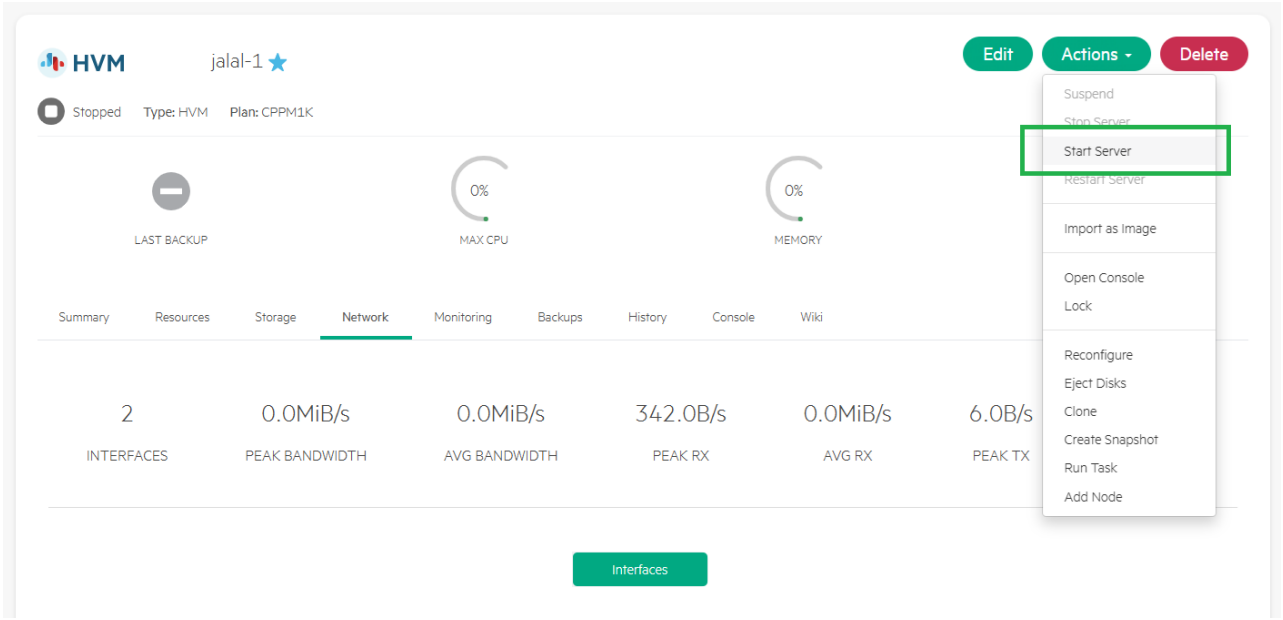
Figure 6. Re-checking MAC Addresses



e. Repeat this reconfigure process (Stop Server > Reconfigure > delete and re-add Compute) until the MGMT interface has the lower MAC address.

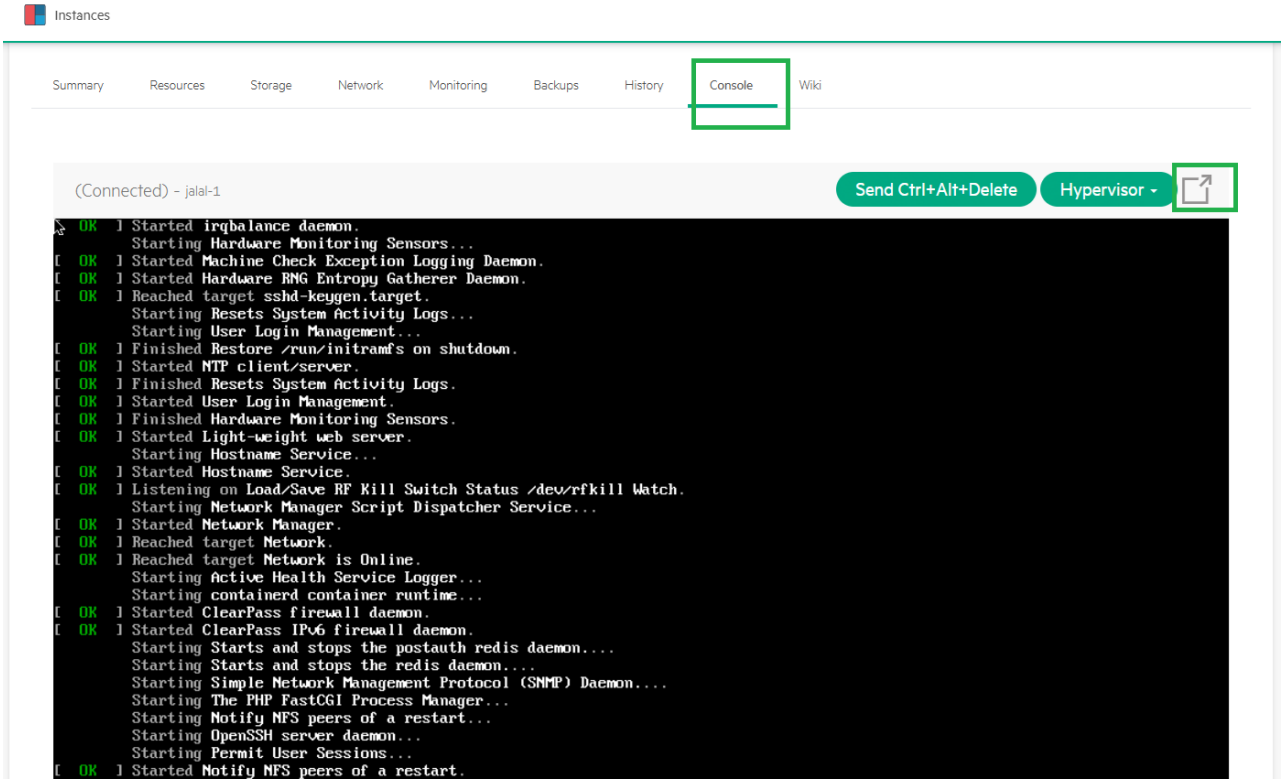
4. Once the MGMT interface has the lower MAC address, click Actions > Start Server.

Figure 7. Starting the server



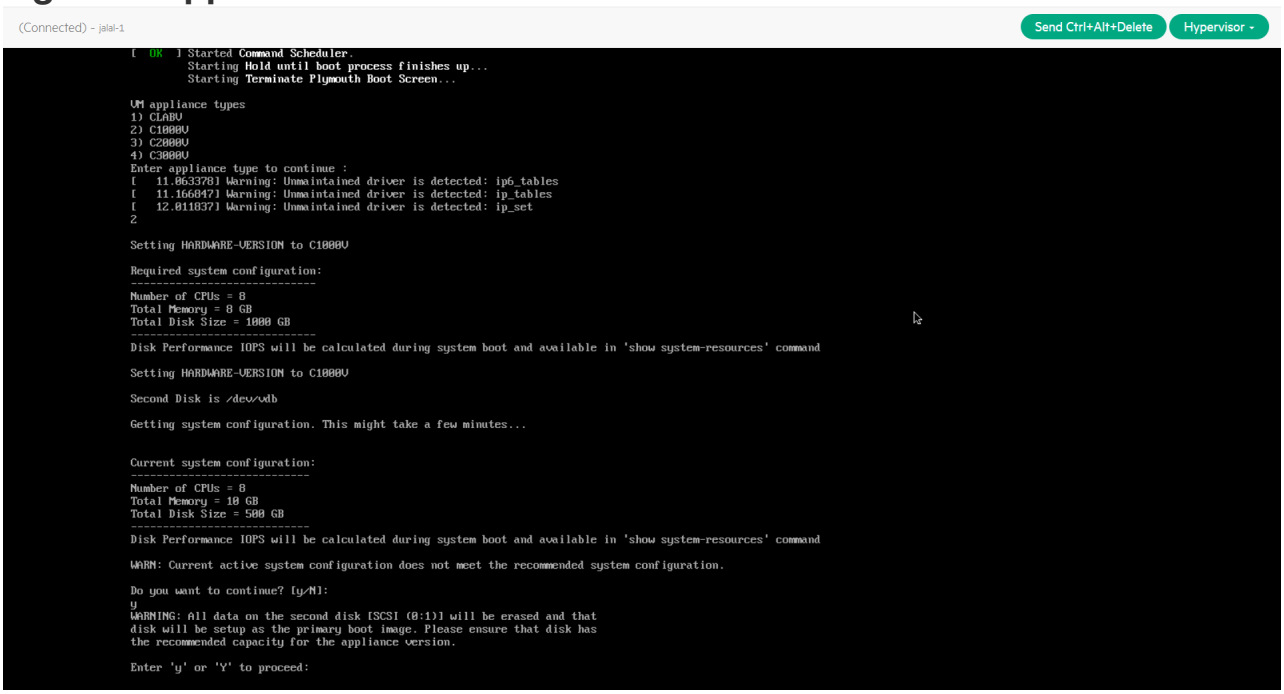
5. Click the Console tab at the bottom of the page and maximize it (use the expand icon on the top right of the console).

Figure 8. Console Tab



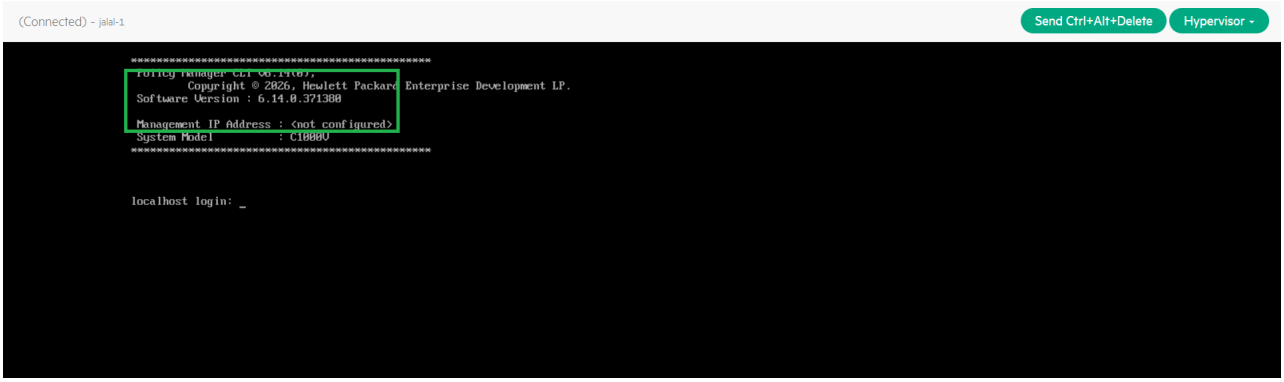
6. In the console, complete the appliance selection and the system pre-checks. Enter the appliance type when prompted and confirm to continue.

Figure 9. Appliance Selection



7. When the appliance finishes initializing, the console displays the login prompt. Log in and complete the bootstrap configuration of the ClearPass appliance.

Figure 10. Login for Bootstrap Configuration



Results

The ClearPass instance is started with the MGMT interface holding the lower MAC address, and the appliance bootstrap configuration can be completed from the console.

Morphing a VME Version

About this task

Morphing changes a ClearPass virtual appliance from one model to a higher model (for example, from C1000V to C2000V). The process stops the instance, adds a new disk sized for the target model, adjusts the CPU and memory, starts the instance, and then runs the system morph-vm command from the appliance console.

NOTE

The illustrations in this section use the example of morphing to a C2000V virtual appliance. Adjust the disk size, CPU, and memory for the model you are morphing to.

NOTE

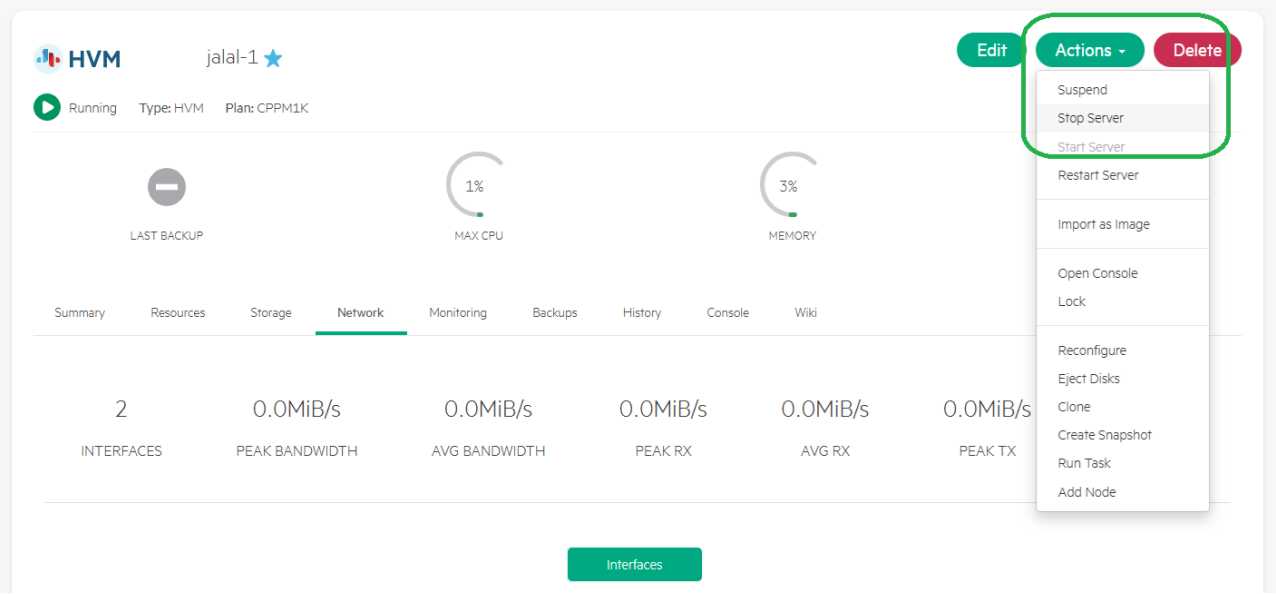
Starting with the 6.11.0 release, after the system morph-vm command is run during a ClearPass upgrade and the new disk is added, the existing hard disk cannot be removed. This is because the existing disk is used to extend the disk space for the server, and is included in the calculations to determine the required capacity of the additional hard disk.

Procedure:

Procedure

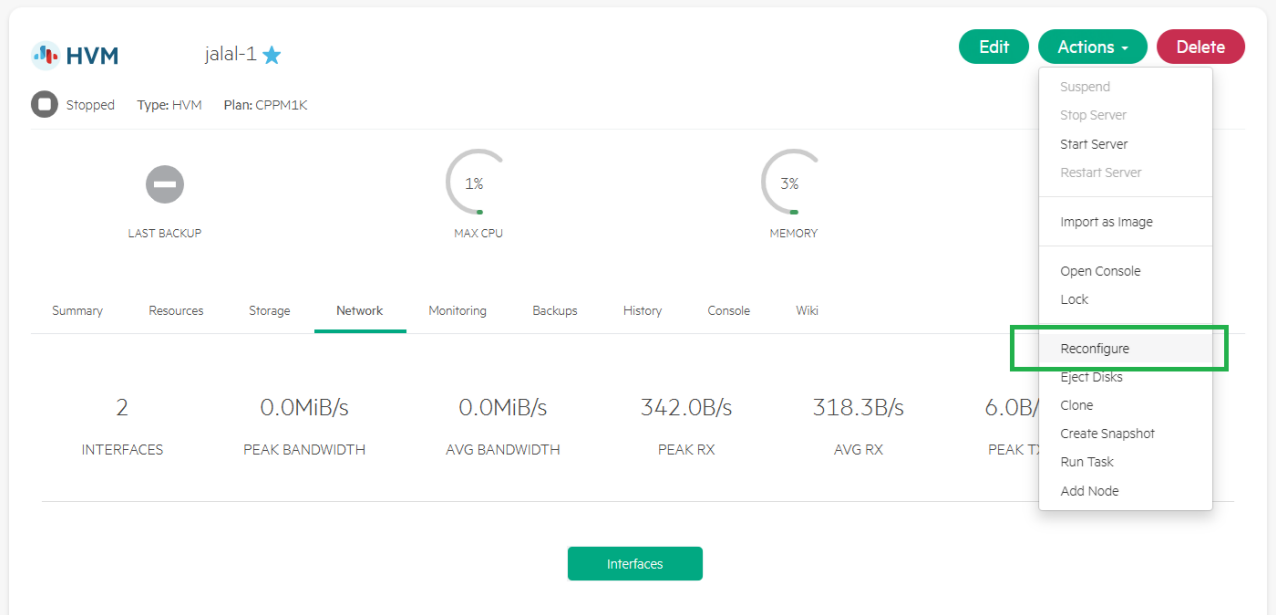
1. Log in to the VME (HPE Morpheus) management console and navigate to Provisioning > Instances. Select the instance you want to morph, then click Actions > Stop Server to stop it.

Figure 1. Stopping the server



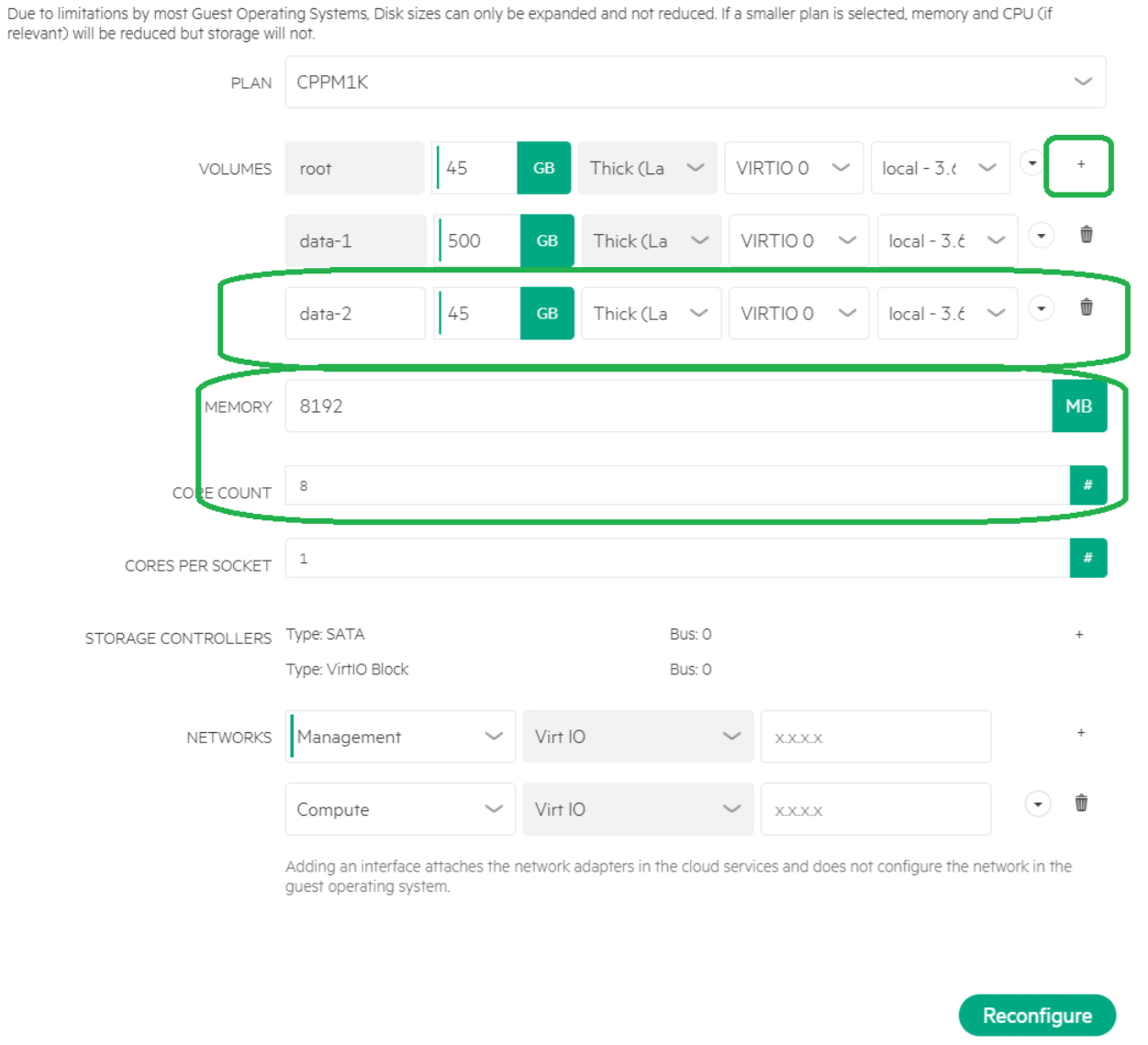
2. Click Actions > Reconfigure.

Figure 2. Reconfigure Menu



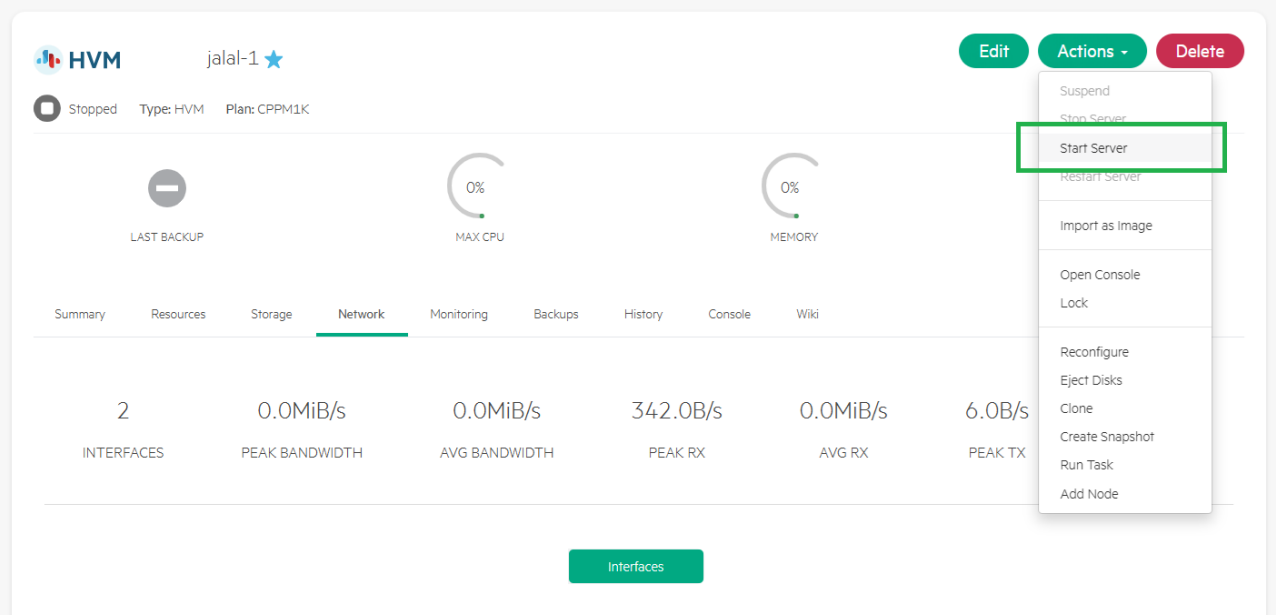
3. In the Reconfigure Instance dialog, add a new disk for the target model:
   a. Click the + option in the Volumes section to add a new disk.
   b. Provide the disk Name, the Size (GB) required for the target model, and set the Storage Type to Thick (Lazy Zero).
   c. Adjust the Memory (RAM) and Core Count (CPU) to match the target appliance model you are morphing to.
   d. Click Reconfigure.

Figure 3. Reconfiguring an instance



4. Click Actions > Start Server to start the instance.

Figure 4. Starting the server



5. Wait for the instance to start. Wait until the instance has fully started before continuing.
6. Open the appliance console (Console tab) or connect via PuTTY/SSH, and log in to the ClearPass CLI.
7. Execute the morph command, specifying the target appliance model. For example, to morph to a C2000V:

   ```
   system morph-vm C2000V
   ```

   (Replace C2000V with the model you are morphing to.)

   Confirm the prompts to proceed. The appliance backs up its configuration, stops all services, and reboots to complete the morph to the target model.

Figure 5. Version Comparison and Warning

```
----------------------------

ERROR: Attach a harddrive with recommended capacity,
       reboot, and attempt the operation again.
[appadmin@Jalal-TEst]#
[appadmin@Jalal-TEst]#
[appadmin@Jalal-TEst]#
[appadmin@Jalal-TEst]#
[appadmin@Jalal-TEst]#
[appadmin@Jalal-TEst]# system morph-vm C2000V
Current Hardware Version - C1000V
New Hardware Version - C2000V

WARN: VM will be morphed to C2000V. Once initiated
you cannot revert to the existing version C1000V.

All services will be stopped and a reboot initiated to complete
this operation.
Press 'y' or 'Y' to proceed: y

Morph VM to C2000V

Required system configuration:
----------------------------
Number of CPUs = 8
Total Memory = 16 GB
Total Disk Size = 1000 GB
Total Disk Write Throughput = 1354 IOPS
Total Disk Read Throughput = 3930 IOPS
----------------------------


Current system configuration:
----------------------------
Number of CPUs = 10
Total Memory = 15.00 GB
Total Disk Size = 745 GB
Total Disk Write Throughput = 3301 IOPS
Total Disk Read Throughput = 9578 IOPS
----------------------------


WARN: All data on harddrive [SCSI ] will be erased.

Press 'y' or 'Y' to proceed: y
Backing up configuration and logs...
INFO - Starting backup. This may take some time...
INFO - Backup databases for AppPlatform
INFO - Backup databases for PolicyManager
```

Results

The ClearPass appliance is morphed to the target model. The additional disk is used to extend the appliance storage and the CPU/memory are updated to the target model's specifications.

After the appliance reboots, the Policy Manager CLI banner shows the updated System Model (for example, C2000V), confirming that the morph completed successfully.

Figure 6. Morphed System Model

```
**********************************************
 Policy Manager CLI v6.14(0),
        Copyright © 2026, Hewlett Packard Enterprise Development LP.
 Software Version : 6.14.0.371380

 Management IP Address : 10.97.38.37
 System Model          : C2000V
**********************************************


Jalal-TEst login: _
```

Virtual Appliance Installation: KVM

This chapter describes procedures for installing and upgrading ClearPass 6.11 on a KVM virtual appliance.

Subtopics

KVM Hypervisor Installation Process Overview
Meeting KVM Server Specifications
Installing ClearPass on a KVM Virtual Appliance
After Installing or Upgrading to ClearPass 6.14 on KVM Servers
Morphing a KVM Version

KVM Hypervisor Installation Process Overview

About this task

The process of installing ClearPass on a KVM virtual appliance is done in four stages:

Procedure

1. ClearPass 6.11 KVM software packages are distributed as Zip files. Download the software image from the **Download Software > ClearPass > Policy Manager > Current Release > KVM** folder on the Support site and unzip it to a folder on your KVM server to extract the files.
2. Import the virtual appliance and choose the import type.
3. Add the hard disk and configure the format, type, and size based on the requirements for your ClearPass virtual appliance. See the System Requirements for ClearPass 6.14 chapter in this Installation Guide.
4. Power on and configure the virtual appliance, and then restart it to complete the update to the latest version.

Instructions for these procedures are provided in Installing KVM Using the virt-manager User Interface .

NOTE

Direct upgrades to ClearPass 6.11.0 are only supported from 6.12.x or any hotfix applied to it. Direct upgrades from any versions below 6.12.x are not supported.

NOTE

When creating a database backup of a lower version of ClearPass to be restored on ClearPass 6.11, best practice is to create separate backups for the configuration (tipsdb) and Insight (insightdb) databases. Backup files can be very large, and creating separate backups ensures that they will be restored correctly without overloading the system and blocking the restore operation.

NOTE

Review the Release Notes for the current release before you upgrade ClearPass.

NOTE

Cloning a virtual appliance to facilitate a ClearPass deployment is not recommended or supported.

NOTE

The **system patch-rollback** command cannot be used after an upgrade to revert to an earlier major version.

Meeting KVM Server Specifications

Before you perform any installations, upgrades, or updates, ensure that your system meets all virtual appliance requirements. Please refer to System Requirements for

ClearPass 6.14 and carefully review all virtual appliance (VA) requirements, including functional IOP ratings, and verify that your system meets these requirements.

Be sure that your system meets the recommended specifications required for the ClearPass virtual appliance. The ClearPass KVM disk image is shipped as a 24 GB hard disk volume. This must be supplemented with additional storage/hard disk through the virtual machine settings by adding a new hard disk. The additional space required depends on the ClearPass virtual appliance version.

> **NOTE**
> If you do not add a new hard disk to the virtual appliance before it is powered on, it will continue to restart with kernel panics.

A KVM version can be morphed to a larger version by using the **system morph-vm** command. For more information, see Morphing a KVM Version in this Installation Guide, and the **Command Line Interface > System Commands > system morph-vm** section in the ClearPass Policy Manager 6.11 User Guide.

Installing ClearPass on a KVM Virtual Appliance

There are two methods available for installing KVM: Through the Virtual Machine Manager (virt-manager) graphical user interface, or through the **virt-install** command line tool.

Subtopics

Installing KVM Using the virt-manager User Interface
Installing KVM Using the virt-install Command Line Interface

Installing KVM Using the virt-manager User Interface

This section describes how to set up a ClearPass virtual appliance using virt-manager. After you download and unzip the ClearPass 6.11 KVM software package Zip files, follow the instructions in this section to deploy the ClearPass files, set the disk bus type, add a new hard disk, and power on and configure the virtual appliance.

Subtopics

About virt-manager
Import the Virtual Appliance
Set the Disk Bus Type
Add a Hard Disk to the KVM Virtual Appliance
Power On and Configure the KVM Virtual Appliance

About virt-manager

The virt-manager application is a desktop user interface for managing KVM virtual appliances.

On RHEL distributions this can be installed using the following command:

#yum install virt-manager

For more information on the virt-manager virtual machine manager, including download instructions, documentation, and supporting tools, refer to **https://virt-manager.org/**.

Import the Virtual Appliance

Procedure

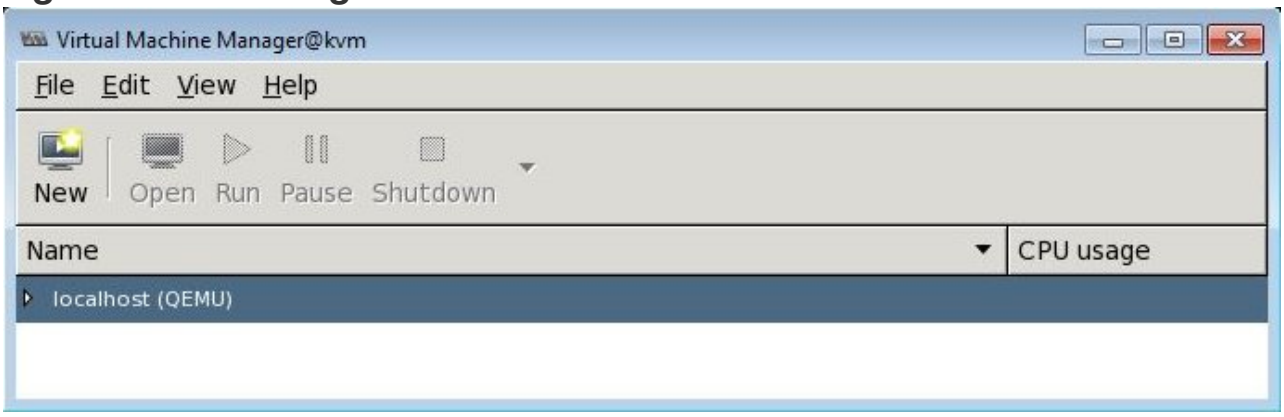
1. To open the virt-manager application window from the shell, enter the following command:

   #virt-manager &

   > **NOTE**
   >
   > If trying to run this command from an ssh session, first make sure to ssh to the KVM server using the **-X** option to enable X11 forwarding. For example: **ssh -X <username>@<host_IP>**

2. Connect to the **localhost**, and then right-click and select **New**.

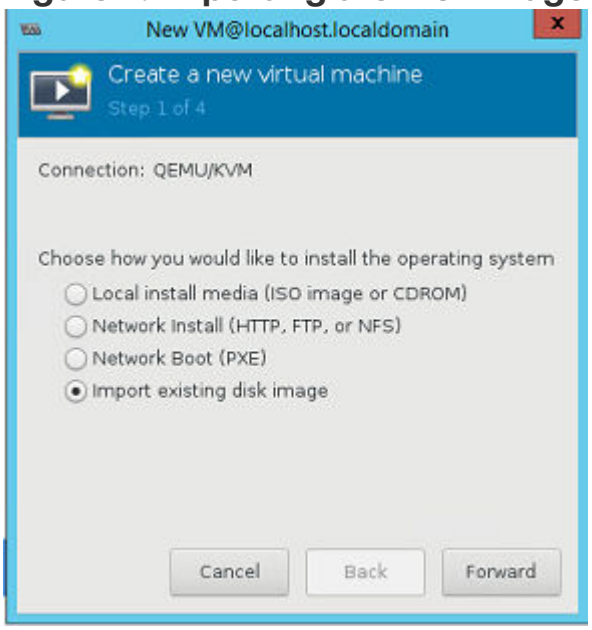
Figure 1. Connecting to the localhost



The **Create a new virtual machine** window opens.

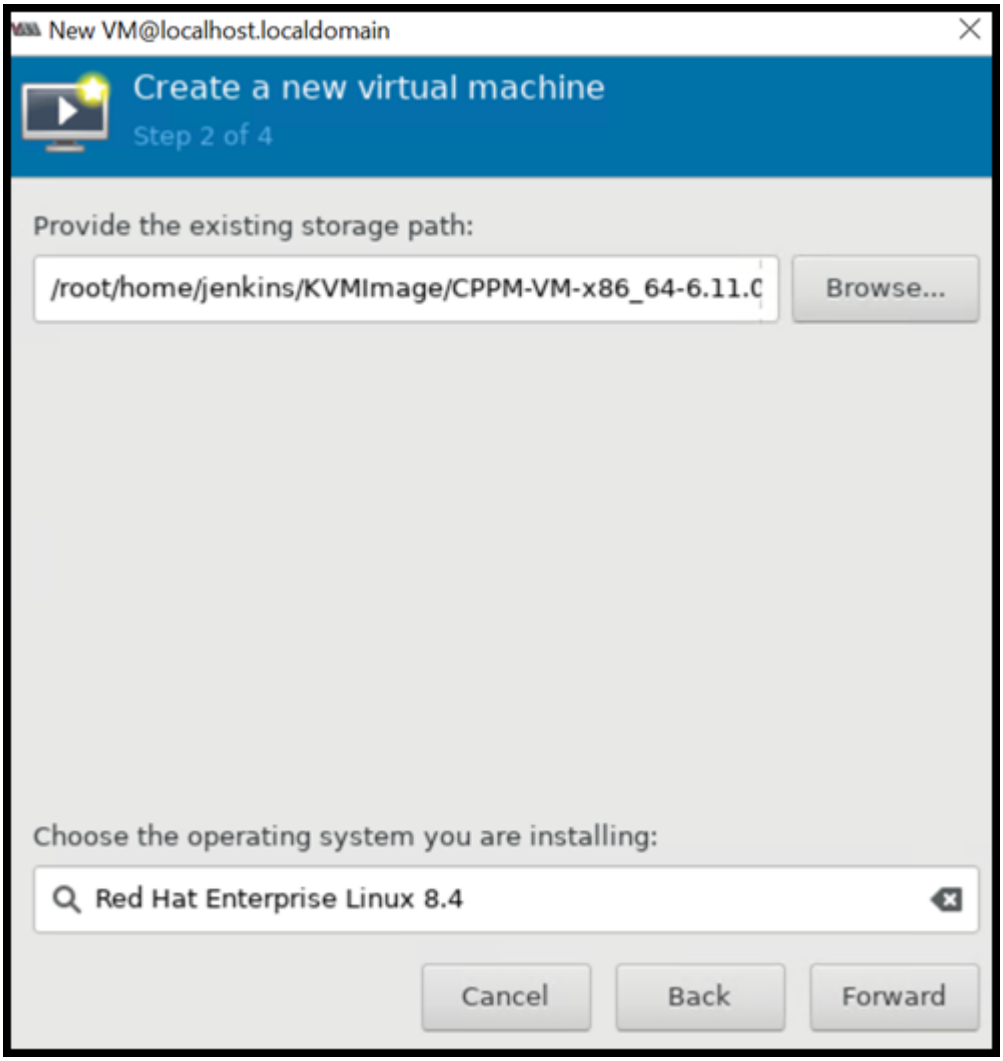
3. On the **Step 1 of 4** page, select the **Import existing disk image** option. Click **Forward**.

Figure 2. Importing the Disk Image



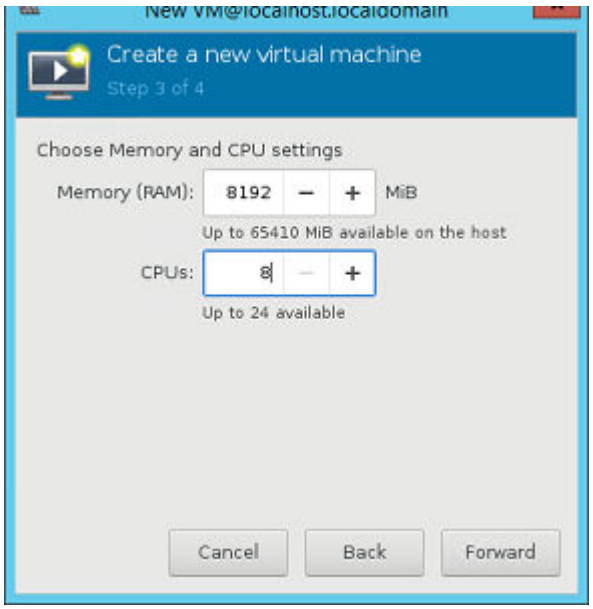
4. On the **Step 2 of 4** page, in the **Provide the existing storage path** field, browse to select the existing disk image that was present in the Zip file. In the **OS type** field select **Linux**, and in the **Version** field select **Red Hat Enterprise Linux 6**. Click **Forward**.

Figure 3. Selecting the Disk Image and Specifying the OS and Version



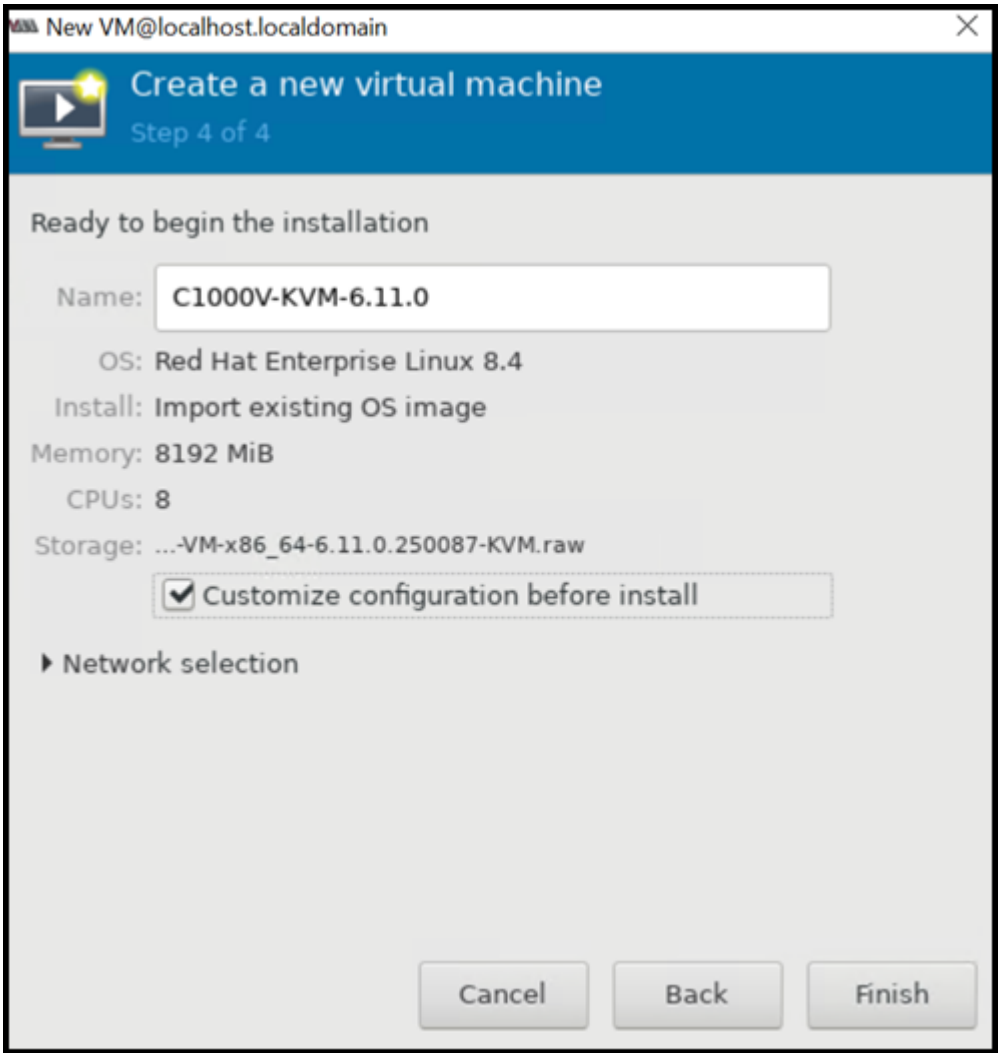
5. On the **Step 3 of 4** page, enter the correct values for **Memory (RAM)** and **CPUs** according to the ClearPass requirements. See Recommended KVM Server Specifications .

Figure 4. Specifying Memory and CPUs



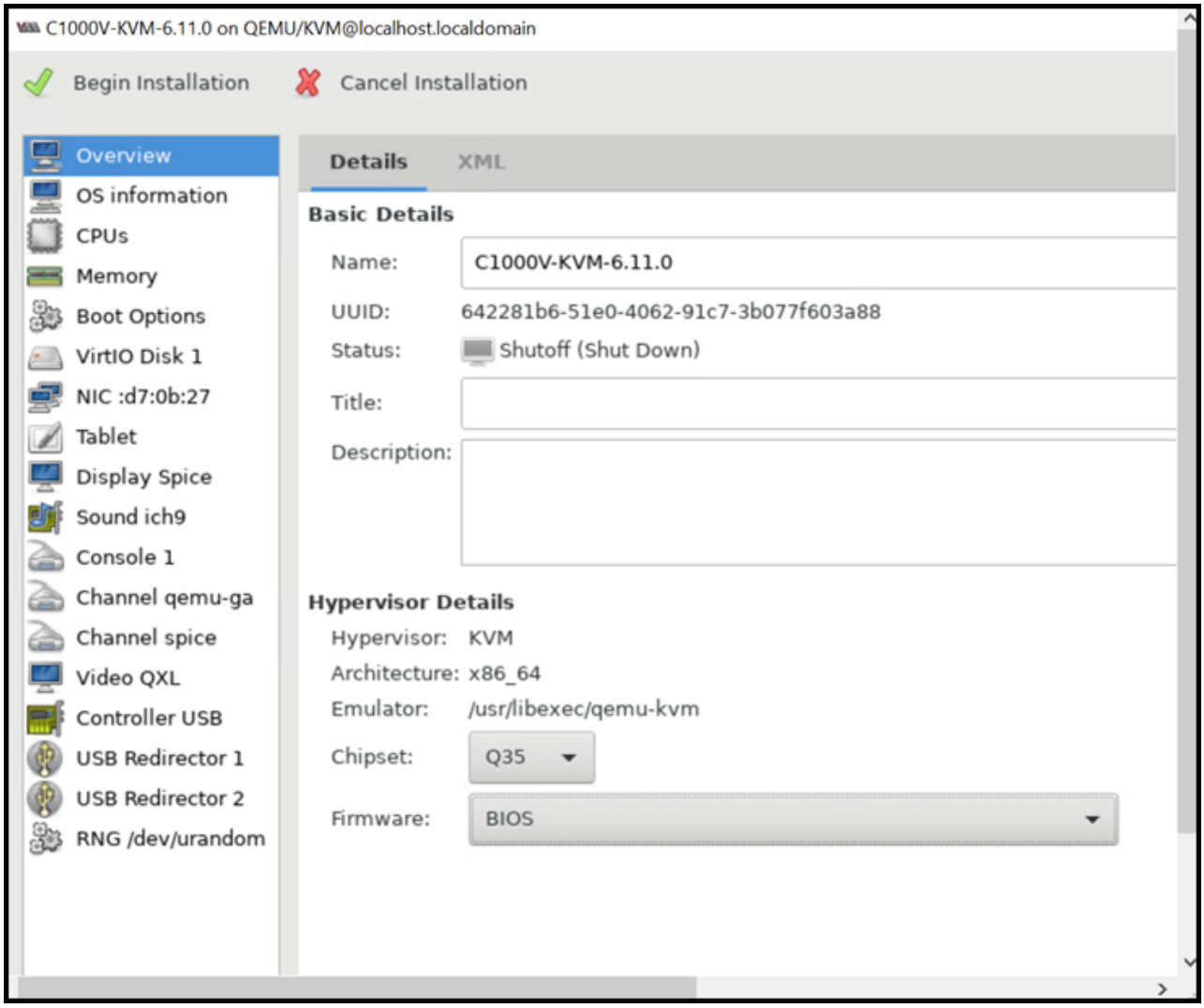
6. On the **Step 4 of 4** page, select the **Customize configuration before install** check box and enter a name for the new virtual appliance in the **Name** field. Under **Network selection**, verify the network interface that the virtual appliance will be connected to. This will be the Bridge interface that was created on the KVM server. Click **Finish**.

Figure 5. Verifying the Network Interface (Assign VM Name)



7. The **Overview** screen is displayed, where you can review a summary of basic details of the new virtual appliance. You may also edit the appliance's name or add a description.

Figure 6. Reviewing Basic Details on the Overview Screen



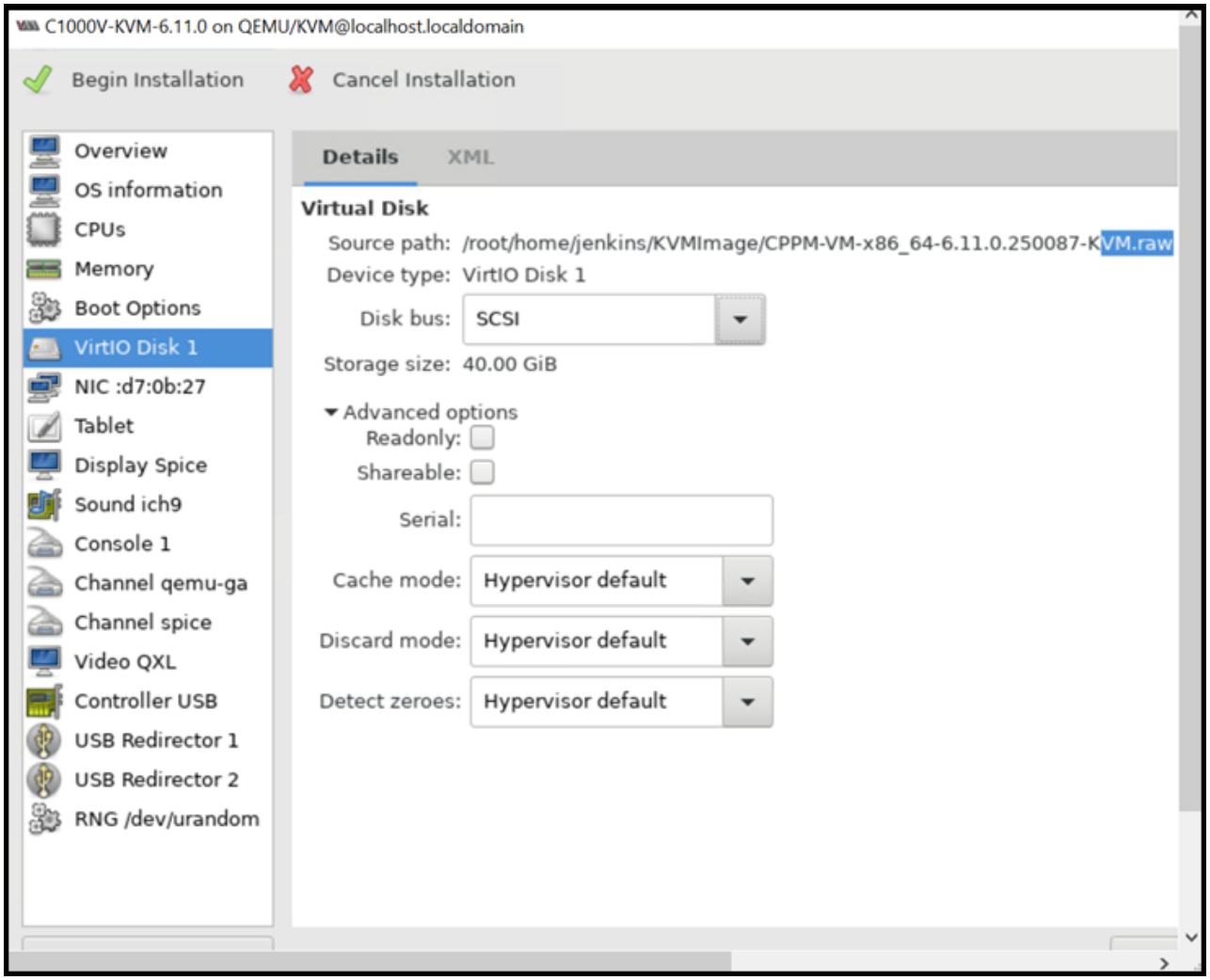
Continue to the steps in the next section, Set the Disk Bus Type .

Set the Disk Bus Type

Procedure

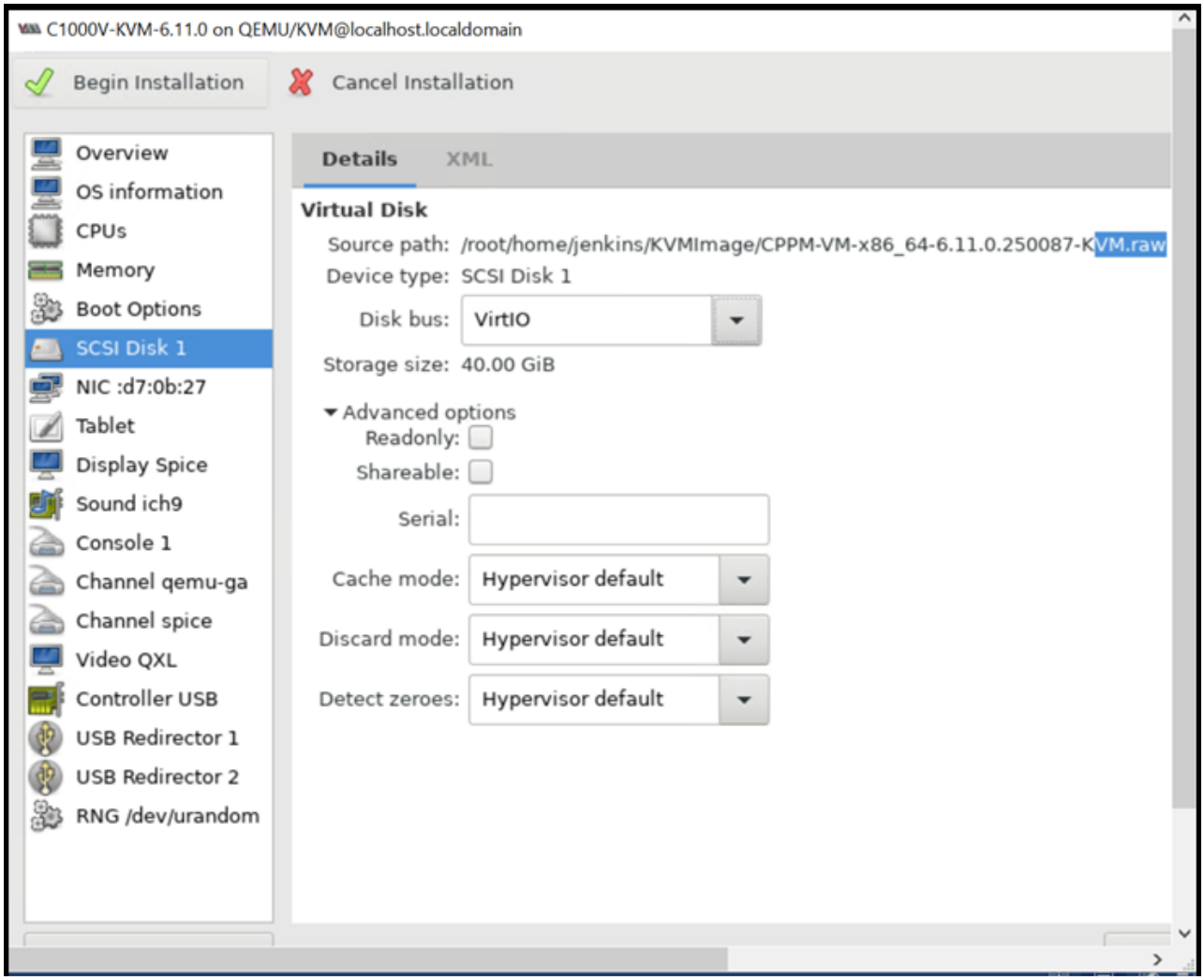
1. In the left menu of the virt-manager interface, select **VirtIO Disk 1**. The **Virtual Disk** page displays.
2. Click **Advanced options**.

Figure 1. Virtual Disk Page, Advanced Options



3. In the **Disk bus** field, change the setting to **SCSI** for the ClearPass 6.11 image.
4. In the **Storage format** drop-down list, select **raw**.

Figure 2. Virtual Disk Page, Disk Bus and Storage Format Options



5. Click **Apply**. The **Overview** window is displayed again, and the disk is included in the left menu. Continue using the steps in the next section, Add a Hard Disk to the KVM Virtual Appliance .

Add a Hard Disk to the KVM Virtual Appliance

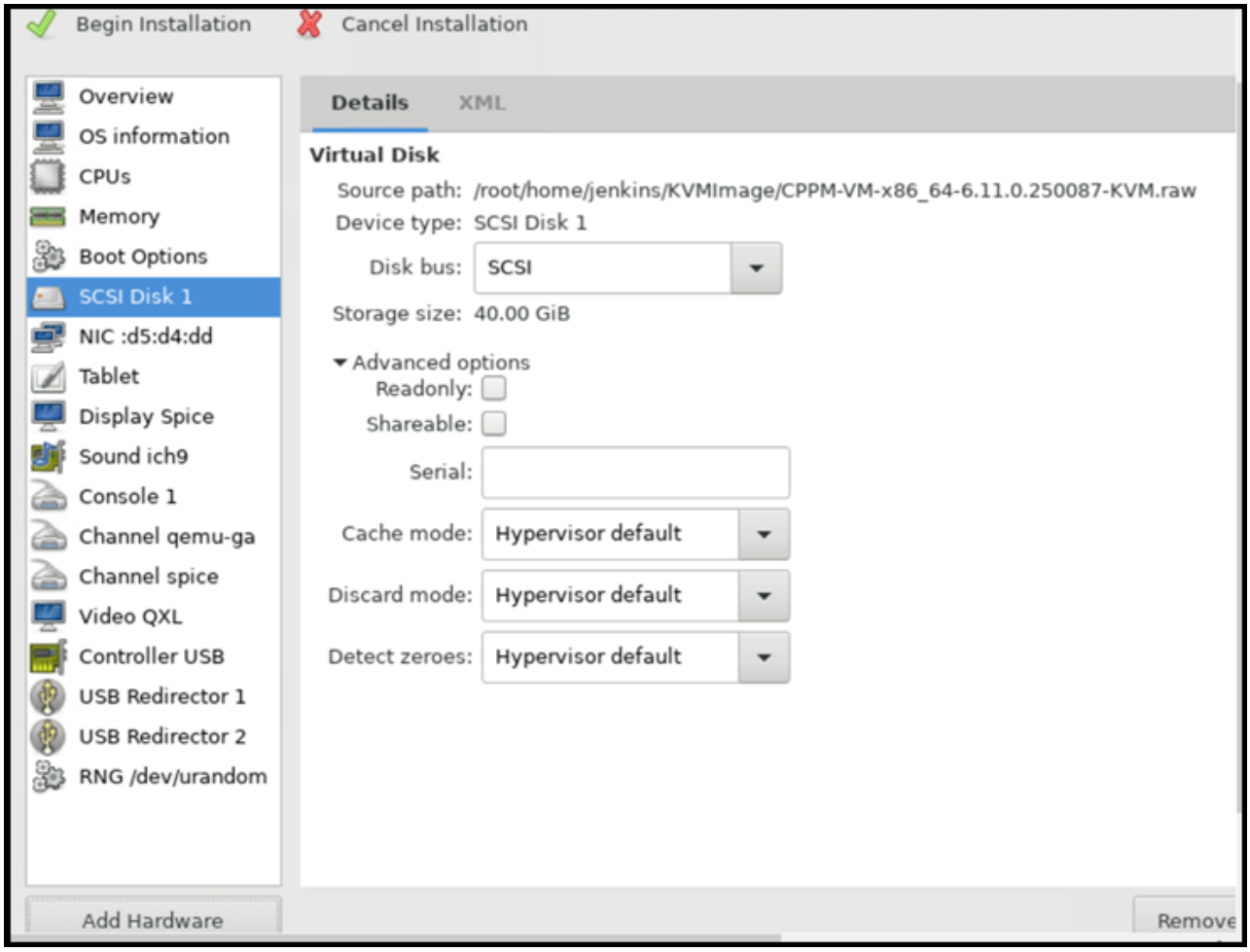
About this task

After the virtual appliance has been imported, add the hard disk.

Procedure

1. On the overview screen, click **Add Hardware**.

Figure 1. The Add Hardware Button

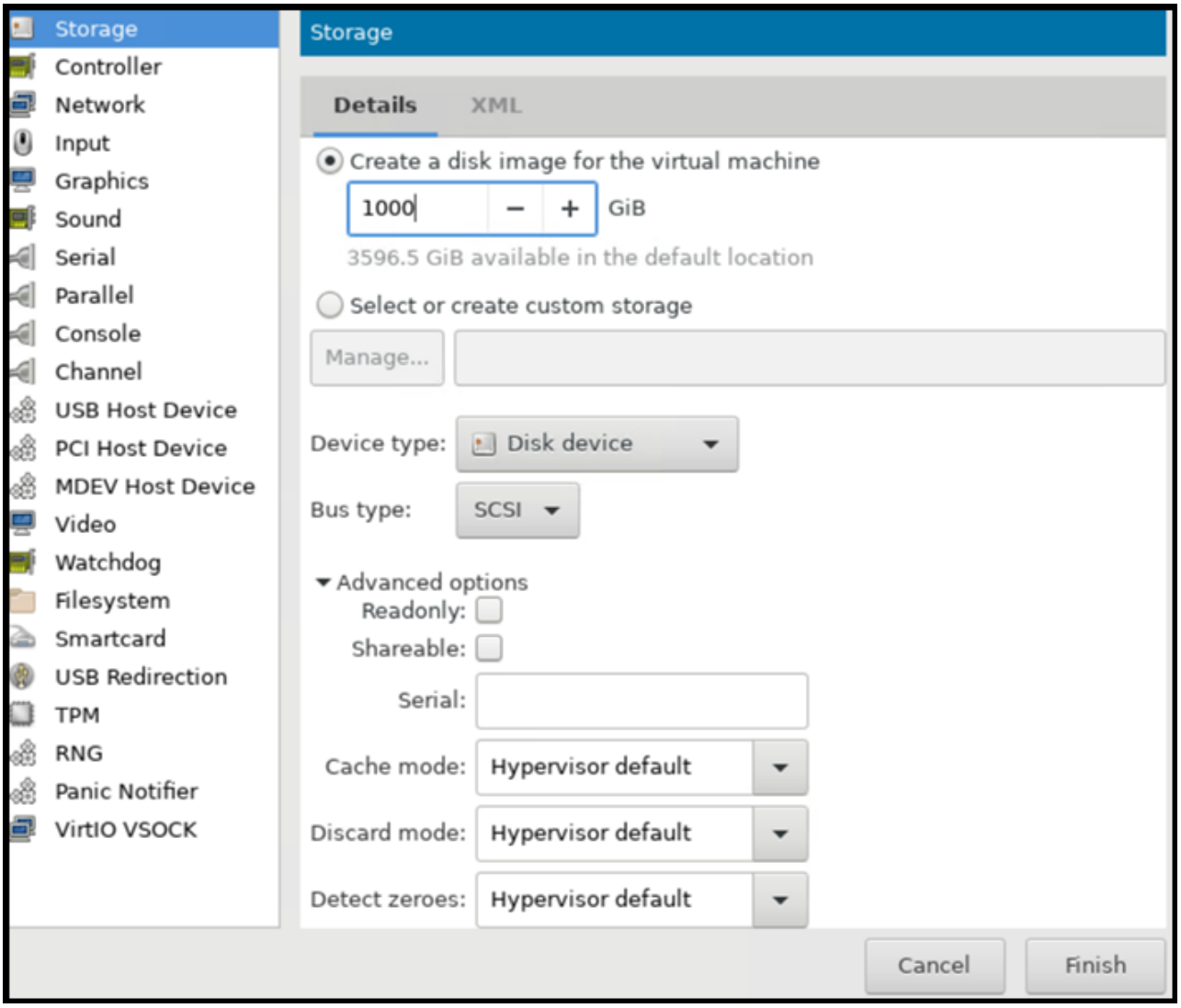


The **Add New Virtual Hardware** window opens with the **Storage** page displayed.

2. On the **Storage** page, make the following selections:

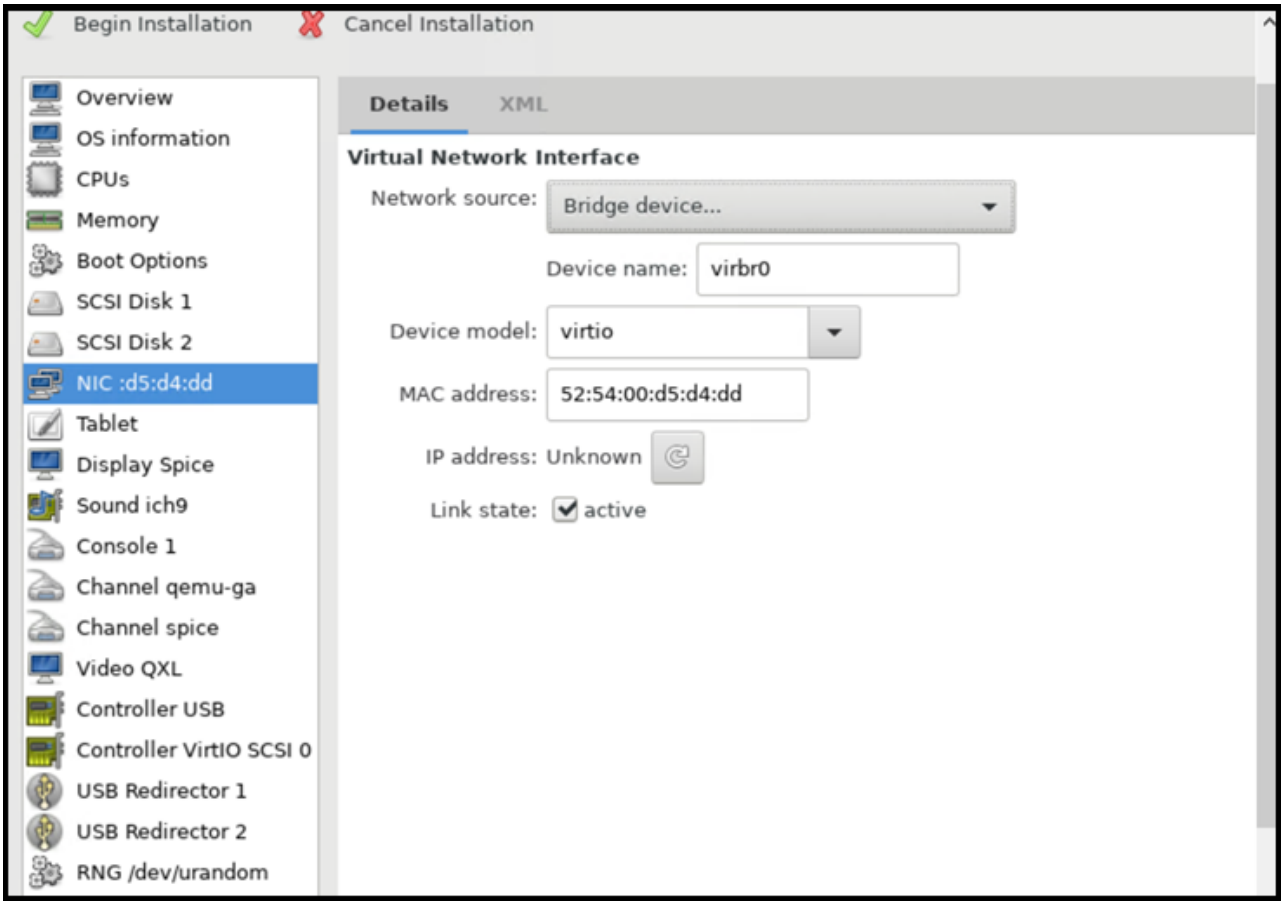
   - In the **Create a disk image on the computer's hard drive** field, enter the disk size required by the ClearPass appliance type. See Meeting KVM Server Specifications .
   - In the **Device type** and **Disk bus** fields, select **IDE**.

Figure 2. Storage Settings



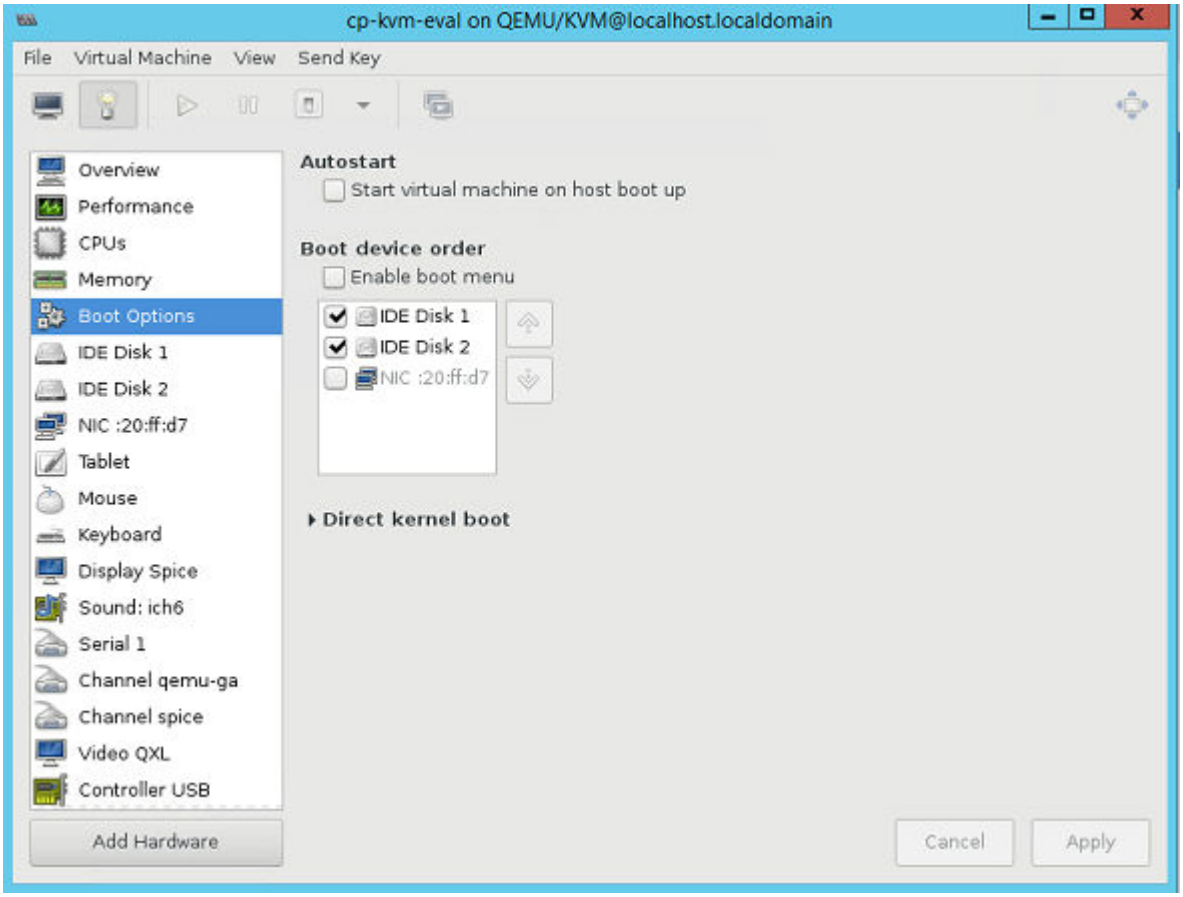
3. In the left menu of the **Add New Virtual Hardware** page, select **Network**. In the **Host device** field, select **Host device vnet7 (Bridge 'br0')**.

Figure 3. Network Settings



4. After specifying the host device on the **Network** page, click **Finish**. The **Add New Virtual Hardware** window closes and the overview window is displayed again.
5. Now that all the hardware has been added, in the left menu of the overview screen select **Boot Options**.
6. In the **Boot device order** area, make sure that **Hard Disk** is selected.
7. Click **Apply**.

Figure 4. Boot Options, Hard Disk



Continue to the steps in the next section, Power On and Configure the KVM Virtual Appliance .

Power On and Configure the KVM Virtual Appliance

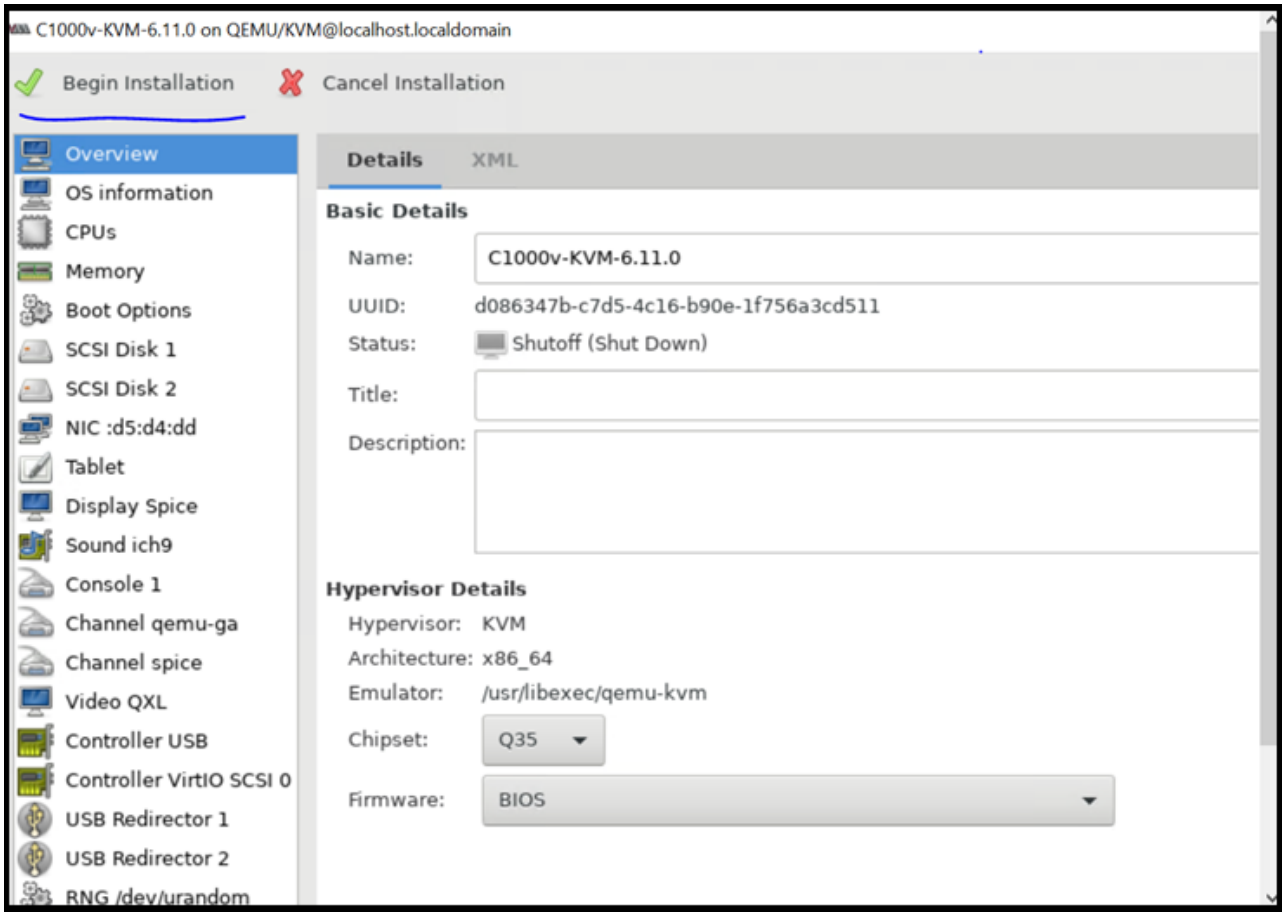
About this task

After you have created the ClearPass KVM virtual appliance and added the hard disk, you will power on the new virtual appliance and start deploying it.

Procedure

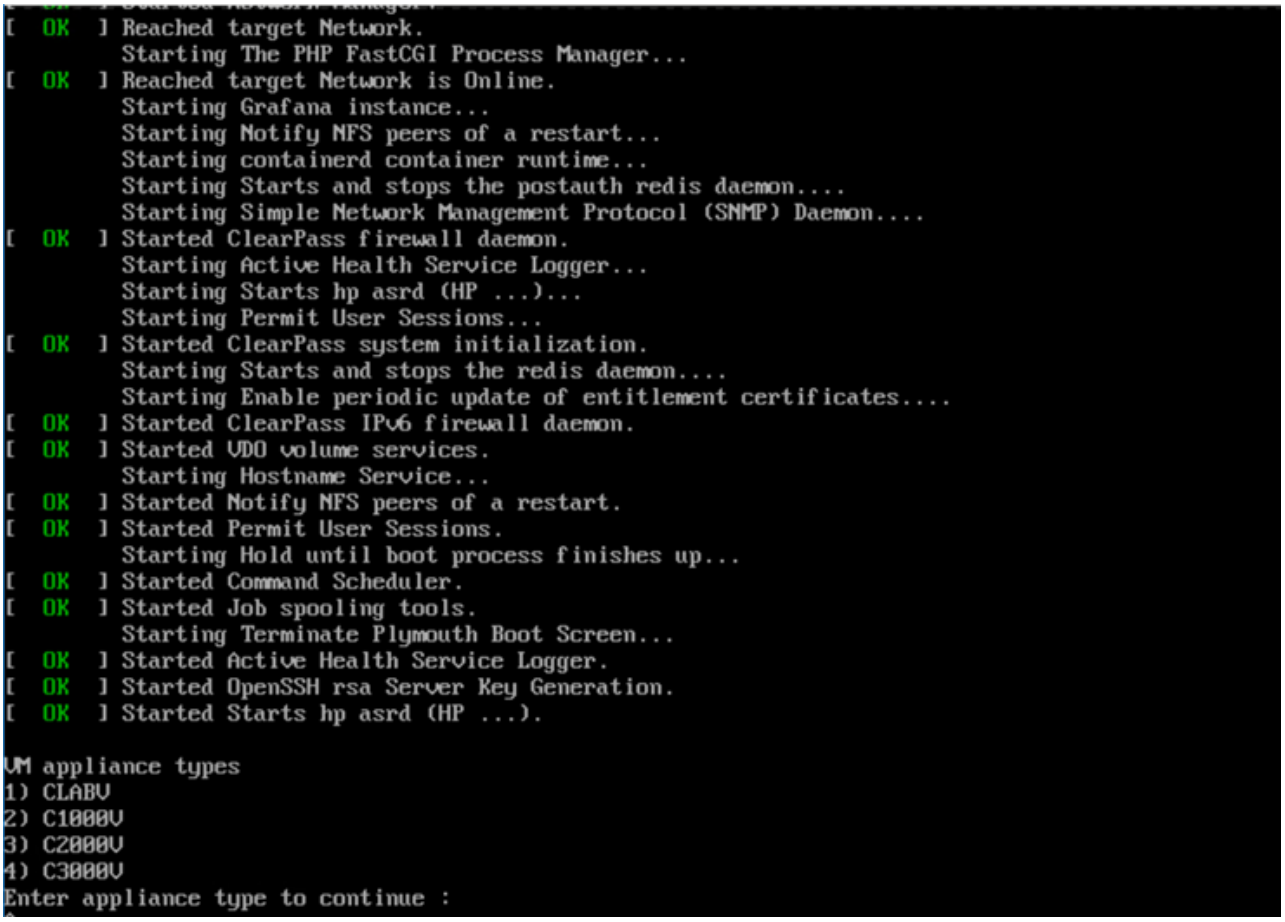
1. On the virt-machine overview window, click **Begin Installation**.

Figure 1. The Begin Installation Link



The command line window opens. You should see the following:

Figure 2. “Enter appliance type to continue”

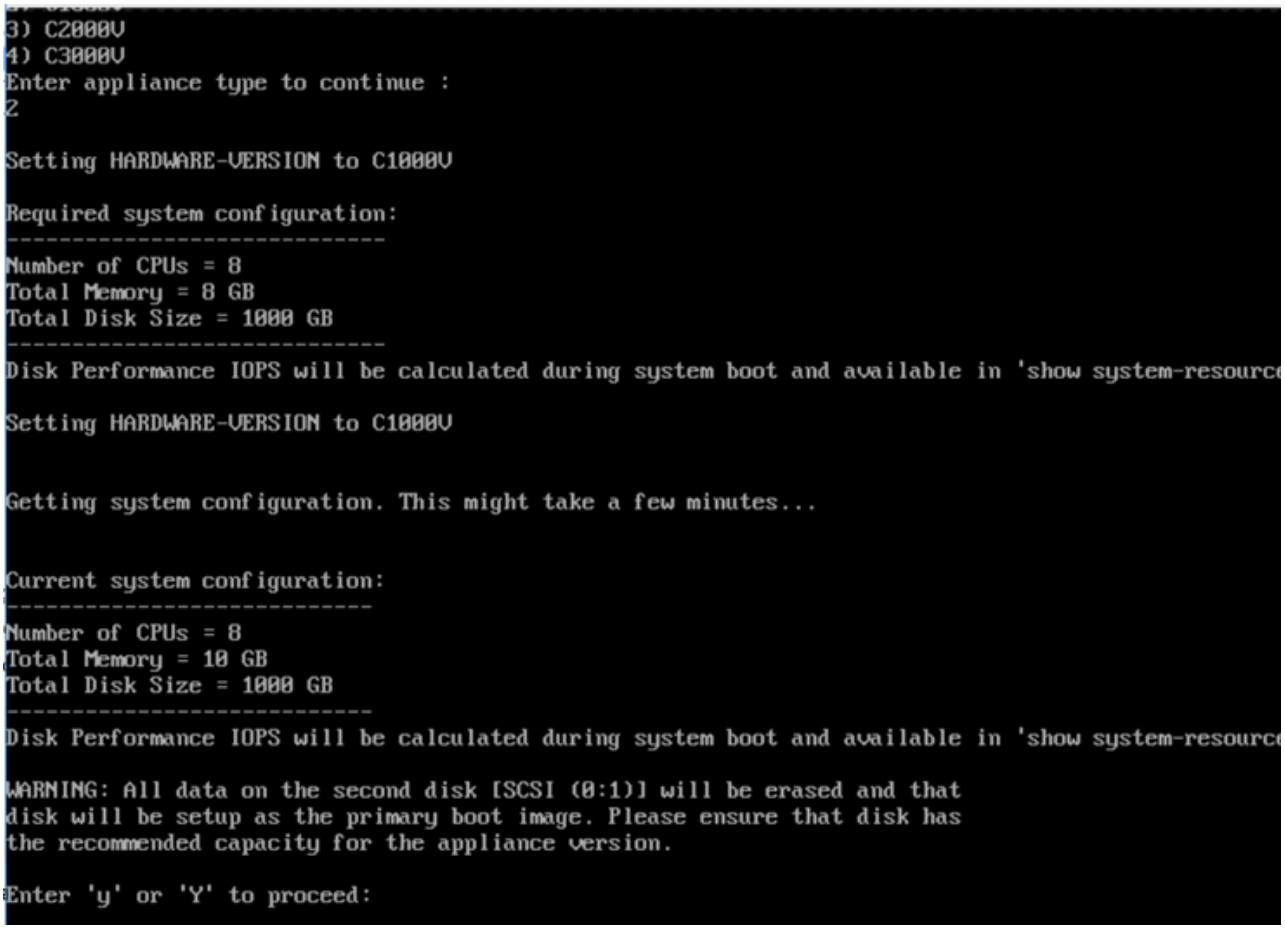


2. Enter the list number for the appropriate appliance type (do not enter the appliance model itself). Options include:

   - **1) CLABV**
   - **2) C1000V**
   - **3) C2000V**
   - **4) C3000V**

   For example, to install a C1000V, enter the number **2**.

Figure 3. Number Entered to Indicate Appliance Option



3. The system requirements are displayed for the appliance model you entered, along with your current system configuration. Compare these to make sure your system meets the new system requirements. For more information, see Meeting KVM Server Specifications .

Figure 4. “Enter y or Y to proceed”

```
Setting HARDWARE-VERSION to C1000V

Required system configuration:
----------------------------
Number of CPUs = 8
Total Memory = 8 GB
Total Disk Size = 1000 GB
----------------------------
Disk Performance IOPS will be calculated during system boot and available in 'show system-resourc

Setting HARDWARE-VERSION to C1000V


Getting system configuration. This might take a few minutes...


Current system configuration:
----------------------------
Number of CPUs = 8
Total Memory = 10 GB
Total Disk Size = 1000 GB
----------------------------
Disk Performance IOPS will be calculated during system boot and available in 'show system-resourc

WARNING: All data on the second disk [SCSI (0:1)] will be erased and that
disk will be setup as the primary boot image. Please ensure that disk has
the recommended capacity for the appliance version.

Enter 'y' or 'Y' to proceed:
```

4. After you have verified that your system meets the new requirements, enter **y**.
5. If desired, enter **y** to encrypt all local data.
6. The ClearPass 6.11.0 setup and installation begins. You should see the following information, and ClearPass will reboot at least once.

Figure 5. Initializing Disk

```
Do you wish to encrypt all local data press 'Y' or 'y'.
Note: Yes (Y/y) is recommended unless virtual system encryption is already enabled.
      This setting cannot be changed after installation.

Press Y/y to encrypt or any other key to skip encryption

Continuing without disk encryption...

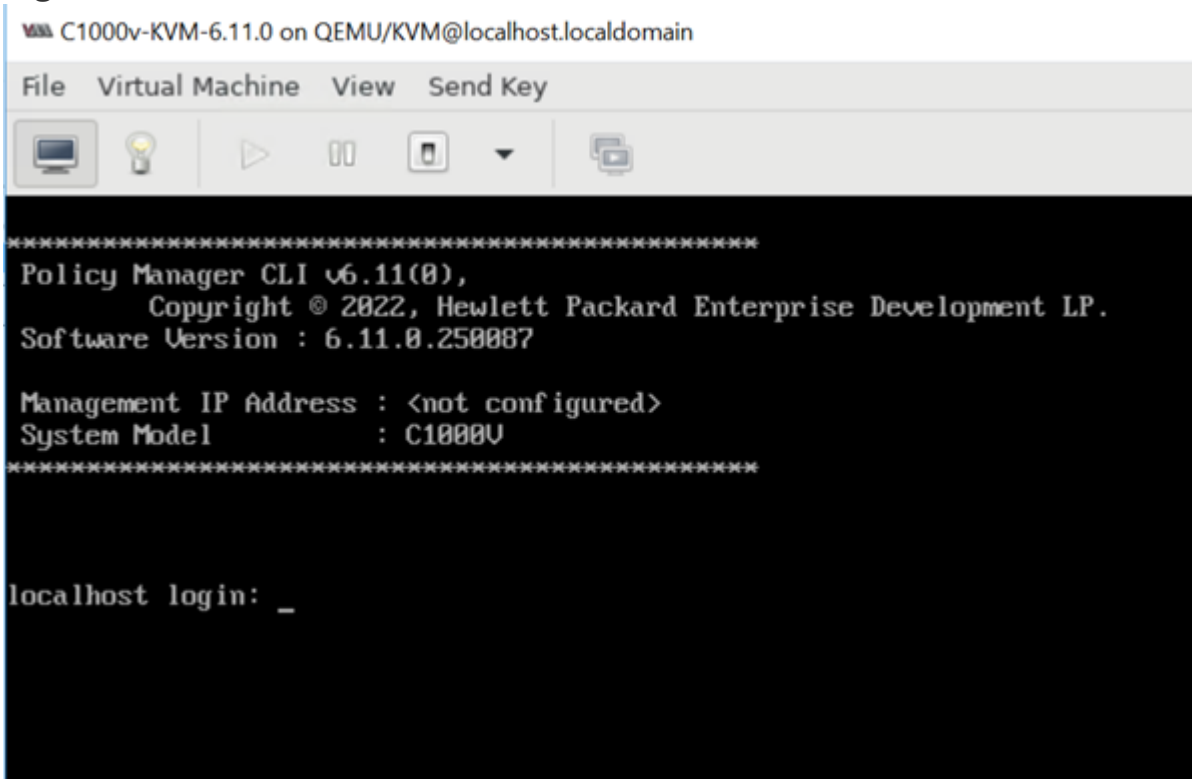

***
*** Initializing disk...
***


Setting up partitions on /dev/sdb...
```

After that reboot, the ClearPass virtual appliance is configured, and will power on and boot up within a couple of minutes. The whole process from deploying the image to the final startup screen should take between 30 and 40 minutes.

7. After the ClearPass virtual appliance launches correctly, you should see the following banner:

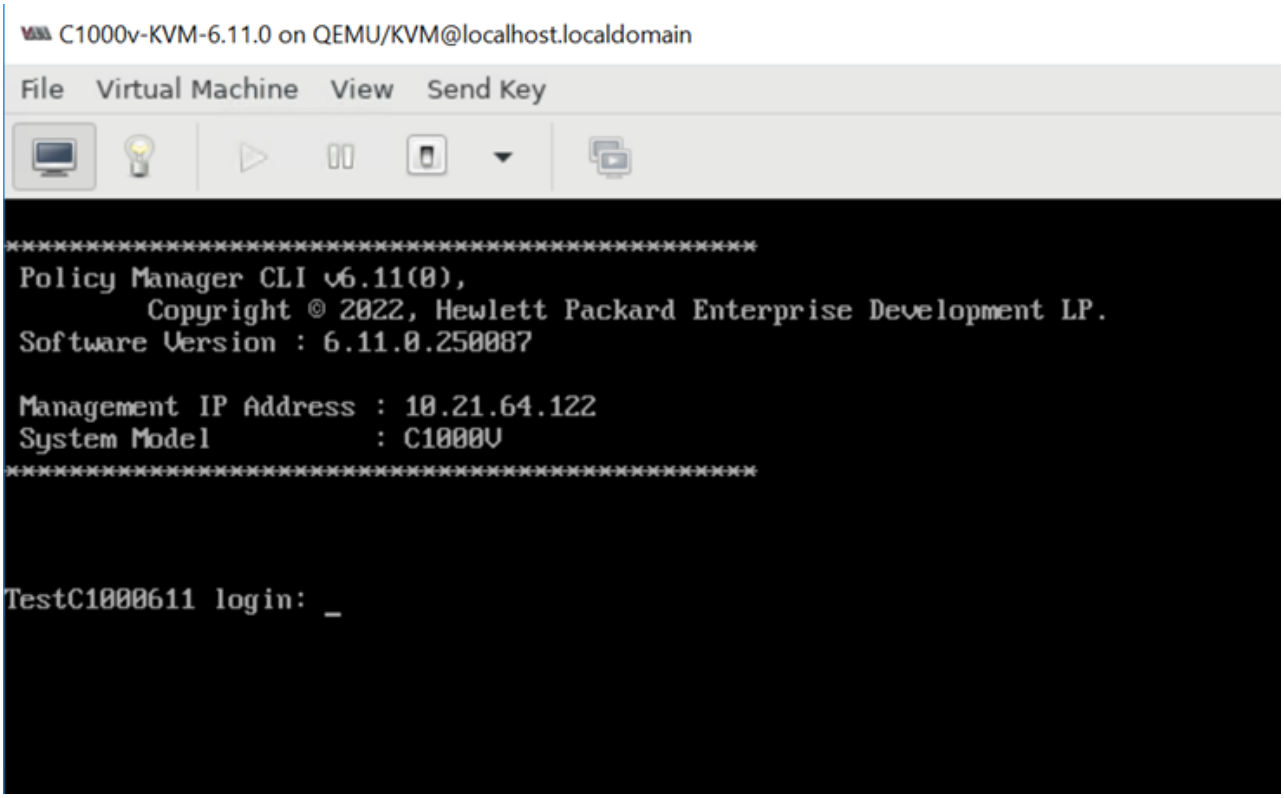
Figure 6. Banner



At this point, the 6.11.0 updates have been downloaded but are not yet installed. When you see the Power-On banner, log in to the local host with the appropriate credentials.

8. After the initial configuration has been completed, you will be prompted to log in to the CLI again using the new password.

Figure 7. Prompt to Log In with the New Password



Installing KVM Using the virt-install Command Line Interface

This section describes how to set up the ClearPass virtual appliance through the virt-install command line tool, with a hard disk image in RAW format.

The virt-install tool is used to create new KVM virtual appliances and uses the "libvirt" hypervisor management library. The create-kvm-cp.sh script shown below incorporates the **virt-install** command. This script can be used to create a virtual appliance by importing an existing disk image. This skips the OS installation step and uses the specified disk image based on the path provided ((/home/KVMImage/CPPM-VM-x86_64-6.11.0.250087--KVM.raw); therefore the disk image must be saved on the KVM server. The second disk is created in the path as specified, and the size of this disk must be specified.

Subtopics

The create-kvm-cp.sh Script
Creating the ClearPass Virtual Appliance with virt-install

The create-kvm-cp.sh Script

```
#!/bin/bash
if [ $# -ne 3 ] ; then
    echo "Usage: $0 <NAME> <RAM_GB> <CPUs>"
    exit 1
fi
NAME=$1
RAM_MB=$(($2*1024))
VCPUS=$3
virt-install                              \
    --connect qemu:///system       \
    --virt-type kvm                \
    --os-type=linux                \
    --os-variant=rhel8.4           \
    --name=$NAME                \
    --ram=$RAM_MB               \
    --arch=x86_64               \
    --disk path=<your local path>/CPPM-VM-x86_64-6.12.0.300732-
KVM.raw,format=raw,bus=scsi,size=40  \
    --disk path=<your local path>/new_612.raw,format=raw,bus=scsi,size=500 \
    --network bridge=br0,model=virtio     \
    --network bridge=br1,model=virtio     \
    --vcpus=$VCPUS              \
    --graphics vnc              \
    --import
                  \
```

Creating the ClearPass Virtual Appliance with virt-install

The virt-install tool provides a number of options that can be passed on the command line. To see a complete list of options, run the following command:

virt-install --help



NOTE

In order for **virt-install** commands to complete successfully, you must have root privileges. The **man virt-install** page also documents each command option and important variables.

Subtopics

Example: Using virt-install to Install a Red Hat Enterprise Linux 5 ClearPass Virtual Appliance

Example: Using virt-install to Install a Red Hat Enterprise Linux 5 ClearPass Virtual Appliance

This example creates a Red Hat Enterprise Linux 5 virtual appliance (in this example, **guest1** is the new virtual appliance):

```
virt-install \
   --name=guest1-rhel5-64 \
   --file=/var/lib/libvirt/images/guest1-rhel5-64.dsk \
   --file-size=8 \
   --nonsparse --graphics spice \
   --vcpus=2 --ram=2048 \
   --location=http://example1.com/installation_tree/RHEL5.6-Server-
x86_64/os \
   --network bridge=br0 \
   --os-type=linux \
   --os-variant=rhel5.4
```

Ensure that you select the correct **os-type** for your operating system when you run the **virt-install** command.

For more examples, refer to **man virt-install**.

After Installing or Upgrading to ClearPass 6.14 on KVM Servers

About this task

For new ClearPass 6.11.0 installations on a KVM hypervisor, or if you are upgrading ClearPass from 6.11.x to 6.11.0 on KVM, **and only** if the MAC address of **Network adapter1** is higher than that of **Network adapter2**, additional steps are required after the upgrade.

After upgrading, follow the steps below in order for ClearPass to have network connectivity:

Procedure

1. After you upgrade to 6.11.0, log in to the console as appadmin and use the CLI command **<system shutdown>** to shut down the ClearPass server. This step must be done only through the console.

2. After the command is executed, wait for the virtual appliance to shut down completely.
3. Edit the ClearPass virtual appliance settings in the virt-manager user interface and remove the two Ethernet adapters that are named **Network adapter1** and **Network adapter2**.
4. Add two new network adapters with the names **Network adapter1** and **Network adapter2** and of type **Ethernet Adapter**:
   - **Network adapter1** should be the management port connected to SwitchManagement. This network adapter should have the lowest MAC address.
   - **Network adapter2** should be the data port connected to SwitchData. This network adapter should have a higher MAC address than **Network adapter1**.

   NOTE

   After you add the two network adapters, verify that **Network adapter 1** has a lower MAC address than **Network adapter 2**. Interfaces map based on the lowest MAC address, such that the lowest MAC address becomes the management port and the second lowest MAC address becomes the data port.
5. Save the new settings and start the ClearPass virtual appliance.
6. Log in to the ClearPass console using the appadmin account, and then run the following CLI command to refresh the network settings:

   system refresh-network
7. After the refresh command is executed, reboot the ClearPass virtual appliance to establish network connectivity.

   NOTE

   When creating a database backup of a lower version of ClearPass to be restored on ClearPass 6.11, best practice is to create separate backups for the configuration (tipsdb) and Insight (insightdb) databases. Backup files can be very large, and creating separate backups ensures that they will be restored correctly without overloading the system and blocking the restore operation.

Morphing a KVM Version

About this task

> **NOTE**
> The illustrations in this section use the example of morphing a CLABV virtual appliance to a C1000V. Adjust your own configuration as needed.

> **NOTE**
> Starting with the 6.11.0 release, after the **system morph-vm** command is run during a ClearPass upgrade and the new disk is added, the existing hard disk cannot be removed. This is because the existing disk is used to extend the disk space for the server, and is included in the calculations to determine the required capacity of the additional hard disk.

A KVM virtual appliance can be morphed into a larger virtual appliance by using the **system morph-vm** command as follows:

Procedure

1. Power off the virtual appliance.
2. Open **Settings** and make the following modifications:
   a. Modify the **RAM** and **CPU** to match the recommended system requirements for the larger VA (see Meeting KVM Server Specifications).

      Figure 1. Hardware, RAM Settings

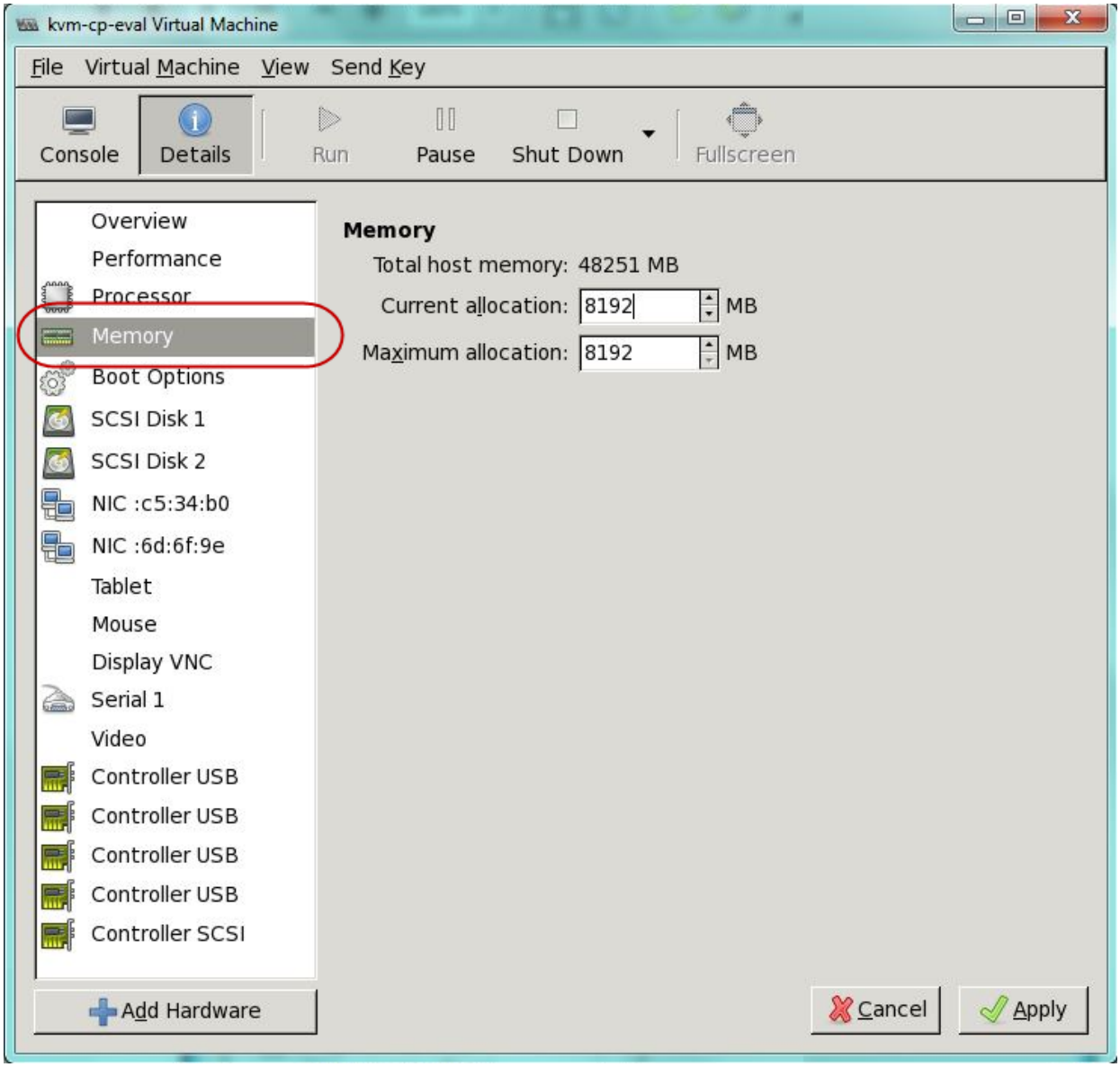
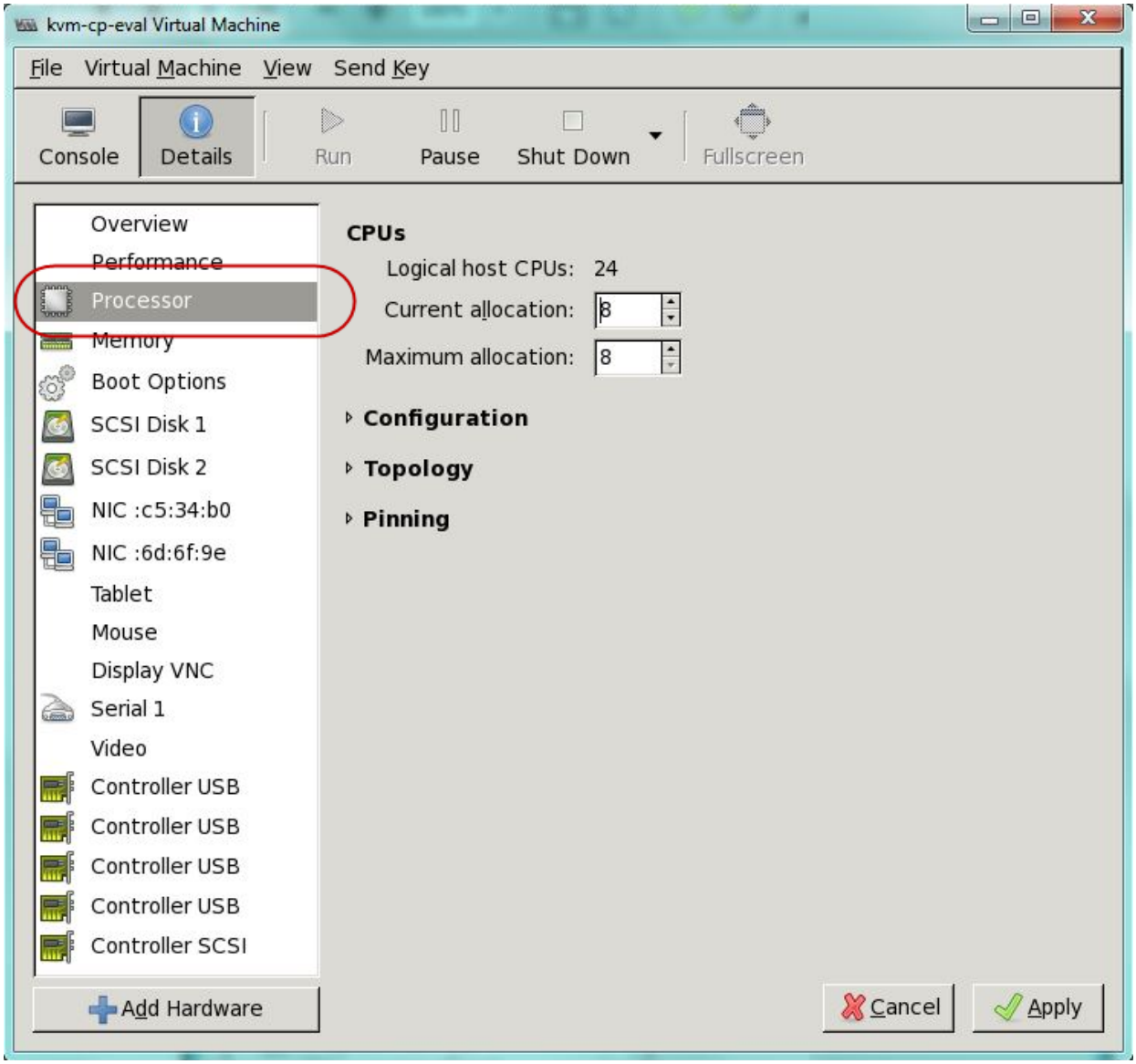


Figure 2. Hardware, CPU Settings



b. Add an additional disk with the recommended disk size for the larger VA:

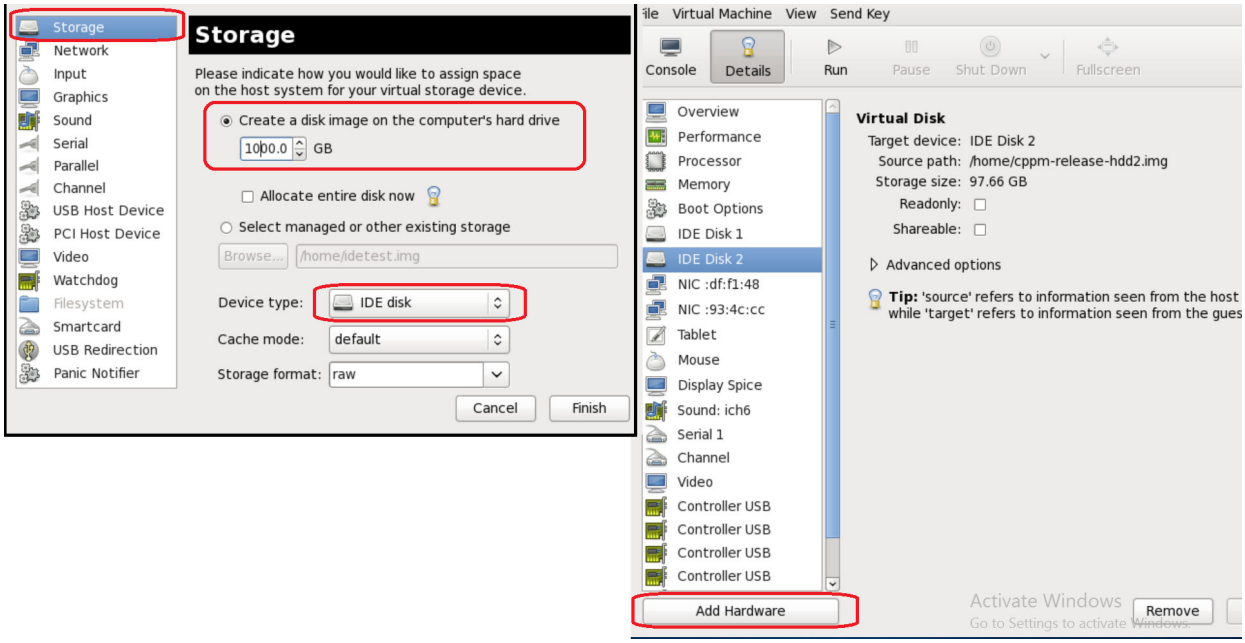
   i. Select **Add Hardware**, and then select **Storage**.

   ii. Create a disk image of the required size on the computer's hard drive.

   iii. In the **Device Type** field, select **IDE** disk.

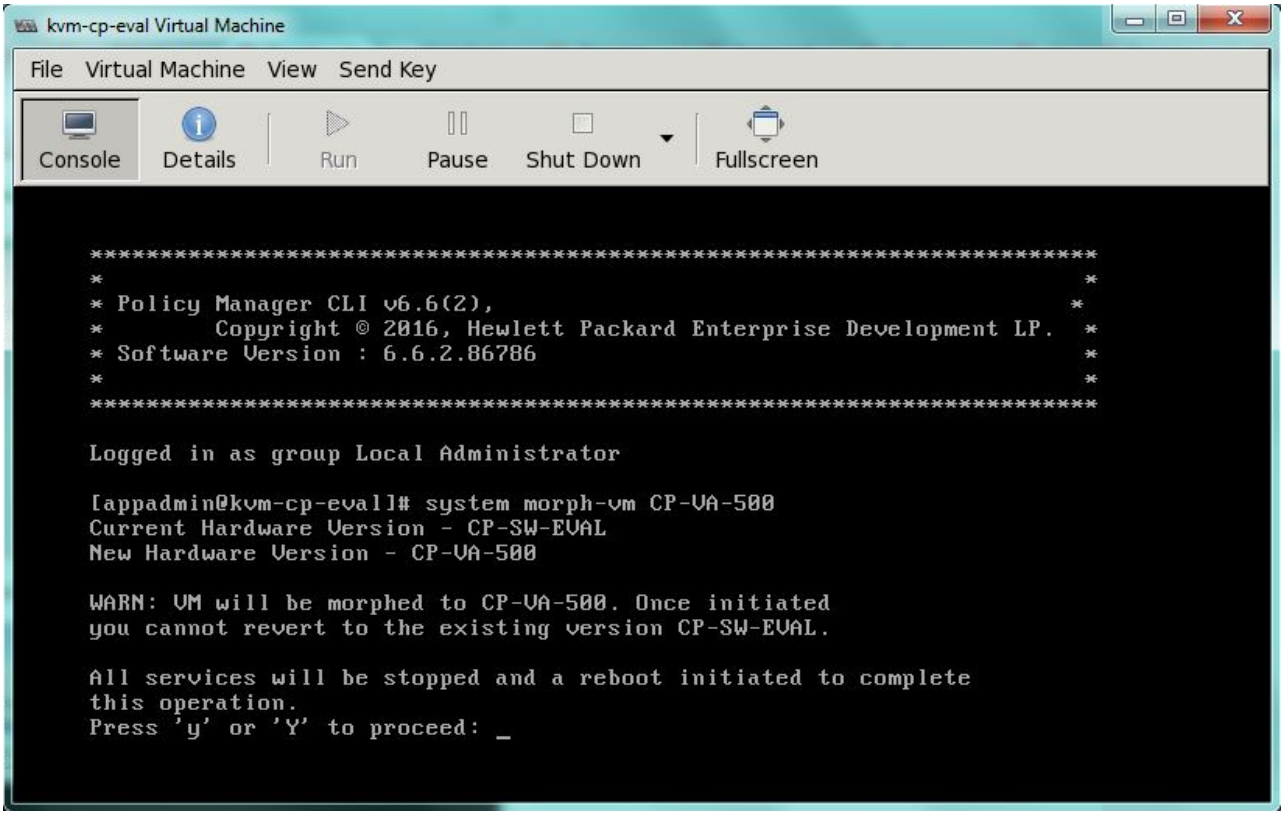
   iv. Click **Finish**.

Figure 3. Choose Disk Type and Size



3. After adding the hard disk, power on the original VA and, using SSH, log in as **appadmin**.
4. Run the command system morph-vm *<CxOOOV>*, replacing the "**x**" shown in this example with the appropriate VA size for your deployment, and then follow the prompts.
5. The existing hardware version and the new hardware version are displayed, along with a warning that you cannot change the setting after installation. When you are ready to proceed, press **y**.

Figure 4. Version Comparison and Warning



6. The system requirements are displayed for the appliance model you entered, along with your current system configuration. Compare these to make sure your system meets the new system requirements. For more information, see Meeting KVM Server Specifications .

   When you have verified that your system meets the new requirements, press **y**. The ClearPass setup and installation begins.

Virtual Appliance Installation: Microsoft Hyper-V

This section describes procedures for installing and upgrading ClearPass 6.11 on a Microsoft Hyper-V virtual appliance.

Subtopics

Hyper-V Hypervisor Installation Process Overview
Meeting Hyper-V Server Specifications
Installing ClearPass on a Hyper-V Virtual Appliance
Morphing a Hyper-V Version
Troubleshooting, Caveats, and Other Information for Hyper-V

Hyper-V Hypervisor Installation Process Overview

About this task

The process of installing ClearPass on a Microsoft Hyper-V virtual appliance (VA) is done in four stages:

Procedure

1. ClearPass 6.11 Hyper-V software packages are distributed as Zip files. Download the software image from the **Download Software > ClearPass > Policy Manager > Current Release > Hyper-V** folder on HPE Networking Support Portal and unzip it to a folder on your server to extract the files.
2. Import the virtual appliance and choose the import type.
3. Add the hard disk and configure the format, type, and size, based on the requirements for your VA. See System Requirements for ClearPass 6.14.
4. Power on and configure the VA.

Instructions for these procedures are provided in Installing Policy Manager on a Hyper-V Virtual Appliance .

NOTE

Review the Release Notes for the current release before you upgrade ClearPass.

NOTE

Starting with the ClearPass 6.11.0 release, ClearPass on Hyper-V is deployed as a Generation 2 virtual appliance. As part of this change, users should be aware that Generation 2 only supports UEFI boot instead of BIOS, so UEFI boot must be enabled.

CAUTION

Avoid using the CLI to update a Hyper-V instance to a higher 6.14.x version, and update it through the user interface instead. If you must use the CLI to update this instance type to a higher 6.14.x version, please increase the **CLI Session Idle Timeout** cluster-wide parameter value to 120 minutes to ensure that the patch update is not interrupted.

NOTE

When creating a database backup of a lower version of ClearPass to be restored on ClearPass 6.11, best practice is to create separate backups for the configuration (tipsdb) and Insight (insightdb) databases. Backup files can be very large, and creating separate backups ensures that they will be restored correctly without overloading the system and blocking the restore operation.

NOTE

Cloning a virtual appliance to facilitate a ClearPass deployment is not recommended or supported.

NOTE

Meeting Hyper-V Server Specifications

Before you perform any installations, upgrades, or updates, ensure that your system meets all virtual appliance requirements. Please refer to the System Requirements for ClearPass 6.14 chapter and carefully review all virtual appliance (VA) requirements, including functional IOP ratings, and verify that your system meets these requirements.

Be sure that your system meets the recommended specifications required for the ClearPass virtual appliance. The ClearPass VMware disk image is shipped as a 40 GB hard disk volume. This must be supplemented with additional storage/hard disk through the VMware settings by adding a new hard disk. The additional space required depends on the ClearPass virtual appliance version.

NOTE

Users should be aware that the maximum transmission unit (MTU) of a ClearPass Policy Manager cluster should not exceed 9000 on Microsoft Hyper-V when Hyper-V is configured for Jumbo Packets and when the cluster nodes are on separate Hyper-V servers.

NOTE

If you do not add a new hard disk to the virtual appliance before it is powered on, it will continue to restart with kernel panics.

A Hyper-V version can be morphed to a larger version by using the morph-vm command. For more information, see Morphing a Hyper-V Version in this Installation Guide, and the **Command Line Interface > System Commands > system morph-vm** section in the ClearPass Policy Manager 6.11 User Guide.

Installing ClearPass on a Hyper-V Virtual Appliance

After you download and unzip the ClearPass 6.11 Hyper-V software package Zip files, follow the instructions in this section to deploy the ClearPass files, add a new hard disk, and power on and configure the virtual appliance.

NOTE

The illustrations in this section use a CLABV appliance.

Subtopics

Import the Virtual Appliance

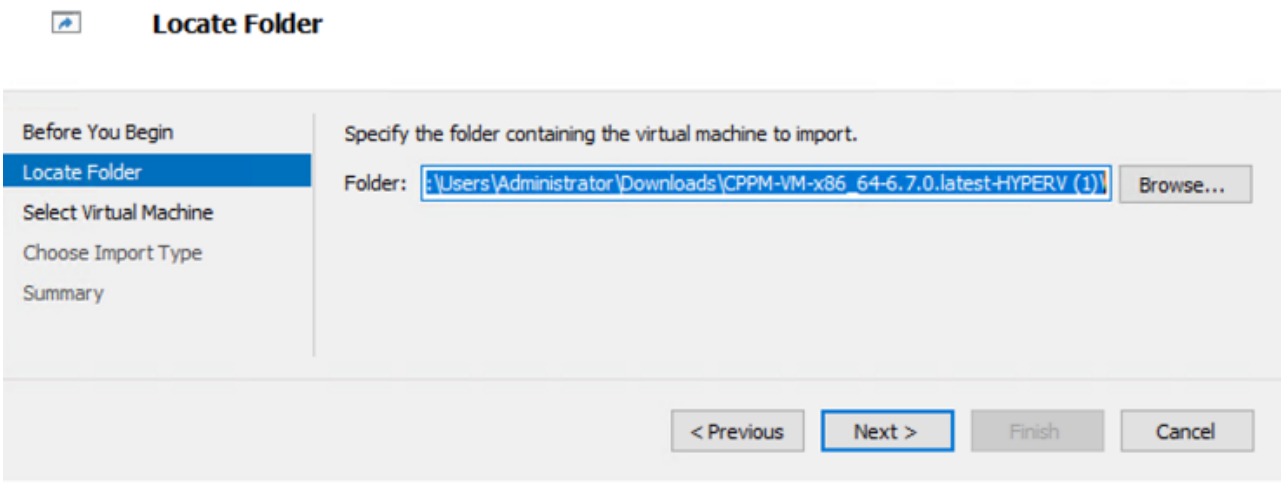
Add a Hard Disk to the Hyper-V Virtual Appliance
Power On and Configure the Hyper-V Virtual Appliance

Import the Virtual Appliance

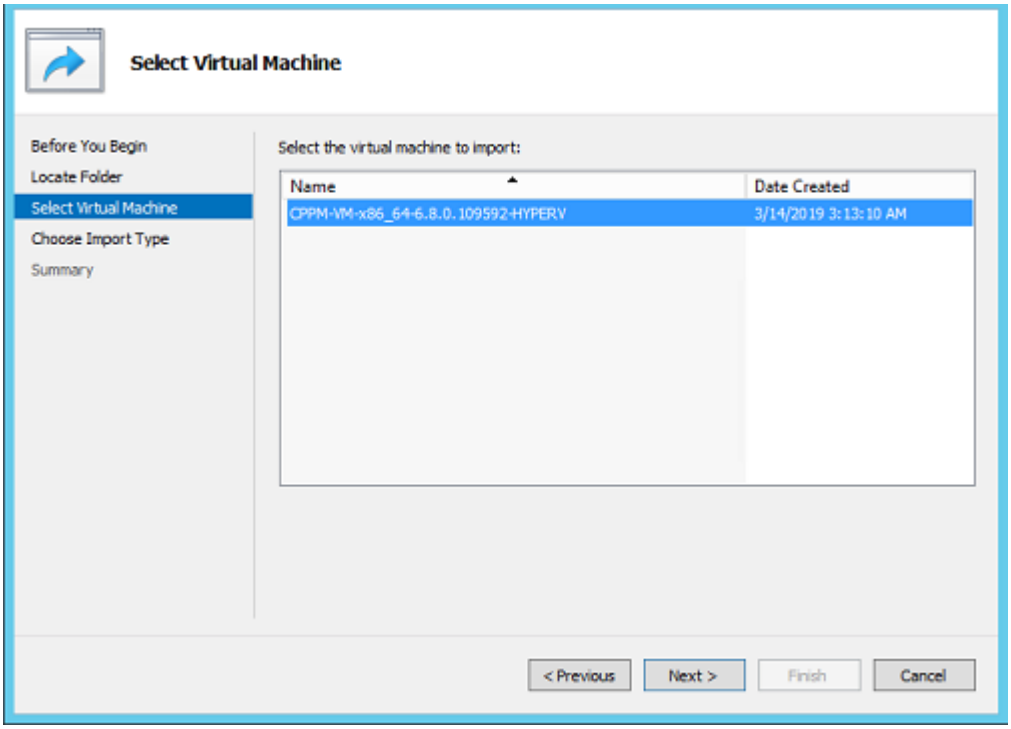
Procedure

1. Download and unzip the Hyper-V package from the **Download Software > ClearPass > Policy Manager > Current Release > Hyper-V** folder on the Support site.
2. From Hyper-V Manager, right-click to select the **Hyper-V server** and select the **Import Virtual Machine** option. The **Import Virtual Machine** window opens.
3. In the **Locate Folder** step, browse to the folder unzipped in step 1, and then click **Next**.

Figure 1. Import Virtual Machine Window, Locate Folder

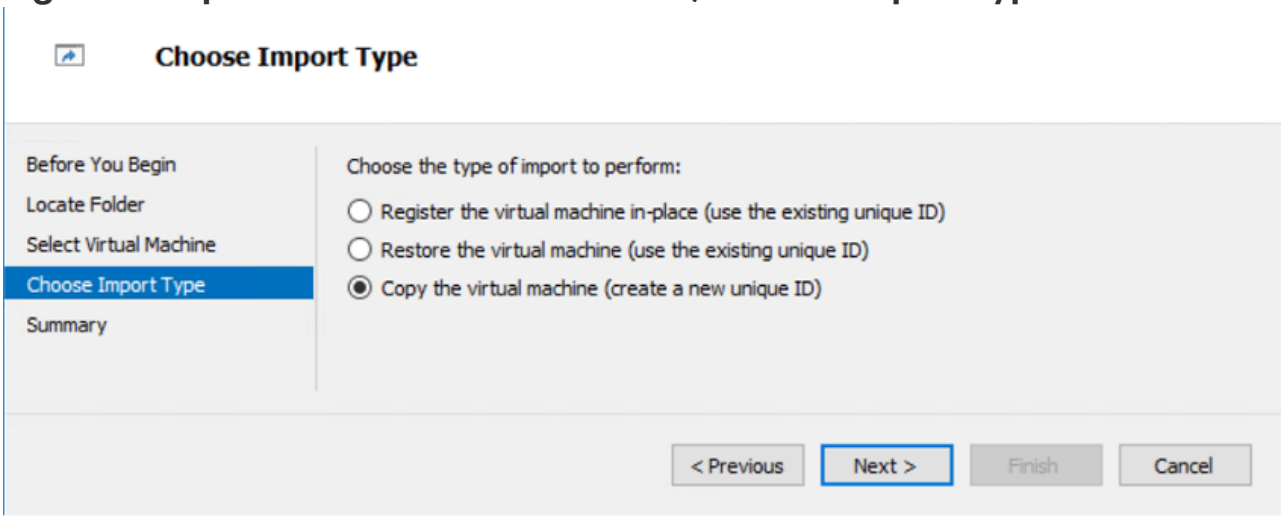


4. In the **Select Virtual Machine** step, select the vhdx file. Verify the summary, and click **Next**.



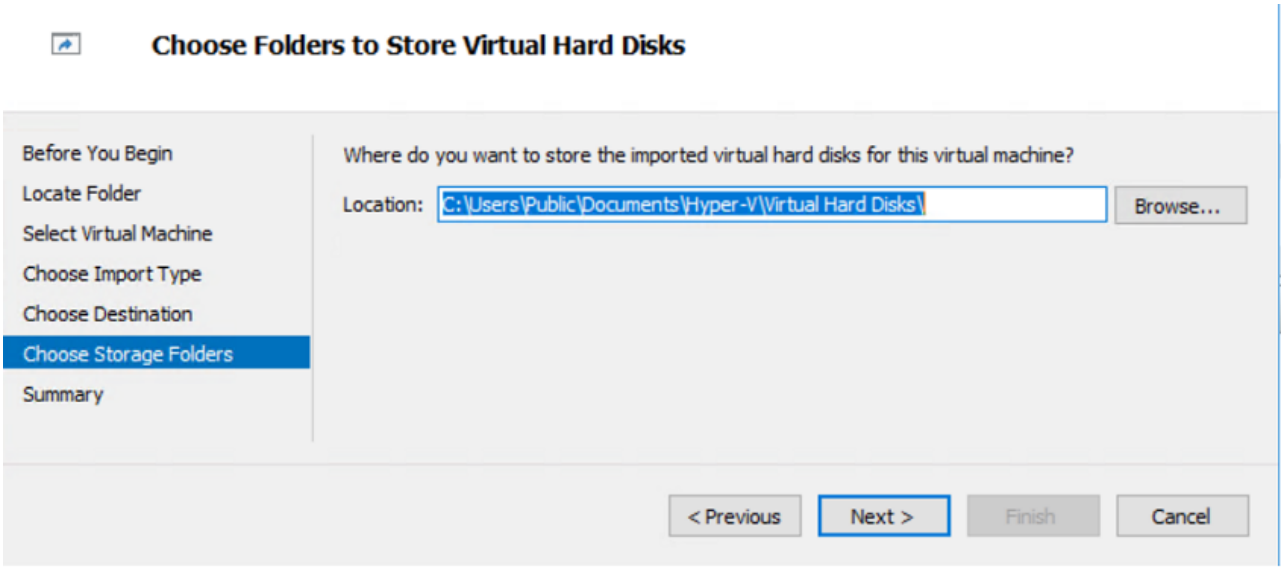
5. In the **Choose Import Type** step, select **Copy the virtual machine** to create a Unique ID.

Figure 2. Import Virtual Machine Window, Choose Import Type



6. Select the location where your HDD needs to be stored and finish the deployment.

Figure 3. Import Virtual Machine Window, Choose Storage Folders



7. Assign network adapters to the management and data port respectively. Make sure they are the “Network adapter” type and not the “Legacy network adapter” type.

Figure 4. Import Virtual Machine Window, Connect Network

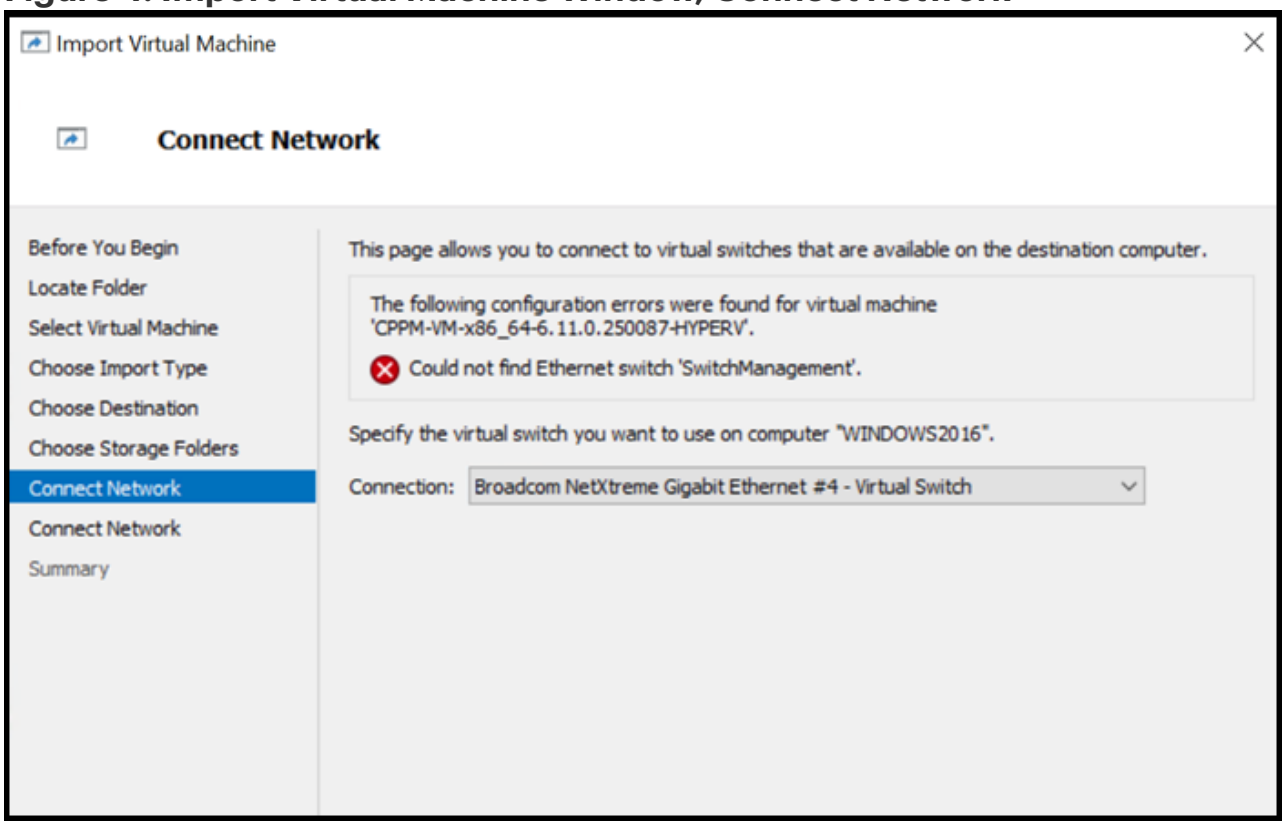
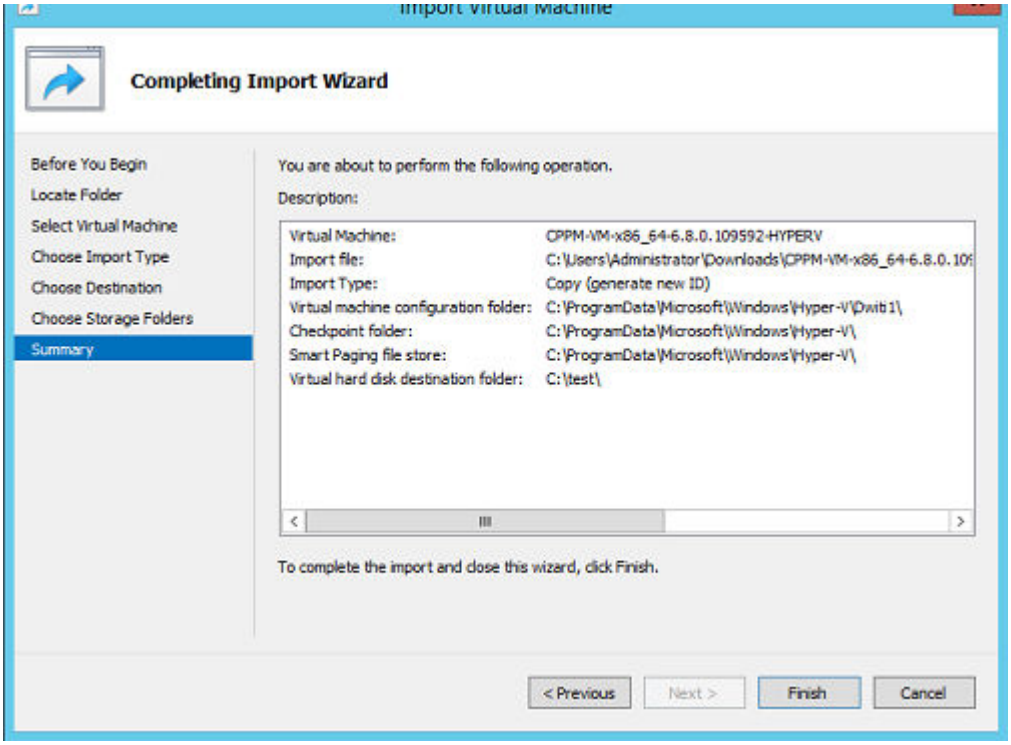


Figure 5. Import Virtual Machine Window, Completing Import Wizard



8. After it is imported, select the VA, right click, and choose properties. The **Settings** configuration window opens, where you will add the hard disk.

Add a Hard Disk to the Hyper-V Virtual Appliance

Procedure

1. Select the **SCSI Controller** option. To install ClearPass, add another disk partition to this deployed VM. Right click on the VA to edit the settings, and add a new hard drive.

Figure 1. Hyper-V Settings, Add New Hard Drive

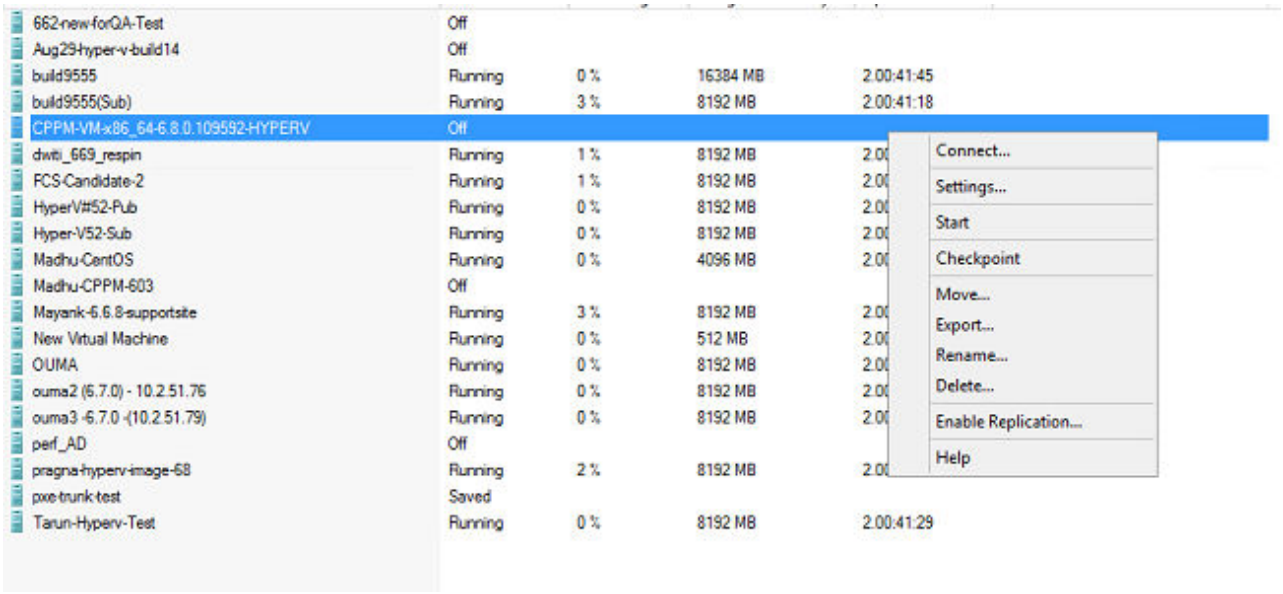


Figure 2. Hyper-V Settings, Select Devices for VA

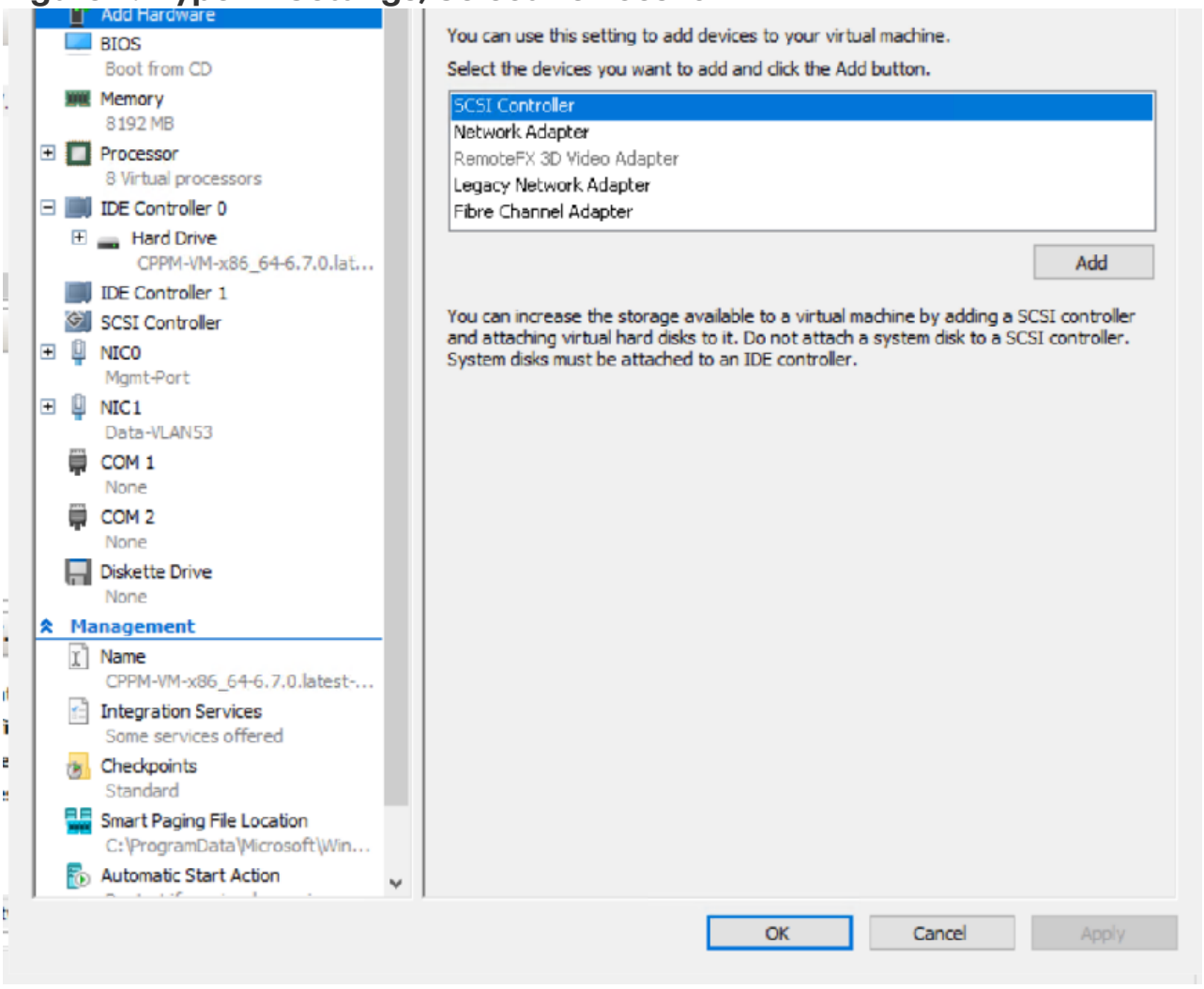
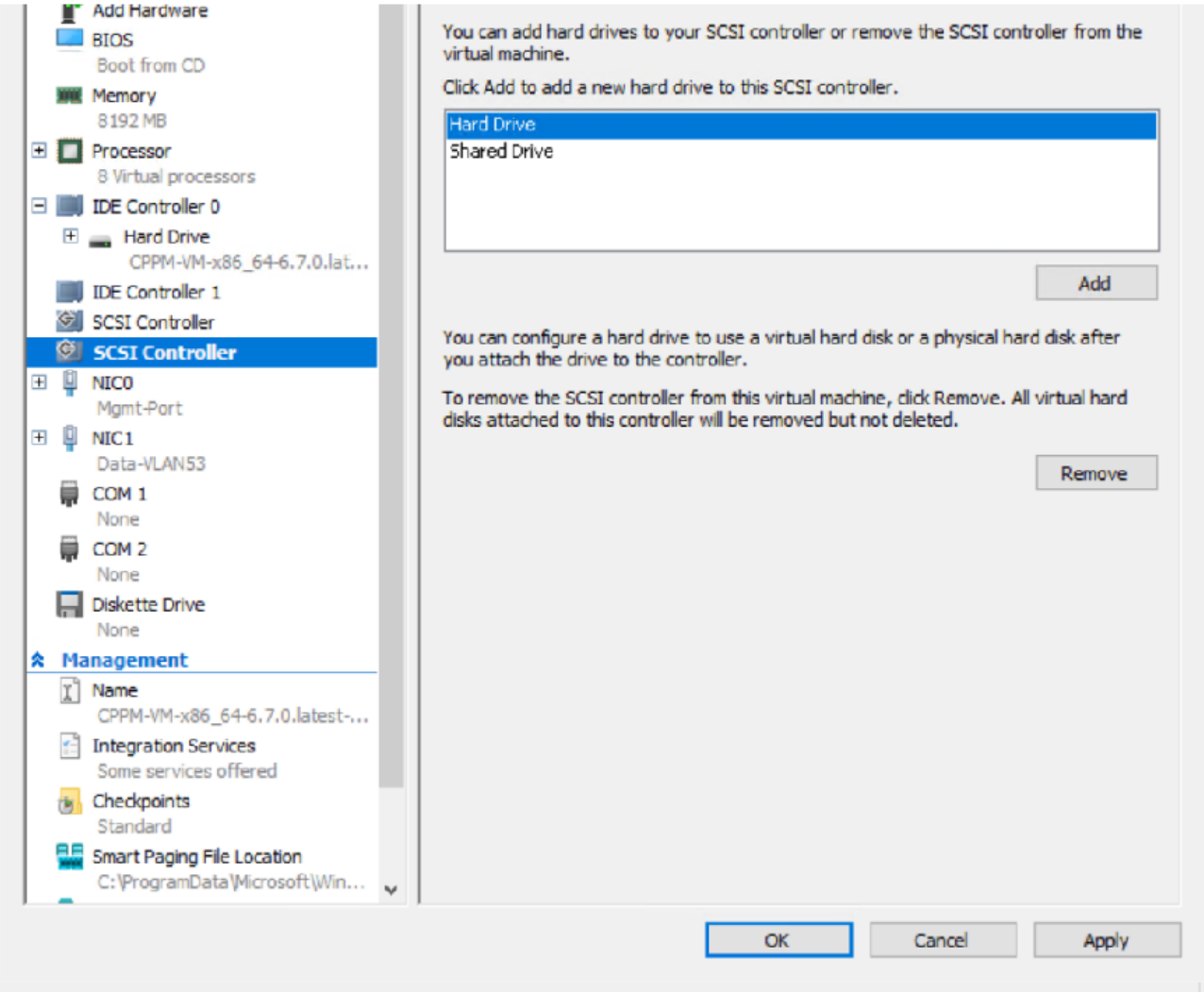
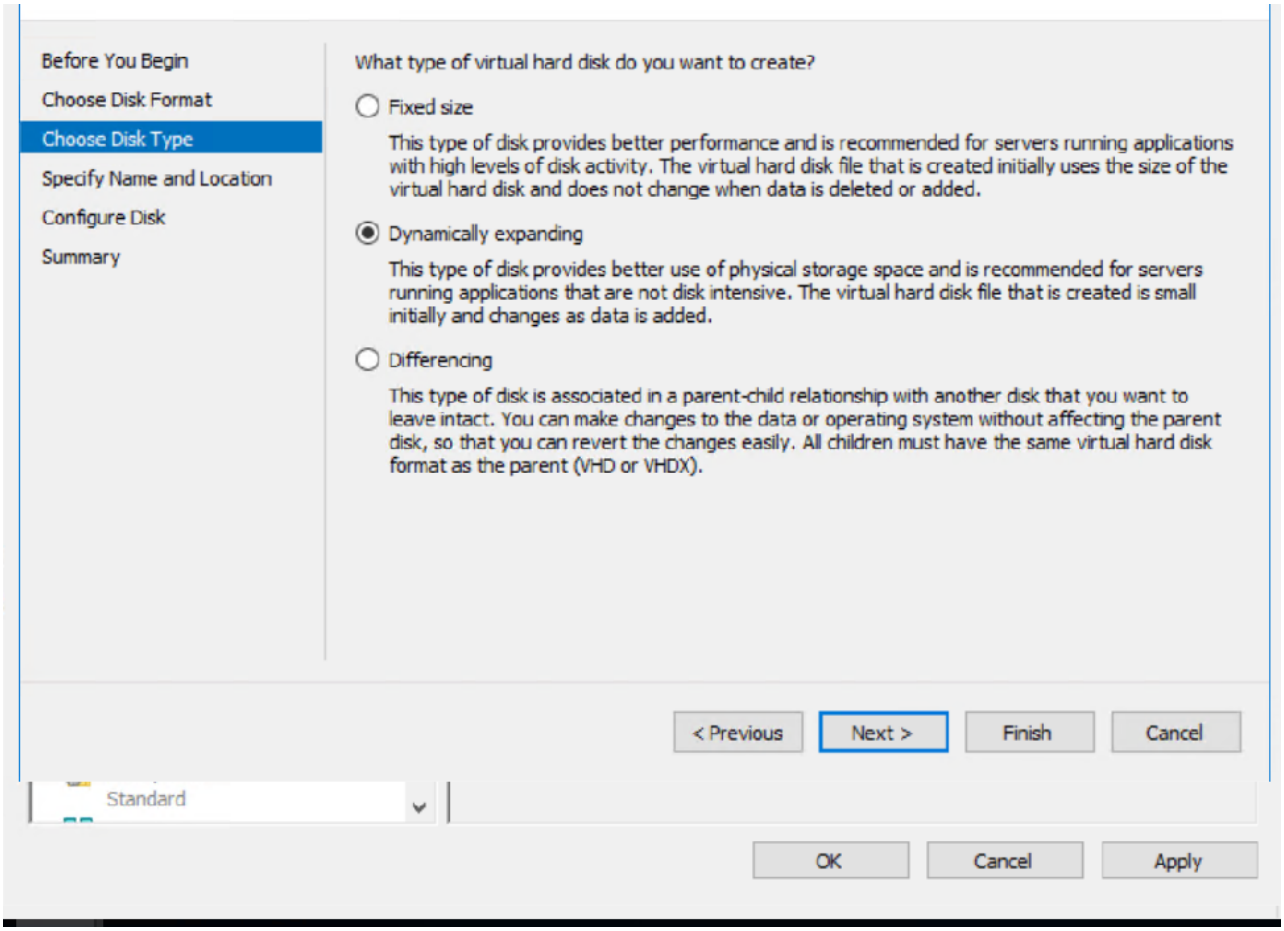


Figure 3. Hyper-V Settings, SCSI Controller Option



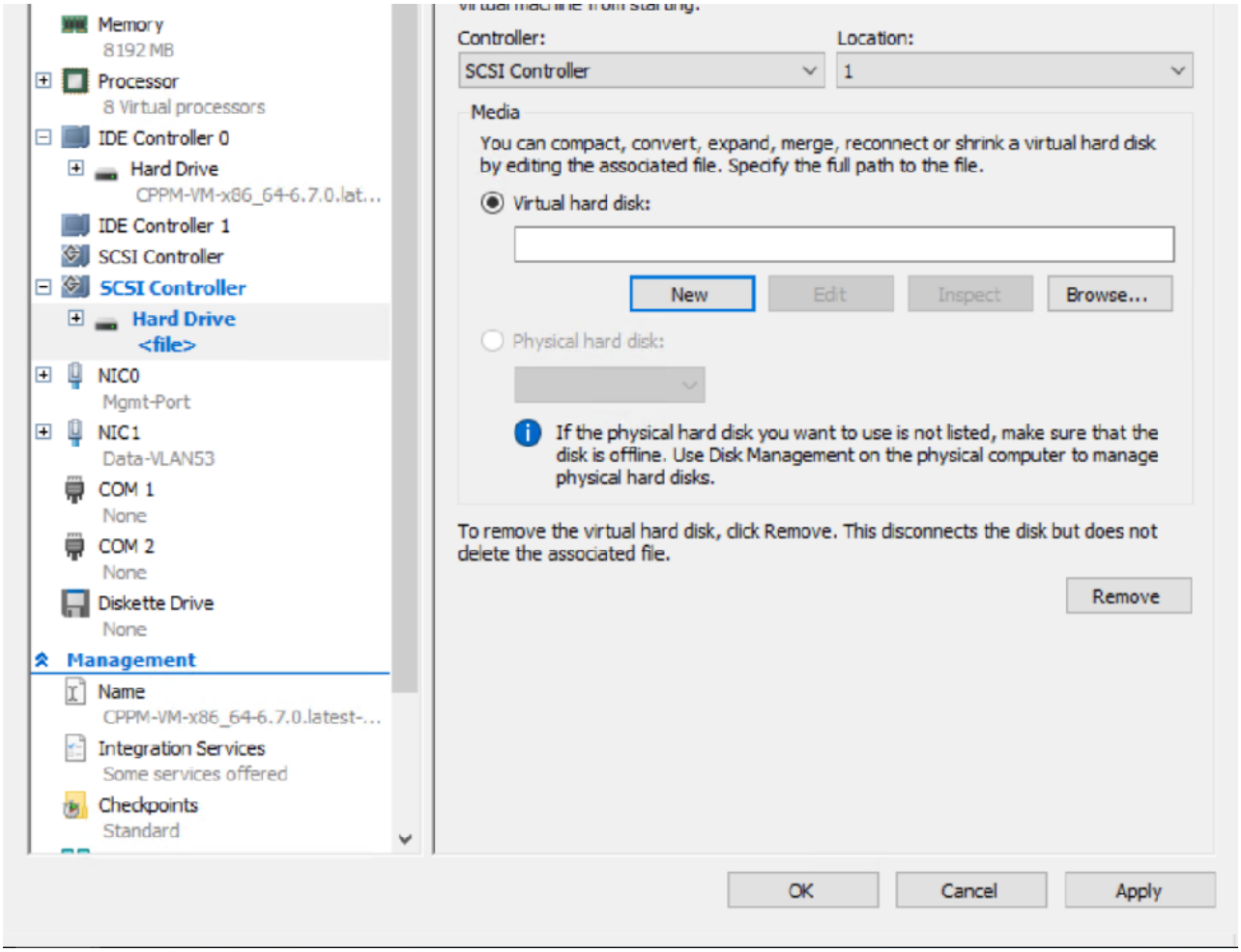
2. Choose the type of HDD and select **Dynamic expanding** and click **Next** to specify the size.

Figure 4. Hyper-V Settings, Choose Disk Type



3. Add a hard drive and verify the following values:
   - Controller = **SCSI Controller**
   - Location = **1**

Figure 5. Hyper-V Settings, Adding Hard Drive and Verifying Values



Disk size required:

- CLABV (evaluation = **400 GB**
- C1000V (500) = **1000 GB**
- C2000V (5K) = **1000 GB**
- C3000V (25K) = **1800 GB**

Using a CLABV as an example, the following images show the name and location, disk configuration, and summary steps. For more information about the correct requirements for your virtual appliance version, see Recommended Hyper-V Server Specifications .

Figure 6. Hyper-V Settings, Choose Storage Folder

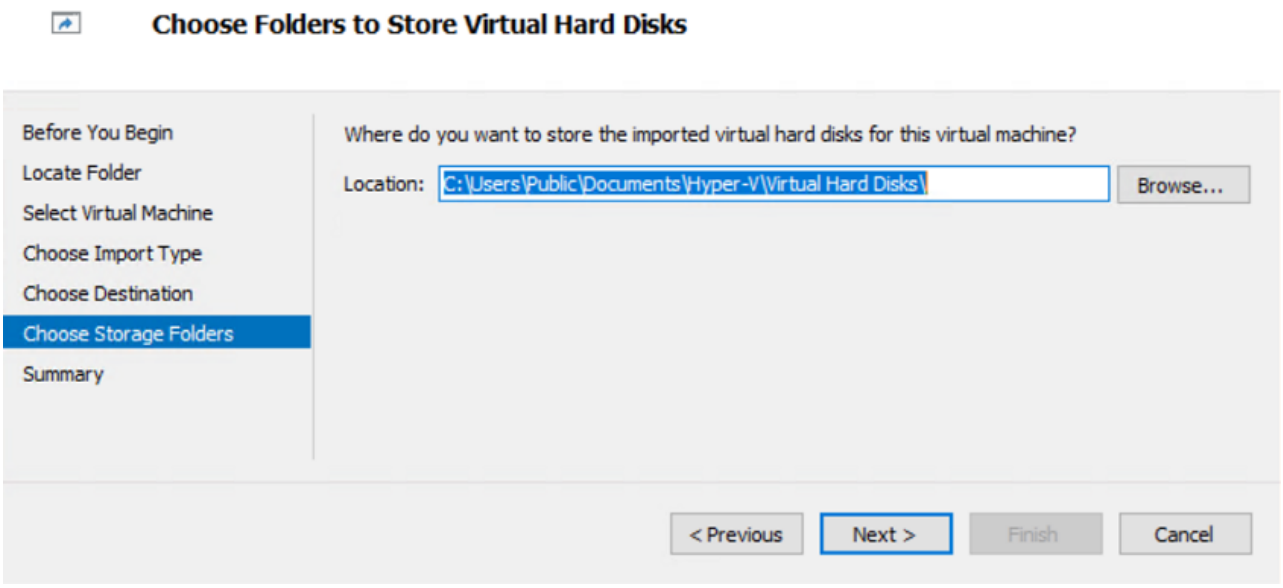
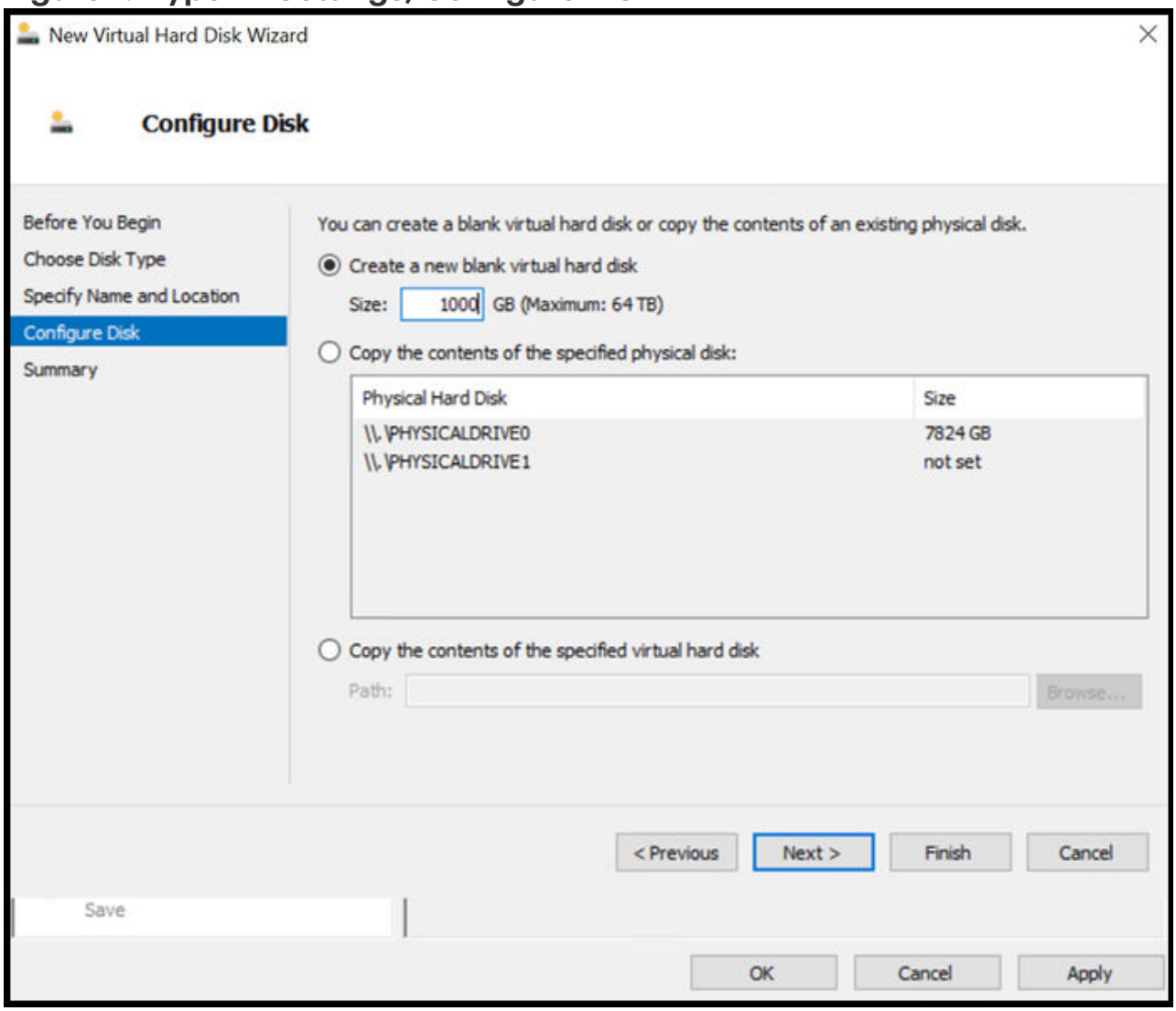
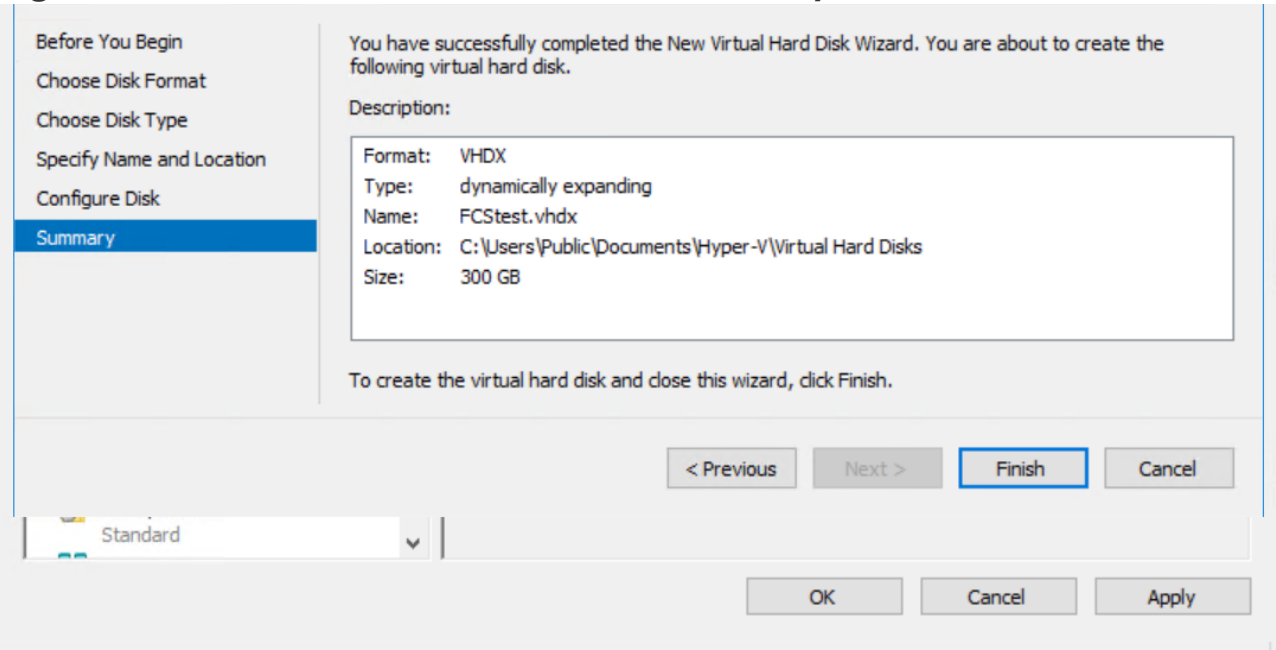


Figure 7. Hyper-V Settings, Configure Disk



4. Click **Apply** in the main window, and then click **OK**.

Figure 8. New Virtual Hard Disk Wizard, Summary

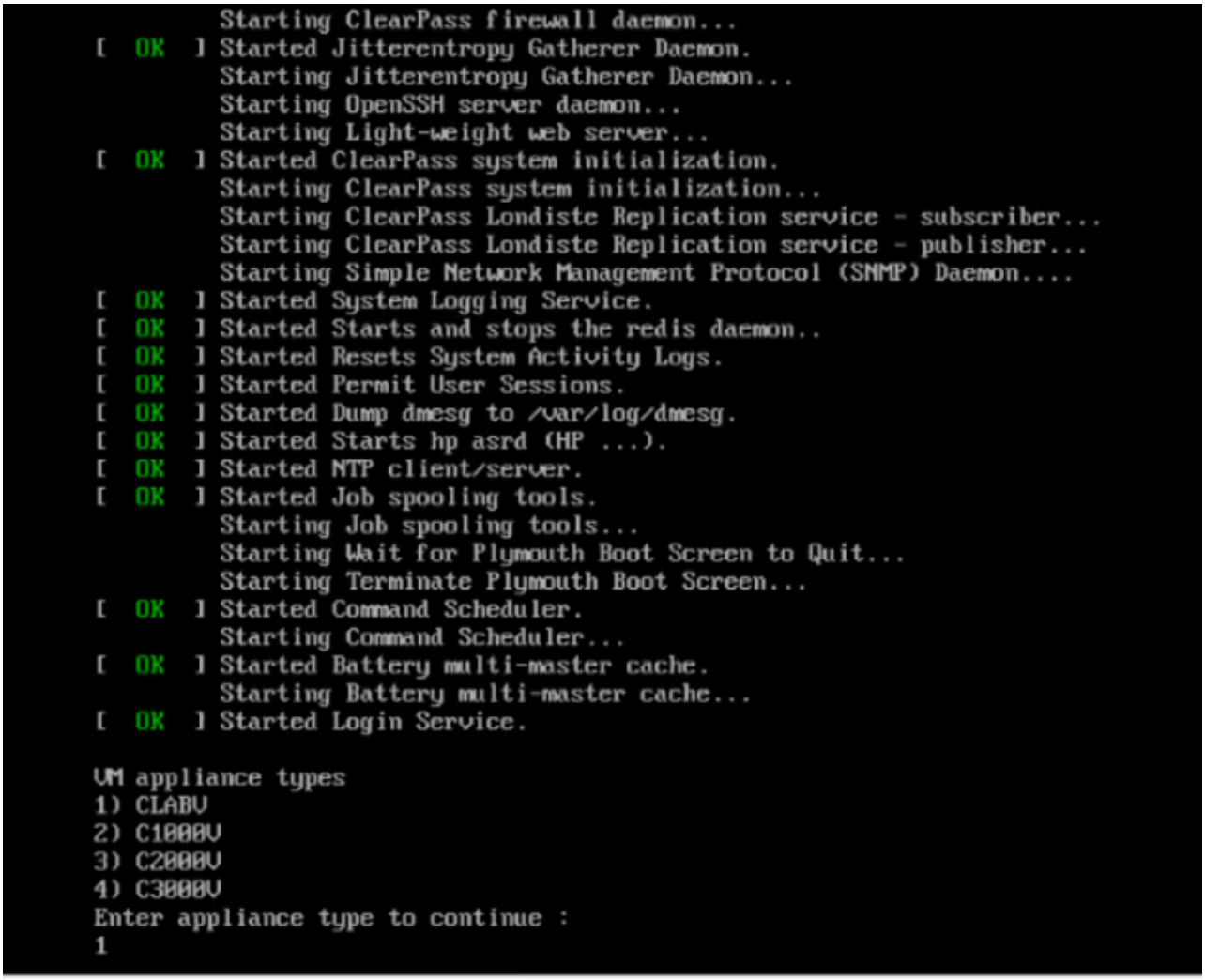


Power On and Configure the Hyper-V Virtual Appliance

Procedure

1. Power on the Hyper-V virtual appliance. You should see the following:

Figure 1. “Enter appliance type to continue”



2. Enter the list number for the appropriate appliance type (do not enter the appliance model itself). Options include:

   - **1) CLABV**
   - **2) C1000V**
   - **3) C2000V**
   - **4) C3000V**

   For example, to install a C1000V, you would enter the number **2**.

3. The system requirements are displayed for the appliance model you entered, along with your current system configuration. Compare these to make sure your system meets the new system requirements. For more information, see Recommended Hyper-V Server Specifications . When you have verified that your system meets the new requirements, press **y**. The ClearPass 6.11.0 setup and installation begins. You should see the following information, and ClearPass will reboot at least once.

Figure 2. “Enter ‘y’ or ‘Y’ to proceed”

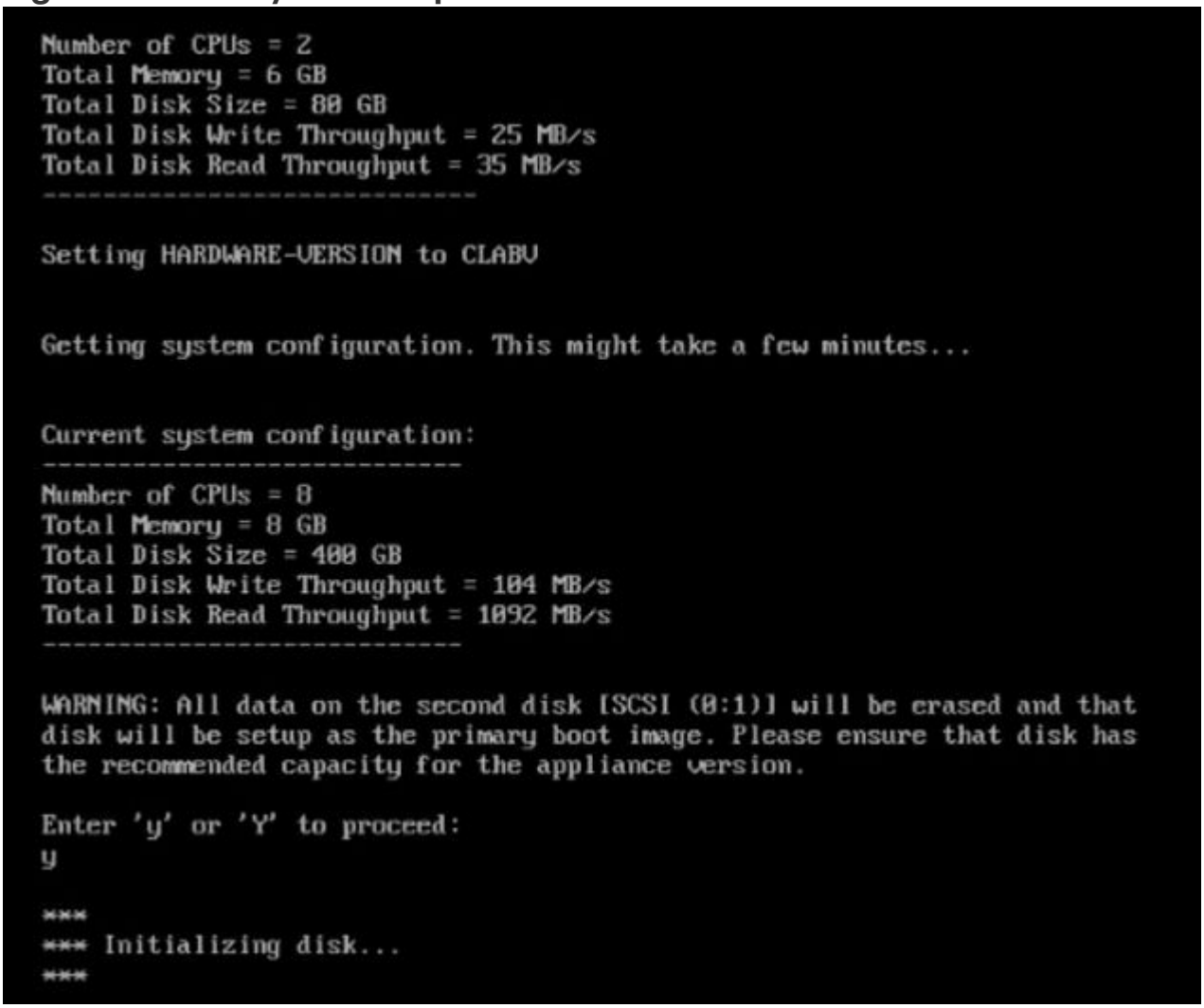
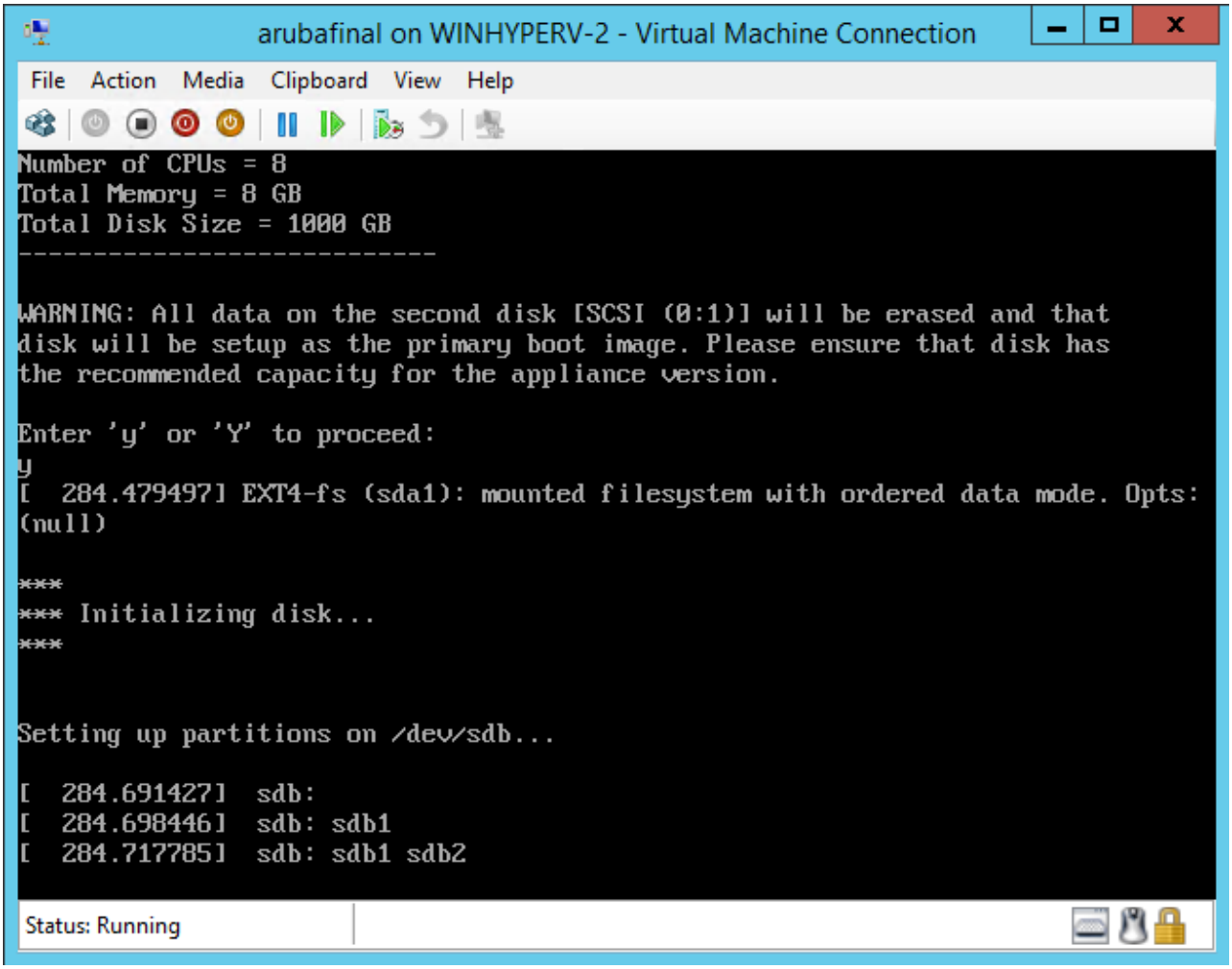
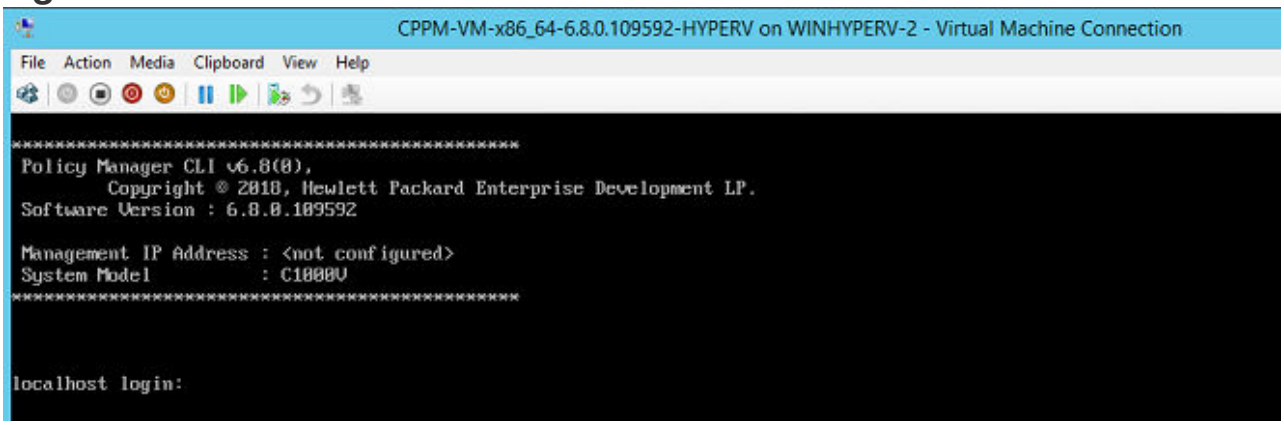


Figure 3. Initializing Disk



4. You will see a LUKS encryption option. Proceed as desired.
5. Afterwards, reboot the ClearPass VA. It will power on and boot up within a couple of minutes. The whole process from deploying the VHDX image to the final startup screen should take between 30 and 40 minutes.
6. After the ClearPass VA launches, you should see the following banner:

Figure 4. Banner



When you see the banner, log in using the instructions in the "Initial Login and Activation of the Policy Manager Platform License" section of the ClearPass Policy Manager 6.11 User Guide.

Morphing a Hyper-V Version

About this task

> **NOTE**
> The illustrations in this section use the example of morphing a C1000V virtual appliance to a C3000V. Adjust your own configuration as needed.

> **NOTE**
> Starting with the 6.11.0 release, after the **system morph-vm** command is run during a ClearPass upgrade and the new disk is added, the existing hard disk cannot be removed. This is because the existing disk is used to extend the disk space for the server, and is included in the calculations to determine the required capacity of the additional hard disk.

A Hyper-V virtual appliance can be morphed to a higher-value virtual appliance using the **system morph-vm** command as follows:

Procedure

1. Power off the virtual appliance.
2. Open **Settings** and make the following modifications:
   a. Modify the **RAM** and **CPU** to match the recommended system requirements for the larger VA (see Recommended Hyper-V Server Specifications).

Figure 1. Hardware, RAM Settings

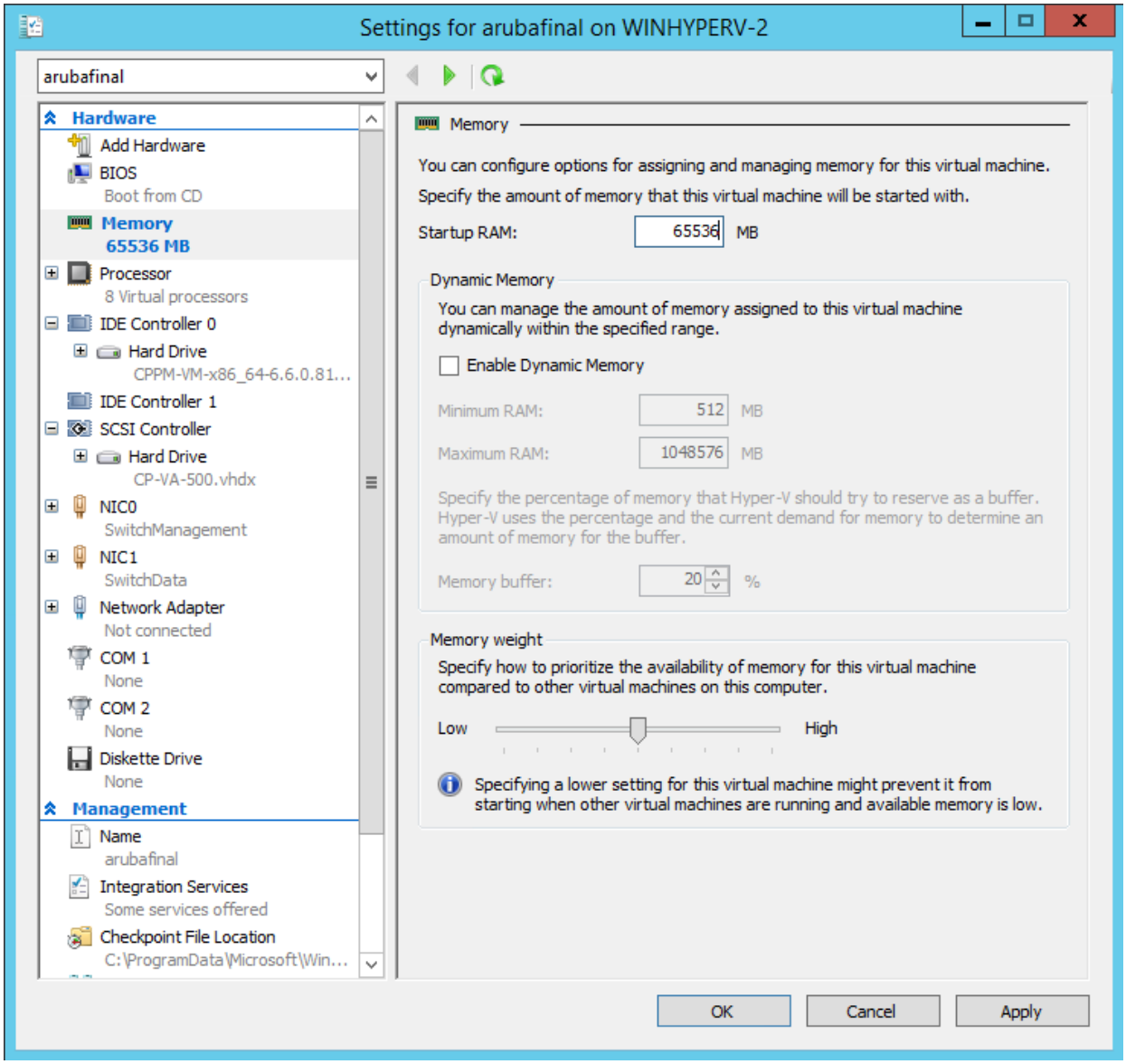
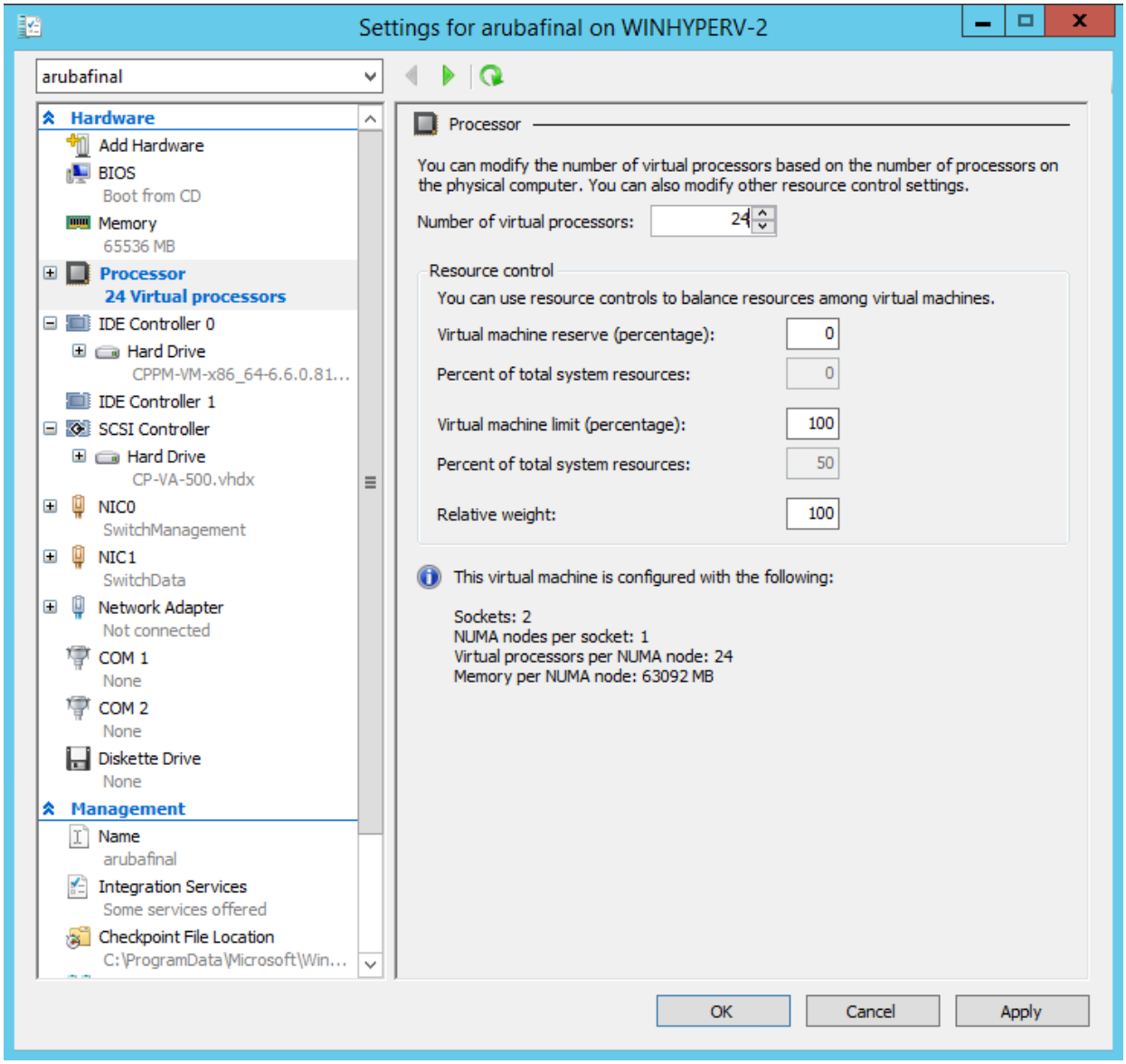


Figure 2. Hardware, CPU Settings



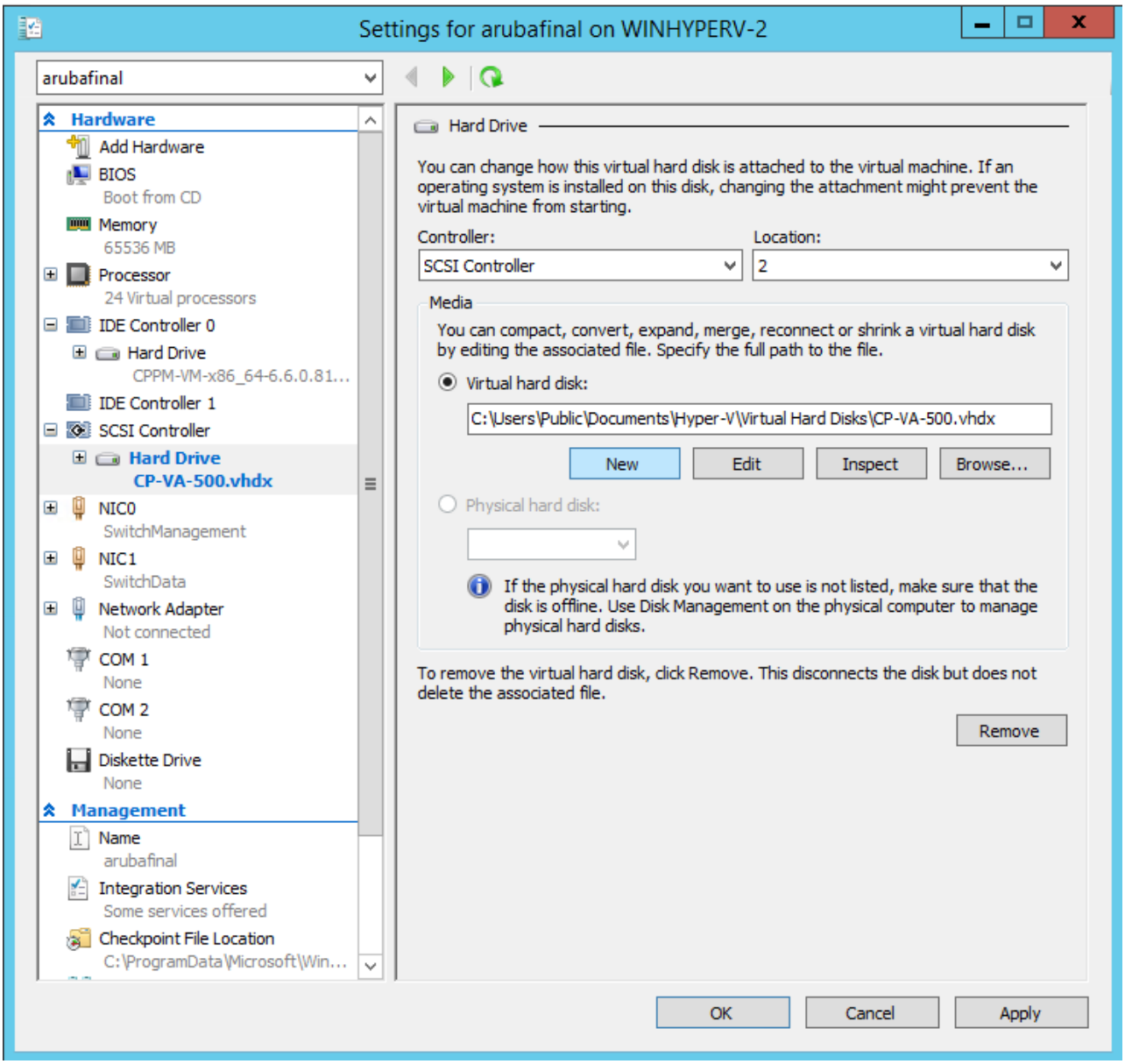
Next, add an additional disk with the recommended disk size for the larger VA:

3. Select **SCSI(0:2) Controller**.
4. Select the **Hard Drive** option and then click **Add**.
5. In the next screen, specify the following values:

   Controller = **SCSI(0:2) Controller 1**

   Location = 2
6. In the **Media** section, click **New**.

Figure 3. Hardware, Controller Settings



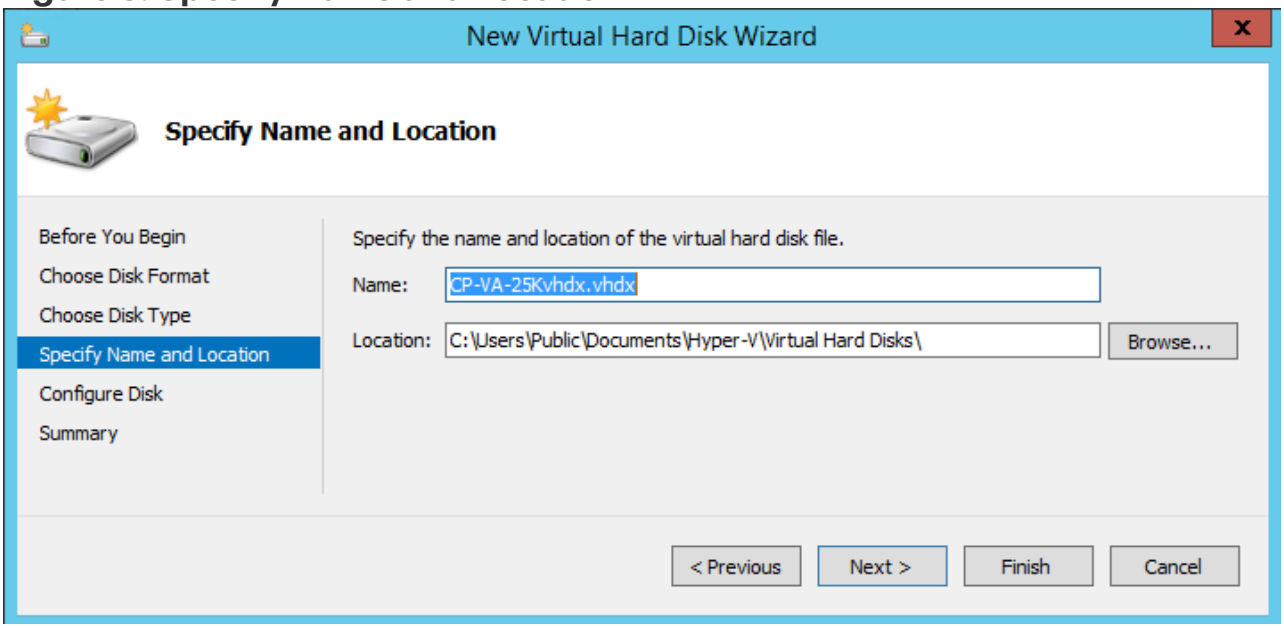
7. Add a new **VHDX** hard disk of **Fixed size**, and size equivalent to the requirements for the larger VA disk size.

Figure 4. Choose Disk Type and Size



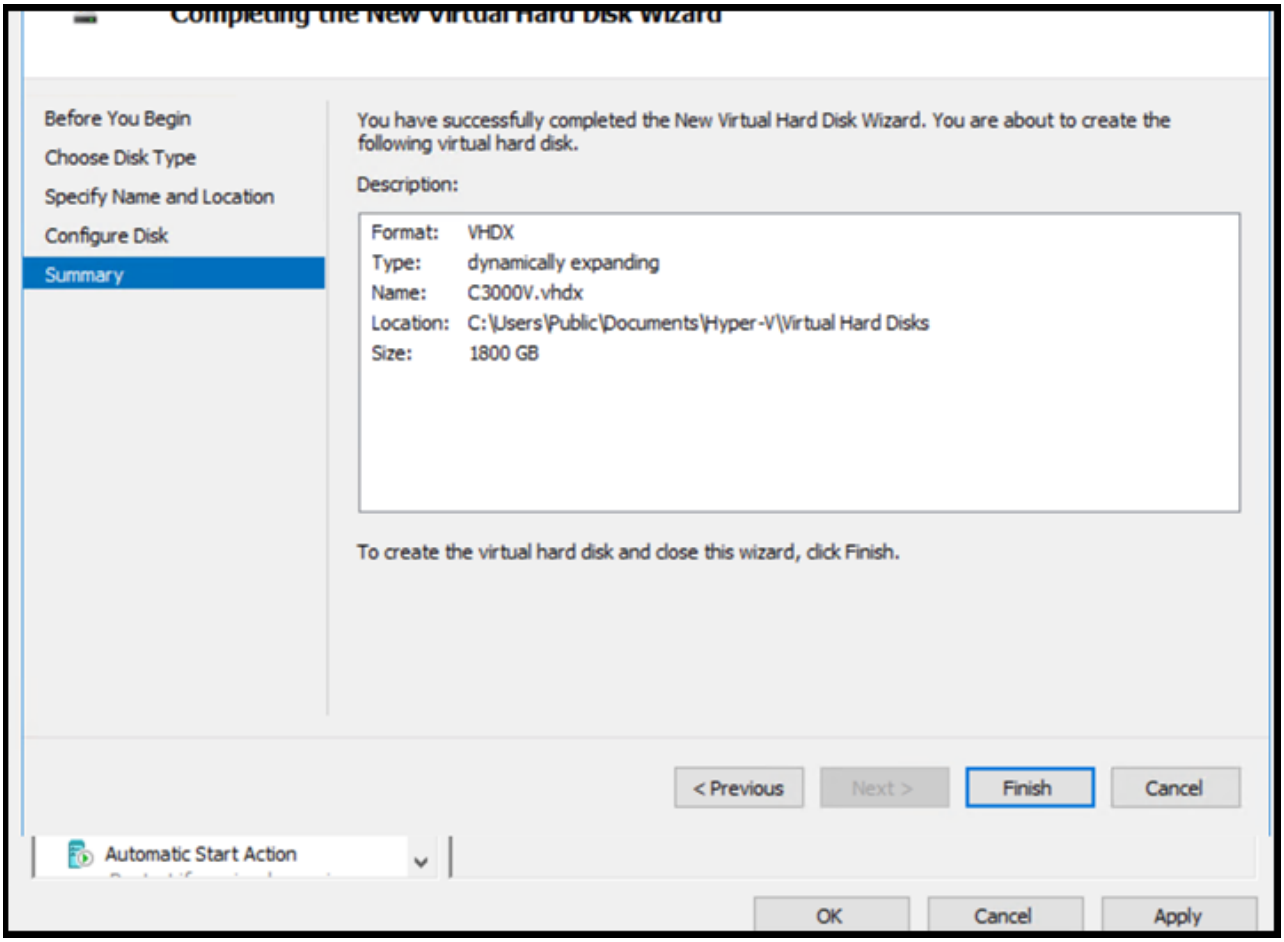
8. Specify the name and location.

Figure 5. Specify Name and Location



9. Configure the disk size and click **Finish**.

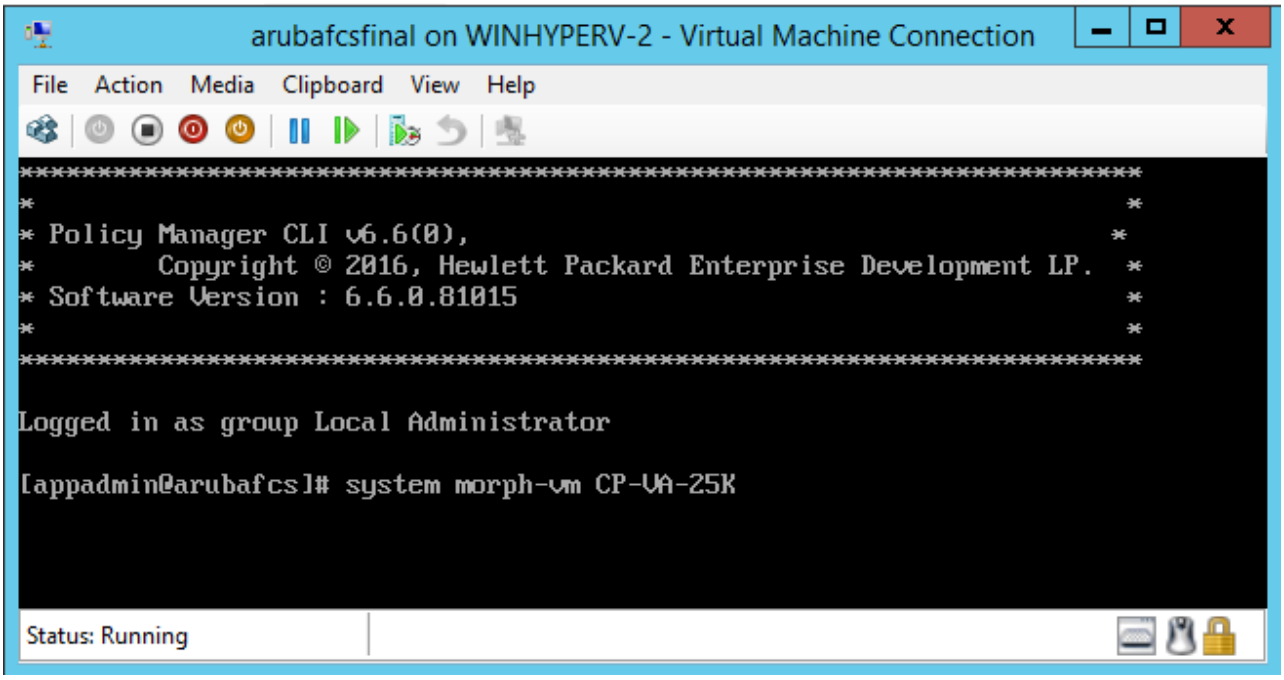
Figure 6. Completing the New Virtual Hard Disk Wizard, Summary



0. After adding the hard disk, power on the original VA and, using SSH, log in as **appadmin**.

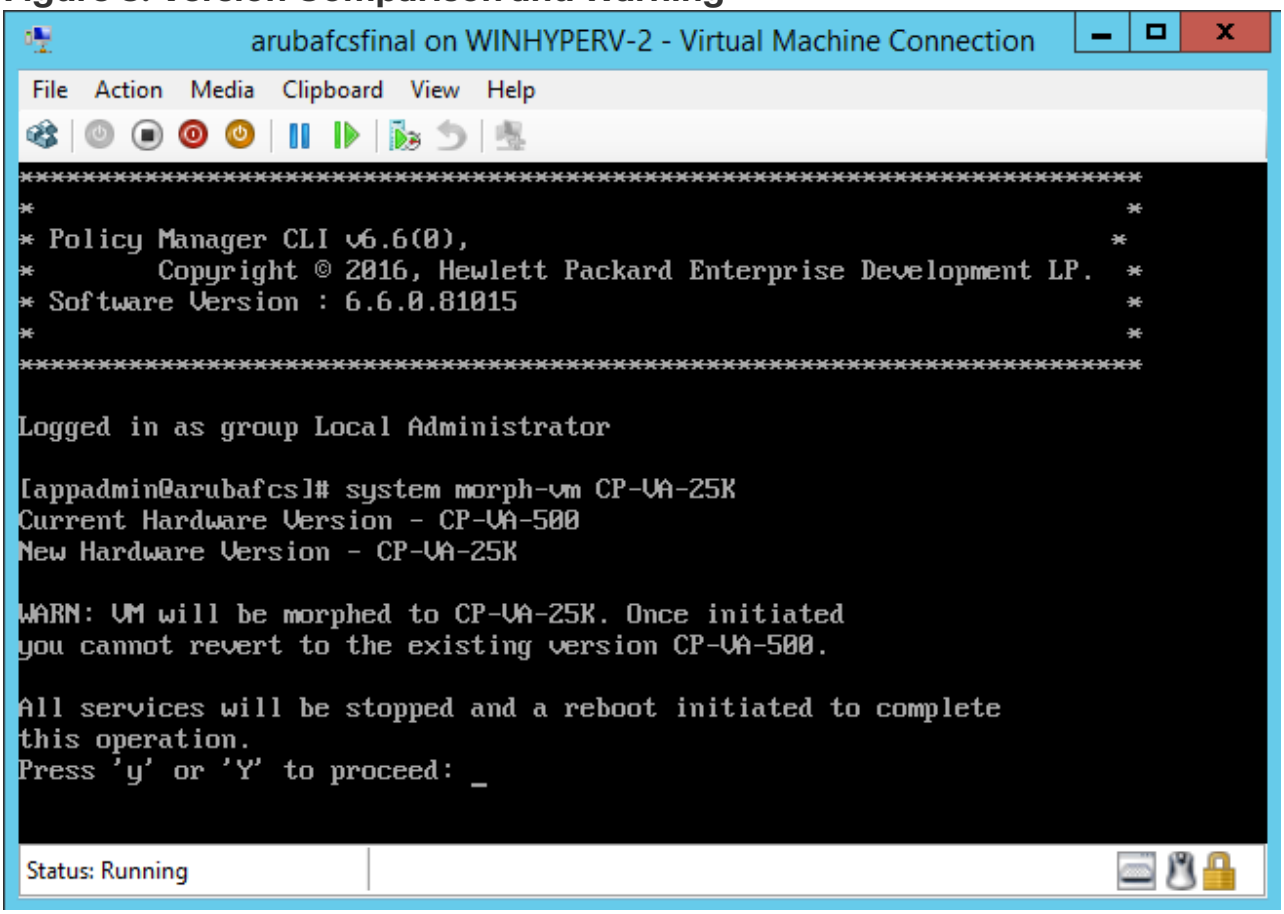
1. Run the **system morph-vm <C3000V>** command and follow the prompts.

Figure 7. System Morph-VM Command



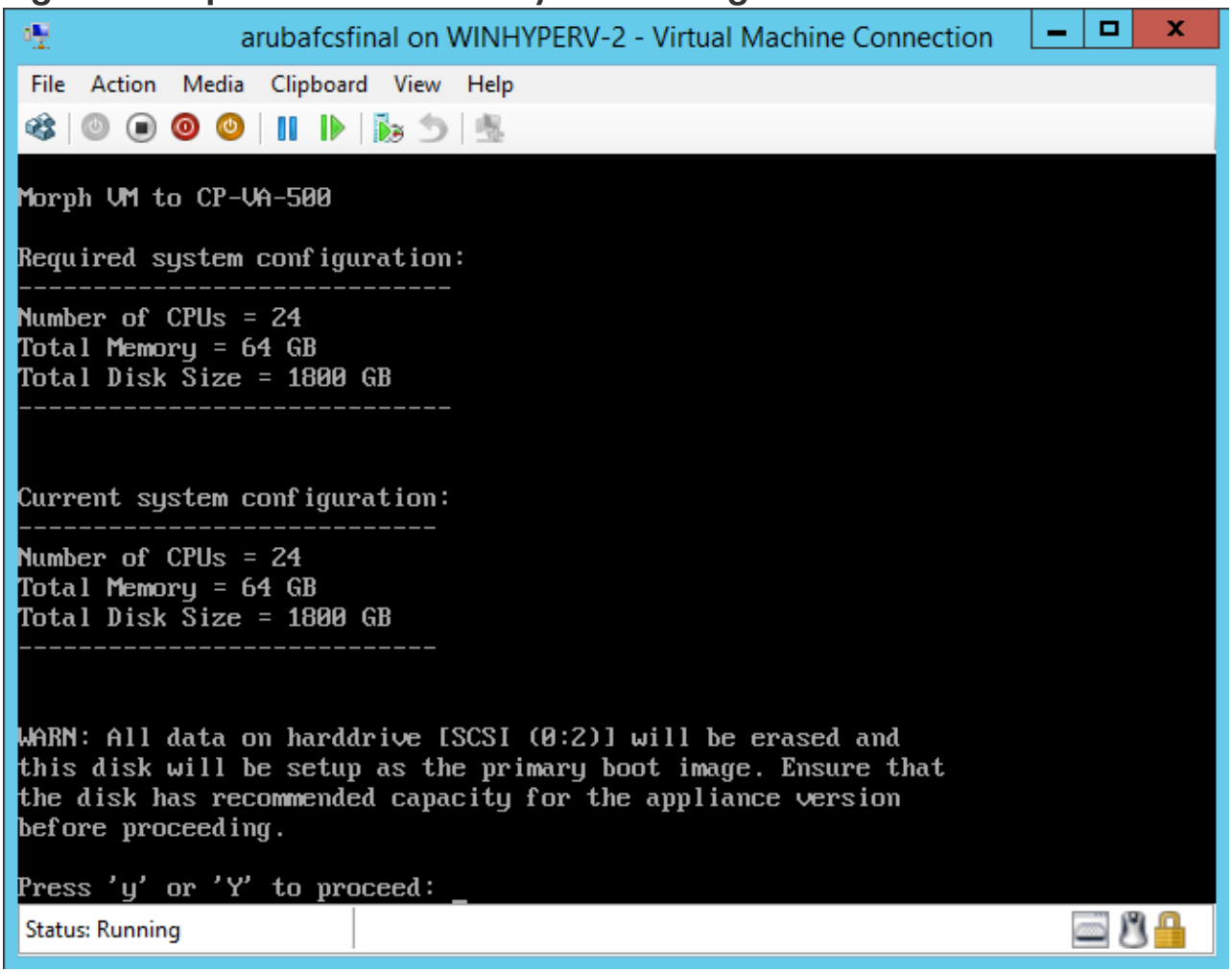
2. The existing hardware version and the new hardware version are displayed, along with a warning that you cannot revert to the existing version after the morphing process is started. When ready to proceed, press **y**.

Figure 8. Version Comparison and Warning



3. The system requirements are displayed for the appliance model you entered, along with your current system configuration. Compare these to make sure your system meets the new system requirements.

Figure 9. Required and Current System Configuration



When you have verified that your system meets the new requirements, press **y**. The ClearPass 6.14 setup and installation begins.

Troubleshooting, Caveats, and Other Information for Hyper-V

This section includes information on how to configure SPAN for Hyper-V, as well as troubleshooting information and caveats for Hyper-V instances.

Subtopics

Configuring SPAN for Hyper-V
Low Network Performance on Hyper-V Due to NIC Cards
Hyper-V Live Migration Caveat

Configuring SPAN for Hyper-V

This section describes how to enable and configure SPAN for Hyper-V.

The default behavior of Hyper-V Manager allows port mirroring between VA instances on the same Hyper-V server. It does not allow users to configure promiscuous mode for a virtual interface on a specific VA instance in order to receive external traffic.

Process Overview for Enabling and Configuring SPAN

About this task

> **NOTE**
> Make sure no ClearPass VA instance is running while you perform the following steps.

Procedure

1. Create a virtual switch.
2. Attach a ClearPass SPAN virtual interface to the virtual switch.
3. Enable Microsoft NDIS capture extensions for the virtual switch.
4. Set the mirroring mode on the external port.
5. Set the local SPAN in a Cisco switch.

> **NOTE**
> Make sure SPAN is enabled only on the data port and not the management port. Also, before you begin the SPAN configuration on the data port, make sure the data port is not configured with an IP address.

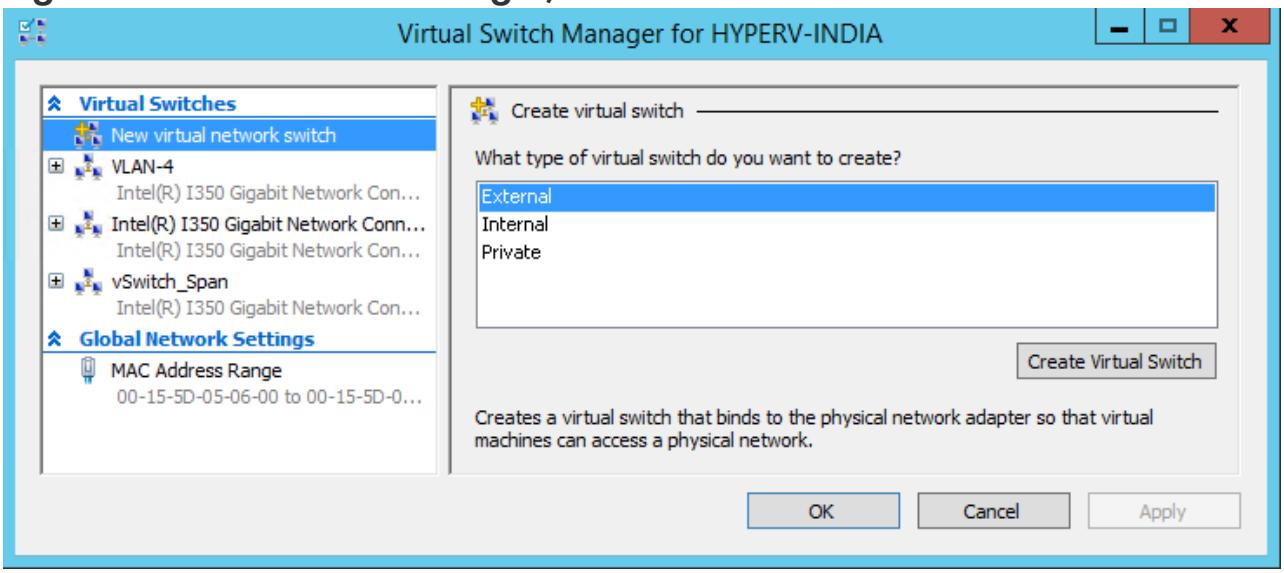
Creating a Virtual Switch

Procedure

1. Open the **Virtual Switch Manager.**

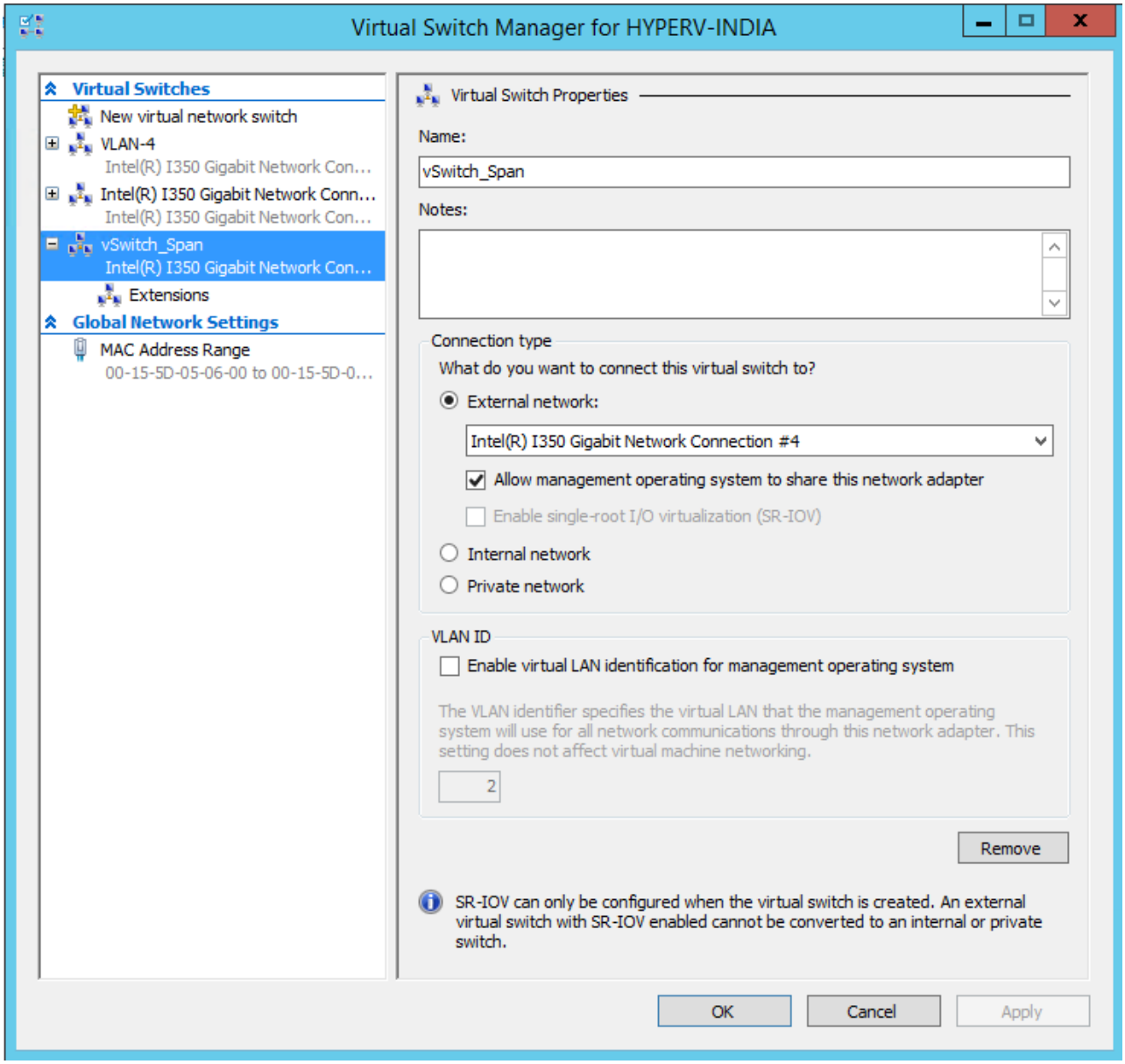
2. In the **Virtual Switches** list, highlight **New virtual network switch** and then select **External** as the dedicated spanned network adapter type. Click **Create Virtual Switch**.

Figure 1. Virtual Switch Manager, New Virtual Network Switch



3. For the switch you just created, select **External Network** as the **Connection type**, and select the **Allow management operating system to share this network adapter** option.

Figure 2. Virtual Switch Manager, Connection Type



Attaching a ClearPass SPAN Virtual Interface to the Virtual Switch

The steps to attach the ClearPass SPAN virtual interface can be performed either from Windows PowerShell or from Hyper-V Manager.

Using PowerShell:

Procedure

1. Add a new network adapter, selecting the newly added SPAN virtual switch. Use the command:

   ADD-VMNetworkAdapter -VMName VK-C1000V-LongRunning-650 -Name Monitor -SwitchName vSwitch_Span

2. Enable port mirroring for the selected interface as the span destination. Use the command:

   Get-VMNetworkAdapter -VMName VK-C1000V-LongRunning-650 | ? Name -eq Monitor | Set-VMNetworkAdapter -PortMirroring Destination

   Where:

   VK-C1000V-LongRunning-650 = CPPM VA name

   vSwitch_Span = Newly added SPAN virtual switch name

   Monitor = Newly added adapter name

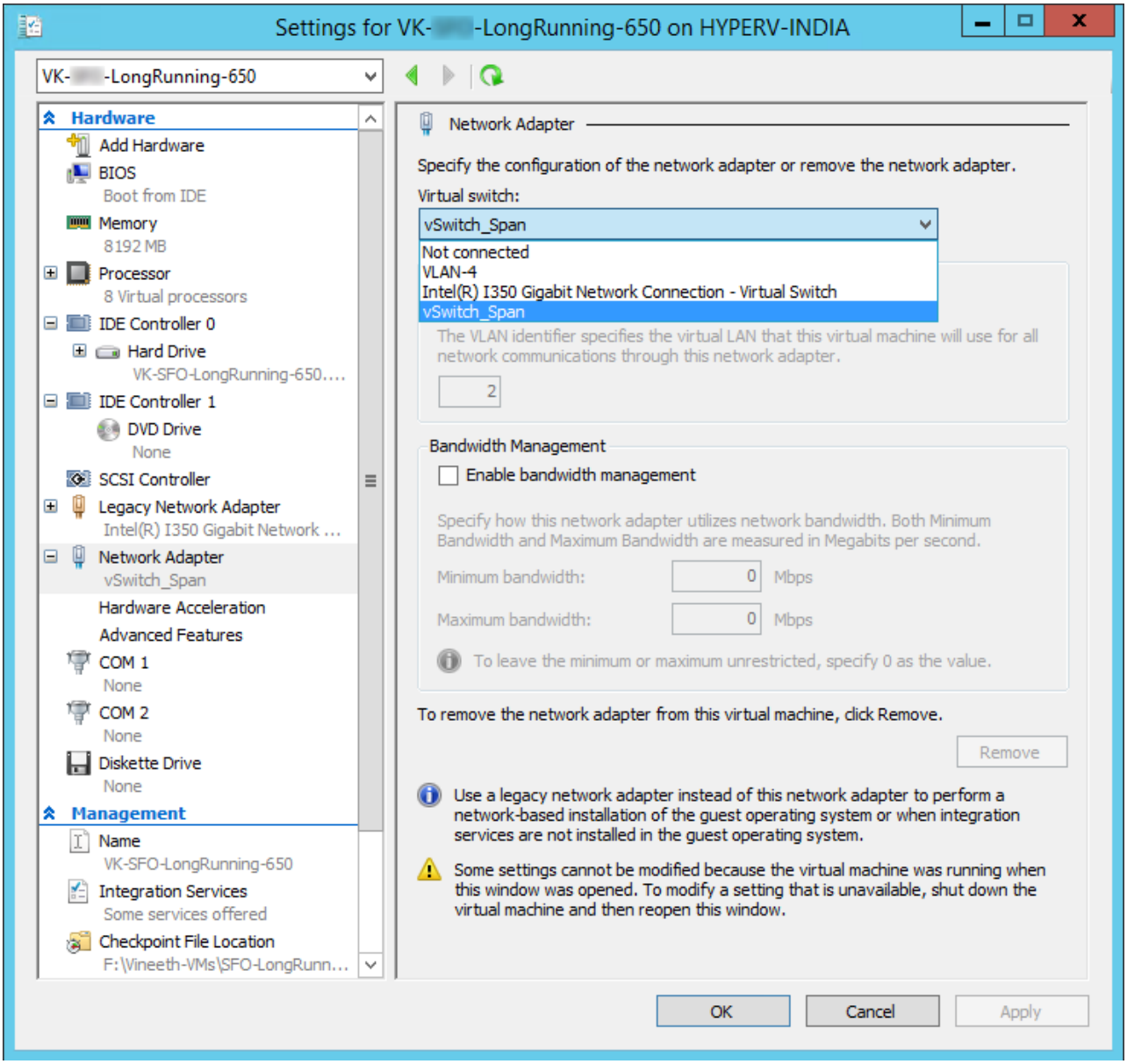
   (The newly added adapter hardware name will be “Monitor” when adding using the above commands, and “Network Adapter” when added using Hyper-V Manager.)

Using Hyper-V Manager:

Procedure

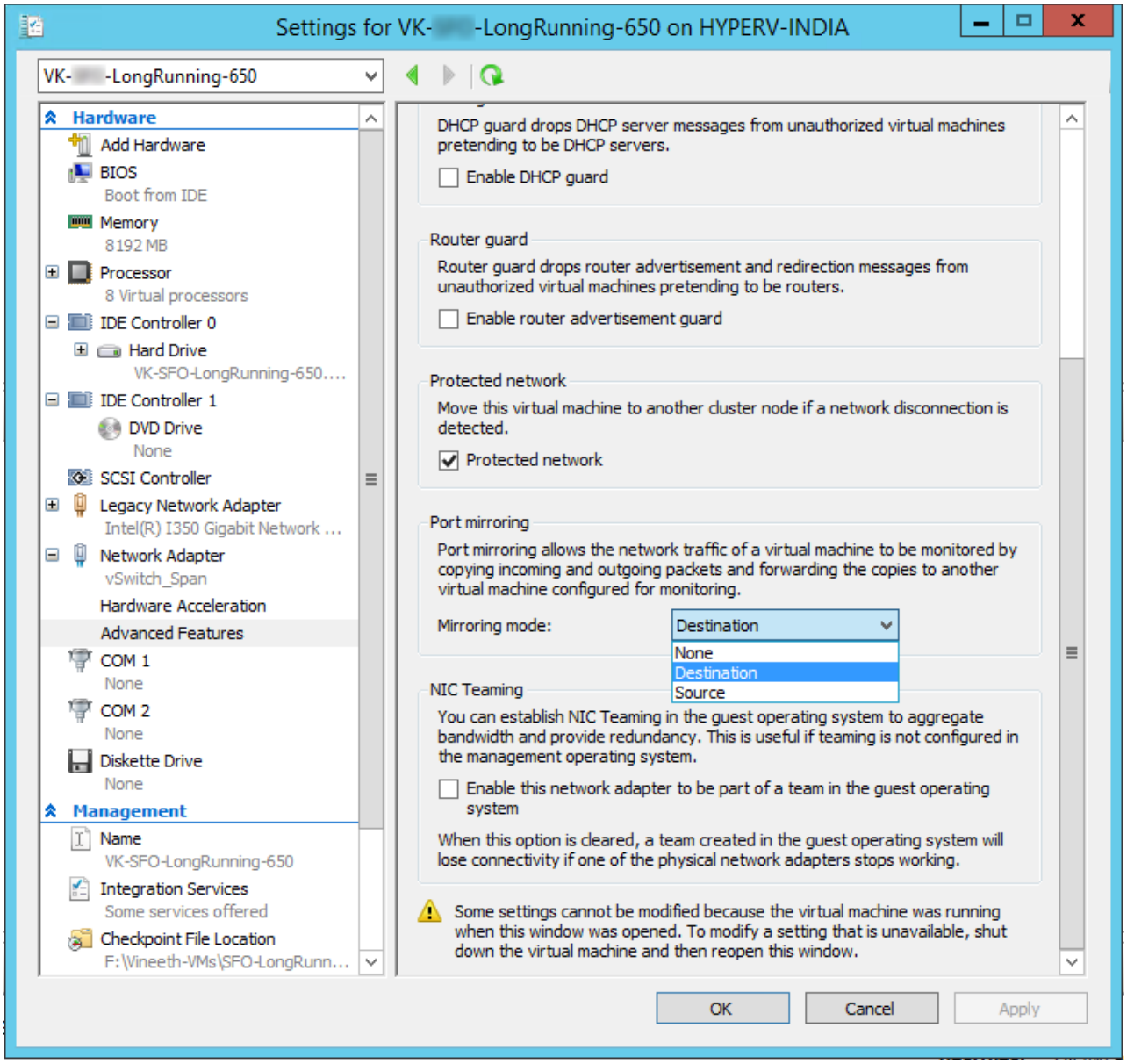
1. Add a new network adapter. In the **Hardware** list, highlight **Network Adapter**.
2. In the **Virtual Switch** field, select **vSwitch_Span**.

Figure 3. Hyper-V Manager, Virtual Switch



3. In the **Hardware** list, expand **Network Adapter** and select **Advanced Features**.
4. In the **Port Mirroring** area, select **Destination** as the **Mirroring mode** for the new virtual interface.

Figure 4. Hyper-V Manager, Mirroring Mode



Enabling Microsoft NDIS Capture Extensions for the Virtual Switch

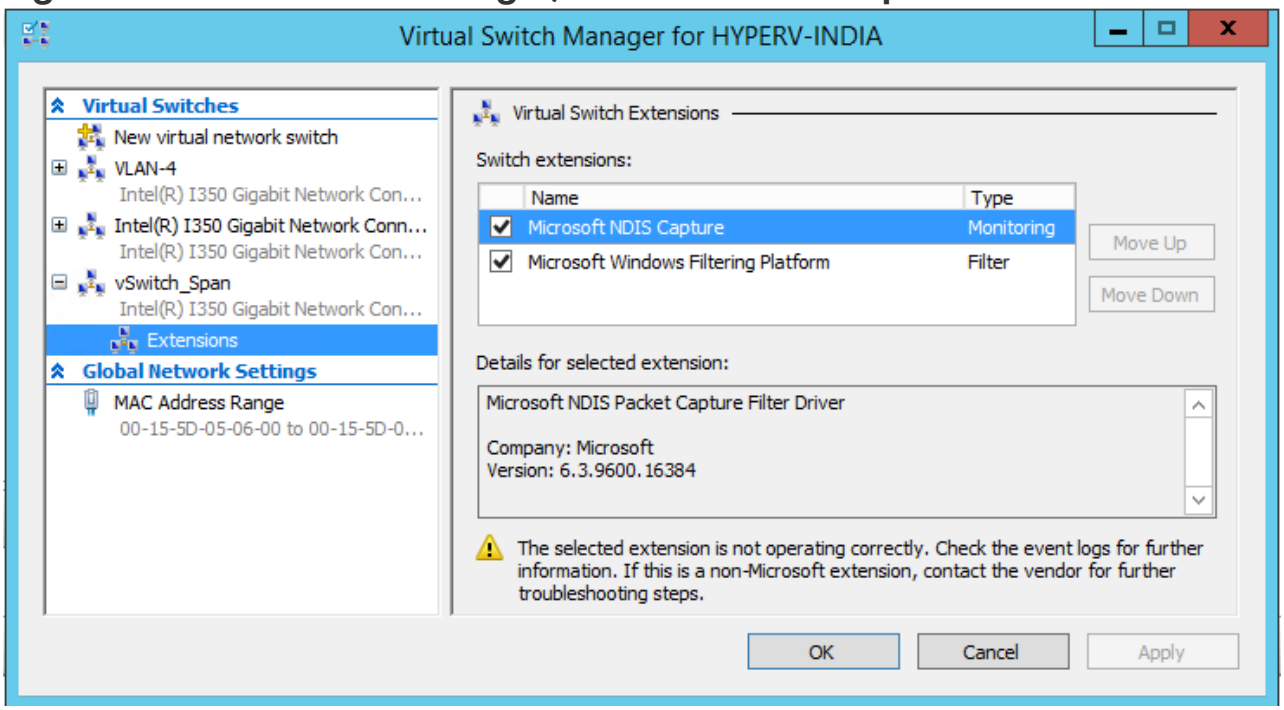
About this task

To enable Microsoft NDIS Capture Extensions for the newly added virtual switch:

Procedure

1. Open the **Virtual Switch Manager** on the Hyper-V host.
2. In the **Virtual Switches** list, expand the virtual switch name **vSwitch_Span** and highlight **Extensions**.
3. In the **Switch Extensions** field, select **Microsoft NDIS Capture**.

Figure 5. Virtual Switch Manager, Microsoft NDIS Capture



4. Click **OK**.

Setting the Mirroring Mode on the External Port

The final part of the procedure is to set the mirroring mode on the external port of the new virtual switch to be the source.

The Hyper-V virtual switch (vSwitch_Span) must be configured so that any traffic that comes to the external source port is forwarded to the virtual network adapter that you configured as the destination.

- The following PowerShell commands can be used to set the external virtual switch port to source mirror mode:

  $ExtPortFeature=Get-VMSystemSwitchExtensionPortFeature -FeatureName "Ethernet Switch Port Security Settings"

  $ExtPortFeature.SettingData.MonitorMode=2

  Add-VMSwitchExtensionPortFeature -ExternalPort -SwitchName vSwitch_Span -VMSwitchExtensionFeature $ExtPortFeature

  Where:

  vSwitch_Span = Newly added SPAN virtual switch name.

  MonitorMode=2 = Source

  MonitorMode=1 = Destination

MonitorMode=0 = None

- The following PowerShell command verifies the monitoring mode status:

  Get-VMSwitchExtensionPortFeature -FeatureName "Ethernet Switch Port Security Settings" -SwitchName vSwitch_Span -ExternalPort | select -ExpandProperty SettingData

  Where:

  vSwitch_Span = Newly added SPAN virtual switch name

Setting the Local SPAN in a Cisco Switch

The following commands can be used to set the local span on a Cisco switch where you plan to test SPAN:

- To add the Source:

  monitor session 1 source interface gigabitEthernet 1/0/1 both

- To add the Destination:

  monitor session 1 destination interface gigabitEthernet 1/0/11

Additional References, SPAN Configuration

The following references provide additional information about Hyper-V and SPAN:

- **http://charbelnemnom.com/2015/01/how-to-deploy-websense-in-stand-alone-mode-on-a-hyper-v-virtual-machine-hyperv-websense/**
- **http://www.cloudbase.it/hyper-v-promiscuous-mode/**

Low Network Performance on Hyper-V Due to NIC Cards

In lab conditions, we noticed that the network latency increases and throughput decreases due to certain features in the NIC not working as expected. This affects network throughput to any OS installed on a Hyper-V server.

In ClearPass, we have noticed the following symptoms when the server is handling authentications:

- Drastic increase in network latency to external servers
- Increase in RADIUS timeout packets
- Increase in RADIUS end-to-end processing of authentication requests

If you notice these symptoms with your ClearPass server running on Hyper-V, please consult with the NIC vendor about compatibility issues with the Microsoft Hyper-V platform, or update to the latest driver version which might resolve network throughput problems.

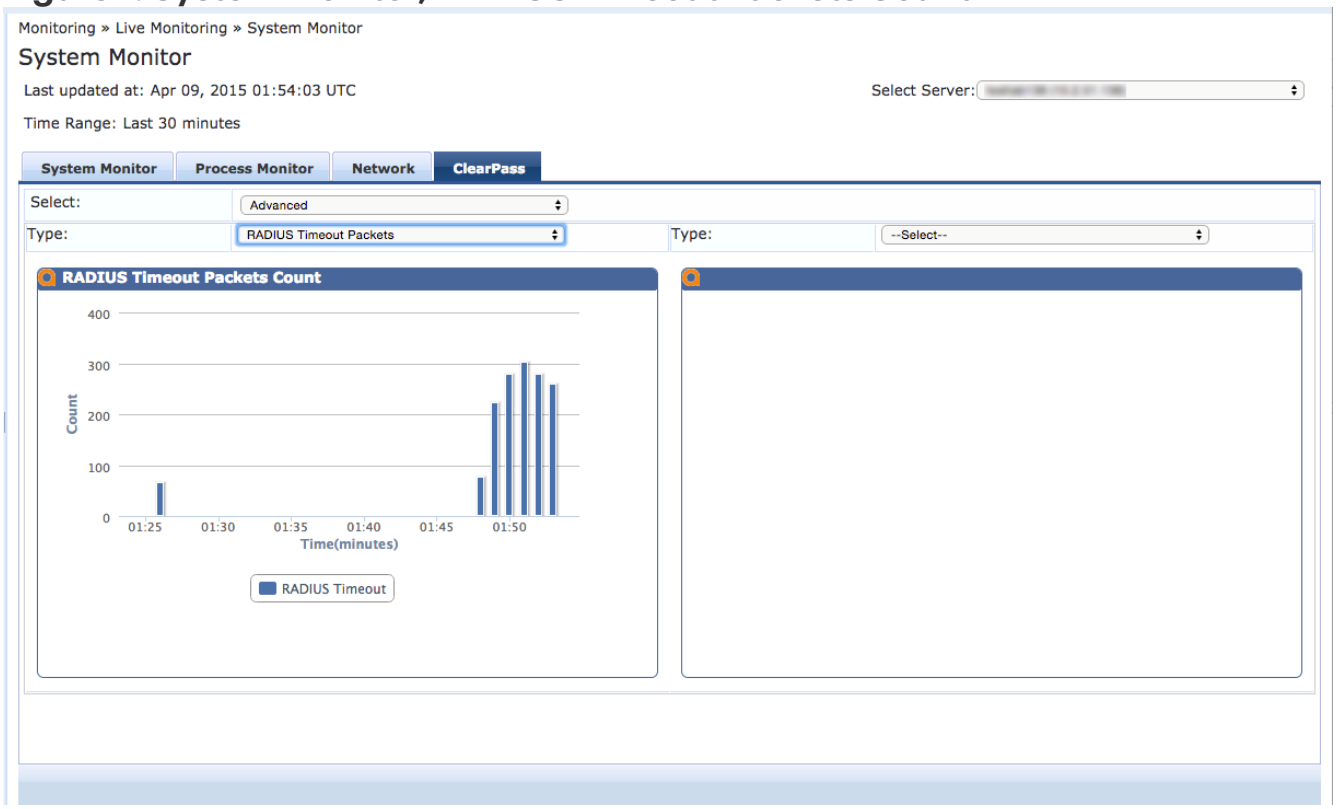
In our test setups, we have found that disabling the Virtual Machine Queue feature on a Broadcom 5720 QP 1Gb Network Daughter Card decreased network latency times; however, this was not true with all Broadcom NICs.

A Microsoft Knowledgebase article detailing the effects of Virtual Machine Queue on Broadcom NICs is available at **http://support.microsoft.com/en-us/kb/2902166**.

The following examples show some sample observations on a ClearPass virtual appliance installed on a Hyper-V server where the Virtual Machine Queue feature was enabled on the NIC card.

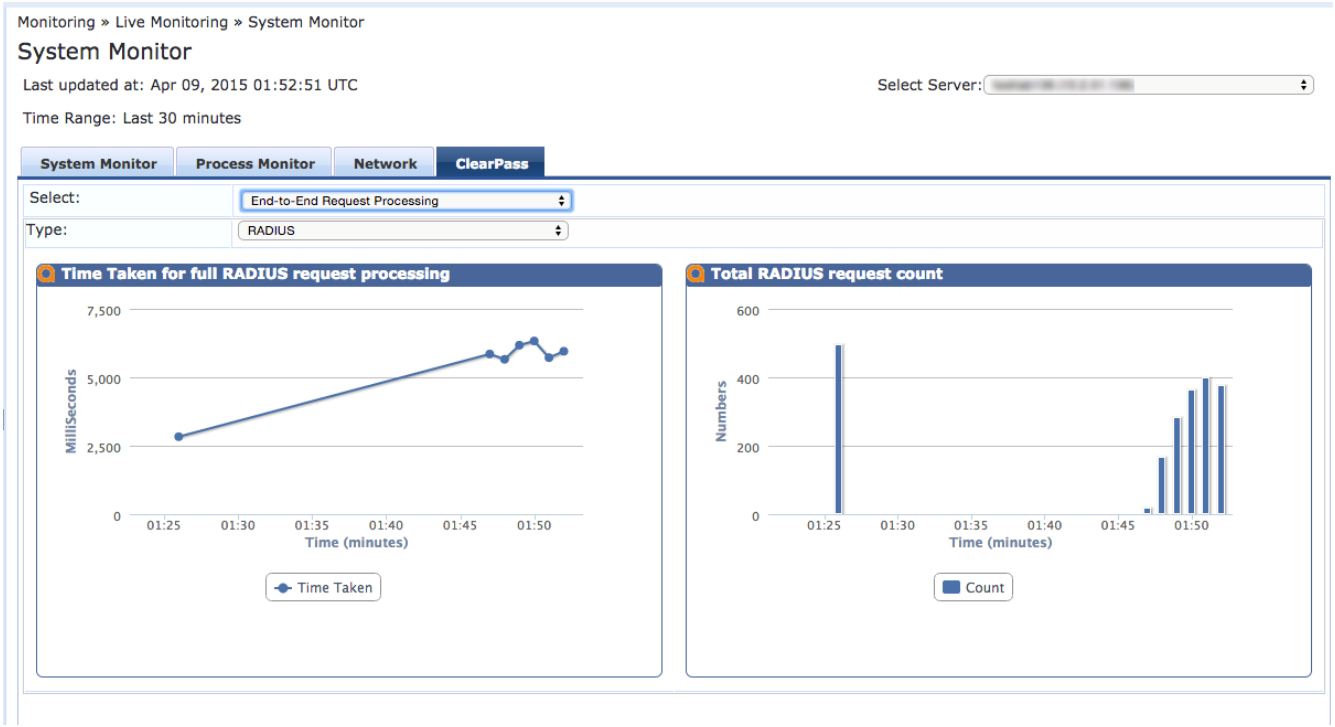
Example 1: Increase in RADIUS Timeout Packets

Figure 1. System Monitor, RADIUS Timeout Packets Count



Example 2: Increase in RADIUS End-to-End Processing

Figure 2. System Monitor, Time for Full RADIUS Request Processing and Total RADIUS Request Count



Example 3: Ping Test

About this task

You can see the network latency increase through a ping test from any VA on the Hyper-V server. In ClearPass, you can perform the ping test as follows:

Procedure

1. Using SSH, log in to ClearPass with the username **appadmin** and your password.
2. Run the command **network ping <ip address>**. The IP address should be a server external to the Hyper-V server.

   The image below shows results of a network ping test to an external server from a ClearPass VA on Hyper-V. Please note that the times shown may vary based on your network:

   Figure 3. Example Results of Network Ping Test to an External Server from a ClearPass VA on Hyper-V

```
64 bytes from 10.2.51.107: icmp_seq=113 ttl=128 time=201 ms
64 bytes from 10.2.51.107: icmp_seq=114 ttl=128 time=17.3 ms
64 bytes from 10.2.51.107: icmp_seq=115 ttl=128 time=70.8 ms
64 bytes from 10.2.51.107: icmp_seq=116 ttl=128 time=74.5 ms
64 bytes from 10.2.51.107: icmp_seq=117 ttl=128 time=94.3 ms
64 bytes from 10.2.51.107: icmp_seq=118 ttl=128 time=109 ms
64 bytes from 10.2.51.107: icmp_seq=119 ttl=128 time=123 ms
64 bytes from 10.2.51.107: icmp_seq=120 ttl=128 time=137 ms
64 bytes from 10.2.51.107: icmp_seq=121 ttl=128 time=162 ms
64 bytes from 10.2.51.107: icmp_seq=122 ttl=128 time=184 ms
64 bytes from 10.2.51.107: icmp_seq=123 ttl=128 time=11.7 ms
64 bytes from 10.2.51.107: icmp_seq=124 ttl=128 time=25.3 ms
64 bytes from 10.2.51.107: icmp_seq=125 ttl=128 time=42.4 ms
64 bytes from 10.2.51.107: icmp_seq=126 ttl=128 time=75.0 ms
```

The image below shows results of a network ping test to an external server from a ClearPass VA after disabling the Virtual Machine Queue feature on a Broadcom 5709c NIC:

Figure 4. Example Results of Network Ping Test to an External Server After Disabling Virtual Machine Queue Feature

```
64 bytes from 10.2.51.107: icmp_seq=26 ttl=128 time=0.249 ms
64 bytes from 10.2.51.107: icmp_seq=27 ttl=128 time=0.328 ms
64 bytes from 10.2.51.107: icmp_seq=28 ttl=128 time=0.370 ms
64 bytes from 10.2.51.107: icmp_seq=29 ttl=128 time=0.451 ms
64 bytes from 10.2.51.107: icmp_seq=30 ttl=128 time=0.382 ms
64 bytes from 10.2.51.107: icmp_seq=31 ttl=128 time=0.460 ms
64 bytes from 10.2.51.107: icmp_seq=32 ttl=128 time=0.327 ms
64 bytes from 10.2.51.107: icmp_seq=33 ttl=128 time=0.412 ms
64 bytes from 10.2.51.107: icmp_seq=34 ttl=128 time=0.278 ms
64 bytes from 10.2.51.107: icmp_seq=35 ttl=128 time=0.306 ms
64 bytes from 10.2.51.107: icmp_seq=36 ttl=128 time=0.362 ms
64 bytes from 10.2.51.107: icmp_seq=37 ttl=128 time=0.302 ms
64 bytes from 10.2.51.107: icmp_seq=38 ttl=128 time=0.299 ms
```

Hyper-V Live Migration Caveat

When a Hyper-V live migration is performed from one ClearPass Hyper-V instance to another while the instance is powered off, then the first time the destination Hyper-V instance is powered on or restarted after the migration, the ClearPass IP address is lost.

This issue only occurs if the migration is done while the Hyper-V instance is powered off. It does not occur if the migration is performed while the instance is powered on.

To avoid this issue when performing a Hyper-V live migration, do one of the following:

- Do not power off the Hyper-V instance before the migration. Instead, do the migration while the instances are powered on.
- Otherwise, if the issue has already occurred, perform a system refresh-network operation in the CLI and then restart the system to restore the IP address to the ClearPass instance.

Virtual Appliance Installation: Nutanix AHV

This chapter describes procedures for installing and upgrading ClearPass 6.14 on a Nutanix Acropolis Hypervisor (AHV) virtual appliance.

Subtopics

Nutanix AHV Installation Process Overview
Morphing a Nutanix AHV Version

Nutanix AHV Installation Process Overview

About this task

The process of installing ClearPass on a Nutanix AHV virtual appliance is done in four stages:

Subtopics

Image upload
Reserved disk
VM creation
Network adapter for MGMT and data interfaces

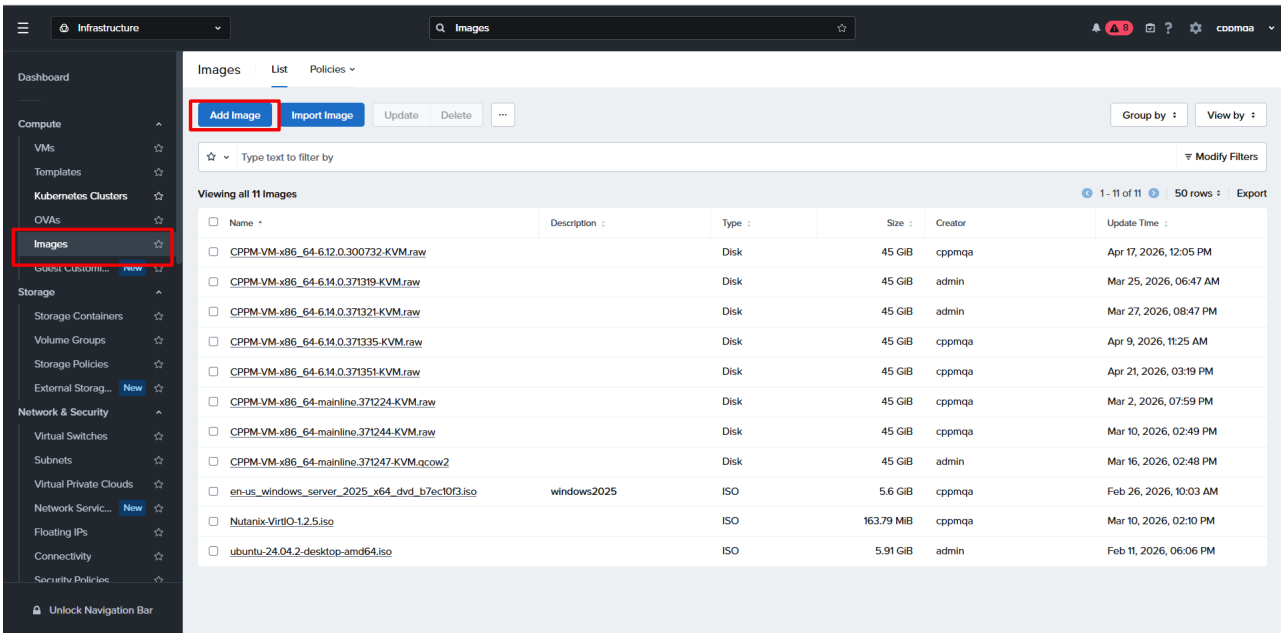
Image upload

Procedure

1. ClearPass 6.14 Nutanix software packages are distributed as Zip files. Download the software image from the **Download Software > ClearPass > Policy Manager > Current Release > Nutanix** folder on the Support site and unzip it to a folder on your client machine to extract the RAW files.

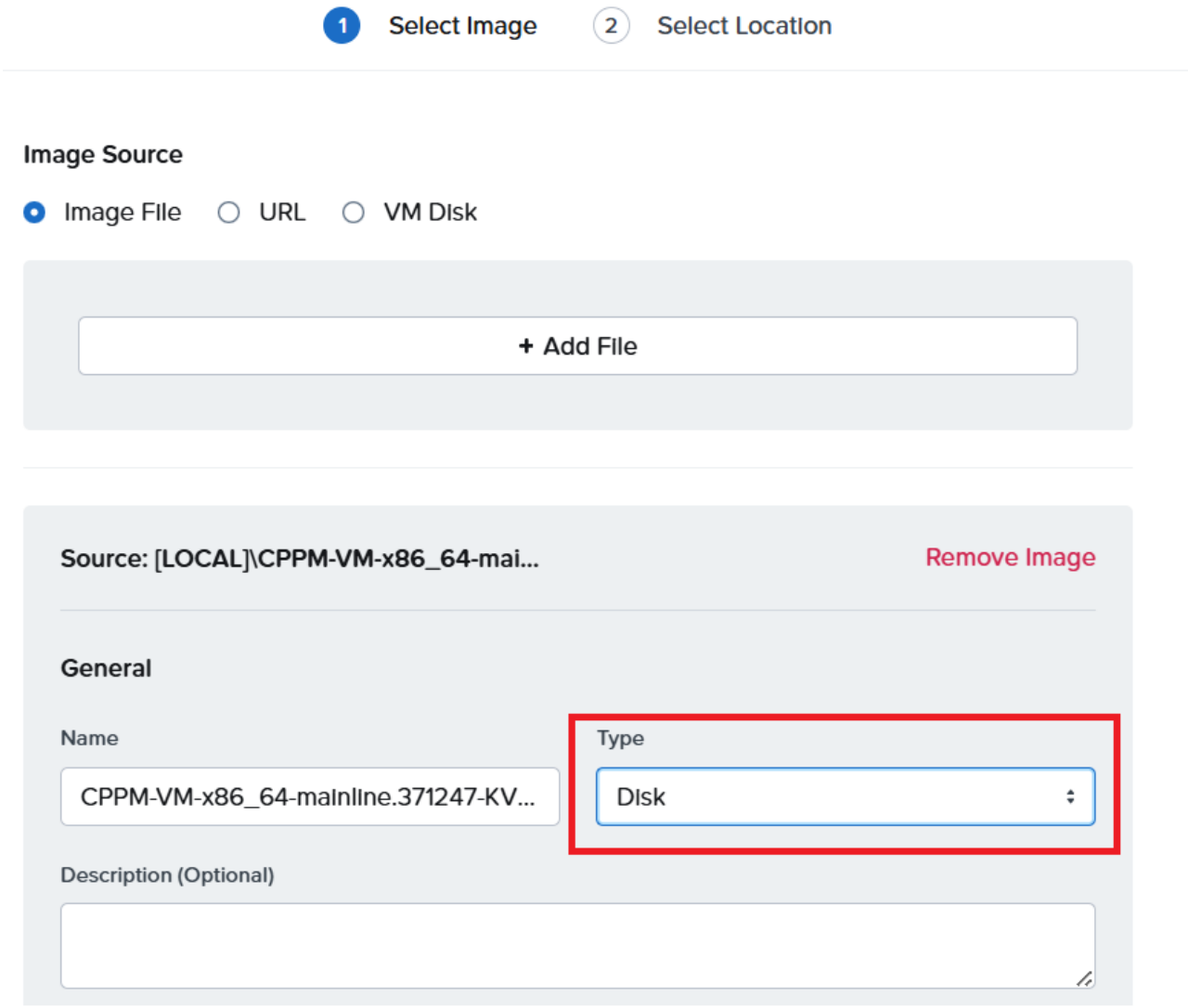
2. Upload the RAW file to Nutanix via **prism central UI > Compute > Images > Add Image** .

Figure 1. AHV upload RAW file



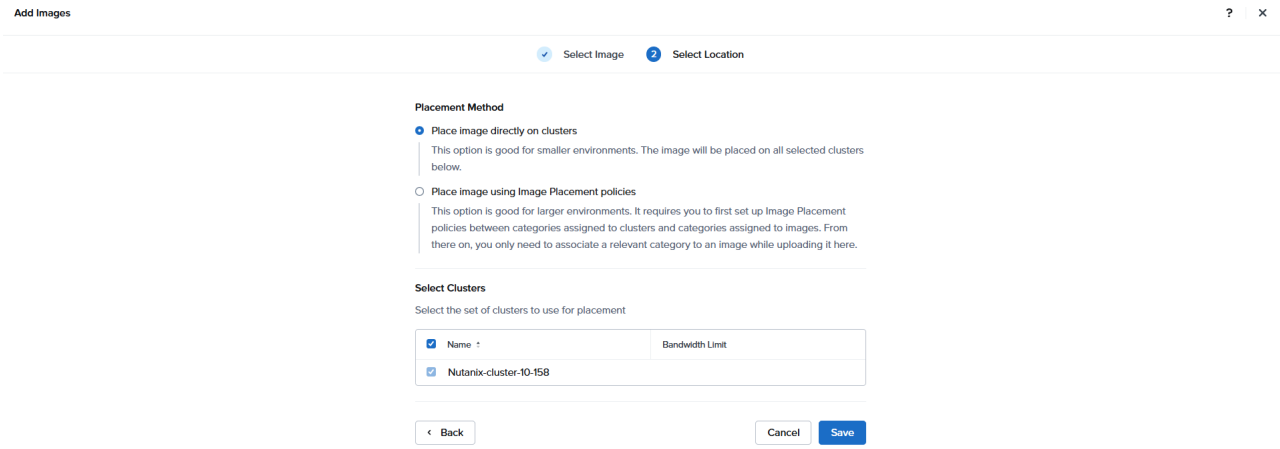
3. Add the file and select the extracted RAW image from the local machine. Select **Disk** from the **Type** drop-down menu.

Figure 2. AHV add RAW image



4. Select **Place image directly on clusters** for the **Placement Method** and click **Save**.

Figure 3. Place image directly on clusters

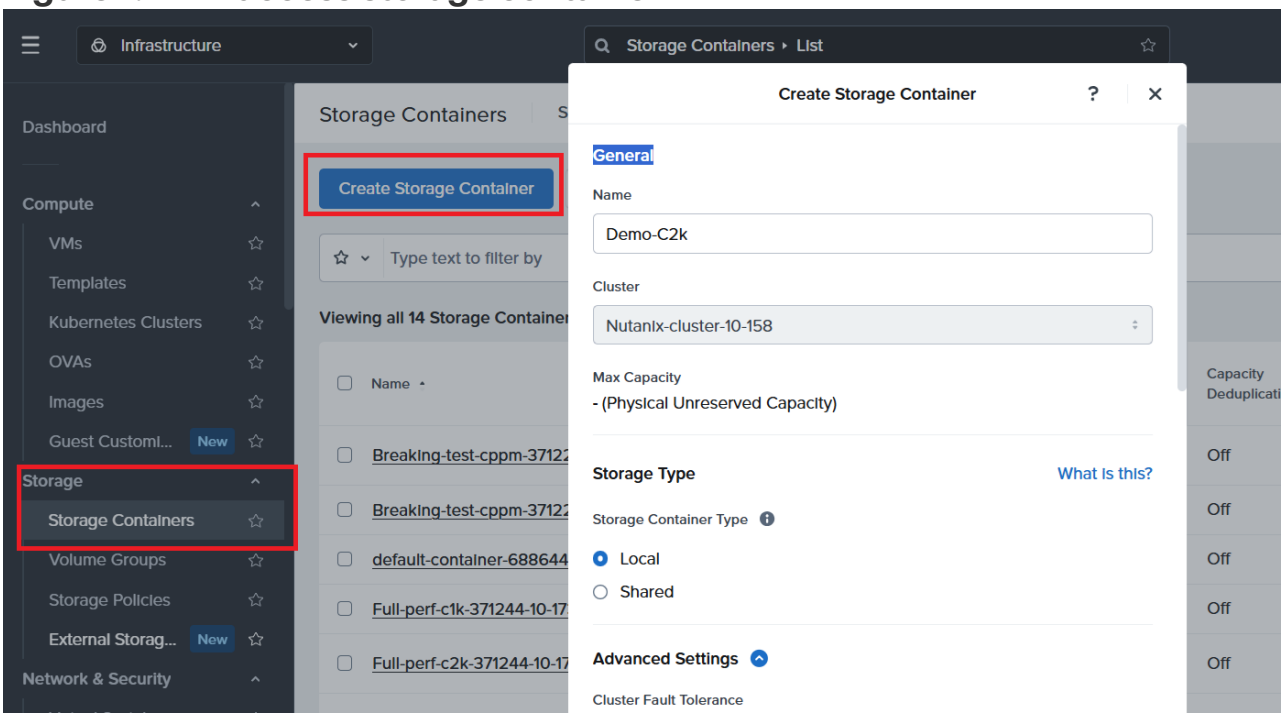


Reserved disk

Procedure

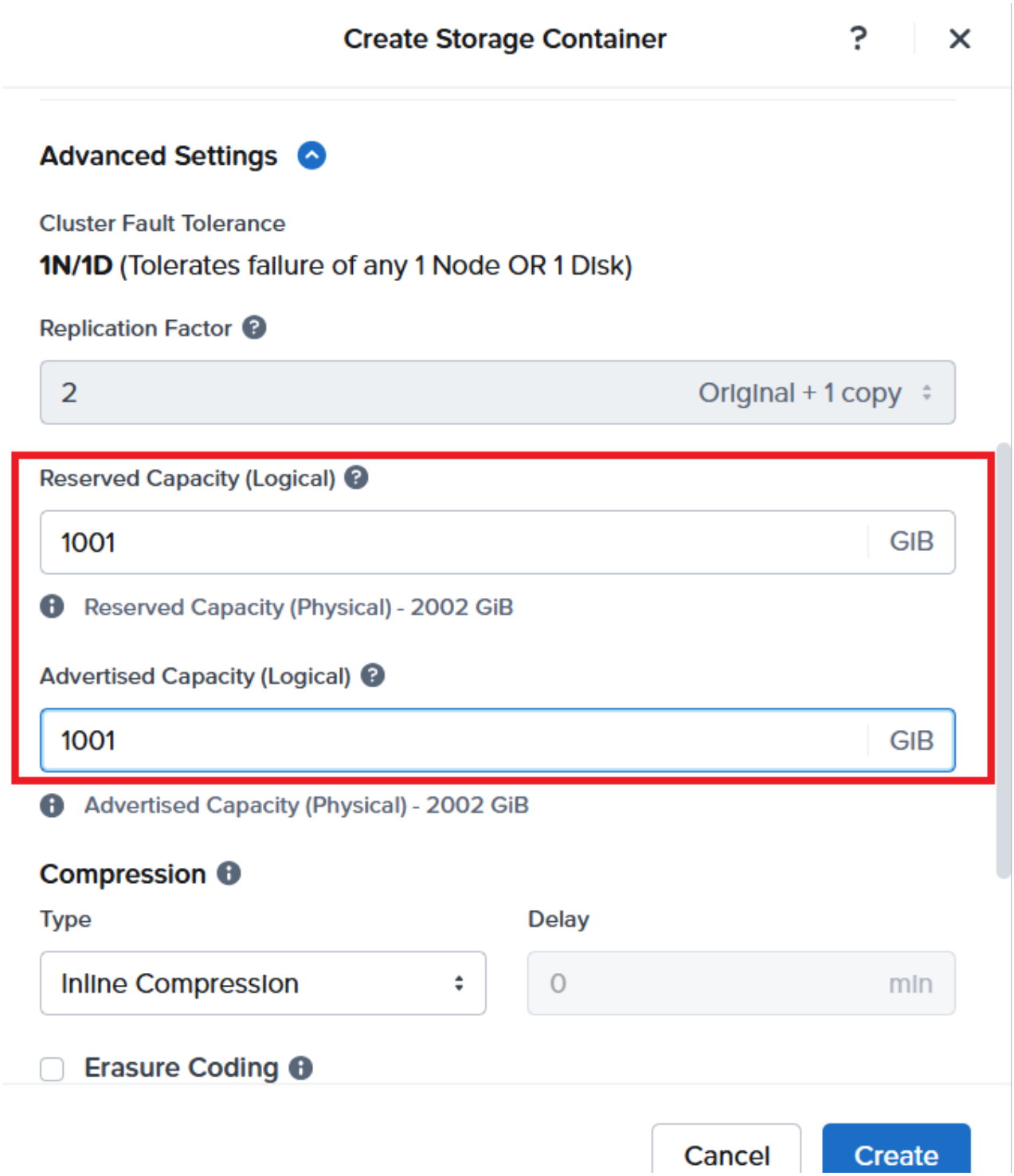
1. Create a reserved storage disk based on Model (C1000V / C2000V / C3000V) in the storage container.
2. Access the storage container from Prism Central via **Infrastructure** > **Storage** > **Storage Containers**.

Figure 1. AHV access storage container



3. In advanced settings, configure the **Reserved Capacity (Logical)** and **Advertised Capacity (Logical)** fields by adding 1GB of extra disk space to the capacity for each model (CLABV,C1000V,C2000V,C3000V) and click **Create**.

Figure 2. AHV configure capacity

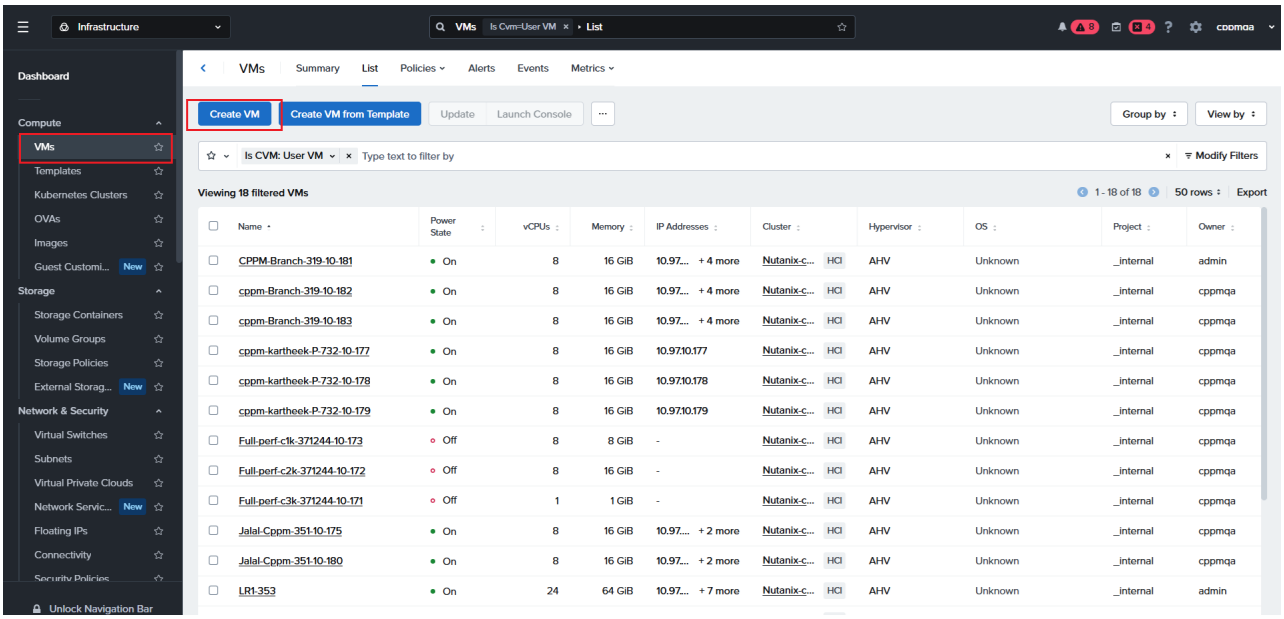

Create Storage Container
Advanced Settings
Cluster Fault Tolerance
1N/1D (Tolerates failure of any 1 Node OR 1 Disk)
Replication Factor
2
Original + 1 copy
Reserved Capacity (Logical)
1001
GiB
Reserved Capacity (Physical) - 2002 GiB
Advertised Capacity (Logical)
1001
GiB
Advertised Capacity (Physical) - 2002 GiB
Compression
Type
Inline Compression
Delay
0
min
Erasure Coding
Cancel
Create

VM creation

Procedure

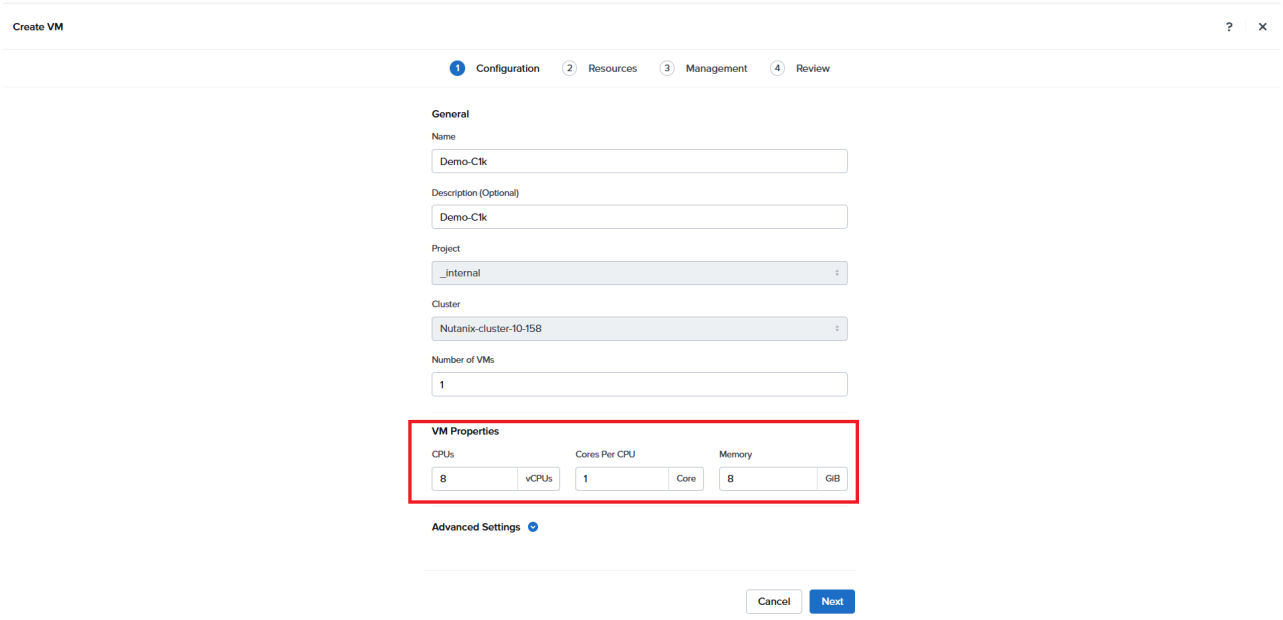
1. Select **Create VM** from **Compute > VMs > Create VM** .

Figure 1. AHV create VM



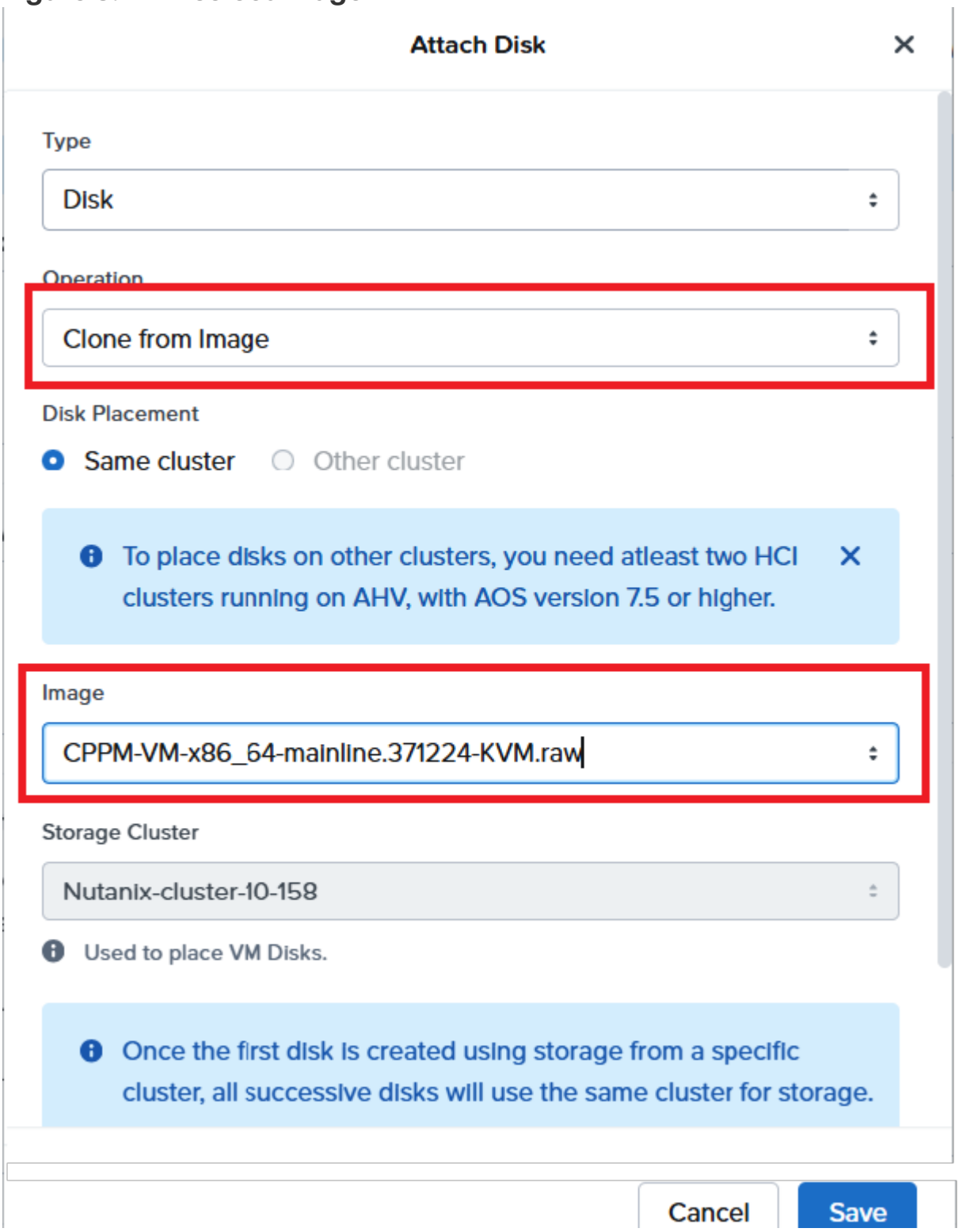
2. Specify the **CPUs** and **Memory** based on the model.

Figure 2. AHV specify CPU and memory



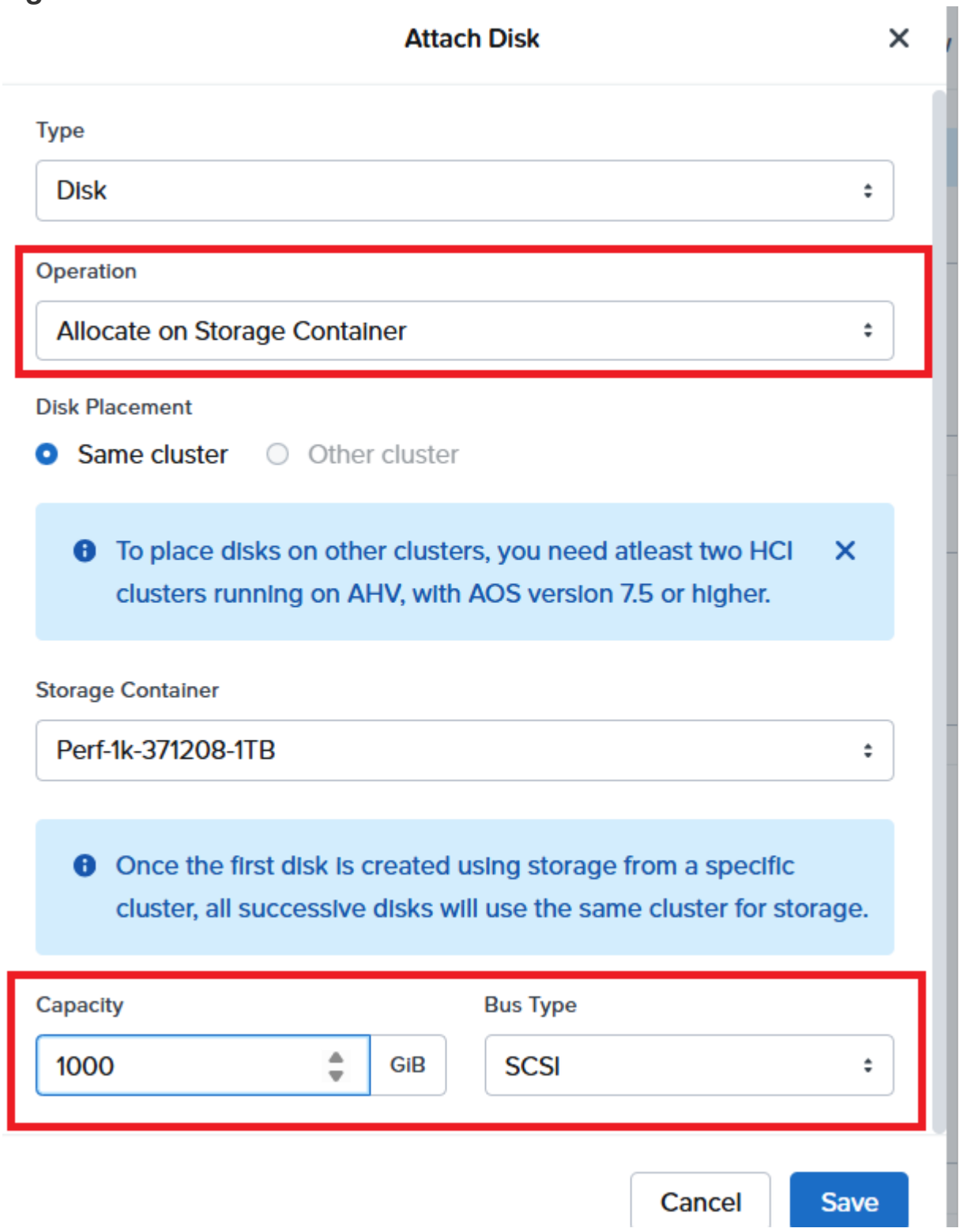
3. Select **Clone from Image** in the **Operation** drop-down menu and selecting the correct image from the **Image** drop-down menu.

Figure 3. AHV select image



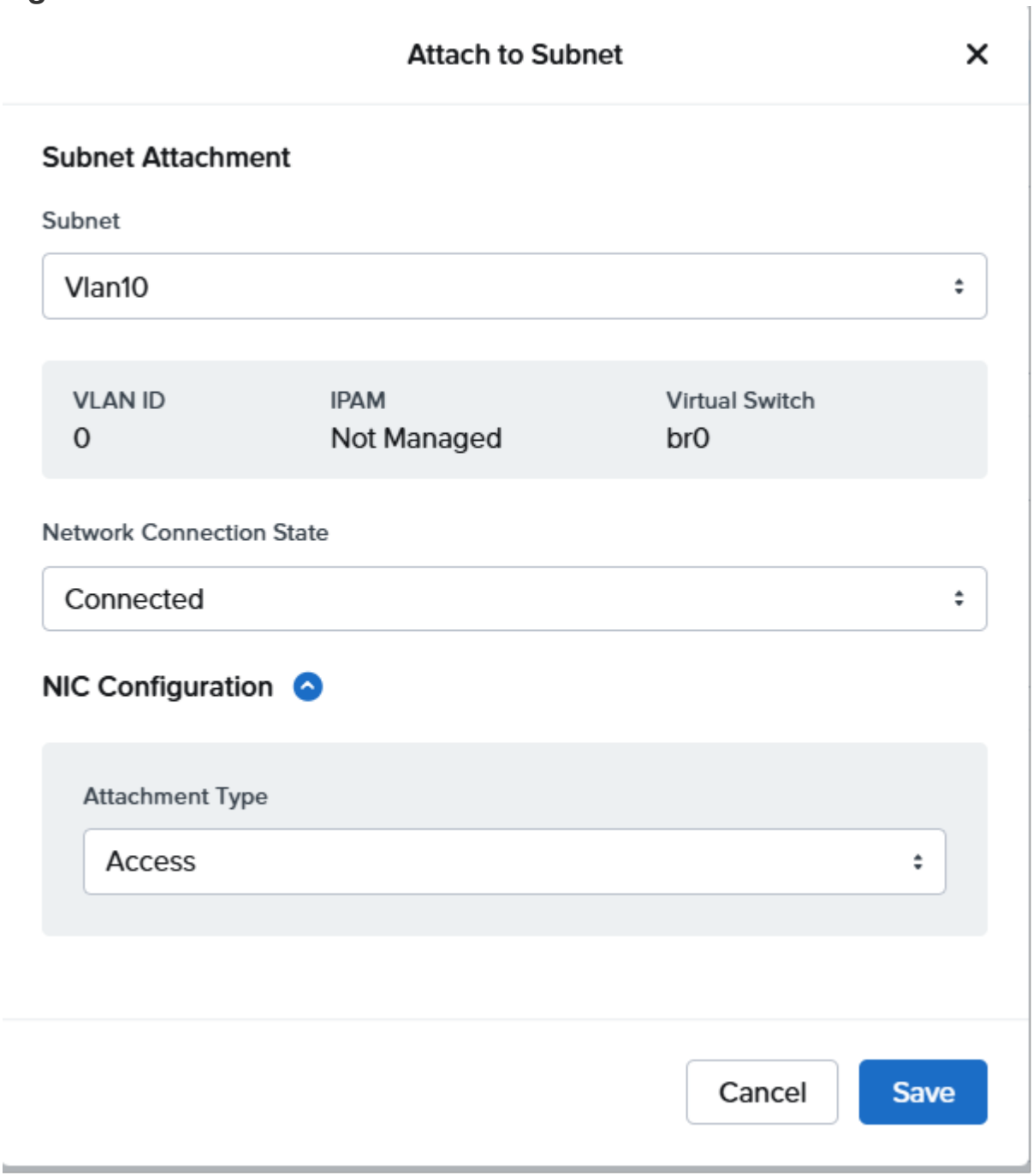
4. Attach a second disk by selecting **Allocate on Storage Container** from the **Operation** drop-down menu, selecting the correct storage container in the **Storage Container** drop-down menu, and configuring the **Capacity** and **Bus Type**.

Figure 4. AHV add second disk



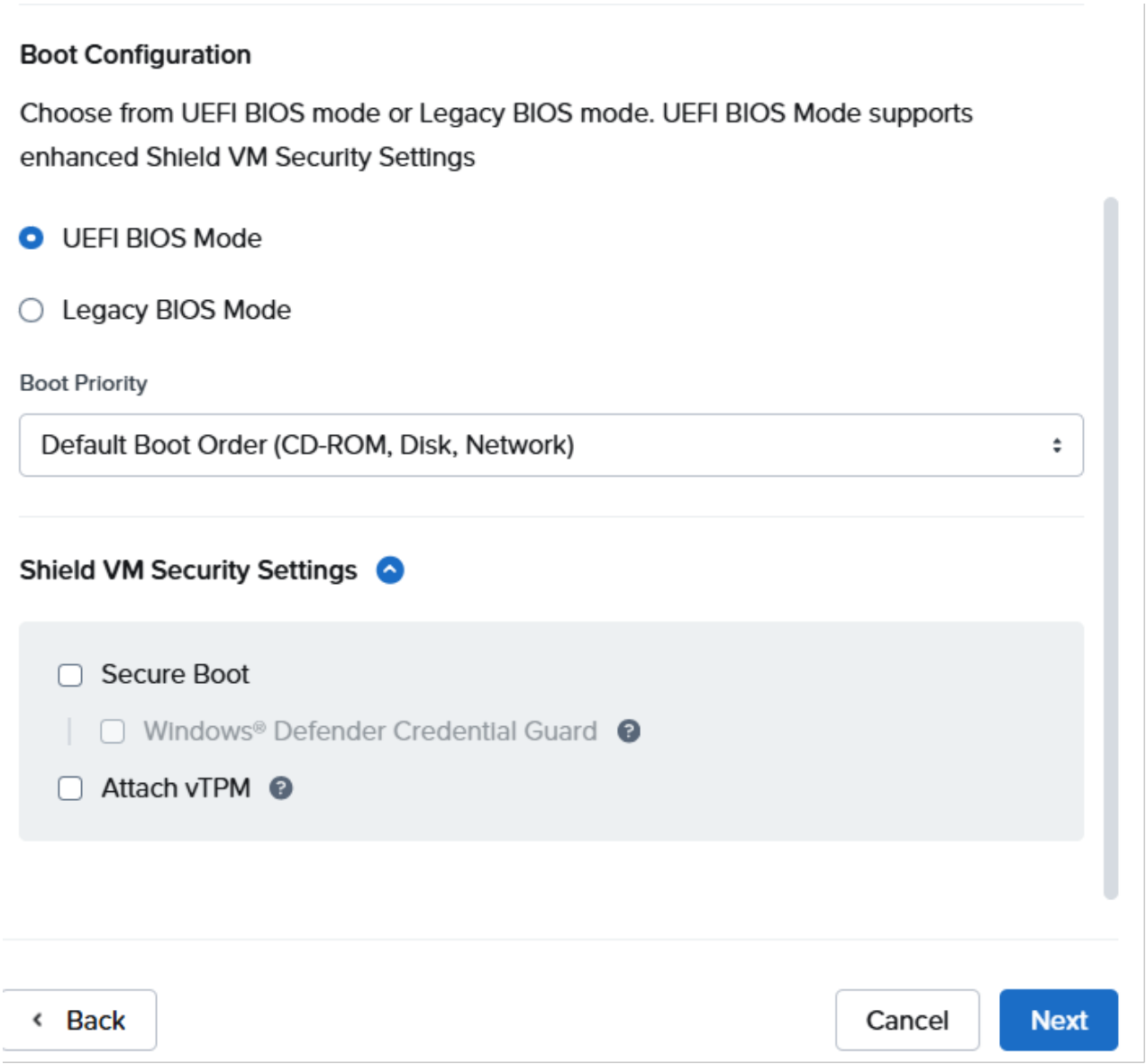
5. Attach two subnets.

Figure 5. AHV add two subnets



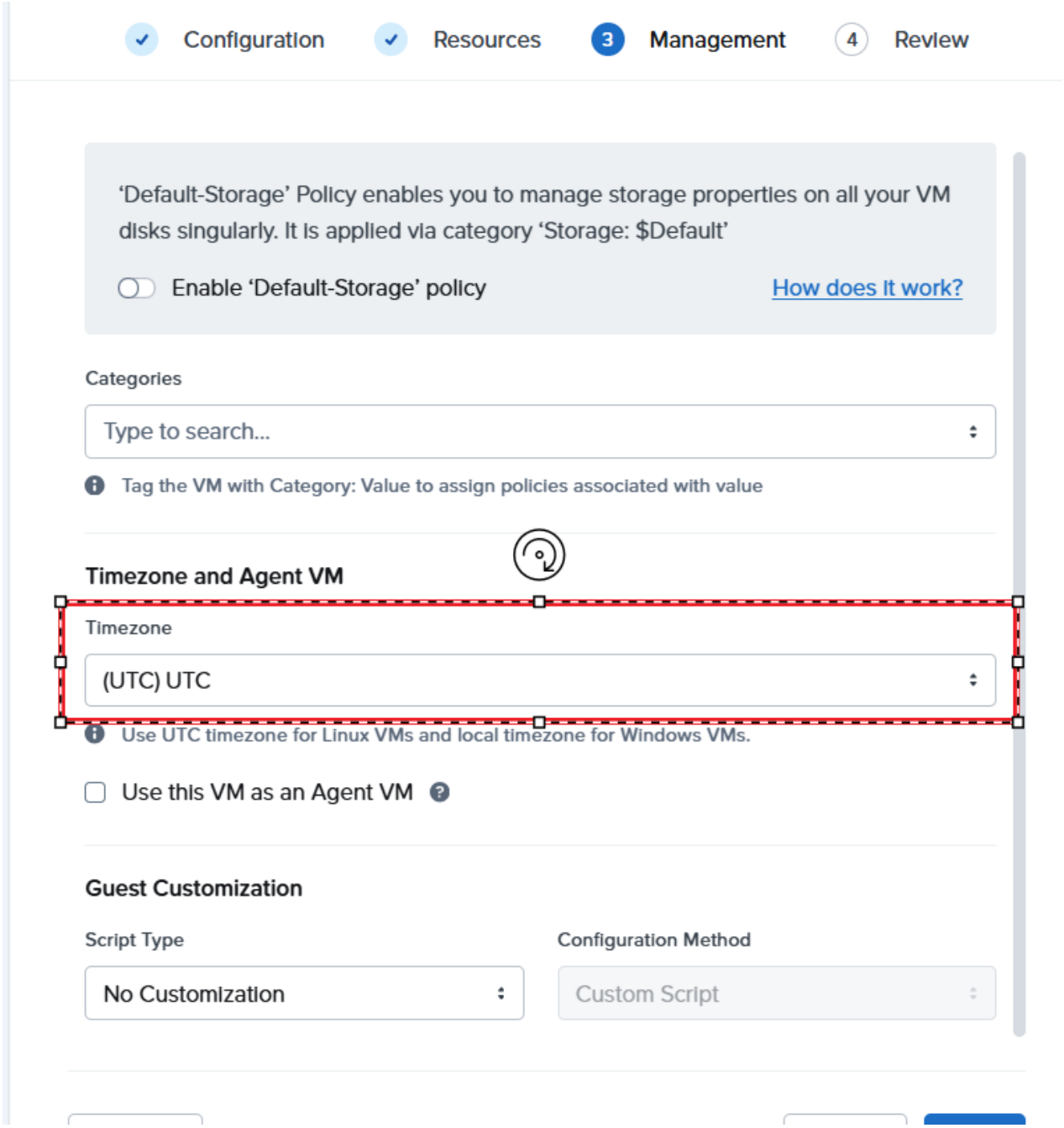
6. Set the **Boot Configuration** to **UEFI BIOS Mode**.

Figure 6. AHV UEFI BIOS mode



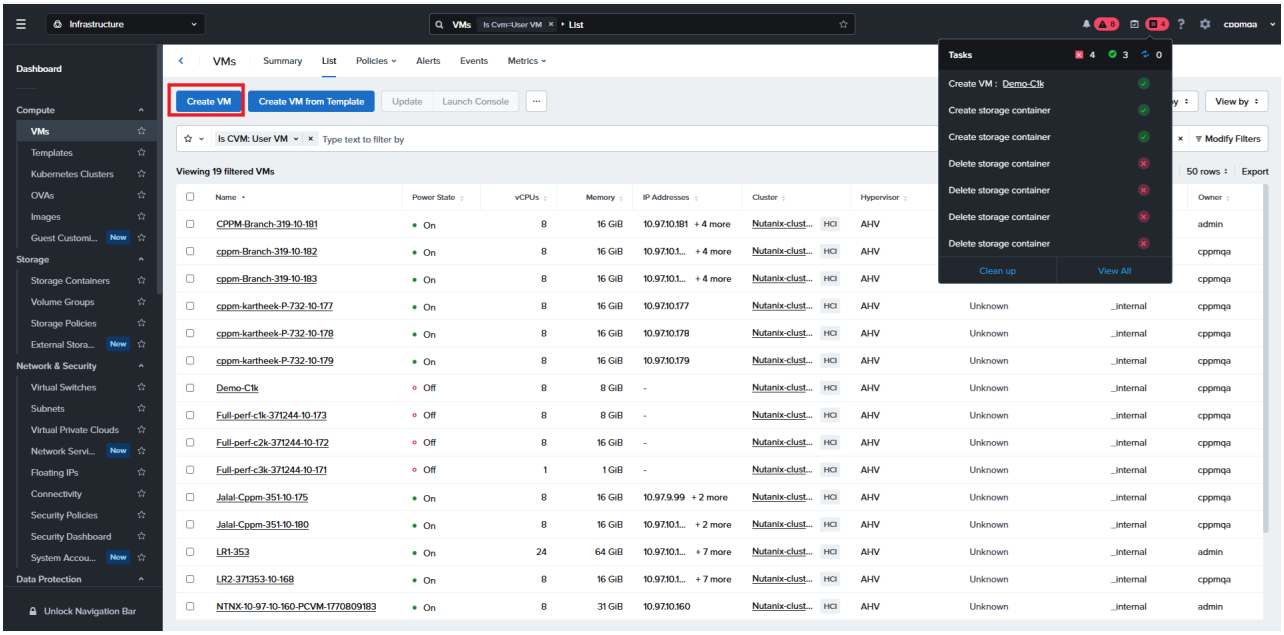
7. Set the **Timezone and Agent VM** to your local timezone.

Figure 7. AHV timezone



8. Review and create the VM.

Figure 8. Review and create VM

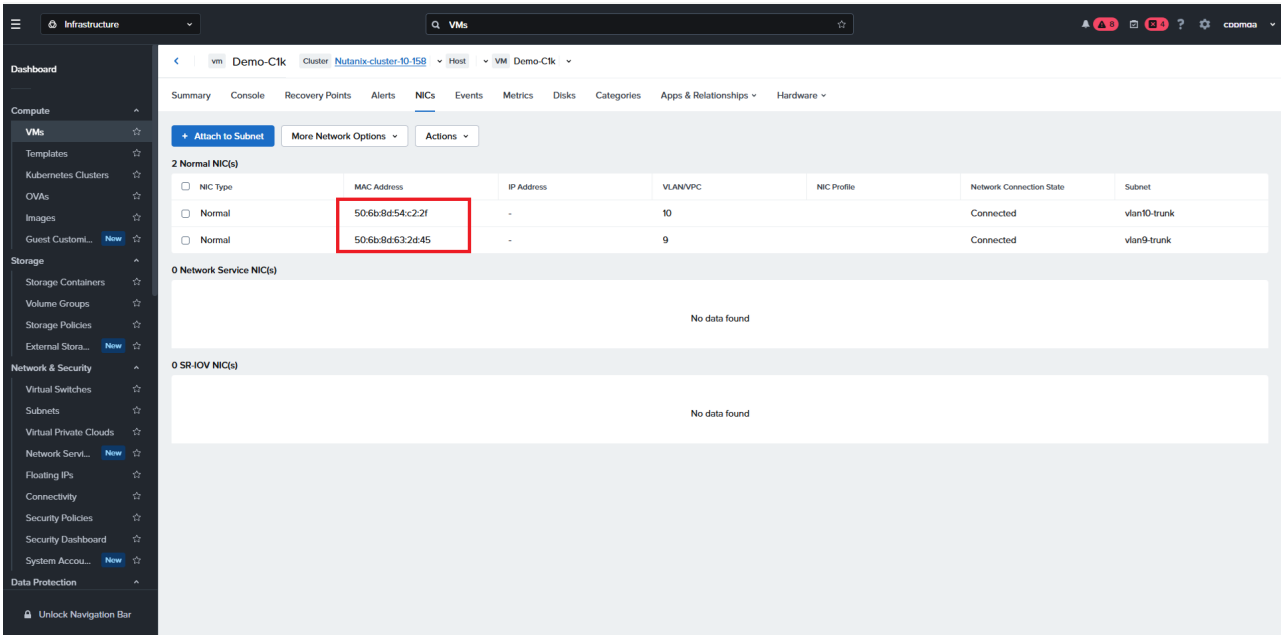


Network adapter for MGMT and data interfaces

Procedure

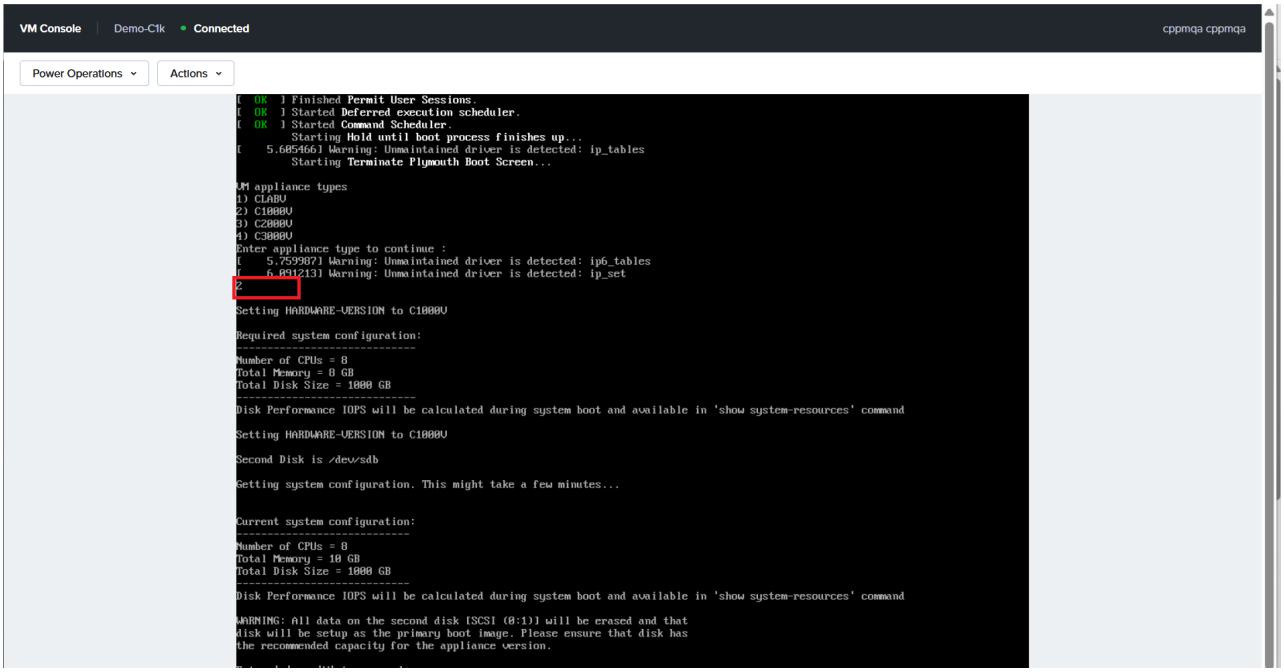
1. After deploying the VM instance and before powering on, check the NIC adapters value by launching the console.
2. Ensure that the first network adapter MAC address is lower than the second. If it is not lower, delete the network adapter and attach another subnet via**Infrastructure > VMs > +Attach to Subnet**. Repeat this process until the first network adapter MAC address is lower than the second.

Figure 1. Compare MAC address



3. Power on the VM and select the appliance model.
4. Enter the list number for the appropriate appliance type. Do not enter the appliance model. Options are as follows: **1. CLABV**, **2. C1000V**, **3. C2000V**, and **4. C3000V**. For example, to install a C1000V, enter **2**.

Figure 2. AHV set appliance type



Morphing a Nutanix AHV Version

About this task

> **NOTE**
> The illustrations in this section use the example of morphing a CLABV virtual appliance to a C1000V. Adjust your own configuration as needed.

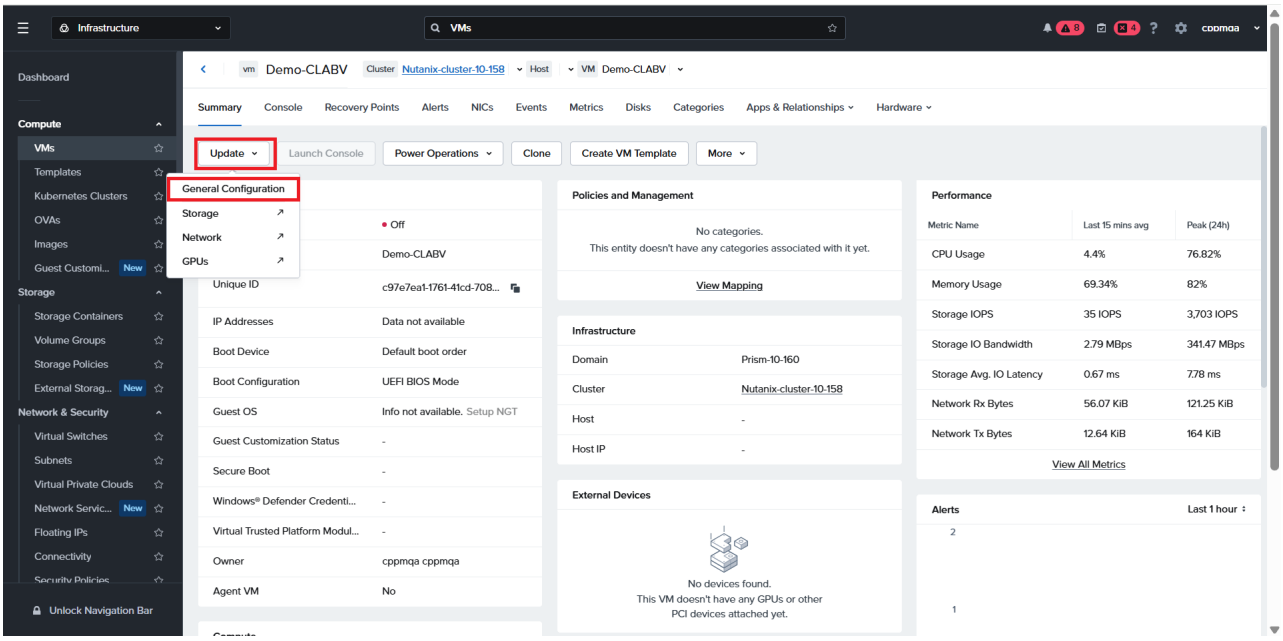
> **NOTE**
> Starting with the 6.11.0 release, after the **system morph-vm** command is run during a ClearPass upgrade and the new disk is added, the existing hard disk cannot be removed. This is because the existing disk is used to extend the disk space for the server, and is included in the calculations to determine the required capacity of the additional hard disk.

Perform the following steps when morphing an Nutanix virtual appliance to a higher model virtual appliance:

Procedure

1. Power off the virtual appliance gracefully.
2. Open the virtual instance that needs to be morphed from **Prism Central**.
3. Launch the VM console.
4. Select **General Configuration** from the **Update** drop-down menu.

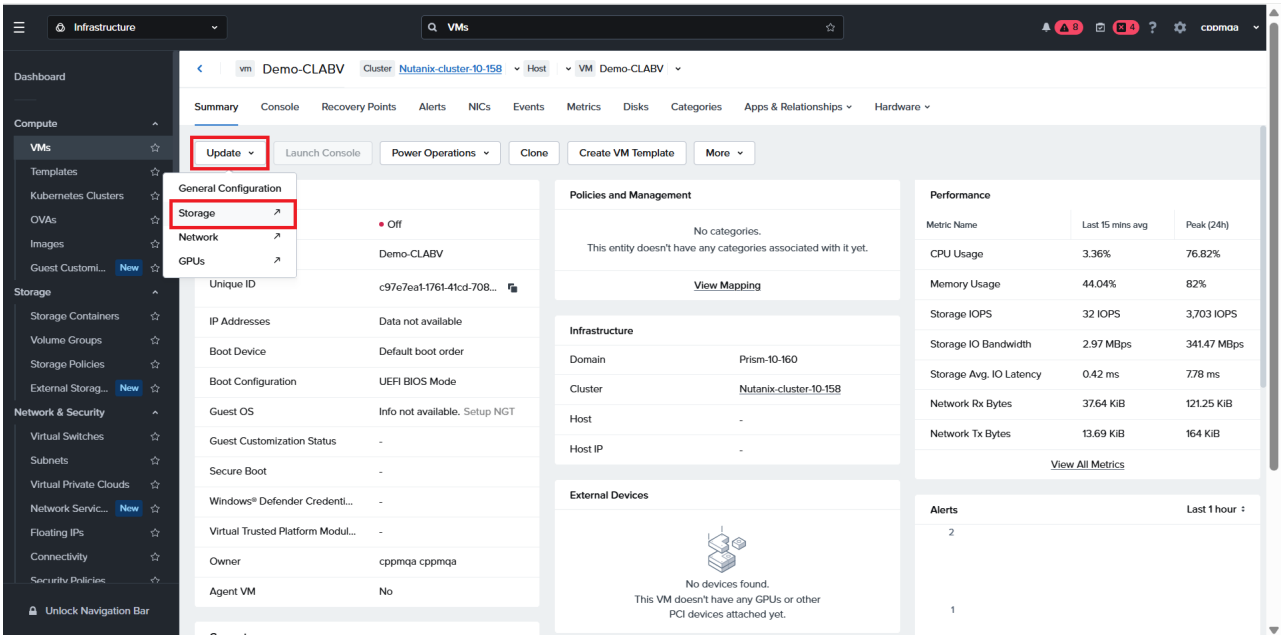
Figure 1. AHV select general configuration



5. Create a storage disk based on the model to be morphed using the steps in Reserved disk.

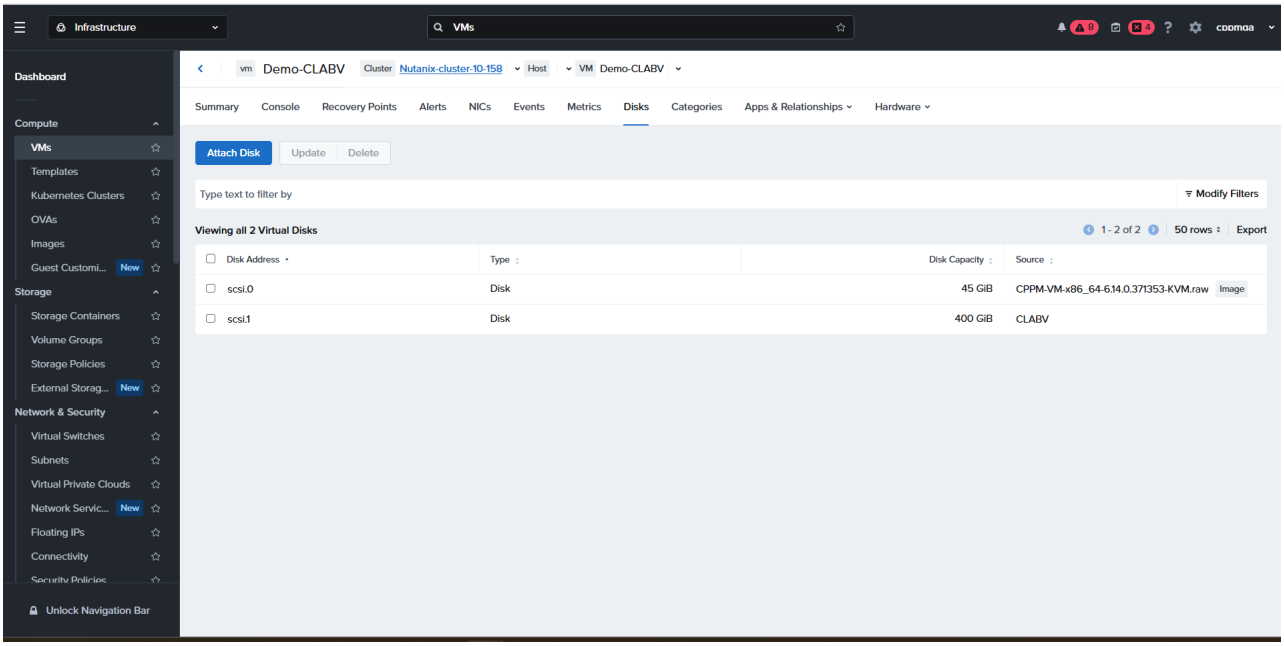
6. Select **VM > VM instance > update > storage**.

Figure 2. AHV create storage disk



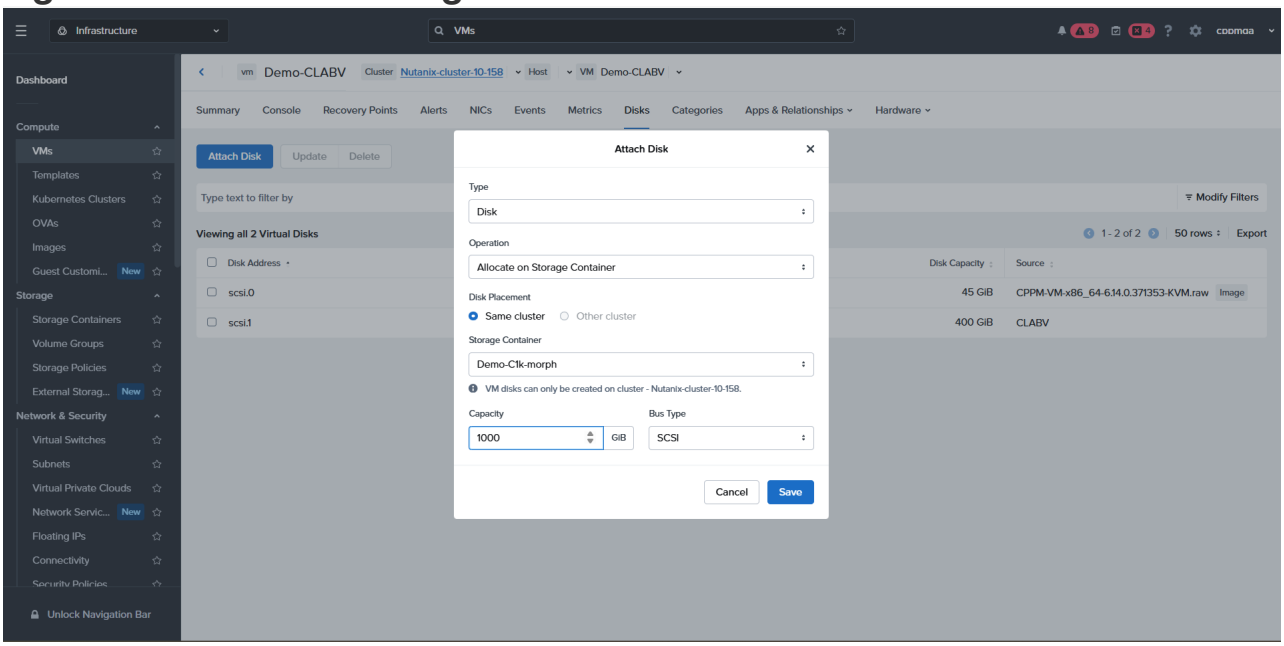
7. Attach the disk.

Figure 3. Attach disk



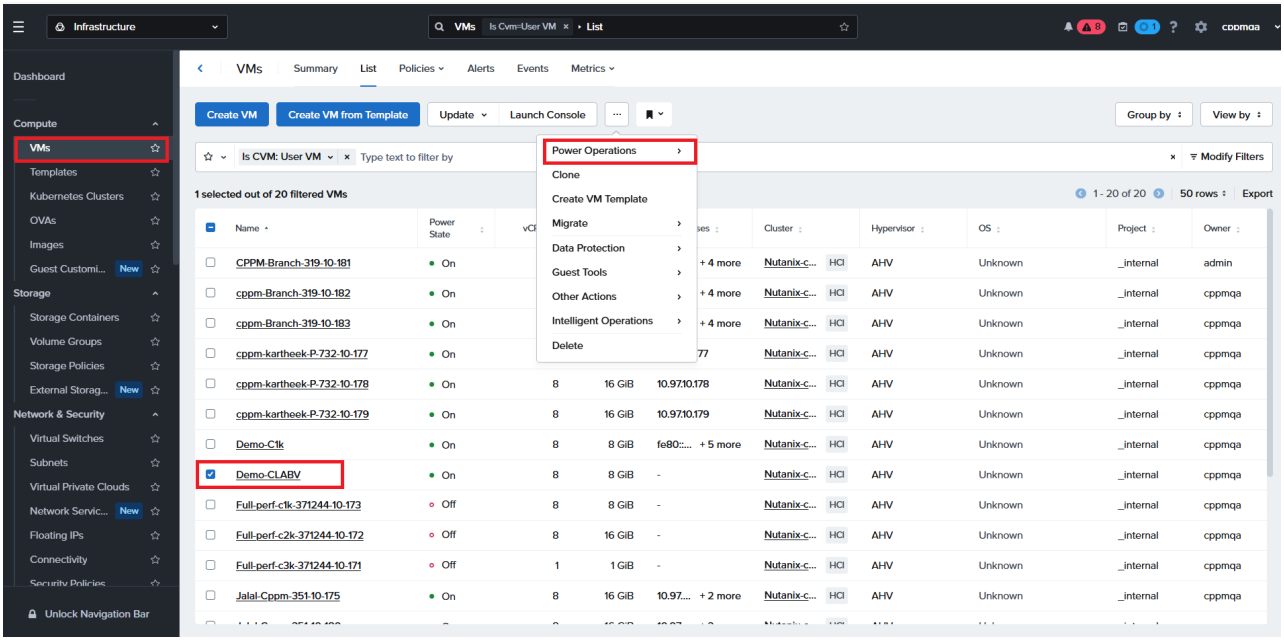
8. Select the storage container created, provide the disk capacity, and save.

Figure 4. AHV select storage container



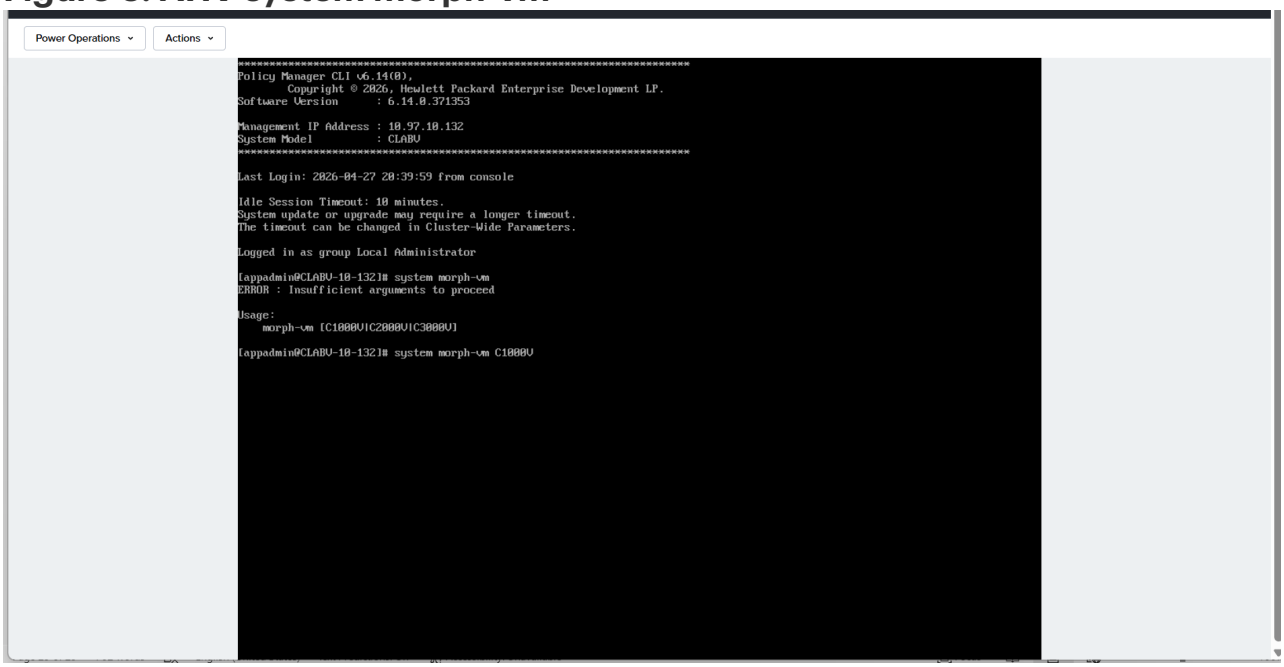
9. Power on the virtual machine from Prism Central by following **Compute > VM > select the VM instance > Power Operations > Power On**.

Figure 5. AHV power on the virtual machine



0. Launch console of the VM instance.
1. Execute **system morph-vm <vm-type>**.

Figure 6. AHV system morph-vm



Virtual Appliance Installation: VMware vSphere (ESXi)

This chapter describes procedures for installing and upgrading to ClearPass 6.11 on a VMware vSphere (ESXi) virtual appliance.

Subtopics

VMware vSphere Hypervisor Installation Process Overview
Meeting VMware vSphere Hypervisor Server Specifications
Installing ClearPass on an ESXi Virtual Appliance
After Upgrading ClearPass to 6.14.0 on ESXi Servers
Morphing ESXi to a Higher Model Virtual Appliance
Manually Upgrading an ESXi Installation

VMware vSphere Hypervisor Installation Process Overview

About this task

The process of installing ClearPass on a VMware vSphere Hypervisor (ESXi) virtual appliance (VA) is done in four stages:

Procedure

1. ClearPass 6.11 VMware software packages are distributed as Zip files. Download the software image from the **Download Software > ClearPass > Policy Manager > Current Release > ESXi** folder on the Support site and unzip it to a folder on your server to extract the files.
2. Follow the steps in the OVF wizard to deploy the OVF files, but do not power on yet.
3. Add a new hard disk, based on the requirements for your type of VA. See the System Requirements for ClearPass 6.14 chapter.
4. Power on and configure the VA.

   Instructions for these procedures are provided in Installing Policy Manager on an ESXi Virtual Appliance .

   Follow the same steps if you are upgrading ClearPass from 6.12.x to 6.11.0 on a VMware ESXi server, **and only** if the MAC address of **Network adapter1** is higher than that of **Network adapter2**. The same steps will also need to be followed if while being on the 6.11.0 OVF deployment, **Network adapter 1** MAC address is still higher

than **Network adapter 2**. See After Upgrading ClearPass to 6.14.0 on ESXi Servers for details.

CAUTION

ESXi 7.0.0 is not supported. Installing or upgrading to ClearPass 6.14.0 on an ESXi 7.0.0 system fails, and the system becomes unreachable and cannot be recovered. If you are on ESXi 7.0 you must apply the latest ESXi patch updates and ensure it is updated to ESXi 7.0 Update 3c or later before upgrading or installing ClearPass. You should also make a backup of your ClearPass 6.12.x system before installing or upgrading ClearPass. (CP-49208)

CAUTION

Avoid using the CLI to update an ESXi 6.14.x instance to a higher 6.14.x version, and update it through the user interface instead. If you must use the CLI to update this instance type to a higher 6.14.x version, please increase the CLI Session Idle Timeout cluster-wide parameter value to 120 minutes to ensure that the patch update is not interrupted.

NOTE

When creating a database backup of a lower version of ClearPass to be restored on ClearPass 6.11, best practice is to create separate backups for the configuration (tipsdb) and Insight (insightdb) databases. Backup files can be very large, and creating separate backups ensures that they will be restored correctly without overloading the system and blocking the restore operation.

NOTE

When Vcenter 7.0 or Vcenter 8.0 is used to deploy the ClearPass 6.14.0 OVF on ESXi 7.0 U3c and later, the error message "CERTIFICATE_INVALID_DIGEST" is displayed.

To deploy through Vcenter 7.0 U3c or Vcenter 8.0, do one of the following: Do not select the .cert file while deploying the OVF, or use an embedded ESXi host instead to deploy the OVF.

Meeting VMware vSphere Hypervisor Server Specifications

Before you perform any installations, upgrades, or updates, ensure that your system meets all virtual appliance requirements. Please refer to the System Requirements for ClearPass 6.14 chapter and carefully review all virtual appliance (VA) requirements, including functional IOP ratings, and verify that your system meets these requirements.

Be sure that your system meets the recommended specifications required for the ClearPass virtual appliance. The ClearPass ESXi disk image is shipped as a 40 GB hard disk volume. This must be supplemented with additional storage/hard disk through the virtual machine settings by adding a new hard disk. The additional space required depends on the ClearPass virtual appliance version.



NOTE

If you do not add a new hard disk to the virtual appliance before it is powered on, it will continue to restart with kernel panics.

An ESXi version can be morphed to a larger version by using the morph-vm command. For more information, see Morphing ESXi to a Higher Model Virtual Appliance in this Tech Note, and the **Command Line Interface > System Commands > system morph-vm** section in the ClearPass Policy Manager 6.11 User Guide.

Installing ClearPass on an ESXi Virtual Appliance

After you download and unzip the ClearPass 6.11 VMware ESXi software package Zip files, follow the instructions in this section to deploy the ClearPass files, add a new hard disk, and power on and configure the VA.

Subtopics

Deploy the ClearPass Image on a VMware ESXi Server
Add a Hard Disk to the ESXi Virtual Appliance
Power On and Configure the ESXi Virtual Appliance

Deploy the ClearPass Image on a VMware ESXi Server

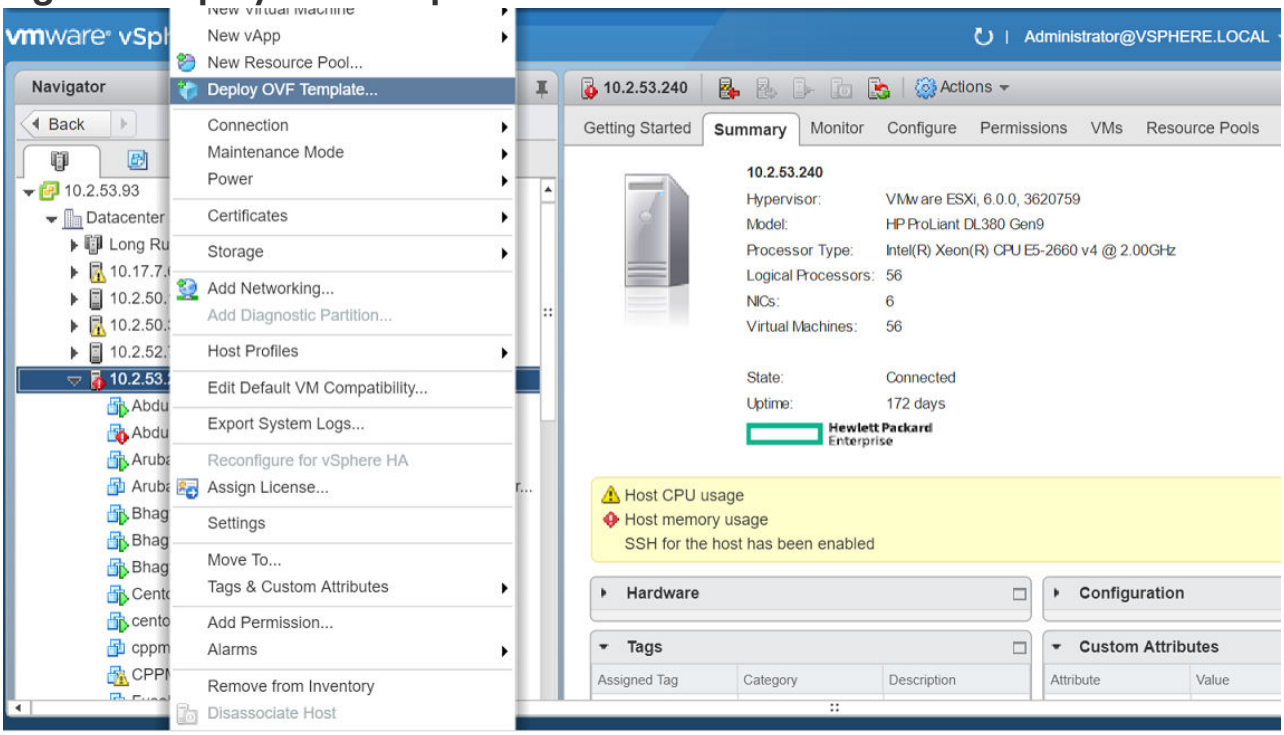
About this task

> **NOTE**
> The illustrations in this section use a CLABV virtual appliance as an example. See the System Requirements for ClearPass 6.14 chapter for the appropriate requirements for your appliance.

Procedure

1. Start the VMware vSphere Web client and connect to the ESXi server. Select **File > Deploy OVF template**.

Figure 1. Deploy OVF Template Wizard



2. Select the OVF and vmdk file from the folder where the ClearPass Zip file was extracted. The Deploy OVF wizard opens with the OVF Template Details page displayed. (OVF, or Open Virtualization Format, is a standard for distributing virtual appliances or software to virtual machines).

Figure 2. Deploy OVF Template Wizard, Select Template Folder

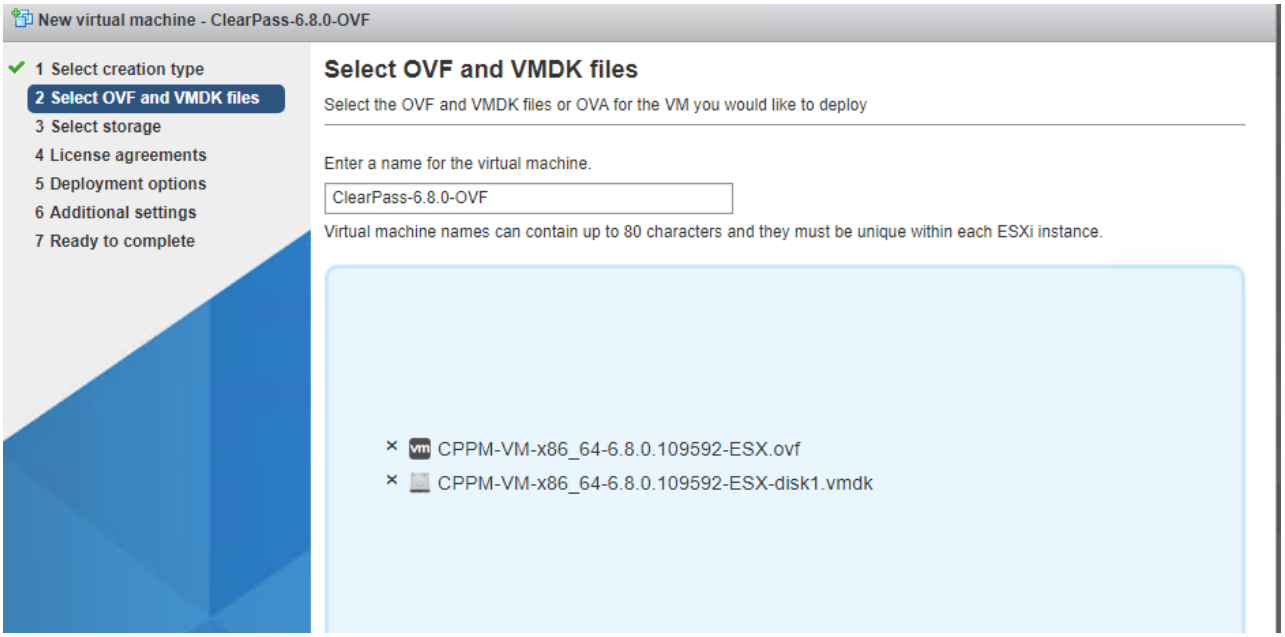
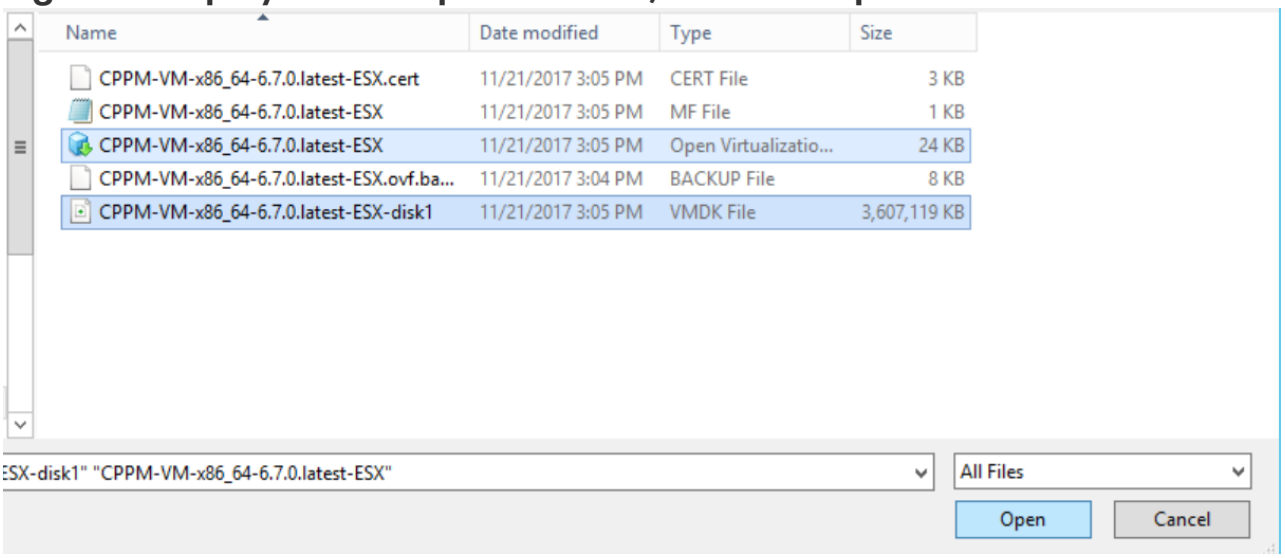
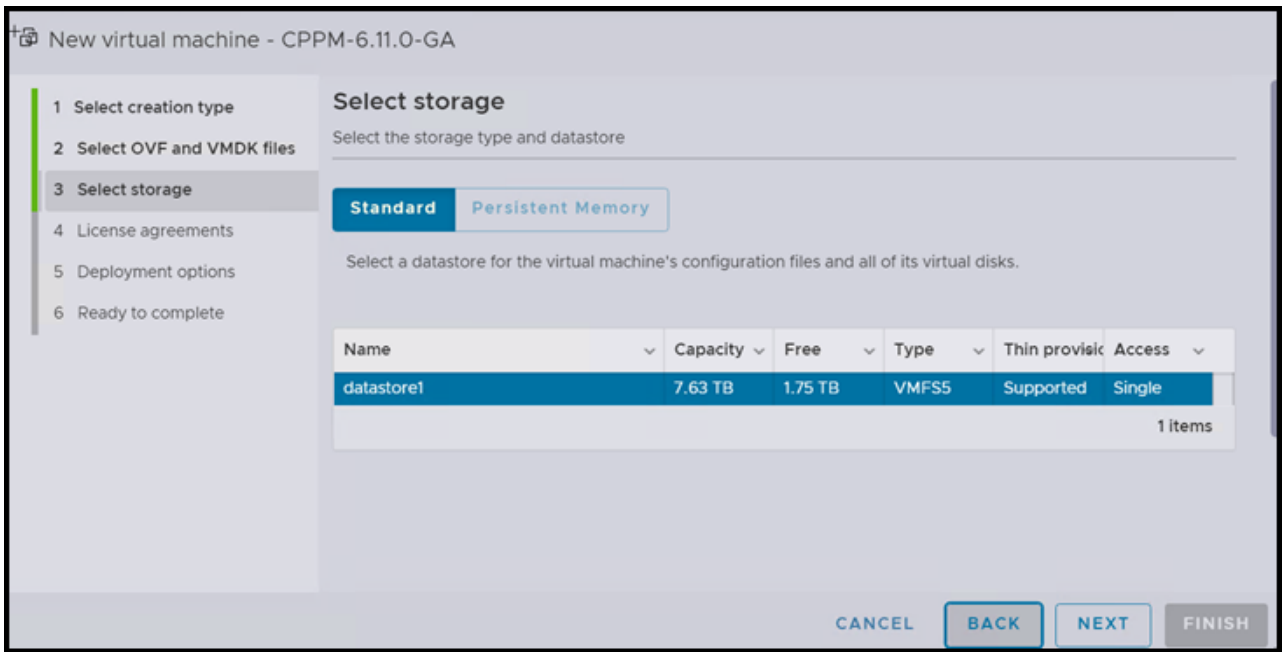


Figure 3. Deploy OVF Template Wizard, Select Template File

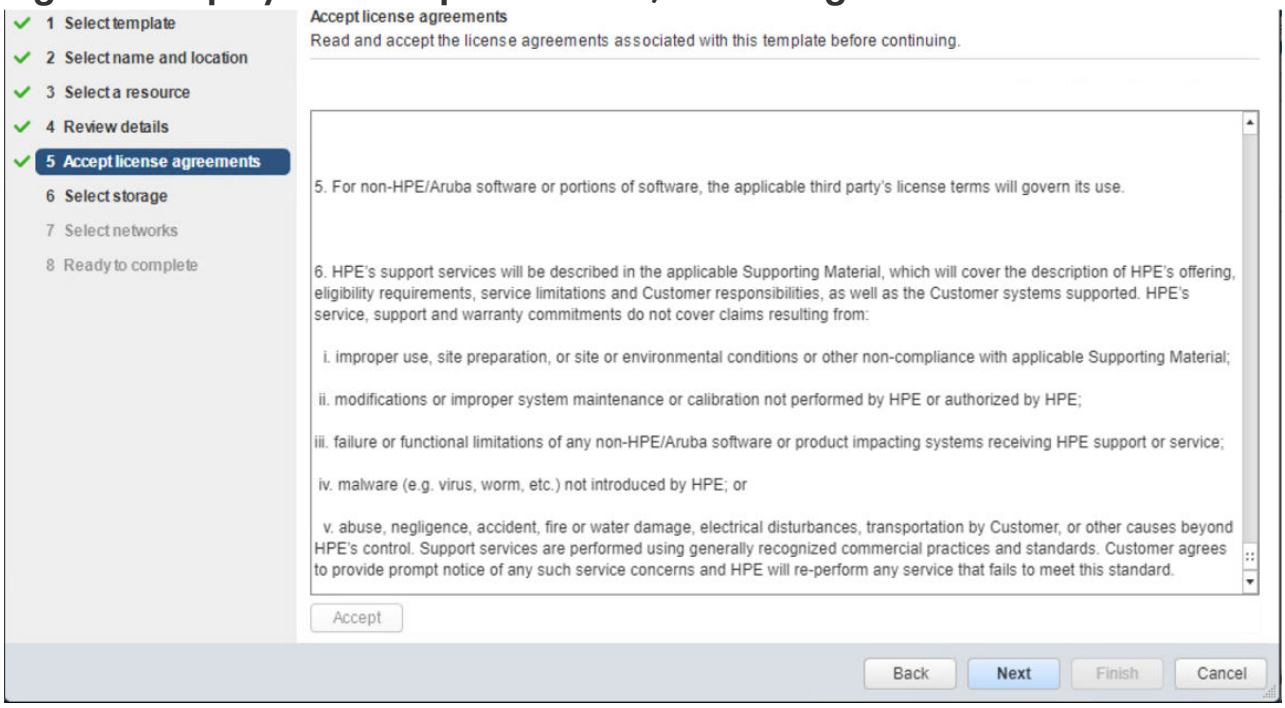


3. Select **Storage** and click **Next**.



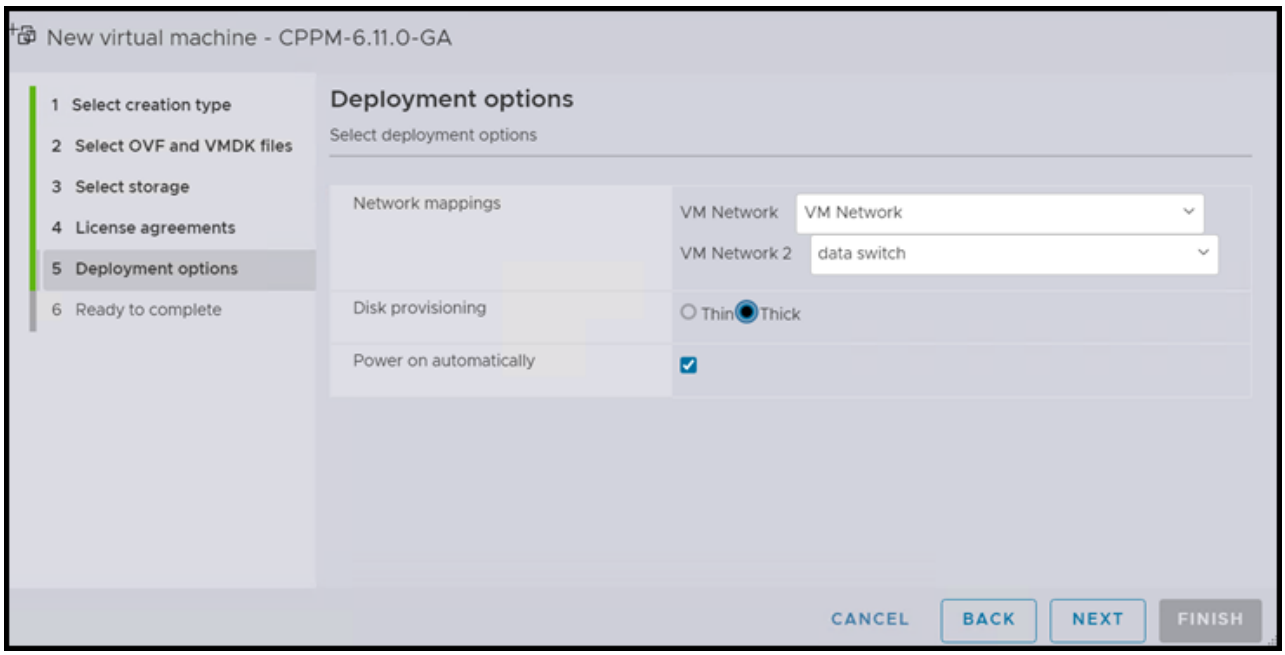
4. On the **End User License Agreement** page, review the information, click **Accept**, and then click **Next**.

Figure 4. Deploy OVF Template Wizard, Review Agreement



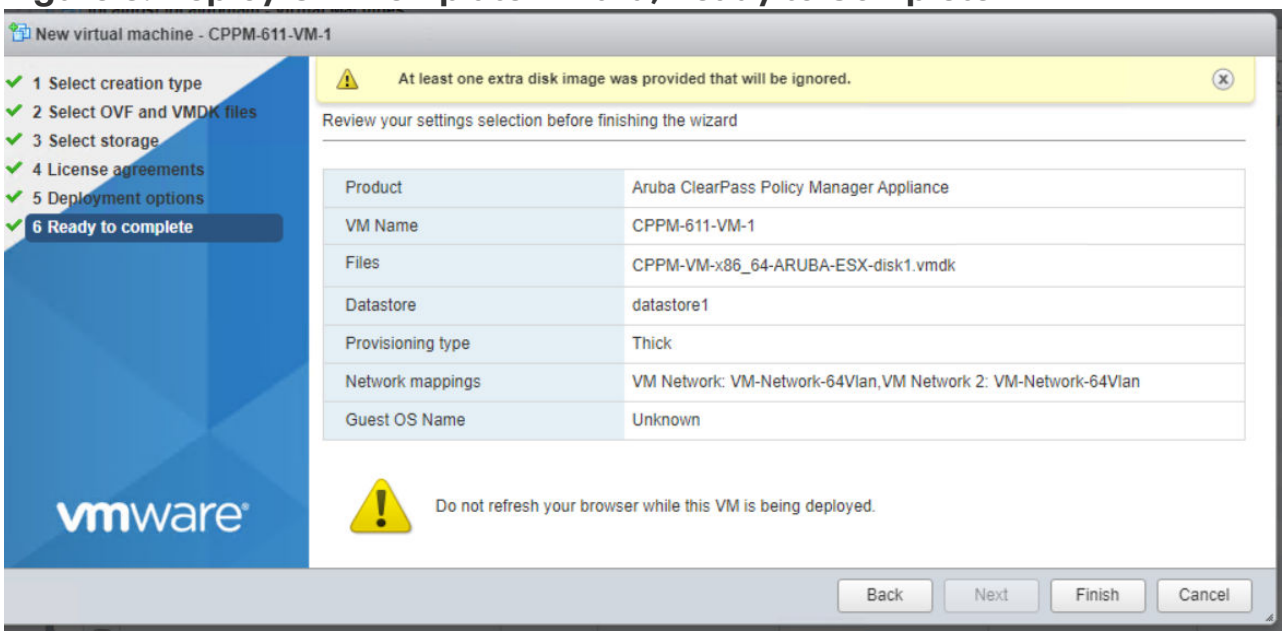
5. In the **Disk Format** page, leave the default **Thick Provision Lazy** option zeroed. Make sure to select **Network mappings** and then click **Next**.

Figure 5. Deploy OVF Template Wizard, Deployment Options



6. Click on **Finish** after reviewing the configuration.

Figure 6. Deploy OVF Template Wizard, Ready to Complete



You will need to reconfigure the VA settings by adding a hard disk before you power on.

> **NOTE**
>
> The error message "At least one extra disk image was provided that will be ignored" is displayed at the top of the Ready to Complete page. This message can be ignored, and the installation will complete correctly. The message refers to an .NVRAM file created as part of the OVF image.

Add a Hard Disk to the ESXi Virtual Appliance

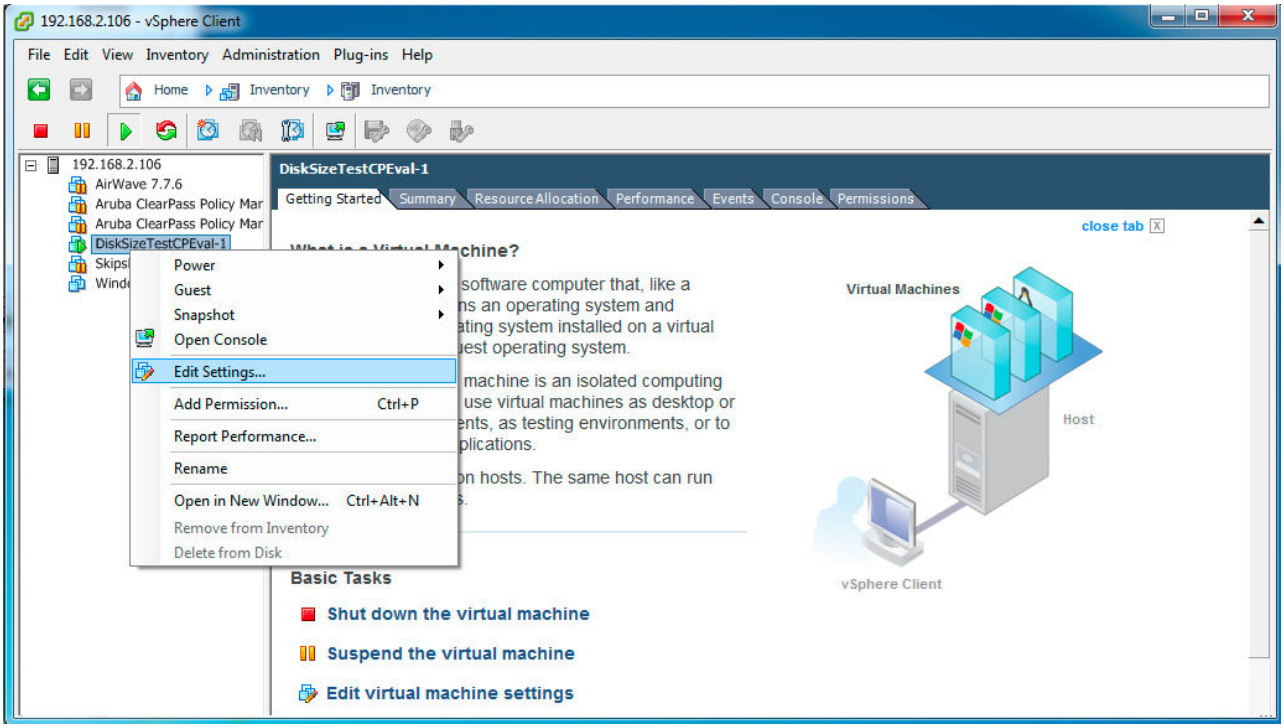
About this task

> **NOTE**
>
> For disk size requirements for the different ClearPass models, see the System Requirements for ClearPass 6.14 chapter.

Procedure

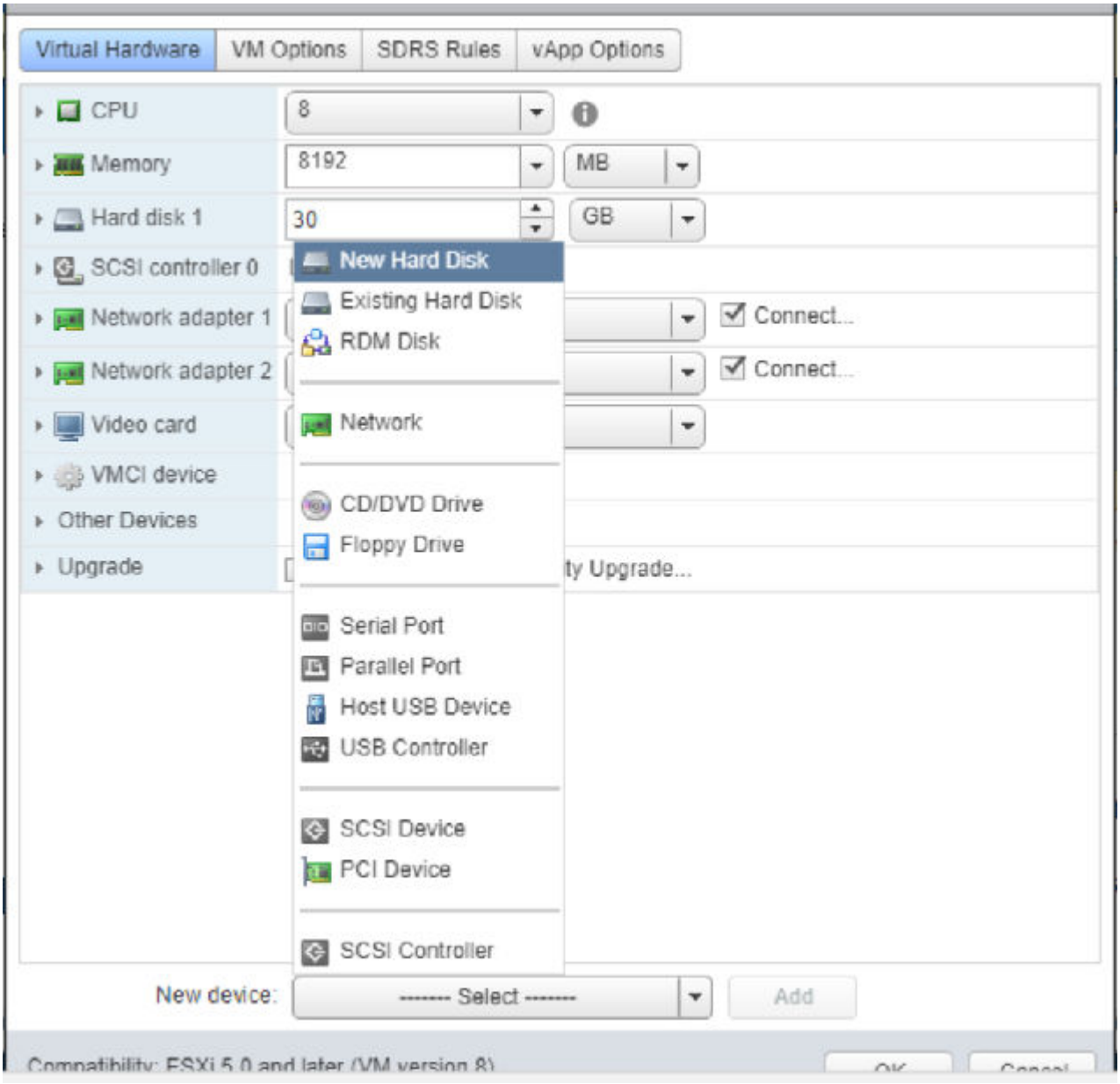
1. On the vSphere Web client, navigate to the deployed virtual appliance, right-click on it, and select **Edit Settings**.

Figure 1. vSphere Web Client, Edit Settings



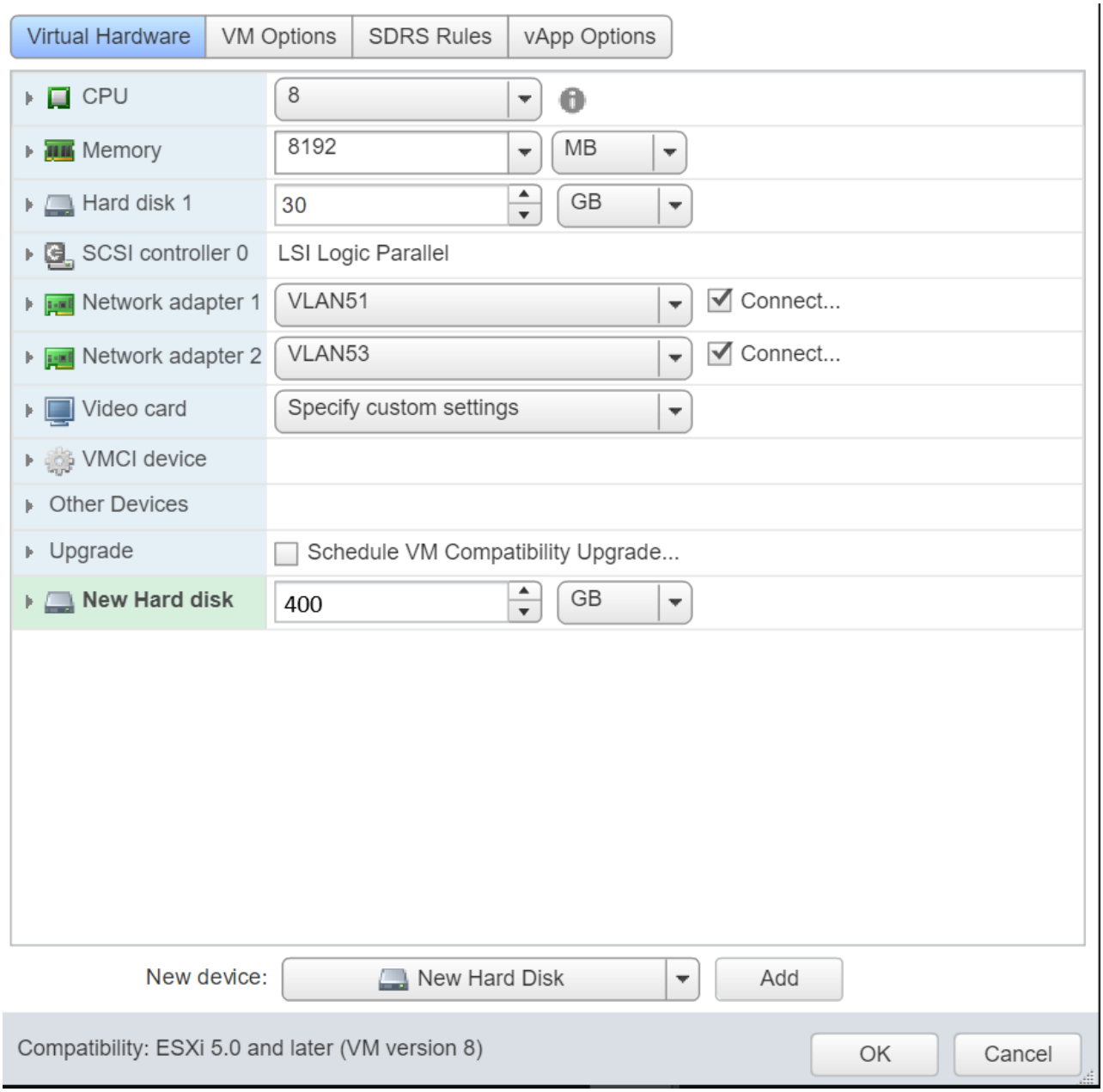
2. Click **New Hard Disk** to add another hard disk. The following example shows the addition of an HDD SCSI(0:1) for 400 GB. Make sure the newly HDD added is pointing to SCSI(0:1).

Figure 2. Virtual Machine Properties, Add Hard Disk



3. Review the system requirements and finish the deployment.

Figure 3. Add Hardware Wizard, New Hard Disk Size



4. Ensure the hard disk node ID is set to SCSI(0:1) and the disk type setting is **Thick Provision Lazy Zeroed**, and then click **Save**.

Power On and Configure the ESXi Virtual Appliance

Procedure

1. Power on the ESXi virtual appliance.

2. Enter the list number for the appropriate appliance type (do not enter the appliance model itself). Options include:

   - **1) CLABV**
   - **2) C1000V**
   - **3) C2000V**
   - **4) C3000V**

   For example, to install a C1000V, enter the number **2**.

Figure 1. Number Entered to Indicate Appliance Option

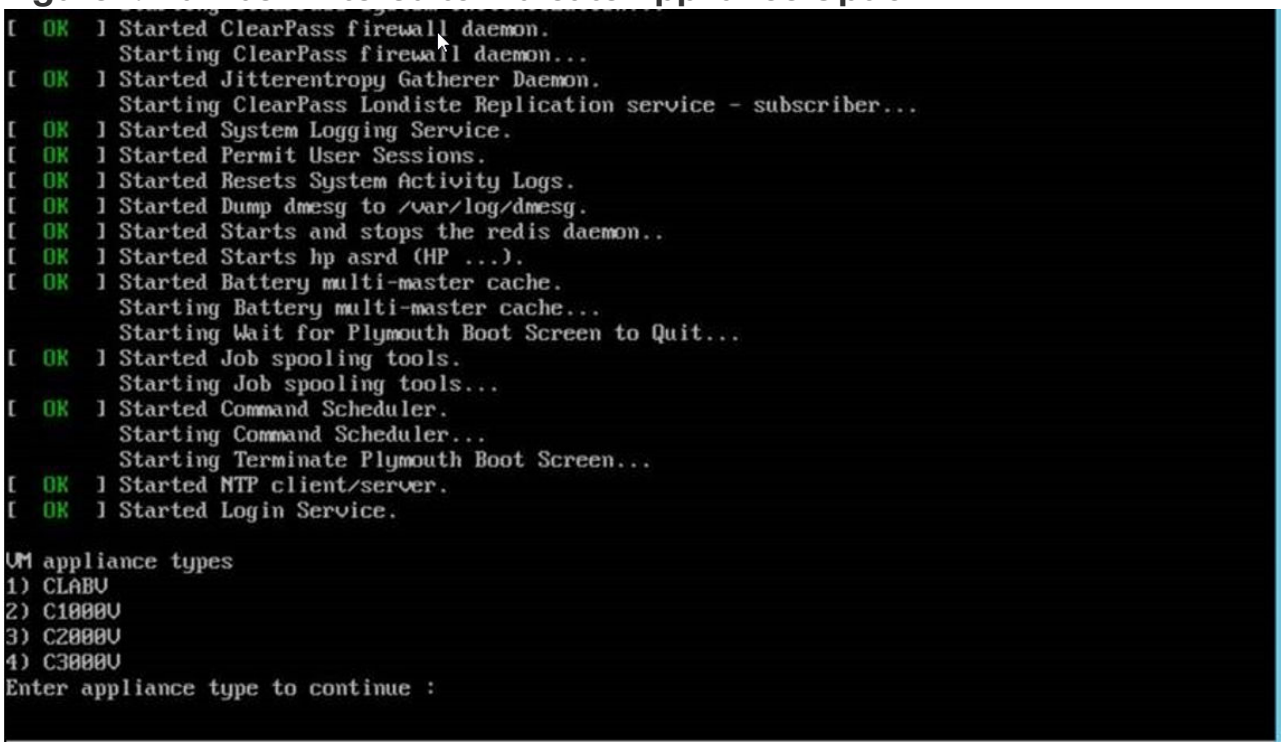
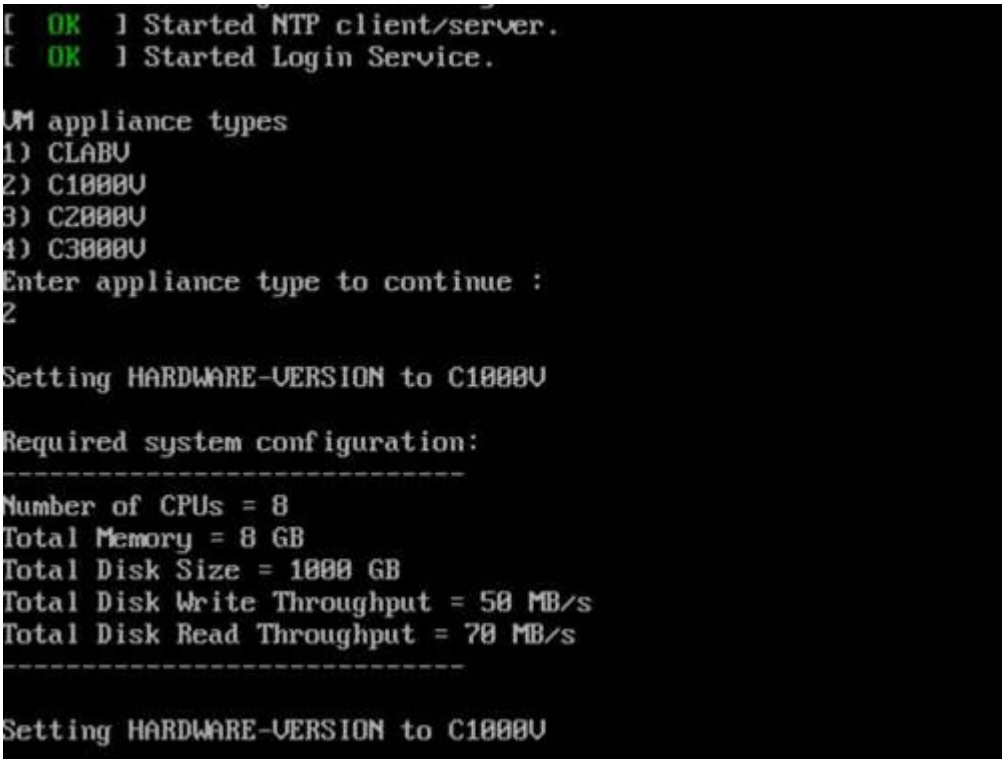
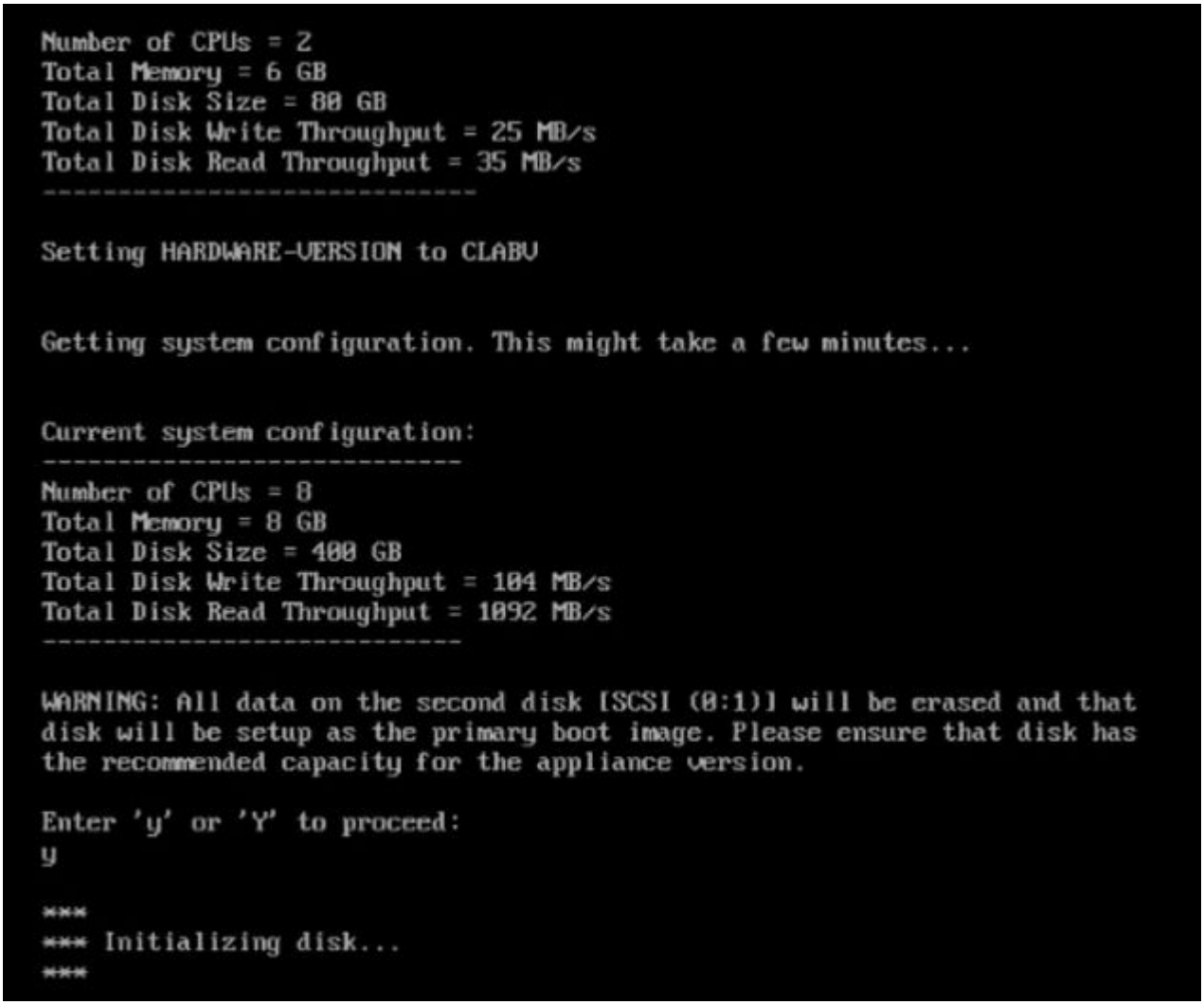


Figure 2. Enter Option to Install



3. The system requirements are displayed for the appliance model you entered, along with the current system configuration. Compare these to make sure your system meets the new system requirements. For more information, see the System Requirements for ClearPass 6.14 chapter.

Figure 3. System Requirements Comparison, and "Enter 'y' or 'Y' to proceed"



4. In the 6.11.0 OVF there is an option to encrypt the new HDD SCSI 0:1 with LUKS encryption. Depending on your system's requirements, select either **Yes** or **No**.
5. When you have verified the system meets the new requirements, press **y**. The ClearPass 6.11.0 setup and installation begins. You should see the following information, and ClearPass will reboot at least once. After that reboot, the ClearPass VA is configured and will power on and boot up within a couple of minutes. The whole process from deploying the OVF image to the final startup screen should take between 30 and 40 minutes.
6. Continue to wait for some time. Once the **Network Configuration Wizard** appears, assign network details to ClearPass. When this is completed, ClearPass should be accessible online.

After Upgrading ClearPass to 6.14.0 on ESXi Servers

About this task

If you are upgrading ClearPass from 6.12.x to 6.11.0 on a VMware ESXi server, **and only** if the MAC address of **Network adapter1** is higher than that of **Network adapter2**, additional steps are required after the upgrade.

After upgrading, follow the steps below in order for ClearPass to have network connectivity:

Procedure

1. After you upgrade to 6.11.0, log in to the console as appadmin and use the CLI command **<system shutdown>** to shut down the ClearPass server. This step must be done only through the console.
2. After the command is executed, wait for the virtual appliance to shut down completely.
3. Edit the ClearPass virtual appliance settings in the vSphere Web client and remove the two Ethernet adapters that are named **Network adapter1** and **Network adapter2**.
4. Add two new network adapters with the names **Network adapter1** and **Network adapter2** and of type **Ethernet Adapter**. Network adapter1 should be the management port connected to SwitchManagement, and Network adapter2 should be the data port connected to SwitchData.
5. Save the new settings and start the ClearPass virtual appliance.
6. Log in to the ClearPass console using the appadmin account, and then run the following CLI command to refresh the network settings:

   system refresh-network
7. After the refresh command is executed, reboot the ClearPass virtual appliance to establish network connectivity.

Morphing ESXi to a Higher Model Virtual Appliance

About this task

> **NOTE**
> The illustrations in this section use the example of morphing a C1000V virtual appliance to a C3000V. Adjust your own configuration as needed.

> **NOTE**
> Starting with the 6.11.0 release, after the **system morph-vm** command is run during a ClearPass upgrade and the new disk is added, the existing hard disk cannot be removed. This is because the existing disk is used to extend the disk space for the server, and is included in the calculations to determine the required capacity of the additional hard disk.

Perform the following steps when morphing an ESXi virtual appliance to a higher model virtual appliance:

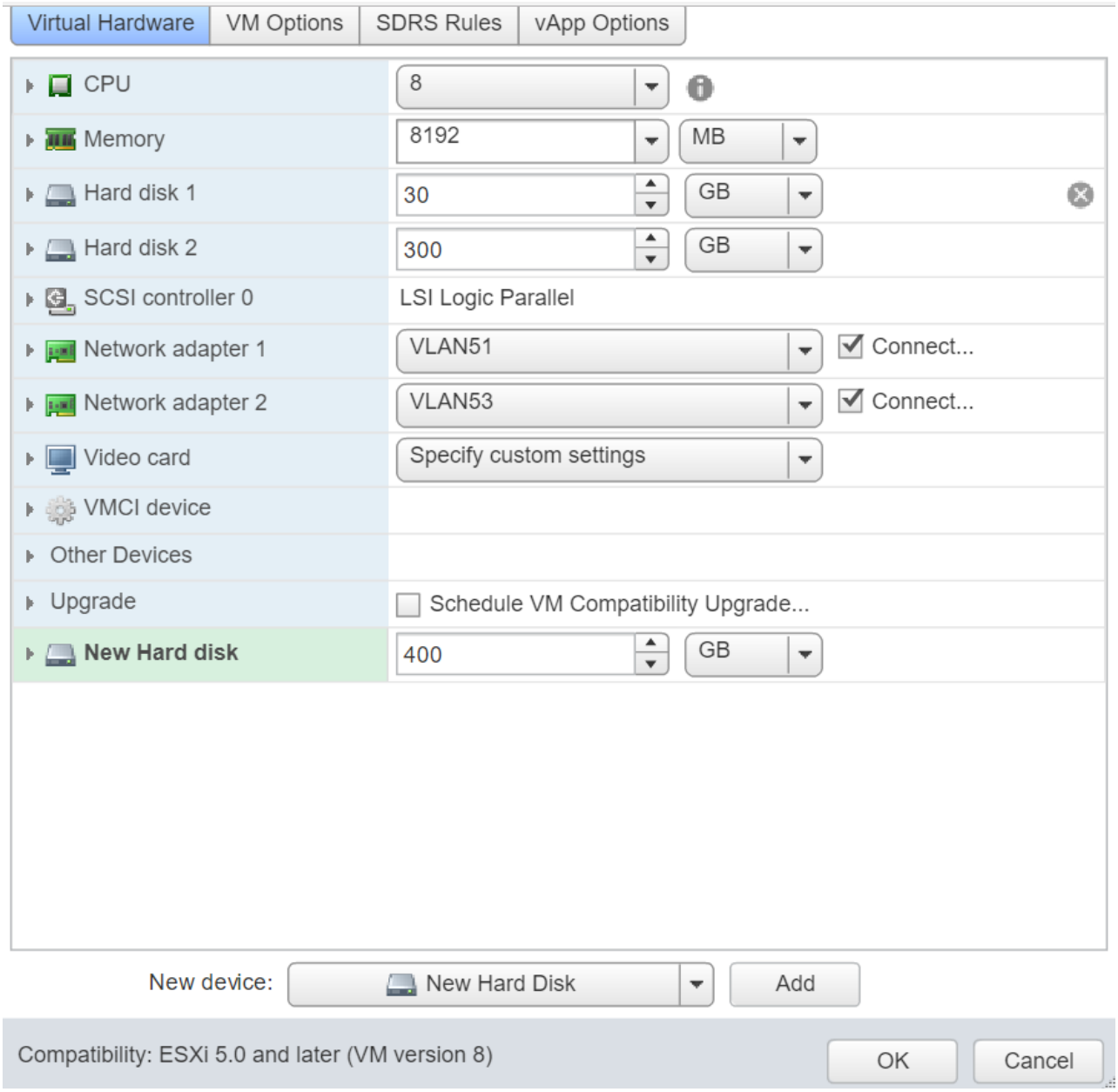
Procedure

1. Power off the ClearPass VMware instance.
2. In VMware, open the ClearPass Virtual Machine Properties.
3. Add a new hard disk to the virtual appliance. The **Virtual Device Node** should be **SCSI(0:2)**.

   Review the VMware disk requirements first. See the System Requirements for ClearPass 6.14 chapter for the appropriate requirements for your appliance.

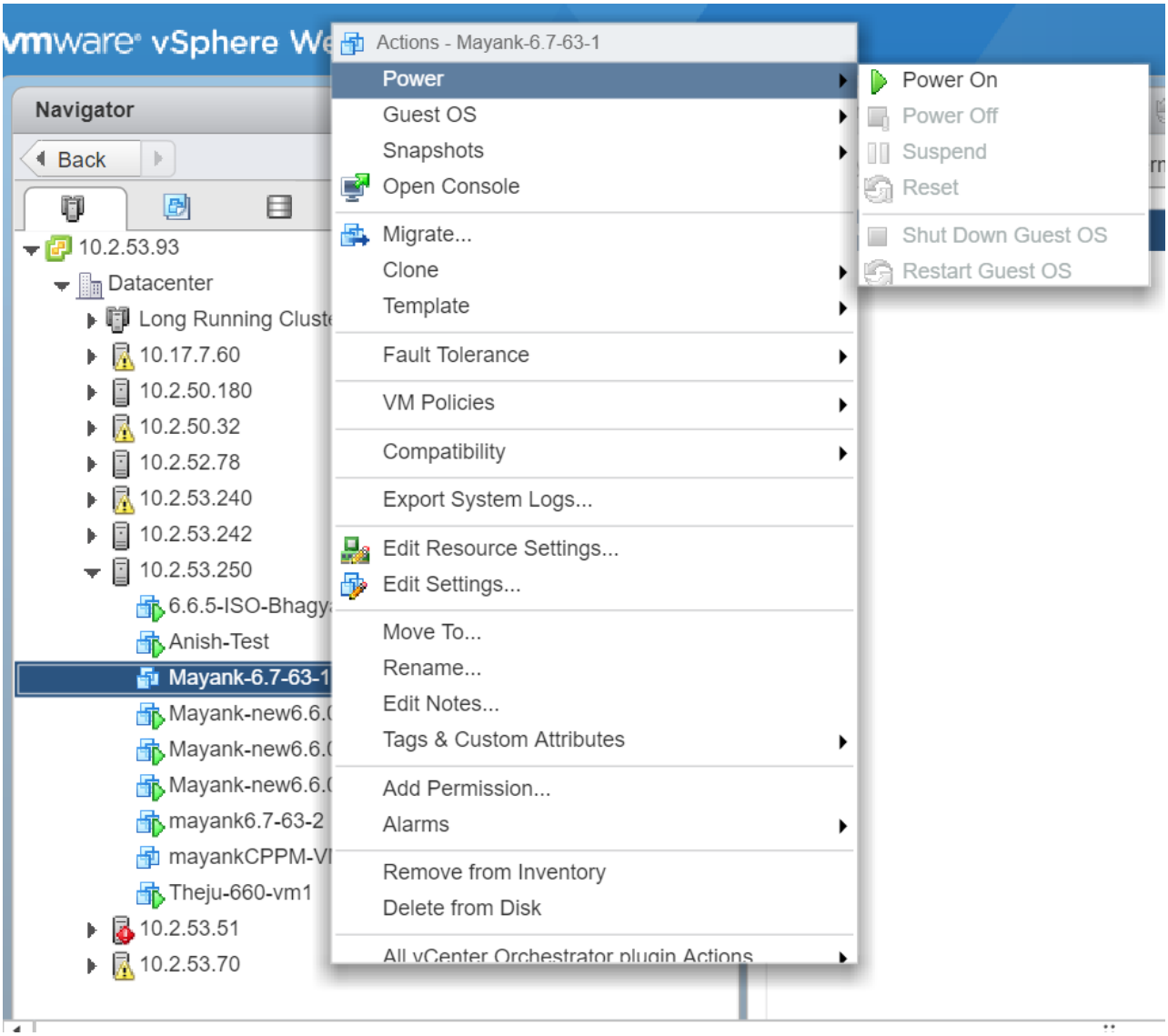
   > **CAUTION**
   > Never remove SCSI 0:0

   Figure 1. Add Hardware, New HDD



4. Power on the VM after adding the new HDD.

Figure 2. Power On VM



5. Enter the command **system morph-vm <vm-type>**. In the example we're using, the information shows that you are now in a C1000V virtual appliance and about to morph to a C3000V.

Figure 3. "Press "y" or "Y" to proceed" (A)

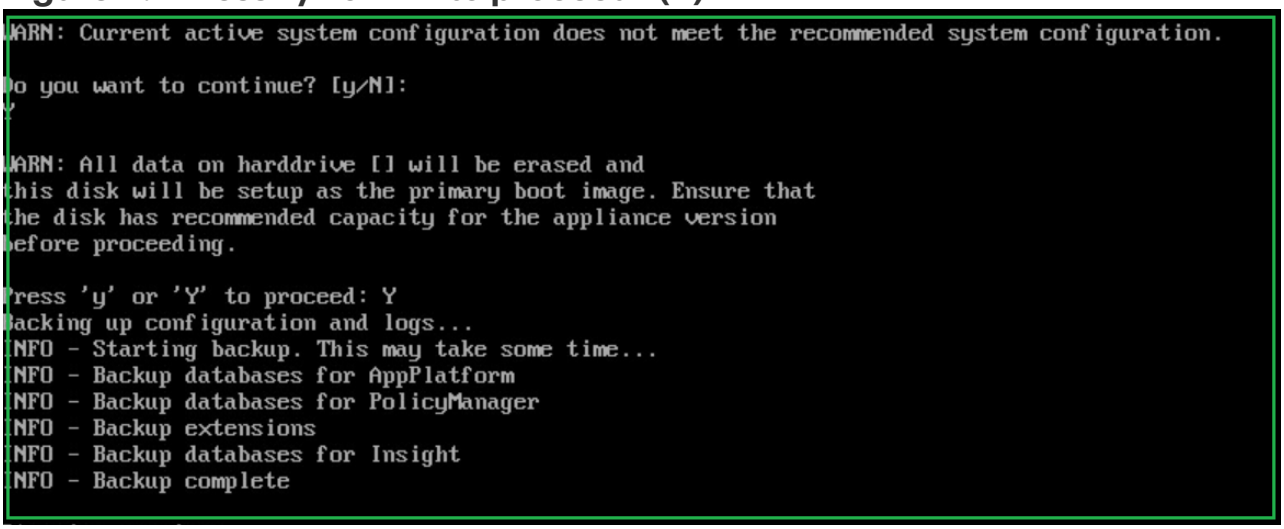
```
[appadmin@PoC-71]# system morph-vm C3000V
Current Hardware Version - C1000V
New Hardware Version - C3000V

WARN: VM will be morphed to C3000V. Once initiated
you cannot revert to the existing version C1000V.

All services will be stopped and a reboot initiated to complete
this operation.
Press 'y' or 'Y' to proceed:
```

6. Review allocated system requirements with required system configuration and enter **Y** to proceed

Figure 4. "Press "y" or "Y" to proceed" (B)



7. The system requirements are displayed for the appliance model you entered, along with your current system configuration. Compare these to make sure your system meets the new system requirements.
8. If desired, encrypt SCSI 0:2 with LUKS.
9. When you have verified that your system meets the new requirements, press **y**. The ClearPass 6.11.0 setup and installation begins.

Manually Upgrading an ESXi Installation

About this task

This section describes how to perform a manual upgrade of a VMware ESXi virtual appliance. This procedure is recommended only if you experience problems when taking snapshots of a virtual appliance on an ESXi version 7.0 update 3 and above.

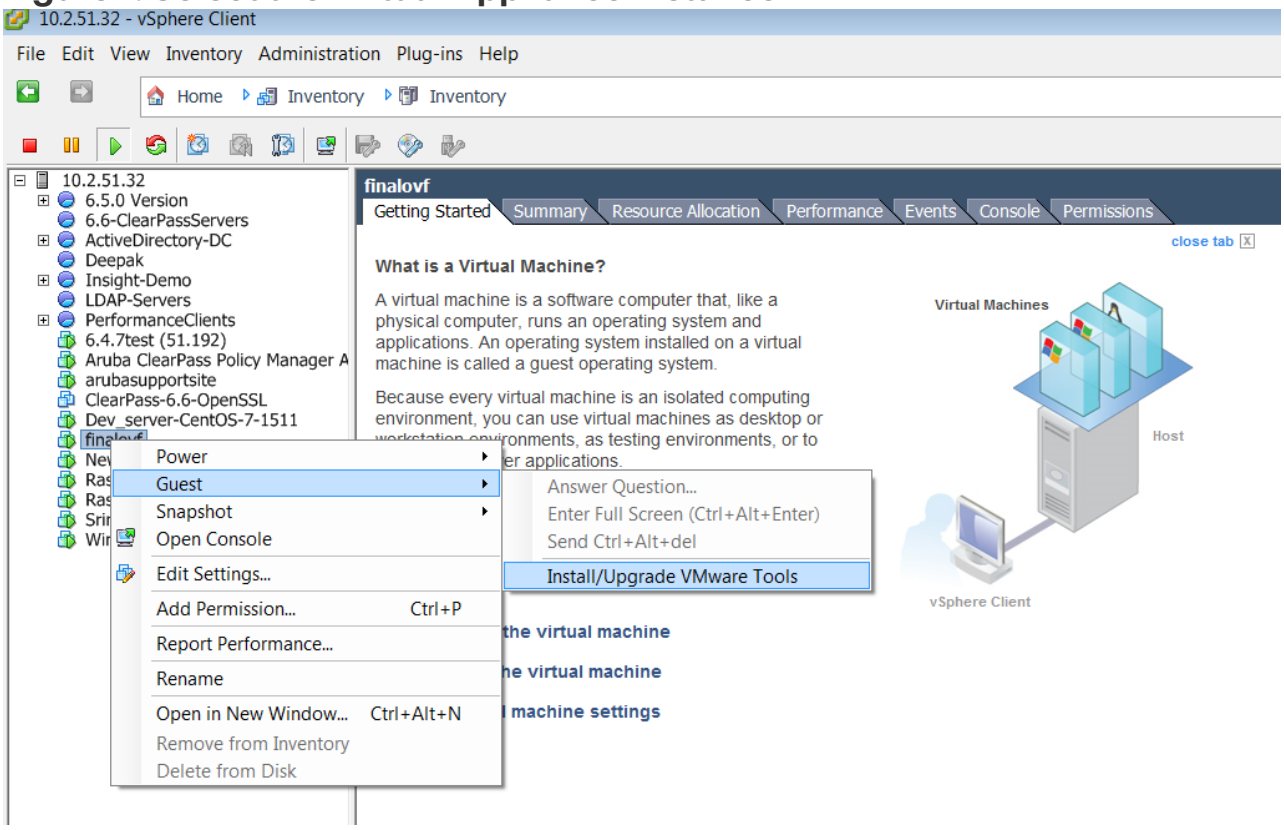
By default, ClearPass comes with VMware Tools version 10.1.5.59732 (build-5055683) installed in it. If you are going to perform a manual VMware Tools upgrade, you must first verify that a version of ClearPass is already installed.

To manually upgrade a VMware installation:

Procedure

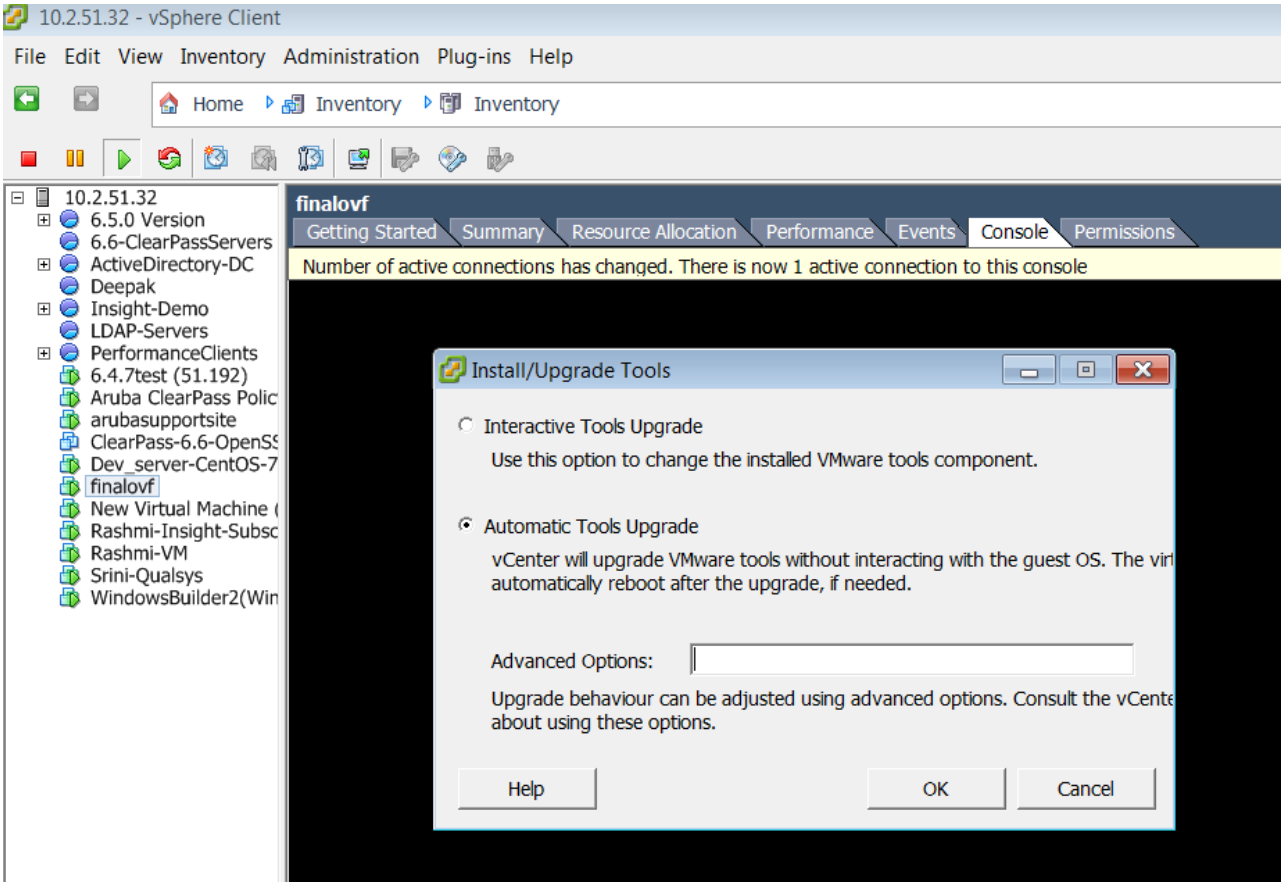
1. Power on the virtual appliance and verify that ClearPass is installed on it. In the vSphere Web client, right-click on the VA instance and select **Guest > Install/ Upgrade VMware Tools**.

Figure 1. Select the Virtual Appliance Instance



2. Select **Automatic Tools Upgrade**. This option ensures that the VA instance is upgraded to the highest supported stable version for the respective version of ESXi server it is hosted on.

Figure 2. Automatic Tool Upgrade Option



3. Click **OK**.
4. The console displays a message that the VMware Tools upgrade has been initiated and is in progress. The process takes approximately five minutes to complete.

> **CAUTION**
>
> Do not make any configuration changes to either ClearPass or the vSphere Web client while the upgrade is in progress.

Hardware Appliance Installation

This section describes the procedures for installing and configuring ClearPass on a hardware appliance, as well as how to complete important administrative tasks such as registering for ClearPass software updates and changing the admin password.

Subtopics

ClearPass Hardware Appliance Specifications and Components
Before Starting the ClearPass Hardware Installation

Configuring the ClearPass Hardware Appliance
Activating ClearPass for a Hardware Appliance
Logging in to the ClearPass Hardware Appliance
Signing Up for Live Software Updates
Changing the Administration Password
Powering Off the ClearPass Hardware Appliance
Resetting the System Passwords to the Factory Defaults

ClearPass Hardware Appliance Specifications and Components

ClearPass provides the C1000, C2010, C2020, C3010, N1000, N3000, and N3001 hardware appliance platforms. Specifications for these hardware appliance types are described in this section.

For additional information and to verify the most recent updates, you may refer to the **Aruba ClearPass Policy Manager Data Sheet** and the **Scaling and Ordering Guide**.

Subtopics

Comparison Specifications, All Appliances
Port Information, All Appliances
ClearPass C1000 Hardware Appliance
ClearPass C2010 Hardware Appliance
ClearPass C2020 Hardware Appliance
ClearPass C3010 Hardware Appliance
ClearPass N1000 Hardware Appliance
ClearPass N3000 and N3001 Hardware Appliances

Comparison Specifications, All Appliances

Table 1. ClearPass Hardware Appliance Specification Comparisons

	C1000 Appliance (JZ508A)	C2010 Appliance (R1V81A)	C2020 Appliance (R1V83A)	C3010 Appliance (R1V82A)	N1000 Appliance (S3N67A)	N3000 Appliance (S3N66A)	N3001 Appliance (S3N65A)
Appliance Specifications							

	C1000 Appliance (JZ508A)	C2010 Appliance (R1V81A)	C2020 Appliance (R1V83A)	C3010 Appliance (R1V82A)	N1000 Appliance (S3N67A)	N3000 Appliance (S3N66A)	N3001 Appliance (S3N65A)
Hardware Model	Unicom S - 1200 R4	HPE DL20 Gen 10	HPE DL360 Gen 10	HPE DL360 Gen 10	Advantech FWA-2012 (customized)	HPE ProLiant DL325 Gen11	HPE ProLiant DL325 Gen11
CPU	(1) Atom 2.40GHz C2758 with Eight Cores (8 Threads)	(1) Xeon 4.0GHz E - 2274G with Four Cores (8 Threads)	(1) Xeon 2.3GHz Gold 5118 with Twelve Cores (24 Threads)	(1) Xeon 2.3GHz Gold 5118 with Twelve Cores (24 Threads)	Intel Atom C3758R with 8 cores (8 threads)	(1) AMD EPYC 9124 16 - Core Processor (32 threads)	(1) AMD EPYC 9124 16 - Core Processor (32 threads)
Memory	8 GB	16 GB	16 GB	64 GB	16 GB	128 GB	128 GB
Hard Drive Storage	(1) SATA (7.2K RPM) 1TB hard drive	(2) SATA (7.2K RPM) 1TB hard drives, RAID - 1 controller	(2) SATA (7.2K RPM) 1TB hard drives, RAID - 1 controller	(6) SAS (10K RPM) 600GB Hot - Plug hard drives, RAID - 10 controller	(1) SSD (1TB) M.2	(3) SSD 960GB SATA, RAID - 5 controller	(3) SSD 960GB SATA, RAID - 5 controller
Out of Band Management	N/A	HPE Integrated Lights - Out (iLO) Standard with Dedicated NIC	HPE Integrated Lights - Out (iLO) Advanced with Dedicated NIC	HPE Integrated Lights - Out (iLO) Advanced with Dedicated NIC	N/A	HPE Integrated Lights - Out (iLO) Advanced with Dedicated NIC	HPE Integrated Lights - Out (iLO) Advanced with Dedicated NIC
Network Interfaces	4 x 1Gbe	4 x 1Gbe ** ** Embedded NICs are disabled.	4 x 1GbE	4 x 1GbE	4 x 1GbE	4 x 1Gbe	4 x 10/25 Gbe SFP28 PCIe

	C1000 Appliance (JZ508A)	C2010 Appliance (R1V81A)	C2020 Appliance (R1V83A)	C3010 Appliance (R1V82A)	N1000 Appliance (S3N67A)	N3000 Appliance (S3N66A)	N3001 Appliance (S3N65A)
Serial Port	Yes (RJ - 45)	Yes (DB - 9)	Yes (DB - 9)	Yes (DB - 9)	Yes (RJ-45)*	Yes (DB - 9)	Yes (DB - 9)
Minimum Software Version	ClearPass Policy Manager 6.6	ClearPass Policy Manager 6.7	ClearPass Policy Manager 6.11	ClearPass Policy Manager 6.7	ClearPass Policy Manager 6.11	ClearPass Policy Manager 6.11	ClearPass Policy Manager 6.11
Form Factor							
Rackmount	Included	1U LFF Easy Install Rail 1U Cable Management Arm	1U SFF Easy Install Rail 1U Cable Management Arm	1U SFF Easy Install Rail 1U Cable Management Arm	Included (mounting ears)	1U Easy Install Rail 2 Kit	1U Easy Install Rail 2 Kit
Dimensions (WxHxD)	17.2″ x 1.7″ x 11.3″	17.11″ x 1.70″ x 15.05″	17.1″ x 1.7″ x 27.8″	17.1″ x 1.7″ x 27.8″	19 x 1.7 x 12.6	17.1" x 1.7" x 25.6"	17.1" x 1.7" x 25.6"
Weight (Max Config)	8.5 Lbs	Up to 19.18 Lbs	Up to 36 lbs	Up to 36 Lbs	Up to 14.5 lb	Up to 34.5 lb	Up to 34.5 lb
Power							
Power Supply	Integrated 200W Low Noise AC - DC Power Supply	HPE 500W Flex Slot Platinum Hot Plug Power Supply	HPE 500W Flex Slot Platinum Hot Plug Power Supply	HPE 500W Flex Slot Platinum Hot Plug Power Supply	(1) 150W	(2) HPE 1000W Flex Slot Titanium Hot Plug Power Supply	(2) HPE 1000W Flex Slot Titanium Hot Plug Power Supply
Power Redundancy	N/A	Optional	Optional	Optional	N/A	Included	Included
AC Input Voltage	100/240 VAC auto - selecting	100/240 VAC auto - selecting	100/240 VAC auto - selecting	100/240 VAC auto - selecting	100/240 VAC auto-selecting	100/240 VAC auto - selecting	100/240 VAC auto - selecting

	C1000 Appliance (JZ508A)	C2010 Appliance (R1V81A)	C2020 Appliance (R1V83A)	C3010 Appliance (R1V82A)	N1000 Appliance (S3N67A)	N3000 Appliance (S3N66A)	N3001 Appliance (S3N65A)
AC Input Frequency	50/60 Hz auto - selecting	50/60 Hz auto - selecting	50/60 Hz auto - selecting	50/60 Hz auto - selecting	50/60 Hz auto-selecting	50/60 Hz auto - selecting	50/60 Hz auto - selecting
Power Cord	C13 - NEMA 5 - 15P US/CA 110V 10Amp Power Cord	C13 - C14 WW 250V 10Amp Jumper Cord	C13 - C14 WW 250V 10Amp Jumper Cord	C13 - C14 WW 250V 10Amp Jumper Cord	(1) C13 -C14 WW 250V 10A Jumper cord	C13 - C14 WW 250V 10Amp Jumper Cord	C13 - C14 WW 250V 10Amp Jumper Cord

***** The N1000 does not include integrated video; only serial interface is supported.

Port Information, All Appliances

Table 1. Functional Description of the ClearPass Hardware Appliance Ports

Port	Description
Data Port (Gigabit Ethernet)	The Data port (ethernet 1) provides a point of contact for RADIUS, TACACS+, Web authentication, and other dataplane requests. This configuration is optional. If this port is not configured, requests are redirected to the Management port.
iLO Port	The iLO (Integrated Lights - Out) port is an Ethernet port that provides out - of - band management facilities. The iLO port makes it possible to perform activities on the ArubaOS switch or an HP server from a remote location. The iLO card has a separate network connection (and its own IP address) to which one can connect via HTTPS. Available on the ClearPass C20X0, C30X0, and N300X hardware appliances.
Management Port	The Management port (ethernet 0) provides access for cluster administration

Port	Description
	and appliance maintenance using the WebUI, CLI, or internal cluster communication. This configuration is mandatory. (Gigabit Ethernet for CXXXX and N3000. For N3001, 10GB/25GB SFP28)
Serial Port	The Serial port is used to initially configure the ClearPass hardware appliance using a hard - wired terminal.
SPAN Ports	A SPAN (Switched Port Analyzer) port is a method of monitoring network traffic. The switch sends a copy of all network packets seen on one port (which is the monitored or source port) to a destination SPAN port, where the packets can be analyzed.
USB Ports	• Four USB v2.0 ports are provided on the ClearPass C1000 hardware appliance. • Three USB 3.1 Gen1 Ports for ClearPass N3000 and N3001 hardware appliances. Users should be aware that the use of mass - storage USB devices is prevented by the operating system.
VGA Connector	You can use the VGA Connector to connect the ClearPass hardware appliance to a monitor and keyboard.

ClearPass C1000 Hardware Appliance

The ClearPass C1000 hardware appliance (SKU: JZ508A) is a RADIUS/ TACACS+ server that provides advanced policy control for up to 1,000 concurrent sessions. The ClearPass C1000 appliance has a single 1 TB SATA disk with no RAID disk protection.

To compare specifications and port information across all appliances, see ClearPass Hardware Appliance Specifications and Components . For additional information and to

verify the most recent updates, you may refer to the **Aruba ClearPass Policy Manager Data Sheet** and the **Scaling and Ordering Guide**.

C1000 Ports and Components

The following figure shows the ports and components on the rear panel of the ClearPass C1000 hardware appliance. The function of each of these ports and components is described in the following table.

Figure 1. Ports and Components on the ClearPass C1000 Hardware Appliance

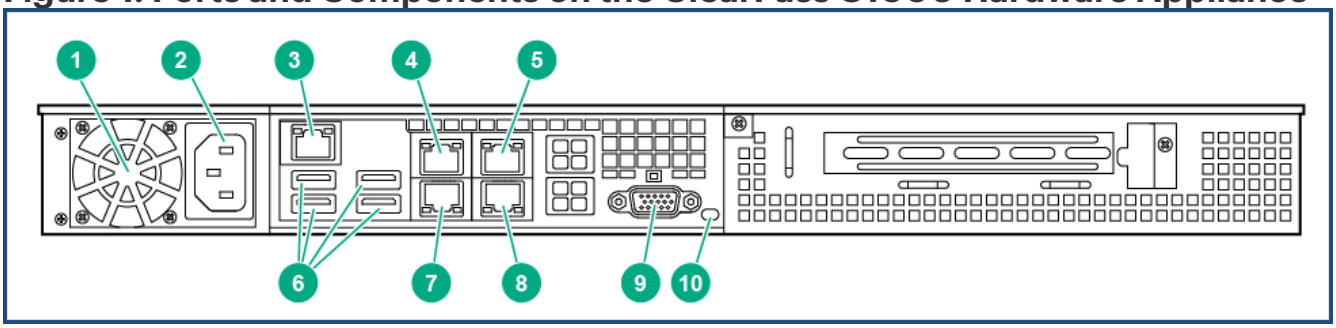


Table 1. ClearPass C1000 Ports and Components

Callout Number	C1000 Port/Component
1	Power supply module fan
2	Power supply AC inlet
3	Serial port
4	Data port (eth1)
5	Port that is optionally available for SPAN - based profiling
6	USB ports (4)
7	Management port (eth0)
8	Port that is optionally available for SPAN - based profiling
9	VGA port
10	Unit Identification (UID) button

NOTE

Callout Number	C1000 Port/Component
	This button is recessed and requires a small tool to access it.

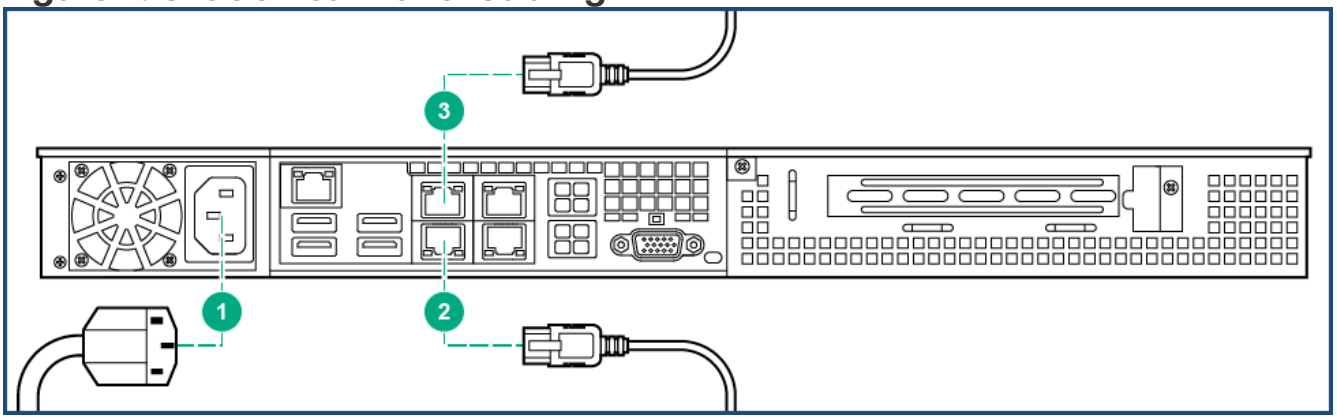
You can also access the ClearPass hardware appliance by connecting a monitor and keyboard to the hardware appliance.

C1000 Rear-Panel Cabling

About this task

The following figure illustrates the rear-panel cabling for the ClearPass C1000 hardware appliance:

Figure 2. C1000 Rear-Panel Cabling



To connect the cables on the ClearPass C1000 hardware appliance:

Procedure

1. Connect the power cord.
2. Connect the Management port.
3. Optionally connect the Data port.
4. Connect any other required cables.

C1000 Serial Port Cable Signaling and Pinouts

Table 2. Serial Port Signaling and Pinouts

Signal	RJ - 45 Pin	DB - 9 Pin	Color
RTS	1	8	Grey

Signal	RJ - 45 Pin	DB - 9 Pin	Color
DTR	2	6	Brown
TxD	3	2	Yellow
GND	4	5	Green
GND	5	3	Red
RxD	6	4	Black
DSR	7	7	Orange
CTS	8	8	Blue

C1000 Hardware Appliance Specifications

Table 5 provides the specifications for the ClearPass C1000 hardware appliance.

Table 3. ClearPass C1000 Appliance Specifications

ClearPass C1000 Appliance	Specifications
Hardware Model	Unicom S - 1200 R4
CPU	(1) Atom 2.40GHz C2758 with Eight Cores (8 Threads)
Memory	8 GB (2 x 4 GB)
Hard Drive Storage	• (1) SATA (7.3K RPM), Serial ATA • 1 TB hard drive
Serial Port	Yes: RJ - 45
Performance & Scale	Please refer to the **ClearPass Scaling & Ordering Guide**
Concurrent Sessions	1000
Form Factor	
Rack Mount	Included
Dimensions (WxHxD)	17.2" x 1.7" x 11.3"

ClearPass C1000 Appliance	Specifications
Weight (Max Configuration)	8.5 lbs
Power	
Power Supply	Single
AC Input Voltage	100/240 VAC auto - selecting
AC Input Frequency	50/60 Hz - selecting
Environmental	
Operating Temperature	5º C to 35º C (41º F to 95º F)
Operating Vibration	0.26 G at 5 Hz to 200 Hz for 15 minutes
Operating Shock	1 shock pulse of 20 G for up to 2.5 ms
Operating Altitude	- 16 m to 3,048 m (- 50 ft to 10,000 ft)

ClearPass C2010 Hardware Appliance

The ClearPass C2010 hardware appliance (SKU: R1V81A) is a RADIUS/ TACACS+ server that provides advanced policy control for up to 10,000 concurrent sessions. The ClearPass C2010 appliance ships with two x 1TB SATA disk drives. These drives are managed by an LSI RAID-1 controller. The drives are configured as a RAID-1 pair. The LSI controller presents to ClearPass a single virtual 1TB drive, masking the two underlying physical drives.



NOTE

The C2010 appliance uses an HPE 12W Smart Storage Battery (3.38Wh) lithium ion battery. For complete information on HPE batteries, refer to **Battery Support Documents** at **https://www.hpe.com/us/en/about/environment/battery-support-documents.html**.

To compare specifications and port information across all appliances, see ClearPass Hardware Appliance Specifications and Components . For additional information and to verify the most recent updates, you may refer to the **Aruba ClearPass Policy Manager Data Sheet** and the **Scaling and Ordering Guide**.

C2010 Ports and Components

The following figure shows the ports and components on the rear panel of the ClearPass C2010 hardware appliance. The function of each of these ports and components is described in the following table.

> **NOTE**
> The image of the ClearPass C2010 hardware appliance shown here includes the optional redundant power supply.

Figure 1. Ports and Components on the ClearPass C2010 Hardware Appliance

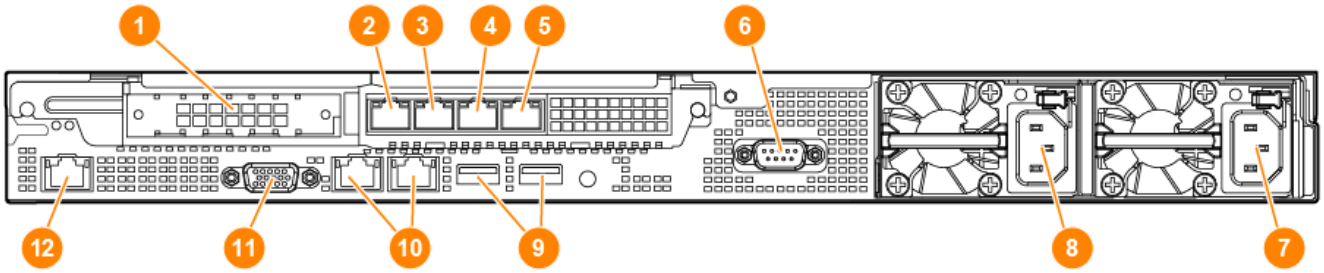


Table 1. ClearPass C2010 Hardware Appliance Ports and Components

Callout Number	C2010 Port/Component
1	PCI Expansion blank
2	MGMT port
3	DATA port **NOTE** If you want your ClearPass deployment to use only the MGMT port, this port can optionally be used as a third SPAN port. This port will only be listed as a span port on the **Administration > Server Manager > Server Configuration** page if it is connected.
4	Destination SPAN ports (2 of 2)
5	Destination SPAN ports (1 of 2)
6	DB - 9 Serial port

Callout Number	C2010 Port/Component
7	Power Supply (500 - watt)
8	(optional) Power Supply (500 - watt)
9	USB ports
10	Embedded NIC ports (disabled)
11	VGA port
12	Dedicated iLO port

You can also access the ClearPass hardware appliance by connecting a monitor and keyboard to the hardware appliance.

C2010 Hardware Appliance Specifications

Table 9 provides the specifications for the ClearPass C2010 hardware appliance.

Table 2. ClearPass C2010 Appliance Specifications

ClearPass C2010 Appliance	Specifications
Hardware Model	HPE DL20 Gen 10
CPU	(1) Intel® Xeon® Processor E - 2274G (8 MB Cache, 4.0GHz)
Memory	16 GB
Hard Drive Storage	• (2) SATA (7.2K RPM) 1TB hard drive • RAID - 1 controller
Out - of - Band Management	HPE Integrated Lights - Out (iLO) Standard with Dedicated NIC
Serial Port	Yes: DB - 9
Performance & Scale	Please refer to the **ClearPass Scaling & Ordering Guide**
Concurrent Sessions	10,000
Form Factor	

ClearPass C2010 Appliance	Specifications
Rack Mount	• 1U LFF Easy Install Rail • 1U Cable Management Arm
Dimensions (WxHxD)	17.11” x 1.70” x 150.5”
Weight (Max Configuration)	Up to 19.18 lbs
Power Specifications	
Power Supply	HPE 500W Flex Slot Platinum Hot Plug Power Supply **NOTE** The optional HPE 900W Redundant Power Supply supports 100 VAC to 240 VAC; this power supply also supports 240 VDC.
Power Redundancy	Optional
AC Input Voltage	100/240 VAC auto - selecting
AC Input Frequency	50/60 Hz auto - selecting
Environmental Specifications	
Operating Temperature	10º C to 35º C (50º F to 95º F)
Operating Vibration	Random vibration at 0.000075 G^2/Hz, 10Hz to 300Hz, (0.15 G’s nominal)
Operating Shock	2 G's
Operating Altitude	3,050 m (10,000 ft)

ClearPass C2020 Hardware Appliance

The ClearPass C2020 hardware appliance (SKU: R1V83A) is a RADIUS/TACACS+ server that provides advanced policy control for up to 10,000 concurrent sessions. The ClearPass C2020 appliance ships with two x 1TB SATA disk drives. These drives are

managed by an LSI RAID-1 controller. The drives are configured as a RAID-1 pair. The LSI controller presents to ClearPass a single virtual 1TB drive, masking the two underlying physical drives.

> **NOTE**
>
> The C2020 appliance uses an HPE 96W Smart Storage Battery (10.8Wh) lithium ion battery. For complete information on HPE batteries, refer to **Battery Support Documents** at **https://www.hpe.com/us/en/about/environment/battery-support-documents.html**.

To compare specifications and port information across all appliances, see ClearPass Hardware Appliance Specifications and Components . For additional information and to verify the most recent updates, you may refer to the **Aruba ClearPass Policy Manager Data Sheet** and the **Scaling and Ordering Guide**.

C2020 Ports and Components

The following figure shows the ports and components on the rear panel of the ClearPass C2020 hardware appliance. The function of each of these ports and components is described in the following tabl.

> **NOTE**
>
> The image of the ClearPass C2020 hardware appliance shown here includes the optional redundant power supply.

Figure 1. Ports and Components on the ClearPass C2020 Hardware Appliance

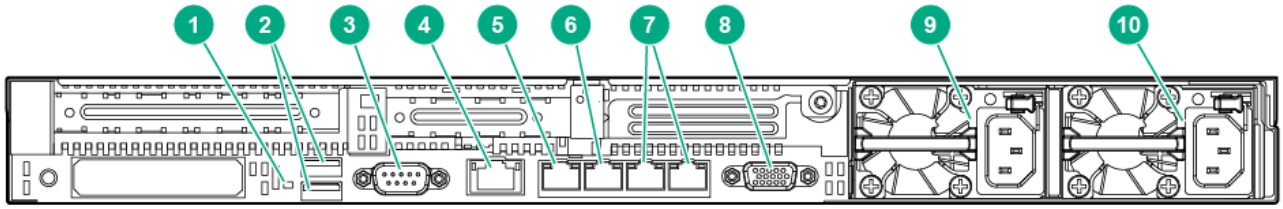


Table 1. ClearPass C2020 Hardware Appliance Ports and Components

Callout Number	C2020 Port/Component
1	UID (Unit ID) LED The UID LED helps you identify and locate a system, especially in high - density rack environments. Additionally, the UID is used to indicate that a critical operation is

Callout Number	C2020 Port/Component
	underway on the host, such as remote console access or ROM flash. **NOTE** This button is recessed and requires a small tool to access it. The "current state" (on or off) of the UID is the last state chosen using one of these methods. If a new state is chosen while the UID is blinking, this new state becomes the current state, and takes effect when the UID stops blinking. **NOTE** The Unit ID Light web page does not automatically refresh itself if the state of the actual light changes after the page is loaded. To ensure the page accurately reflects the state of the UID Light, click on the Virtual Indicators link to update the page
2	USB ports (2)
3	Serial port
4	iLO (Integrated Lights - Out) port
5	Management port (eth0)
6	Data port (eth1)
7	Destination SPAN ports (2)
8	VGA Connector
9	Fan and power supply AC inlet
10	Optional redundant fan and power supply

You can also access the ClearPass hardware appliance by connecting a monitor and keyboard to the hardware appliance.

C2020 Hardware Appliance Specifications

Table 11 provides the specifications for the ClearPass C2020 hardware appliance.

Table 2. ClearPass C2020 Appliance Specifications

ClearPass C2020 Appliance	Specifications
Hardware Model	HPE DL360 Gen 10
CPU	(1) Intel® Xeon® Processor 2.3GHz Gold 5118 with Twelve Cores (24 Threads)
Memory	16 GB
Hard Drive Storage	(2) SATA (7.2K RPM) 1TB hard drive (RAID - 1 controller)
Out - of - Band Management	HPE Integrated Lights - Out (iLO): Advanced
Serial Port	Yes: DB - 9
Performance & Scale	Please refer to the **ClearPass Scaling & Ordering Guide**
Concurrent Sessions	10,000
Form Factor	
Rack Mount	• 1U LFF Easy Install Rail • 1U Cable Management Arm
Dimensions (WxHxD)	17.1" x 1.7" x 27.5"
Weight (Max Configuration)	Up to 36 lbs
Power Specifications	
Power Supply	HPE 500W Flex Slot Platinum Hot Plug Power Supply
Power Redundancy	Optional

ClearPass C2020 Appliance	Specifications
AC Input Voltage	100/240 VAC auto - selecting
AC Input Frequency	50/60 Hz auto - selecting
Environmental Specifications	
Operating Temperature	10º C to 35º C (50º F to 95º F)
Operating Vibration	Random vibration at 0.000075 G^2/Hz
Operating Shock	2 G's
Operating Altitude	3,050 m (10,000 ft)

ClearPass C3010 Hardware Appliance

The ClearPass C3010 (SKU: R1V82A) hardware appliance is a RADIUS/ TACACS+ server that provides advanced policy control for up to 50,000 concurrent sessions. The ClearPass C3010 hardware appliance ships with six Serial-Attach SCSI (SAS) (10K RPM) 600 GB Hot-Plug hard drives (RAID-10 controller). The LSI controller presents to ClearPass a single virtual 1.675 TB drive, masking the underlying two physical drive groups (two groups of three mirrored drives).

NOTE

The C3010 appliance uses an HPE 96W Smart Storage Battery (10.8Wh) lithium ion battery. For complete information on HPE batteries, refer to **Battery Support Documents** at **https://www.hpe.com/us/en/about/environment/battery-support-documents.html**.

To compare specifications and port information across all appliances, see ClearPass Hardware Appliance Specifications and Components . For additional information and to

verify the most recent updates, you may refer to the **Aruba ClearPass Policy Manager Data Sheet** and the **Scaling and Ordering Guide**.

> **NOTE**
> Gen9 C2000 and C3000 hardware appliances cannot be upgraded to ClearPass 6.14.0, as these appliances have already reached end of support (EOS/EOSL). If an upgrade is attempted on these appliances, the error message "ClearPass upgrade is not supported on Gen9 hardware (C2000/C3000)" is displayed. (CP-56976)

C3010 Ports and Components

The following figure shows the ports and components on the rear panel of the ClearPass C3010 hardware appliances. The function of each of these ports and components is described in the following table.

> **NOTE**
> The image of a ClearPass hardware appliance shown here includes the optional redundant power supply.

Figure 1. Ports and Components on the ClearPass C3010 Hardware Appliance

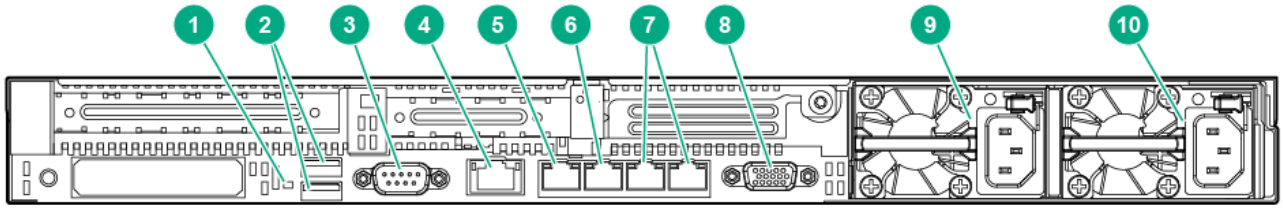


Table 1. ClearPass C3010 Ports and Components

Callout Number	Port or Component
1	UID (Unit ID) LED The UID LED helps you identify and locate a system, especially in high - density rack environments. Additionally, the UID is used to indicate that a critical operation is underway on the host, such as remote console access or ROM flash. **NOTE**

Callout Number	Port or Component
	This button is recessed and requires a small tool to access it. The "current state" (on or off) of the UID is the last state chosen using one of these methods. If a new state is chosen while the UID is blinking, this new state becomes the current state, and takes effect when the UID stops blinking. **NOTE** The Unit ID Light web page does not automatically refresh itself if the state of the actual light changes after the page is loaded. To ensure the page accurately reflects the state of the UID Light, click on the Virtual Indicators link to update the page
2	USB ports (2)
3	Serial port
4	iLO (Integrated Lights - Out) port
5	Management port (eth0)
6	Data port (eth1)
7	Destination SPAN ports (2)
8	VGA Connector
9	Fan and power supply AC inlet
10	Optional redundant fan and power supply

C3010 Hardware Appliance Specifications

This topic provides the specifications for the ClearPass C3010 hardware appliance.

Table 2. ClearPass C3010 Hardware Appliance Specifications

ClearPass Appliance	Specifications
Hardware Model	C3010: HPE DL360 Gen 10
CPUs	C3010: (1) Intel® Xeon® Processor 2.3GHz Gold 5118 with Twelve Cores (24 Threads)
Memory	64 GB Memory
Hard Drive Storage	(6) 600 GB Serial - Attach SCSI (SAS) (10K RPM) 60 GB Hot - Plug hard drives (RAID - 10 controller)
Out - of - Band Management	HPE Integrated Lights - Out (iLO): Advanced
Serial Port	Yes: DB - 9
Performance & Scale	Please refer to the **ClearPass Scaling & Ordering Guide**
Concurrent Sessions	50,000
Form Factor	
Rack Mount	• 1U SFF Easy Install Rail • 1U Cable Management Arm
Dimensions (WxHxD	17.1" x 1.7" x 27.5"
Weight (Max Configuration)	C3010: Up to 36 Lbs
Power Specifications	
Power Supply	HPE 500W Flex Slot Platinum Hot Plug Power Supply
Power Cord	C3010: C13 - C14 WW 250V 10Amp Jumper Cord
Power Redundancy	Optional
AC Input Voltage	100/240 VAC auto - selecting
AC Input Frequency	50/60 Hz auto - selecting

ClearPass Appliance	Specifications
Environmental Specifications	
Operating Temperature	10º C to 35º C (50º F to 95º F)
Operating Vibration	Random vibration at 0.000075 G^2/Hz
Operating Shock	2 Gs
Operating Altitude	3050 m (10,000 ft)

ClearPass N1000 Hardware Appliance

The ClearPass N1000 (SKU: (S3N67A) hardware appliance is a RADIUS/ TACACS+ server that provides advanced policy control for up to 1000 concurrent sessions. The ClearPass N1000 hardware appliance ships with one SSD (1TB) M.2 hard drive with a 150W power supply. The N1000 has a 4-port copper network interface.

To compare specifications and port information across all appliances, see ClearPass Hardware Appliance Specifications and Components . For additional information and to verify the most recent updates, you may refer to the **Aruba ClearPass Policy Manager Data Sheet** and the **Scaling and Ordering Guide**.

N1000 Ports and Components

The following figure shows the ports and components on the rear panel of the ClearPass N1000 hardware appliance. The function of each of these ports and components is described in the following table.

Figure 1. ClearPass N1000 Ports and Components

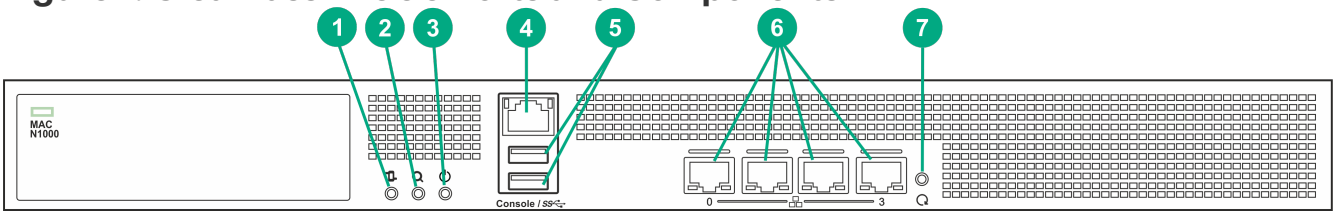


Table 1. ClearPass N1000 Ports and Components

Callout Number	N1000 Port or Component
1	Status LED. Amber. Indicates if a fault is detected (such as fans not working, chassis open, etc.).

Callout Number	N1000 Port or Component
2	Unit ID (UID) LED, also called the Location LED. Blue. The Unit ID LED helps you identify and locate a system, especially in high - density rack environments, by blinking if "locate" is triggered from the out - of - band management of the server. Additionally, the Unit ID LED is used to indicate that a critical operation is underway on the host, such as remote console access or ROM flash. **NOTE** This button is recessed and requires a small tool to access it. The "current state" (on or off) of the UID LED is the last state chosen using one of these methods. If a new state is chosen while the UID LED is blinking, this new state becomes the current state, and takes effect when the UID LED stops blinking. **NOTE** The Unit ID Light web page does not automatically refresh itself if the state of the actual light changes after the page is loaded. To ensure the page accurately reflects the state of the UID Light, click on the Virtual Indicators link to update the page.
3	Power LED. Green. Indicates the appliance is powered on.
4	Serial management port.
5	USB ports (2)
6	Left to right: Management Port (eth0), Data Port (eth1), two Destination SPAN Ports.

Callout Number	N1000 Port or Component
7	Ethernet card reset button. Can be used to initiate a hardware reset on the ethernet card.

N1000 Hardware Appliance Specifications

Table 15 provides the specifications for the ClearPass N1000 hardware appliance.

Table 2. ClearPass N1000 Hardware Appliance Specifications

ClearPass Appliance	Specifications
Hardware Model	Advantech FWA-2012 (customized)
CPUs	Intel Atom C3758R with 8 cores (8 threads)
Memory	16 GB
Hard Drive Storage	(1) SSD (1TB) M.2
Out - of - Band Management	N/A
Serial Port	Yes (RJ-45) *
Performance & Scale	Please refer to the **ClearPass Scaling & Ordering Guide**
Concurrent Sessions	1000
Form Factor	
Rack Mount	1U Easy Install Rail 2 Kit
Dimensions (WxHxD	19 x 1.7 x 12.6
Weight (Max Configuration)	Up to 14.5 lb
Power Specifications	
Power Supply	(1) 150W
Power Cord	(1) C13 -C14 WW 250V 10A Jumper cord
Power Redundancy	N/A
AC Input Voltage	100/240 VAC auto - selecting

ClearPass Appliance	Specifications
AC Input Frequency	50/60 Hz auto - selecting
Environmental Specifications	
Operating Temperature	10° to 35°C (50° to 95° F)
Operating Vibration	0.5 Grms, 5-500Hz, 1hr/axis
Operating Shock	1.5 Gs
Operating Altitude	3050 m (10,000 ft)

***** The N1000 does not include integrated video; only the serial interface is supported.

ClearPass N3000 and N3001 Hardware Appliances

The ClearPass N3000 (SKU: (S3N66A) and the N3001 (SKU: (S3N65A) hardware appliances are RADIUS/ TACACS+ servers that provide advanced policy control for up to 100,000 concurrent sessions. The N3000 and N3001 hardware appliances ship with three x 960GB SATA SSD hard drives (RAID 5 controller) with a 1000W Flex Slot Titanium Hot Plug power supply. The N3000 has a 4-port copper network interface. The N3001 has a 4-port 10/25 Gbe SFP28 PCIe network interface and requires transceivers, which are not included and must be purchased separately. For more information, see the "Transceiver guide" sections of the **ClearPass Scaling and Ordering Guide**.

NOTE

The N3000 and N3001 appliances use an HPE 96W Smart Storage lithium-ion Battery (HPE part number P01366-B21) For complete information on HPE batteries, refer to **Battery Support Documents** at **https://www.hpe.com/us/en/about/environment/battery-support-documents.html**.

To compare specifications and port information across all appliances, see ClearPass Hardware Appliance Specifications and Components . For additional information and to verify the most recent updates, you may refer to the **Aruba ClearPass Policy Manager Data Sheet** and the **Scaling and Ordering Guide**.

N3000 and N3001 Ports and Components

The following figure shows the ports and components on the rear panel of the ClearPass N3000 and N3001 hardware appliances. The function of each of these ports and components is described in the following table.

Figure 1. Ports and Components on the ClearPass N3000 and N3001 Hardware Appliance

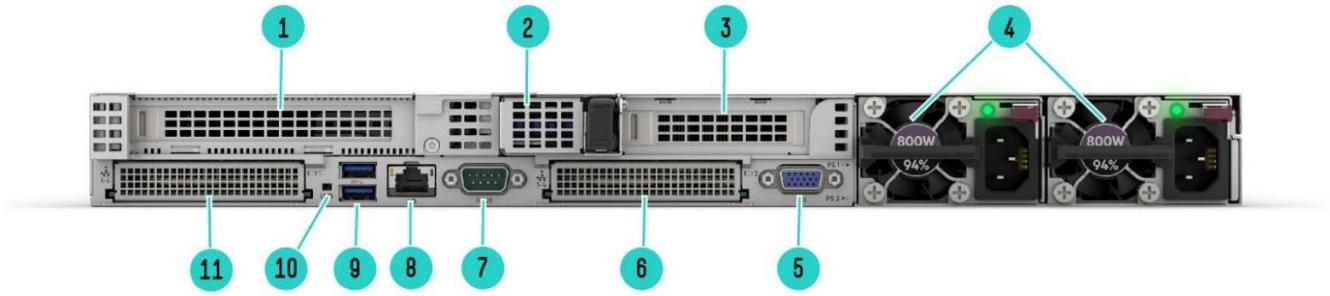


Table 1. ClearPass N3000 and N3001 Ports and Components

Callout Number	N3000 and N3001 Port or Component
1	Primary PCIe 5.0 Riser Used by the N3001 for the 10GB/25GB 4 - port SFP28 NIC
2	Optional NS204i - u hot - plug NVMe boot device
3	Secondary PCIe 5.0 Riser
4	Hot - plug Power Supply 1 and 2
5	Video (VGA) port
6	OCP 3.0 slot 22 Used by the N3000 for the 1GB 4 - port NIC
7	Optional Serial port
8	Dedicated iLO (Integrated Lights - Out) management port
9	USB 3.1 Gen1 Ports (2)
10	UID (Unit ID) LED The UID LED helps you identify and locate a system, especially in high - density rack environments. Additionally, the UID is used to indicate that a critical operation is

Callout Number	N3000 and N3001 Port or Component
	underway on the host, such as remote console access or ROM flash. **NOTE** This button is recessed and requires a small tool to access it. The "current state" (on or off) of the UID is the last state chosen using one of these methods. If a new state is chosen while the UID is blinking, this new state becomes the current state, and takes effect when the UID stops blinking. **NOTE** The Unit ID Light web page does not automatically refresh itself if the state of the actual light changes after the page is loaded. To ensure the page accurately reflects the state of the UID Light, click on the Virtual Indicators link to update the page.
11	OCP 3.0 slot 21

N3000 and N3001 Hardware Appliance Specifications

Table 17 provides the specifications for the ClearPass N3000 and N3001 hardware appliances.

Table 2. ClearPass N3000 and N3001 Hardware Appliance Specifications

ClearPass Appliance	Specifications
Hardware Model	HPE ProLiant DL325 Gen11
CPUs	(1) AMD EPYC 9124 16 - Core Processor (32 threads)
Memory	128 GB Memory

ClearPass Appliance	Specifications
Hard Drive Storage	(3) x 960GB SATA SSD hard drives (RAID 5 controller) with a 1000W Flex Slot Titanium Hot Plug power supply.
Out - of - Band Management	HPE Integrated Lights - Out (iLO) Advanced with Dedicated NIC
Serial Port	Yes: DB - 9
Performance & Scale	Please refer to the **ClearPass Scaling & Ordering Guide**
Concurrent Sessions	100,000
Form Factor	
Rack Mount	1U Easy Install Rail 2 Kit
Dimensions (WxHxD	17.1" x 1.7" x 25.6"
Weight (Max Configuration)	Up to 34.5 lb
Power Specifications	
Power Supply	(2) HPE 1000W Flex Slot Titanium Hot Plug Power Supply
Power Cord	C13 - C14 WW 250V 10Amp Jumper Cord
Power Redundancy	Included
AC Input Voltage	100/240 VAC auto - selecting
AC Input Frequency	50/60 Hz auto - selecting
Environmental Specifications	
Operating Temperature	10° to 35°C (50° to 95° F)
Operating Vibration	Random vibration at 0.000075 G/Hz, 10Hz to 300Hz (0.15 G's nominal)
Operating Shock	2 Gs
Operating Altitude	3050 m (10,000 ft)

Before Starting the ClearPass Hardware Installation

This section provides information you should be aware of before you perform a new hardware installation.

NOTE

We recommend that customers ensure the latest hardware appliance firmware is applied prior to initial appliance deployment. Customers may need to update the firmware to the latest custom Service Pack for Proliant (SPP) image for their HPE CXXXX hardware model, or to the latest firmware image for their NXXXX hardware model. The SPP and firmware images are available in the downloads section of the **HPE Aruba Networking Support Portal (NSP)**.

NOTE

Gen9 C2000 and C3000 hardware appliances cannot be upgraded to ClearPass 6.14.0, as these appliances have already reached end of support (EOS/EOSL). If an upgrade is attempted on these appliances, the error message "ClearPass upgrade is not supported on Gen9 hardware (C2000/C3000)" is displayed.

NOTE

ClearPass now supports Classless Inter-Domain Routing (CIDR) notation to configure IPv4 and IPv6 addresses on the management and data ports when the system is bootstrapped during installation or upgrade. As part of this change, the prefix length of the address is required as part of the input, and the input prompt displays an example of this address format. Netmask format is no longer used to acquire IP addresses.

NOTE

In ClearPass 6.11.0 and later versions, the maximum single cluster size is limited to 32 servers. This includes the publisher, standby publisher, subscribers, dedicated Insight server, and standby Insight servers.

NOTE

Direct upgrades to ClearPass 6.14.0 are only supported from 6.12.x versions. Direct upgrades from any versions below 6.12.x are not currently supported, and database migrations from configuration backups taken on versions below 6.12.x might fail.

NOTE

Before starting the ClearPass installation and configuration procedures for the hardware appliance, determine the following information for the ClearPass server on your network, note the corresponding values for the parameters listed in Table 1, and keep it for your records.

Table 1. ClearPass Server Configuration Values

Required Information	Value for Your Installation
Host name (ClearPass server)	
Management port IP address	
Management port subnet mask	
Management port gateway	
Data port IP address (optional)	Make sure that the Data port IP address is *not* in the same subnet as the Management port IP address.
Data port subnet mask (optional)	
Data port gateway (optional)	
Primary DNS	
Secondary DNS	
NTP server (optional)	

Subtopics

Battery Charging Behavior Changes

Battery Charging Behavior Changes

This section describes changes in 2025 that affect battery behaviors during setup of new hardware appliances. This information applies to all HPE server models used with ClearPass:

- All N300x hardware
- C3010 hardware (RMA shipments)
- C2020 hardware (RMA shipments)

- C2010 hardware (RMA shipments)

Subtopics

Previous Behavior
Current Behavior
Customer Effect
What to Expect

Previous Behavior

Prior to August 2025, ClearPass hardware units were shipped to customers with their RAID batteries fully charged. This ensured the units would work immediately out of the box without errors when customers booted the appliances.

Current Behavior

Beginning in August 2025, an EU regulatory change requires that lithium ion (LiON) batteries have no more than a 30% State of Charge (SoC) when they are shipped. To comply with this requirement, we no longer fully charge RAID batteries before shipment, and may actually need to partially discharge batteries prior to some shipments to ensure they do not exceed the partial-charge restrictions.

Customer Effect

To our customers, this means that in most cases the units will arrive with low or no charge on the battery. Unlike some earlier generations of appliances, these batteries can be recharged from a 0% SoC, but it will take some time. We recommend that customers plan to recharge the battery for 4 - 6 hours when they receive it and before powering on the appliance. However, the RAID battery for the controller is a backup battery in case of power outages, and the appliance will continue to work correctly if it is plugged in but will display an error message.

What to Expect

If you boot the system before you recharge the battery, it will display an error message similar to the following, with slight differences depending on the appliance type:

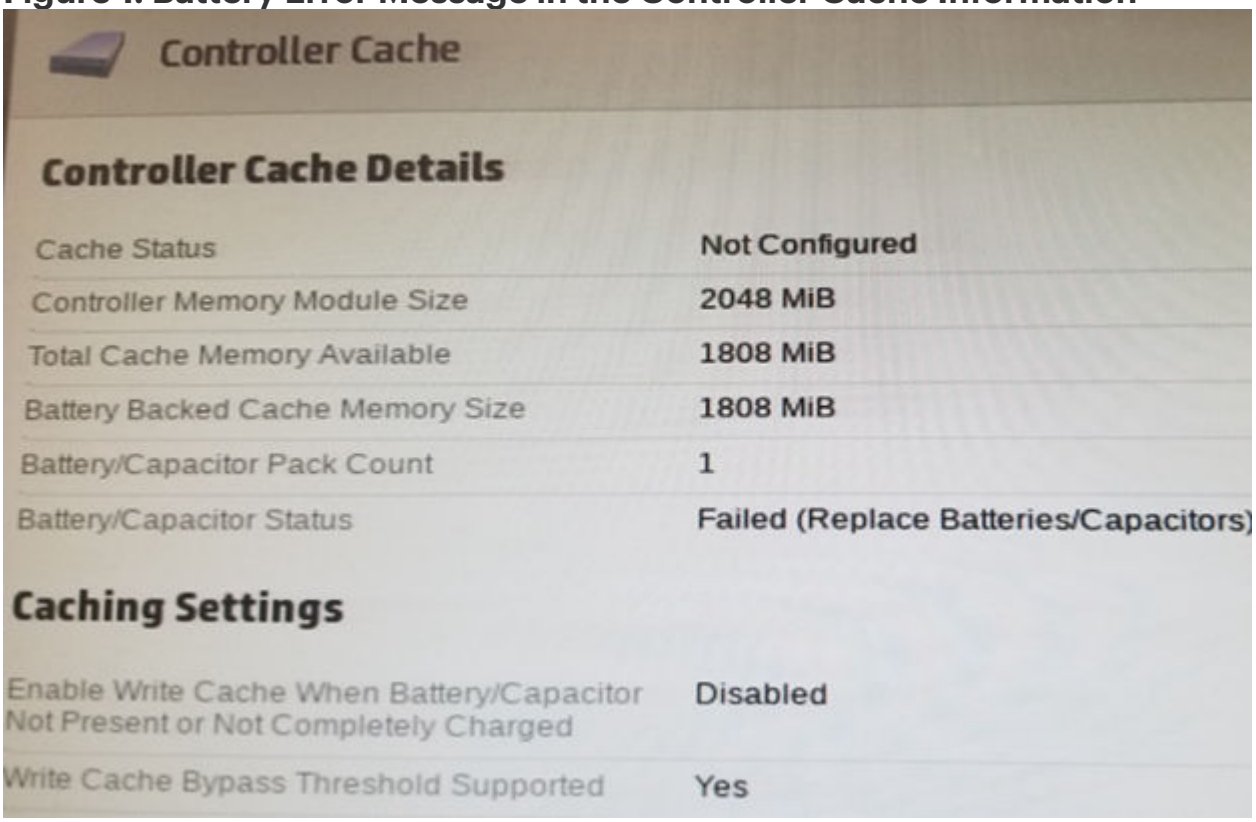
```
   Embedded RAID 1: HPE Smart Array P408i-a SR Gen10 - Configuration
Required
   1932 - Slot 0 Smart Array - Cache Status: Disabled (error Code: Energy
Pack Failure.)
   Action: Do one of more of the following:
   1. Update system ROM and BMC firmware
   2. Replace energy pack
   3. Replace the controller
```

No action is needed. This error is the expected behavior if you've powered on while the battery is charging. Allow the system to continue to boot normally and the battery will recharge correctly within 4-6 hours. The subsequent reboot will no longer display the message.

Remember that this error condition can be avoided by connecting the appliance to power for 4-6 hours prior to booting the first time. This will ensure that the appliance has had sufficient time to fully charge the RAID battery prior to power-on.

The UEFI on an appliance may also show the following information:

Figure 1. Battery Error Message in the Controller Cache Information



Again, this is a notification about the RAID battery having insufficient charge at initial boot. Allow 4-6 hours for the appliance to recharge to normal levels and the message will not be displayed on the subsequent reboot.

If the error does not clear after being connected to power for more than eight hours (it does not need to be powered on the entire time) then the customer may investigate the RMA process. At this time it is not recommended to perform a full-system RMA due

to the fact that the RAID battery on the next unit will also be in a discharged state. The appropriate SPN should be ordered to replace the battery exclusively.

Users should be aware that the RAID battery RMA will also likely ship a battery that has a 30% or lower SoC to comply with these new laws.

Configuring the ClearPass Hardware Appliance

About this task

The initial setup dialog starts when you connect a terminal, PC, or laptop running a terminal emulation program to the Serial port on the ClearPass hardware appliance.

> **NOTE**
> RAID Batteries in new appliances arrive with a low charge and must be fully charged before booting. See Battery Charging Behavior Changes before starting setup.

To configure the ClearPass hardware appliance:

Procedure

1. **Connect the Serial port**.
   a. Connect the Serial port to a terminal using the null modem cable (not provided).
   b. Power on the hardware appliance. The hardware appliance is now available for configuration.
2. **Configure the Serial port**.
   Apply the following parameters for the Serial port:
   - **Bit Rate**: 9600
   - **Data Bits**: 8
   - **Parity**: None
   - **Stop Bits**: 1
   - **Flow Control**: None
3. **Log in**.
   Use the following preconfigured credentials to log in to the hardware appliance. (You will create a unique appliance/cluster administration password in Step 5.)

- `login: appadmin`
- `password: eTIPS123`

This initiates the ClearPass Configuration wizard.

4. **Configure the ClearPass hardware appliance**.

   Follow the prompts, replacing the placeholder entries with the information you entered in Table 1 in Before Starting the Policy Manager Installation:

   - Enter hostname:
   - Enter Management Port IP Address:
   - Enter Management Port Subnet Mask:
   - Enter Management Port Gateway:
   - Enter Data Port IP Address:
   - Enter Data Port Subnet Mask:
   - Enter Data Port Gateway:
   - Enter Primary DNS:
   - Enter Secondary DNS:
   - Do you want to enable SLAAC mode? [y|n]

   NOTE

   When a new 6.7.0 C1000 hardware appliance is first powered on, it already shows a management IP address (10.2.48.74). This IP address is used during the manufacturing process. It is replaced when you enter your management IP address during appliance configuration.

5. **Specify the cluster password**.

   NOTE

   Setting the cluster password also changes the password for the CLI user **appadmin**, as well as the Administrative user **admin**. If you want the **admin** password to be unique, see Changing the Administration Password .

a. Enter any string with a minimum of six characters, then you are prompted to confirm the cluster password.

b. After this configuration is applied, use this new password for cluster administration and management of the ClearPass virtual appliance.

6. **Configure the system date and time**.

   a. Follow the prompts to configure the system date and time.

   b. To set the date and time by configuring the NTP server, use the primary and secondary NTP server information you entered in Table 1 in Before Starting the Policy Manager Installation.

7. **Apply the configuration**.

   a. To apply the configuration, press **Y**.

      - To restart the configuration procedure, press **N**.
      - To quit the setup process, press **Q**.

      Configuration on the hardware appliance console is now complete. The next task is to activate ClearPass.

Activating ClearPass for a Hardware Appliance

Licensing requirements and procedures vary slightly between ClearPass virtual appliances and ClearPass hardware appliances. For a ClearPass hardware appliance, the **Platform Activation Key** activates a **Platform License Key** .

You must have a Platform Activation Key for every hardware appliance. Platform Activation Keys are available for purchase as a **Permanent** license (with no expiration date), or as an **Evaluation** license with a 90- or 180-day duration.

To activate a ClearPass hardware appliance and apply the ClearPass license, follow the steps described below.



NOTE

These activation instructions are specific to hardware appliances. For more information, and for information about activating ClearPass Policy Manager on a virtual appliance (VA), see the "License Management" section of the *Policy Manager User Guide*.

Subtopics

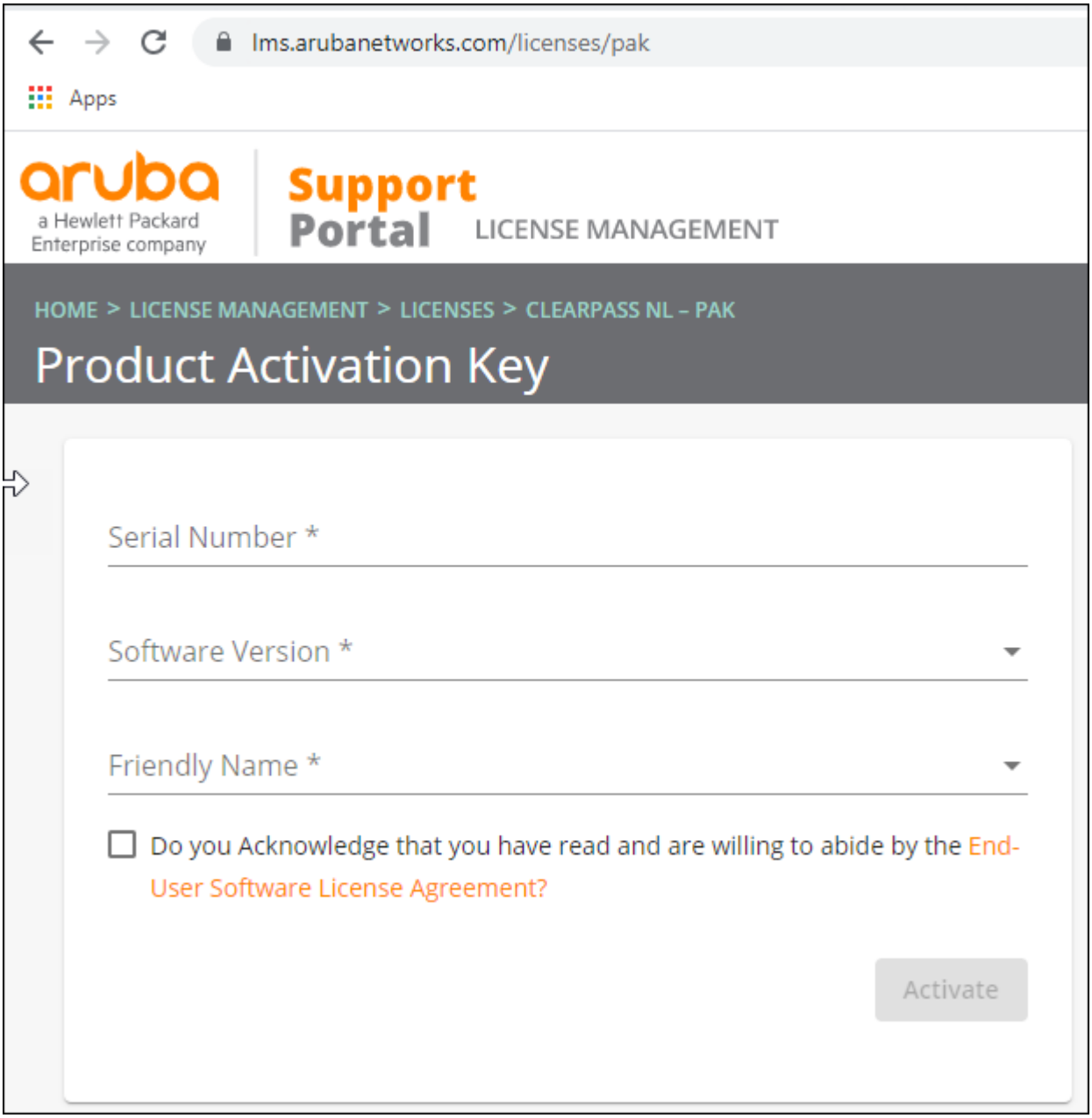
Receiving the Platform Activation Key in the HPE Networking Support Portal
Activating the License in ClearPass Policy Manager

Receiving the Platform Activation Key in the HPE Networking Support Portal

Procedure

1. After the configuration for the hardware appliance is completed as described in the previous section, open the HPE Networking Support Portal at **https://asp.arubanetworks.com/**.
2. In the top navigation, go to **Service Management > License Management**, and then go to **Licenses > ClearPass NL – PAK**. The **Product Activation Key** form opens (**https://lms.arubanetworks.com/licenses/pak**).

Figure 1. The Product Activation Key Form in License Management Services (LMS)



3. Enter the information for this hardware appliance in the **Serial Number** and **Software Version** fields.
4. In the **Friendly Name** field, enter a name that will clearly identify this Platform Activation Key as belonging to this hardware appliance. This is included as a column in the list of licenses in the Support Portal.
5. Mark the check box to accept the End User Software License Agreement (EULA), and then click **Activate**. The Platform Activation Key (PAK) for the appliance is displayed.
6. Copy and paste the PAK into a text file. Be sure to include the "begin" and "end" lines, including all the hyphen characters (for example, "-----BEGIN ACCESS LICENSE KEY-----"); however, do not include any leading or trailing space characters. Keep the

key displayed in the HPE Networking Support Portal while you do the next steps, and save the text file for future reference.

Activating the License in ClearPass Policy Manager

Procedure

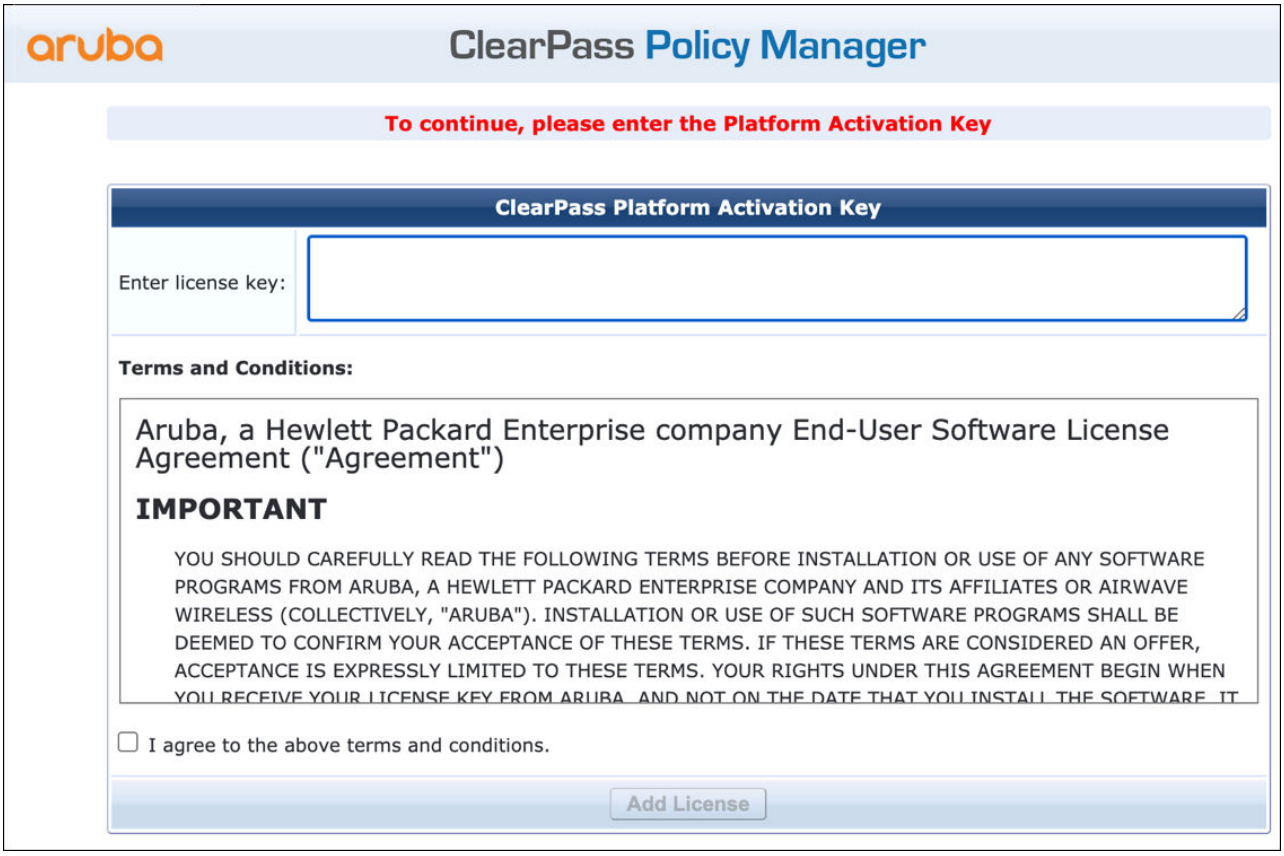
1. In a separate tab of your web browser, go to the ClearPass management interface:

   `https://x.x.x.x/tips/`, where **x.x.x.x** is the IP address of the management port defined for the ClearPass server you entered in Table 1 in Before Starting the Policy Manager Installation.

2. Accept any security warnings from your browser regarding the self-signed SSL certificate, which comes installed in ClearPass by default.

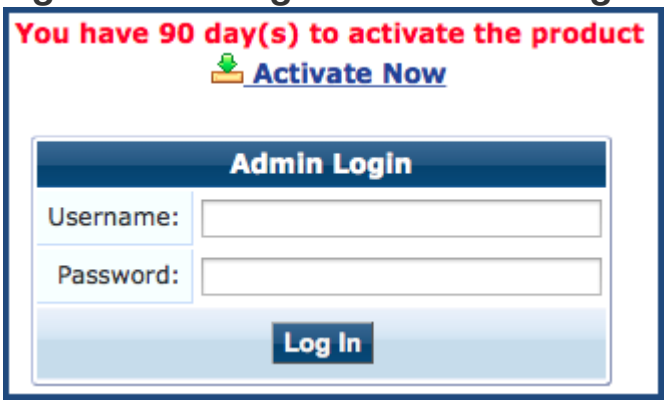
3. The **ClearPass Platform Activation Key** form opens and asks you to enter your license key. Paste in the Platform Activation Key you copied earlier, including the "-----BEGIN <license type>-----" and "-----END <license type>-----" lines. Make sure there are no empty space characters before or after those lines or the license key will not be accepted.

   Figure 1. The ClearPass Platform Activation Key Form: Enter PAK



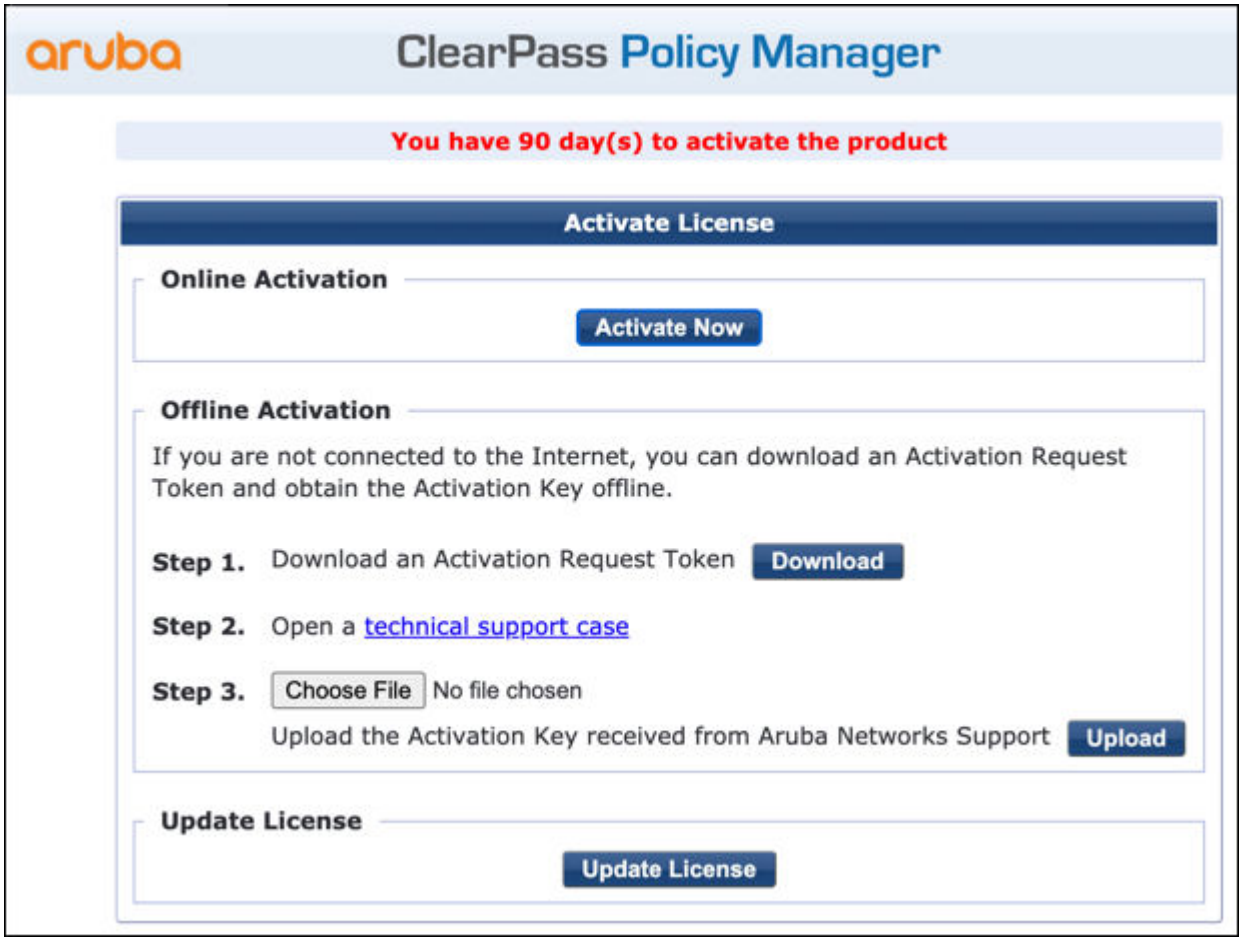
4. Mark the check box to accept the terms and conditions, and then click **Add License**. The ClearPass Policy Manager **Admin Login** page opens. It displays a message indicating that you have 90 days to activate the product, and includes an **Activate Now** link.

Figure 2. The Login Form: Activating ClearPass



5. Click the **Activate Now** link in the message on the login page. The **Activate License** form opens.

Figure 3. The Activate License Form for Online or Offline Activation



- If you are connected to the internet, click the **Activate Now** button. ClearPass attempts to activate the product over the Internet with Aruba license activation servers.
- Otherwise, if the ClearPass hardware appliance does not have Internet access, follow the steps shown in the **Offline Activation** area of the **Activate License** form.

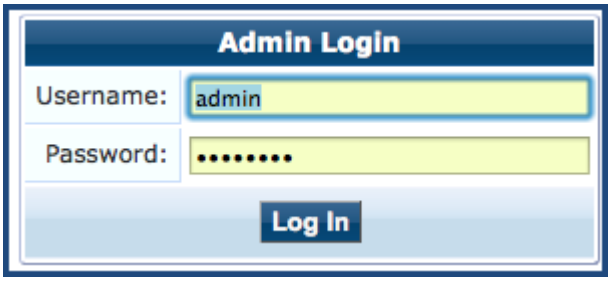
6. After the license is successfully activated, the **Admin Login** form is displayed again and the banner message indicates that the product has been successfully activated. Enter your credentials and log in to ClearPass Policy Manager.

Logging in to the ClearPass Hardware Appliance

About this task

After a successful activation, the **Admin Login** dialog opens.

Figure 1. Logging in to the ClearPass Hardware Appliance

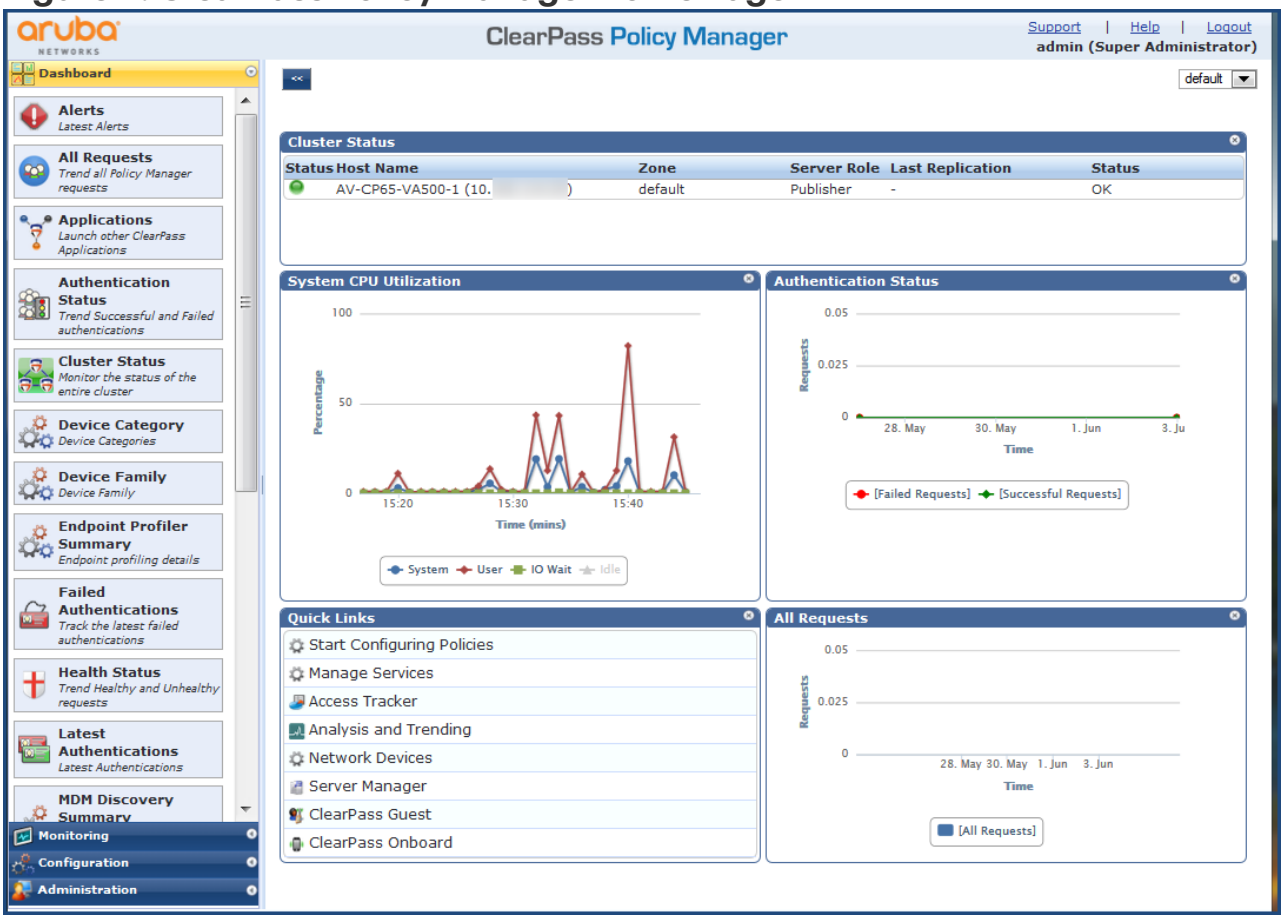


Procedure

1. Log in to the ClearPass hardware appliance with the following credentials:
   - **Username**: admin
   - **Password**: Enter the cluster password defined in Configuring the ClearPass Hardware Appliance .
2. Click **Log In**.

   The Policy Manager home page is displayed.

Figure 2. ClearPass Policy ManagerHome Page



Signing Up for Live Software Updates

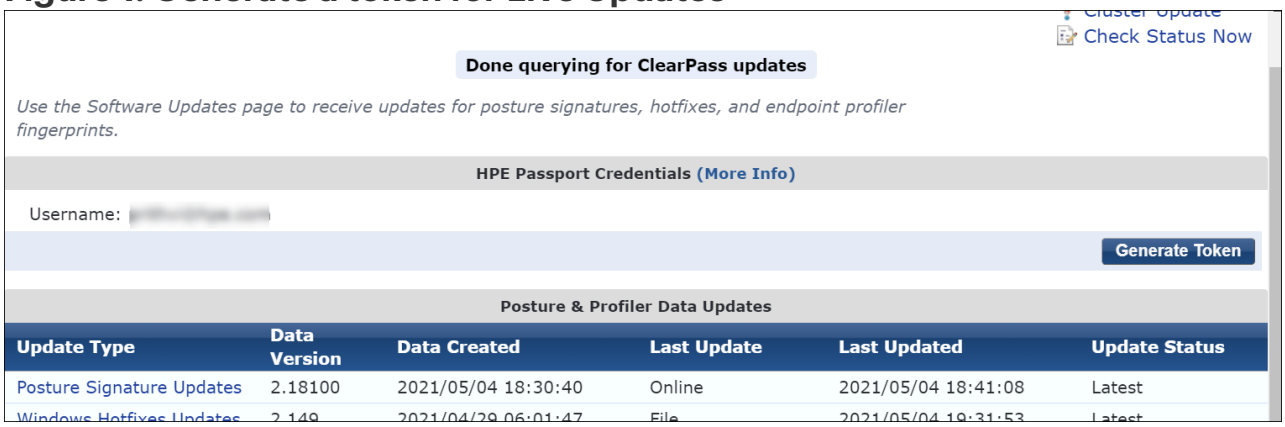
About this task

Upon your initial login to ClearPass, register for software updates.

Procedure

1. Go to the **Administration > Agents and Software Updates > Software Updates** page. A message is displayed indicating that the ClearPass hardware appliance is not signed up for live updates and that you must generate a token.

Figure 1. Generate a token for Live Updates



2. If the ClearPass server has Internet access, click **Generate Token**. The single sign-on (SSO) login page opens. After successful authentication, you will see a message that Policy Manager is processing updates from the ClearPass Webservice.

   Note that **Posture & Profile Data Updates** are downloaded and installed automatically, while **Firmware & Patch Updates** are displayed only.

Changing the Administration Password

About this task

When the cluster password for this ClearPass server is set upon initial configuration, the administration password is also set to the same password (see Configuring the ClearPass Hardware Appliance). If you wish to assign a unique **admin** password, use this procedure to change it.

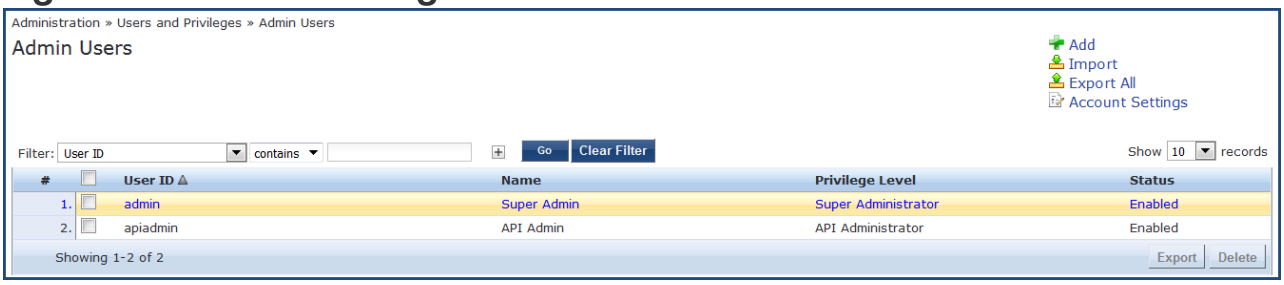
To change the administration password:

Procedure

1. In Policy Manager, go to **Administration > Users and Privileges > Admin Users**.

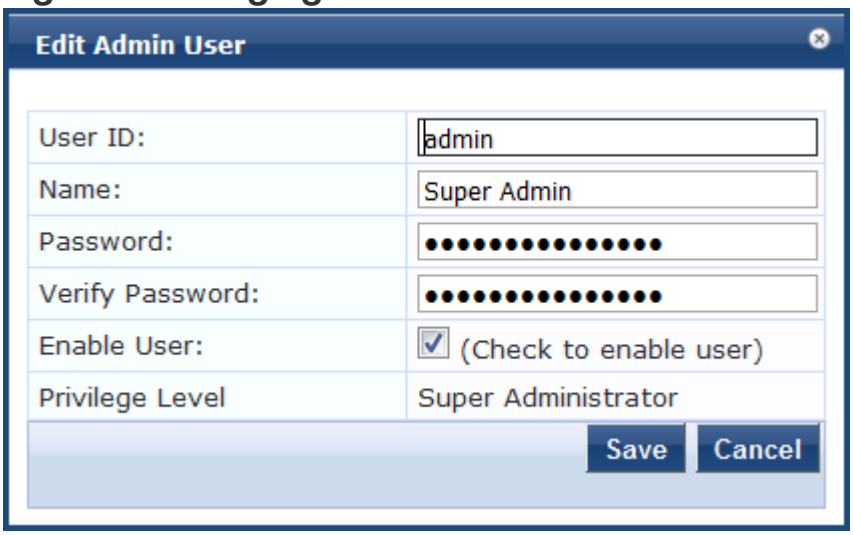
   The **Admin Users** page appears.

Figure 1. Admin Users Page



2. Select the appropriate **admin** user. The **Edit Admin User** dialog appears.

Figure 2. Changing the Administration Password



3. Change the administration password, verify the new password, then click **Save**.

Powering Off the ClearPass Hardware Appliance

About this task

This procedure gracefully shuts down the hardware appliance without having to log in. To power off the ClearPass hardware appliance:

Procedure

1. Connect to the CLI from the console using the serial port.

2. Enter the following commands:
   - **login: poweroff**
   - **password: poweroff**

   The ClearPass hardware appliance shuts down.

> **NOTE**
>
> You can also power off from the WebUI and the appadmin prompt.

Resetting the System Passwords to the Factory Defaults

To reset the system account passwords in ClearPass to the factory defaults, you must first generate a password recovery key, then log in as the apprecovery user to reset the system account passwords.

Generating the Password Recovery Key

About this task

To generate the password recovery key:

If you are employing a hardware connection, connect to the ClearPass hardware appliance using the serial port (using any terminal program). See Configuring the ClearPass Hardware Appliance for details.

1. If you are employing a virtual appliance, refer to the instructions for the appropriate hypervisor in the "Virtual Appliance Installation" sections in this Installation Guide.

Reboot the system using the **restart** command.

Procedure

1. After the system reboots, the following prompt is displayed for ten seconds:

   Generate support keys? [y/n]:

2. At the prompt, enter **y**. The system prompts you with the following choices:

   Please select a support key generation option.

1) Generate password recovery key

2) Generate a support key

3) Generate password recovery and support keys

Enter the option or press any key to quit.

3. To generate a password recovery key, select option **1**.
4. After the password recovery key is generated, open a support case with Aruba Technical Support.

   A unique password will be dynamically generated from the recovery key and sent to you.

Resetting the System Account Passwords to the Factory Defaults

About this task

To reset the administrator password:

Procedure

1. Log in as the **apprecovery** user with the password recovery key provided by Aruba Technical Support.
2. Enter the following command at the command prompt:

```
[
apprecovery
]
app reset-passwd
   ********************************************************
   * WARNING: This command will reset the system account *
   * passwords to factory default values                  *
   ********************************************************
Are you sure you want to continue? [y/n]: y

INFO - Password changed on local node

INFO - System account passwords have been reset to factory default
values
```

3. To reset the system account passwords to the factory default values, enter **y**.
4. You can now log in with the new administrator password sent to you by Aruba Technical Support.

Cloud Deployments: Amazon Web Services (AWS)

This document describes the necessary steps to successfully deploy ClearPASS 6.11.0 on a hosted cloud service such as Amazon Web Services (AWS). This document assumes that you have an AWS account, a ClearPass 6.11.0 Amazon Machine Image (AMI) or similar image for your cloud service, and a subnet that contains Network Access Devices (or NADs).

This document gives an example of the AWS ClearPass topology with a NAD, then describes the required prerequisite steps you must perform before launching your AWS ClearPass deployment, such as configuring a customer gateway and virtual private gateway, and setting up a secure VPN connection between the VPC and the NAD network. Then, it shows how to launch an AWS instance with an Amazon Machine Image (AMI). Then it describes additional tasks that must be performed after launching an instance like attaching a network interface, creating an elastic IP address, defining an EBS volume, and configuring the ClearPass CLI.

ClearPass supports hybrid clusters with both on-premises ClearPass servers and instances of ClearPass deployed using AWS EC2. AWS EC2 cluster members can be in different regions. Hybrid clusters can be configured in FIPS mode, standard (non-FIPS) mode, and in Common Criteria (CC mode). The mode will appear in the footer in both on-premises and AWS EC2 instances of ClearPass.

Hybrid deployments with cluster members deployed both on-premises and in AWS EC2 have the following limitations:

- A hybrid cluster does not support Virtual IPs (VIPs).
- Hybrid clusters support ClearPass in AWS EC2 and on premise ClearPass instances only. It doesn't support clustering between ClearPass in AWS EC2 and Azure ClearPass instances."
- Only the MGMT interface is supported for both on-premises and AWS-EC2 instances of ClearPass.
- Hybrid cloud deployment is not supported for EC2 instances in China regions Ningxia and Beijing, as Site-Site VPN is not supported.

Subtopics

Behavior Differences in an AWS Deployment
Example Topology
Before You Begin
Launching ClearPass on AWS from an AMI
Post-Launch Configuration Tasks in AWS
Configuring ClearPass
Upgrading an M5 Instance Type to an M7 Instance Type

Behavior Differences in an AWS Deployment

When ClearPass is hosted on AWS, users should be aware of the following differences:



NOTE

Beginning in ClearPass 6.11 the use of m4 Amazon Web Services is no longer supported. Amazon indicates that all customers must move to M5 or M5a instances to be able to support operating ClearPass.

- In server configuration, editing is disabled for the management port IP address, subnet mask, default gateway, and data port IP address.
- On an Amazon Web Services (AWS) instance, after the AWS restore service is used to restore a backup and it has assigned a new IP address to the restored instance, the new IP address is not updated on the **Administration > Server Manager > Server Configuration** page. In the CLI, the **system refresh-network** command must be run for ClearPass to be refreshed with the new IP address.
- AWS virtual private cloud (VPC) deployments do not support virtual IP (VIP) addresses. Virtual IP settings and SPAN port settings are hidden.
- Using more than one cloud service provider in a single ClearPass cluster is not supported. Within each cluster, AWS instances can be combined (hybrid cluster) with virtual appliances and hardware appliances, but not with different cloud providers. To work with more than one cloud service provider, they must be in separate clusters.
- All servers in the cluster should have the same FIPS mode setting — either all enabled or all disabled.
- In the footer of the ClearPass Policy Manager user interface, the indicator “[Cloud]” is included after the version number.
- In ClearPass 6.8.1 and earlier releases, HTTP proxy settings were hidden for ClearPass servers hosted on AWS. However, starting from ClearPass 6.8.2, there is no behavior difference from a standard deployment, and users can configure an HTTP proxy for ClearPass servers deployed in AWS.

- The CLI commands **system factory-reset** and **system install-image** are not available in ClearPass running on AWS.

> **CAUTION**
>
> While deploying ClearPass on AWS, before you execute the system morph-vm command you must attach a fresh additional disk as an extended volume. Do not attach any volume that was detached from an existing ClearPass instance, and do not re-use any data volume that was previously morphed. Attaching a previously used volume could cause the server to fail to boot after the morph operation.

> **CAUTION**
>
> Avoid using the CLI to update an AWS instance to a higher 6.11.x version, and update it through the user interface instead. If you must use the CLI to update this instance type to a higher 6.11.x version, please increase the **CLI Session Idle Timeout** cluster-wide parameter value to 120 minutes to ensure that the patch update is not interrupted.

> **NOTE**
>
> ClearPass does not support auto-scaling (elastic scaling) in AWS. All appliances in the cluster are expected to be on and connected at all times in order to maintain the cluster. Load balancers may be used instead to balance traffic. For more information, see the "**Deploying CPPM with F5 BIG-IP Local Traffic Manager (LTM)**" Tech Note.

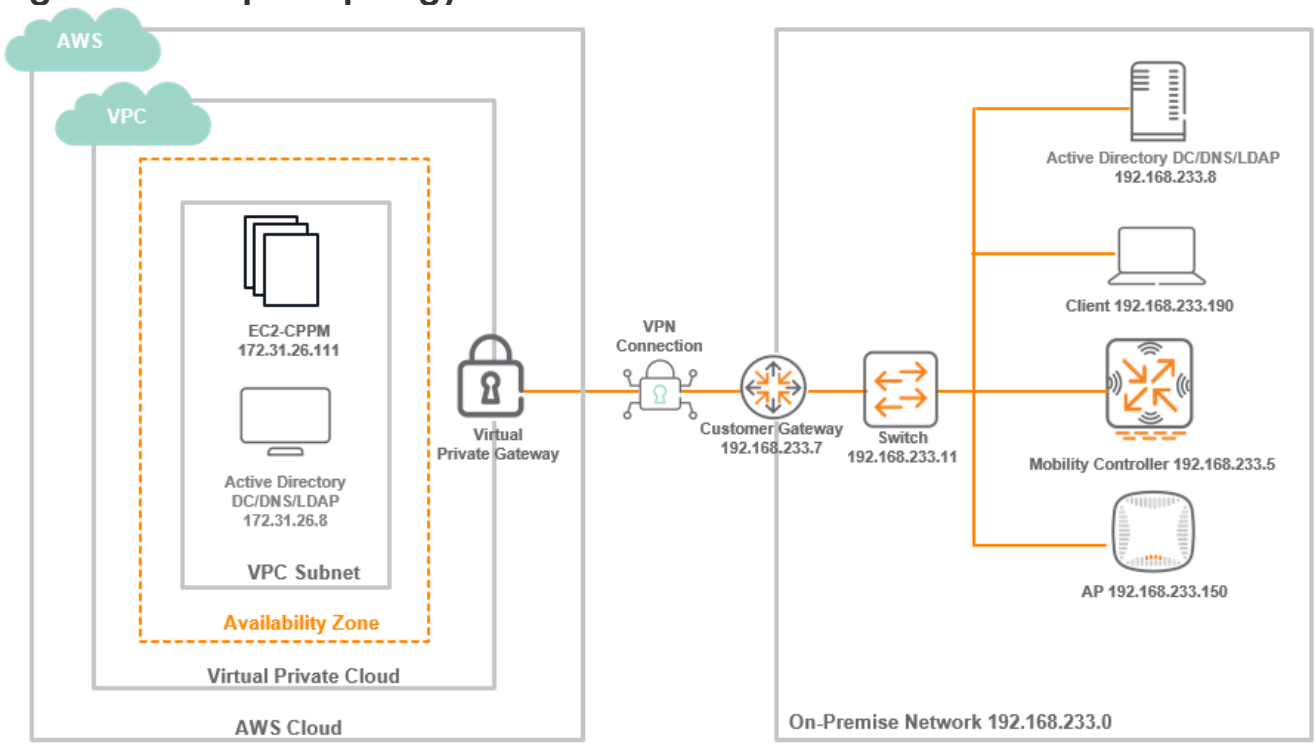
> **CAUTION**
>
> The **system morph-vm** command should be used only once for each instance of a cloud deployment. If this command is used multiple times in the same cloud instance, it can cause disk corruption.

Example Topology

ClearPass is hosted on an AWS Virtual Private Cloud instance. Communication between the controller and ClearPass takes place through a secure VPN connection (IPsec tunnel) that is initiated by the controller. The figure below shows the test network setup that we used to test the ClearPass services on AWS.

In the example below, the network access device (NAD), which in this case is the Aruba controller, sits in a private network in the datacenter along with other APs, clients, DNS and a switch. The private gateway IP of this network is 192.168.233.7. On the other side is a Virtual Private Cloud (VPC) of subnet 172.31.0.0/16. This topology establishes a secure IPsec connection between the two networks to enable communication between the Aruba controller and ClearPass on AWS. ClearPass is communicating with all authentication sources, NADs and clients over a VPN connection (IPSec), and the VPN tunnel is terminated on a Virtual Private Gateway in the AWS Virtual Private Cloud (VPC) and on an on-premise Aruba wireless mobility controller (WLC).

Figure 1. Example Topology for ClearPass on AWS



Before You Begin

The example topology for deploying ClearPass on AWS has the following requirements:

- A customer gateway: a configuration object in the virtual private cloud (VPC) that defines the on-premises VPN termination point.

- A virtual private gateway: a virtual router object that acts as the VPN termination point in the VPC.
- A VPN connection: a configuration object in the VPC that defines the VPN settings to be used when communicating with the Customer Gateway.
- A Site-to-Site IPSec map: a configuration object in the Aruba Mobility Controller that defines the customer gateway in the VPC as well as the VPN settings to be used when communicating with the VPC.
- Define your interfaces with two different subnets in the same availability zone in your VPC. ClearPass requires its eth0 and eth1 interfaces to be connected to different subnets.

  Before you launch an instance of ClearPass using Amazon Web Services (AWS), you must perform the following tasks using the AWS management console.

Subtopics

Create a Customer Gateway on AWS
Add a Virtual Private Gateway to a VPC
Create a VPN Connection on AWS
Configure the Site-To-Site VPN Connection on the Gateway

Create a Customer Gateway on AWS

About this task

To create a customer gateway:

Procedure

1. Log in to the AWS console, and open the **VPC** dashboard.
2. Navigate to the **VPN Connections** section and select **Customer Gateways**.
3. Click **Create Customer Gateway**.

Figure 1. Create a Customer Gateway



4. Enter a name and a static public IP for the Customer VPN gateway.
5. Click **Create Customer Gateway.**

Add a Virtual Private Gateway to a VPC

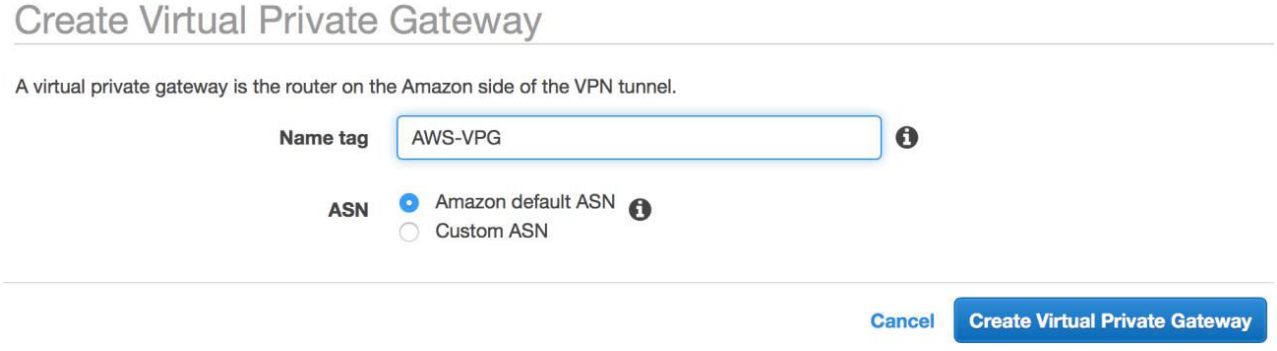
About this task

Use the following procedure to create a Virtual Private Gateway and attach it to a Virtual Private Cloud (VPC).

Procedure

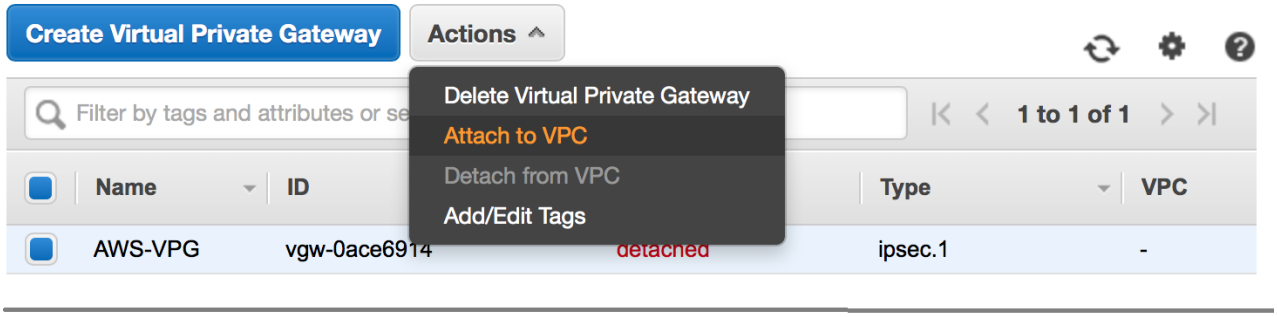
1. in the **VPC** dashboard on the AWS console, navigate to the **VPN Connections** section and select **Virtual Private Gateway.**
2. Click **Create Virtual Private Gateway**.

Figure 1. Create a Virtual Private Gateway



3. Enter a name for the gateway in the Name tag field.
4. Click the **Create Virtual Private Gateway** button.

Figure 2. Steps to Attach a Virtual Private Gateway to a VPC





Next, you must attach the Virtual Private Gateway to the VPC.

5. Select the virtual private gateway that you want to attach.
6. Click **Actions** and select **Attach to VPC**. The **Attach to VPC** window opens.
7. Click the **VPC** drop-down menu and select a VPC.
8. Click **Yes, Attach**.

Create a VPN Connection on AWS

About this task

To add a secure VPN connection between the Customer Gateway and the Virtual Private Gateway:

Procedure

1. in the **VPC** dashboard on the AWS console, navigate to the **VPN Connections** section and select **VPN Connections**.

2. Click **Create VPN Connection**.
3. Enter a name in the **Name tag** field.
4. Click the **Virtual Private Gateway** drop down menu and select a virtual private gateway from the list of gateways.
   - To add the customer gateway you defined in the previous steps, select the **Existing** radio button and select the customer gateway ID from the **Customer Gateway ID** dropdown menu.
   - To define and use a new customer gateway, select the **New** radio button and enter the static IP of the customer gateway.
5. In the **Routing Options** field, select **Dynamic** or **Static**. If you selected the **Static** optionthen fill in the IP static routes by adding IP prefixes and subnets, for example, 192.168.10.0/24.
6. Click **Create VPN Connection**.

Configure the Site-To-Site VPN Connection on the Gateway

About this task

The procedure to configure a Site-To-Site VPN Connection on the Aruba controller or Mobility Master varies slightly, depending upon the version of ArubaOS currently running on that device.

Procedure

1. On an Aruba controller running ArubaOS 6.x, navigate to **Configuration > Advanced Services > VPN Services**.
2. On a Mobility Master or standalone controller running ArubaOS 8.x, navigate to **Configuration > Services > VPN**.
3. Select **Site-to-Site**.
4. in the **IPsec Maps** section click **Add** (for ArubaOS 6.x) or **+** (For ArubaOS 8.x.)
5. Next, return to **VPC dashboard** in the AWS console.
6. Navigate to the **VPN Connections** section and select **VPN Connections**
7. Select the VPN connection that you want to configure on the Aruba controller.

8. Click **Download Configuration** and select the following options:
    - Vendor = Generic
    - Platform = Generic
    - Software = Vendor Agnostic
9. Click **Download** and open the downloaded file as a text file.The file contains configuration details such as pre-shared key, transform set type etc. which have to be entered on the site-to-site VPN page of the Aruba controller or Mobility Master. Also, if required, the specific transform set has to be created on the controller.
0. Save you controller or Mobility Master settings.
    - On an Aruba controller running ArubaOS 6.x, click **Done**.
    - On Mobility Master or a standalone controller runnnig ArubaOS 8.x, click **Submit**, click **Pending Changes**, then in the **Pending changes** window, click **Deploy Changes**.

Launching ClearPass on AWS from an AMI

The following section describes the procedure to launch an instance of ClearPass from an Amazon Machine Image (AMI) using the Amazon Web Services (AWS) console. These steps must be performed in the order shown in this document.

Subtopics

Obtain the ClearPass AMI
Choose an Instance Type
Configure Instance Details
Add Storage
Add Tags
Configure a Security Group
Review Instance Launch

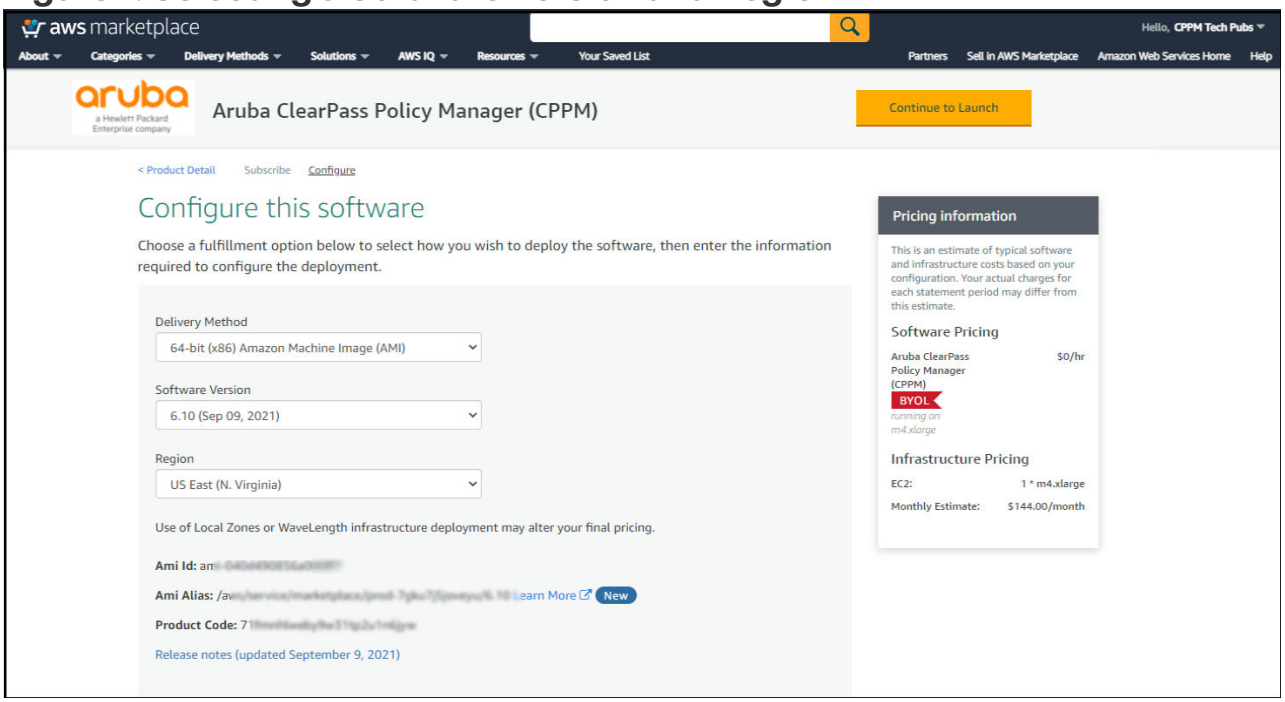
Obtain the ClearPass AMI

Procedure

1. Log in to the AWS console, and navigate to the **My Account** page.

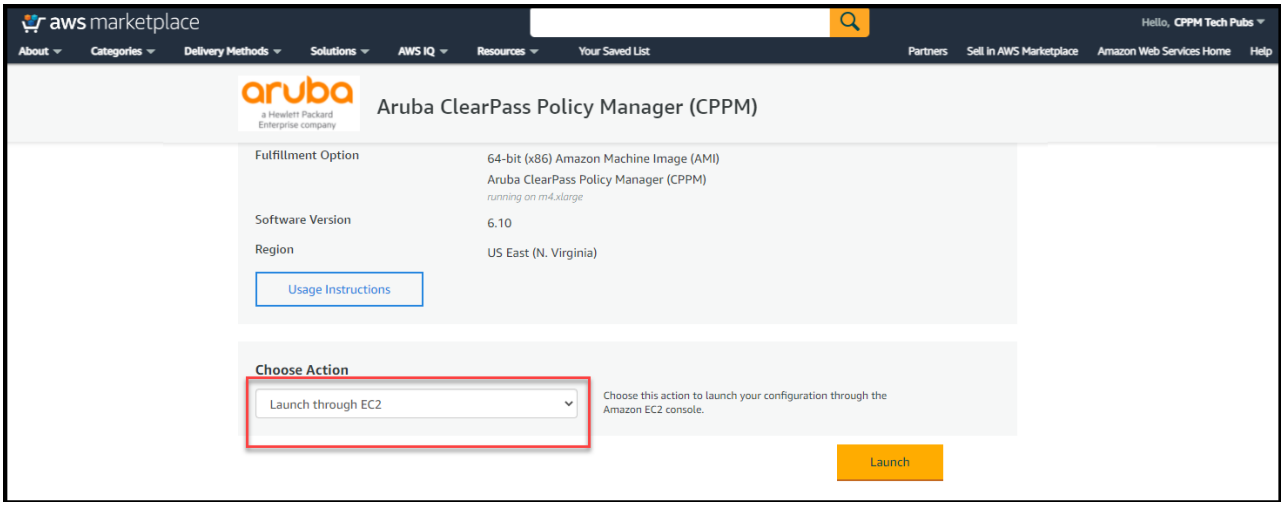
2. Make a note of your AWS account ID in the **Account Settings** field at the top of the page.
3. Return to the AWS console home page and select the region drop-down list in the upper toolbar to identify your AWS region. The region drop down list appears in the upper toolbar, by your account name.
4. Navigate to the URL **https://aws.amazon.com/marketplace/pp/prodview-mra4typhsk3zm?sr=0-1&ref_=beagle&applicationId=AWSMPContessa** to view the ClearPass AMI overview and pricing details per region.
5. Click **Continue to Subscribe**.
6. Review the Terms and conditions, then click **Continue to Configuration**.

Figure 1. Selecting a Software Version and Region



7. In the **Software Version** menu, select a version of ClearPass software.
8. Click the **Region** menu and select the region where you want to launch this instance.
9. Make a note of the AMI ID, AMI alias and Product code for the version of software you selected.
0. Click **Continue to Launch**.

Figure 2. Launch AMI Through EC2 Console



1. In the **Choose Action menu**, select **Launch through EC2**.
2. Click **Launch** to configure and launch your AMI in the EC2 Console.

Choose an Instance Type

About this task

The next step of the launch process requires you to select the appropriate instance type for your ClearPass model.

NOTE

ClearPass 6.11.x and later supports either M5, M7i, or M7a instances. M4 instance types are not supported.

NOTE

This section uses the M5 instance types as an example. See the Amazon Web Services (AWS) Requirements section for details.

NOTE

Users should be aware that an existing M5 instance type cannot be directly upgraded to an M7 instance type, and additional steps are necessary. See the Upgrading an M5 Instance Type to an M7 Instance Type section for details.

Procedure

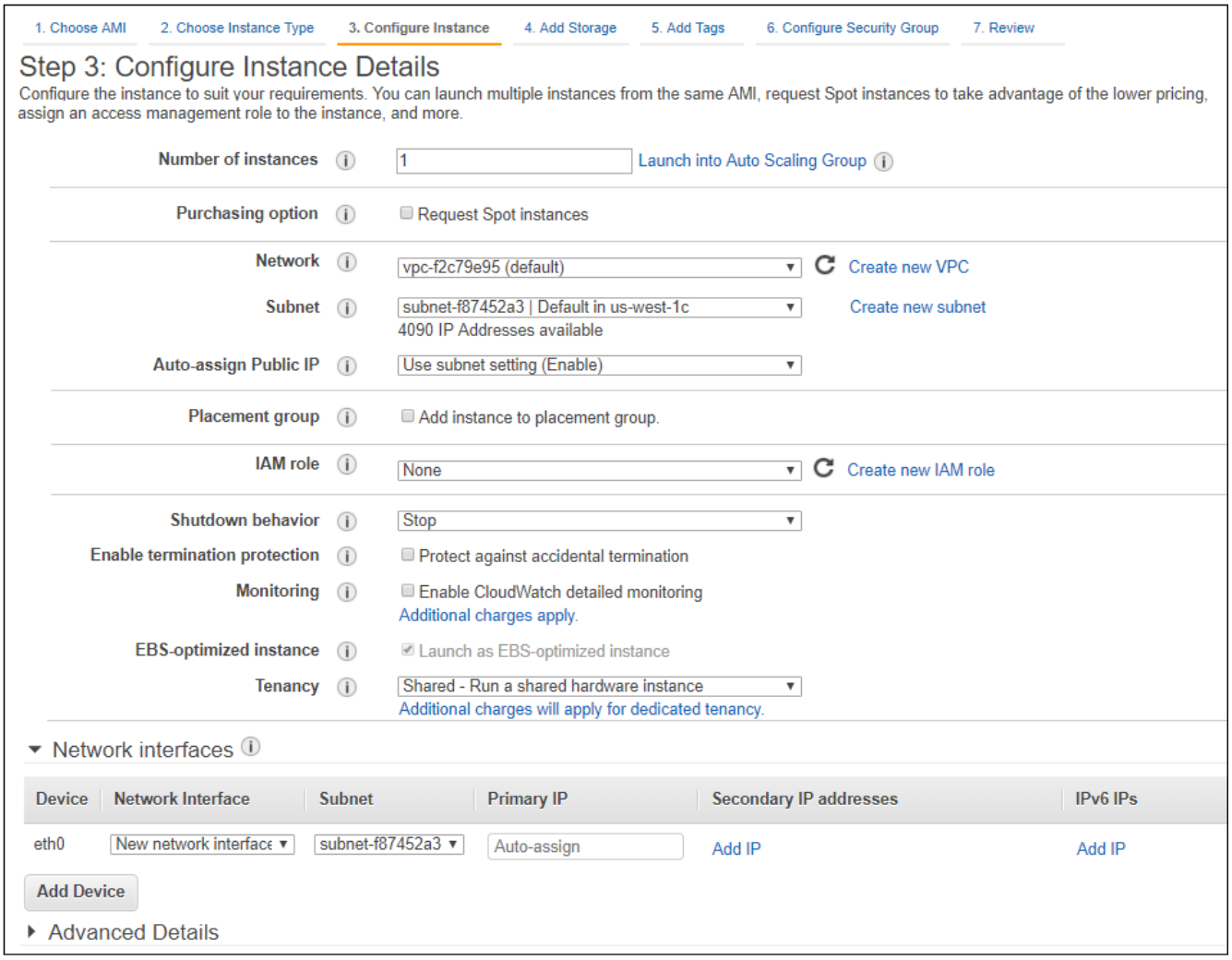
1. In the **Application and OS Images (Amazon Machine Image)** area, click in the **Search** field to display the drop-down list of AMI types and scroll to the M5 and M5a options. Select the appropriate instance type. Using M5 as an example:
   - For C1000V, select **m5.large**. The recommended AWS instance size is 4CPU and 8GB RAM.
   - For C2000V, select **m5.xlarge**. The recommended AWS instance size is 4CPU and 16GB RAM.
   - For C3000V, select **m5.4xlarge**. The recommended AWS instance size is 16CPU and 64GB RAM.
2. Click **Next: Configure Instance Details**.

Configure Instance Details

About this task

Configure the following settings in the **Configure Instance Details** window.

Figure 1. Configure Instance Details



Procedure

1. In the **Number of Instances** field, enter **1**.
2. Click the **Network** drop-down menu and select the name of your VPC network, for example, **vpc-f90dac9d**.

3. Click the **Subnet** drop-down menu, select the network subnet, for example, **us-west-2a**.

> **NOTE**
>
> All cluster members must meet the same intra-node latency requirements required for regular clusters to ensure that the cluster communication is maintained. Clusters in both AWS and on-premise data centers are not supported. If your deployment uses authorization sources such as Active Directory, you can prevent performance issues by ensuring that the authorization source is also located in the same AWS region.

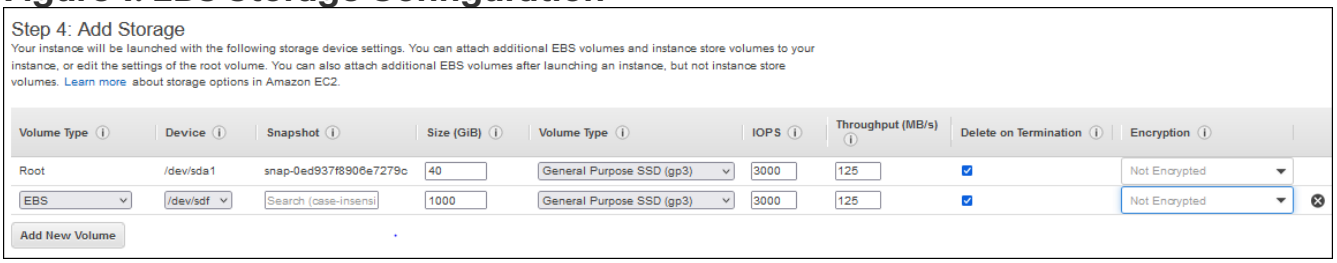
4. Click the **Auto-assign Public IP** drop-down list and select **Enable**.
5. Click **Next: Add storage**.

Add Storage

About this task

When configuring Elastic Block Store (EBS) storage device settings, do not modify the size or snapshot ID of the **Root** volume (/dev/sda1).

Figure 1. EBS Storage Configuration



Procedure

1. In the **Root** volume's row, select the **General Purpose SSD (GP3)** option.
2. Click on **Add New Volume** and enter the following values:
   a. In the **Volume** type field, enter **EBS**.
   b. Click the **Device** drop-down menu and select **/dev/sdf**. This name is required; the additional volume must be named **/dev/sdf**.

c. Leave the **Snapshot** field empty.

d. In the **Size** field, enter a minimum of **1000** (GB) for C1000V and C2000V models, and a minimum of **1800** (GB) for C3000V models.

e. In the second **Volume Type** field, select the **General Purpose SSD (GP3)**option**.**

3. Click **Next: Add Tags**.

Add Tags

About this task

Note that tags are not required; this step is optional.

Procedure

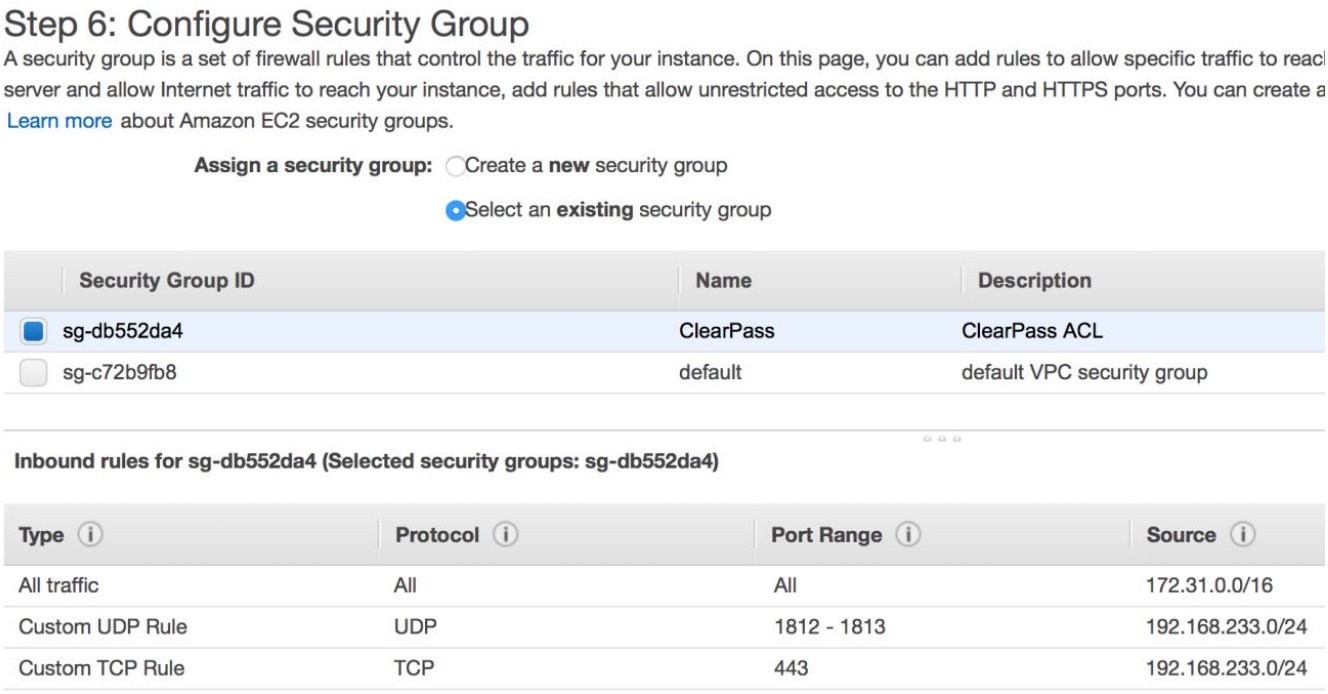
1. To add a tag, click **Add Tag**.
2. In the **Key** field, enter **Name**.
3. In the **Value** field, enter the name of your ClearPass instance.
4. If you want to add additional tags, click Add another tag and configure other tags as needed.
5. Click **Next: Configure Security Group**.

Configure a Security Group

About this task

A security group is a set of firewall rules that allow or restrict specific traffic types to your instance. In this next configuration step, you must select an existing security group or a create a new security group. For additional instructions on creating a security group outside of the procedure to create and launch a ClearPass instance from an AMI, see also Post Launch Configuration Tasks in AWS.

Figure 1. Security Group Configuration



To create a new security group:

Procedure

1. Click **Create a new security group.**
2. Enter a security group name and a description of the group.
3. Click **Add Rule** to add new rules to your security group.
4. Click **Review and Launch**.

   To select an existing security group:

5. Click **Select an existing security group.**
6. Select a security group.

7. Click **Review and Launch**.

> **NOTE**
>
> ClearPass will not use this key while accessing the EC2 via SSH. You must enter a username and password for SSH. The Aruba support team can access ClearPass via an Aruba support account for debugging.
>
> The administrator password for the ClearPass user interface is not changed when the password is configured during the initial configuration but instead retains the default value. This issue occurs during onboot because not all services are available yet when the HTTP request for the password update is sent. After AWS is deployed, wait 15 minutes before submitting the password change.

Review Instance Launch

About this task

Use the review window to review all your configurations. You can return to a previous window to make any changes, or click **Launch** when the instance is configured correctly. After you click **Launch**, a pop-up window prompts you to select or create a new key-pair:

Procedure

1. Select a key pair option.
   - If you already have a key pair, then select **Choose an existing key pair** from the drop-down menu and select a key pair.
   - If you want to create a new key pair, select **Create a new key pair** from the drop-down menu, enter a name for your key pair then click **Download Key Pair**.
   - If you do not want a key pair, then select **Proceed without a key pair**.
2. Click the **I acknowledge** check box.
3. Click **Launch Instances**.

Once the instance gets launched it takes around 30 minutes for all the services to be up and running. In the meantime, you can go to the EC2 Dashboard and verify that your instance has been created and that the instance configuration is correct.

Use EC2 serial console / SSH to check that all the services are up and running (Ingress services will be in stopped state by default).

After the instance is launched and configured, the SSH Public Key can be managed in Policy Manager on the **Administration > Server Manager > Server Configuration > Network** tab for the instance. The initial appadmin and admin passwords for AWS are configured with an eth0 MAC address.

Post-Launch Configuration Tasks in AWS

The following section describes the tasks that can be performed after you launch your ClearPass instance.



NOTE

The procedure to create and attach a network interface or create and attach an EBS storage volume can be performed either as part of the launch setup, or after the ClearPass instance has already been launched.

Subtopics

Create and Attach an Elastic IP
Create and Attach an Elastic Block Store (EBS) Volume
Create a Security Group
Define and Configure Network Interfaces
Attach a Network Interface to an Instance

Create and Attach an Elastic IP

About this task

An elastic IP address is a public IPv4 address, which is reachable from the Internet. If you wish to assign a public IP address to port eth0, use the following procedure to create an elastic IP address and attach it to a network interface.

To create an elastic IP address:

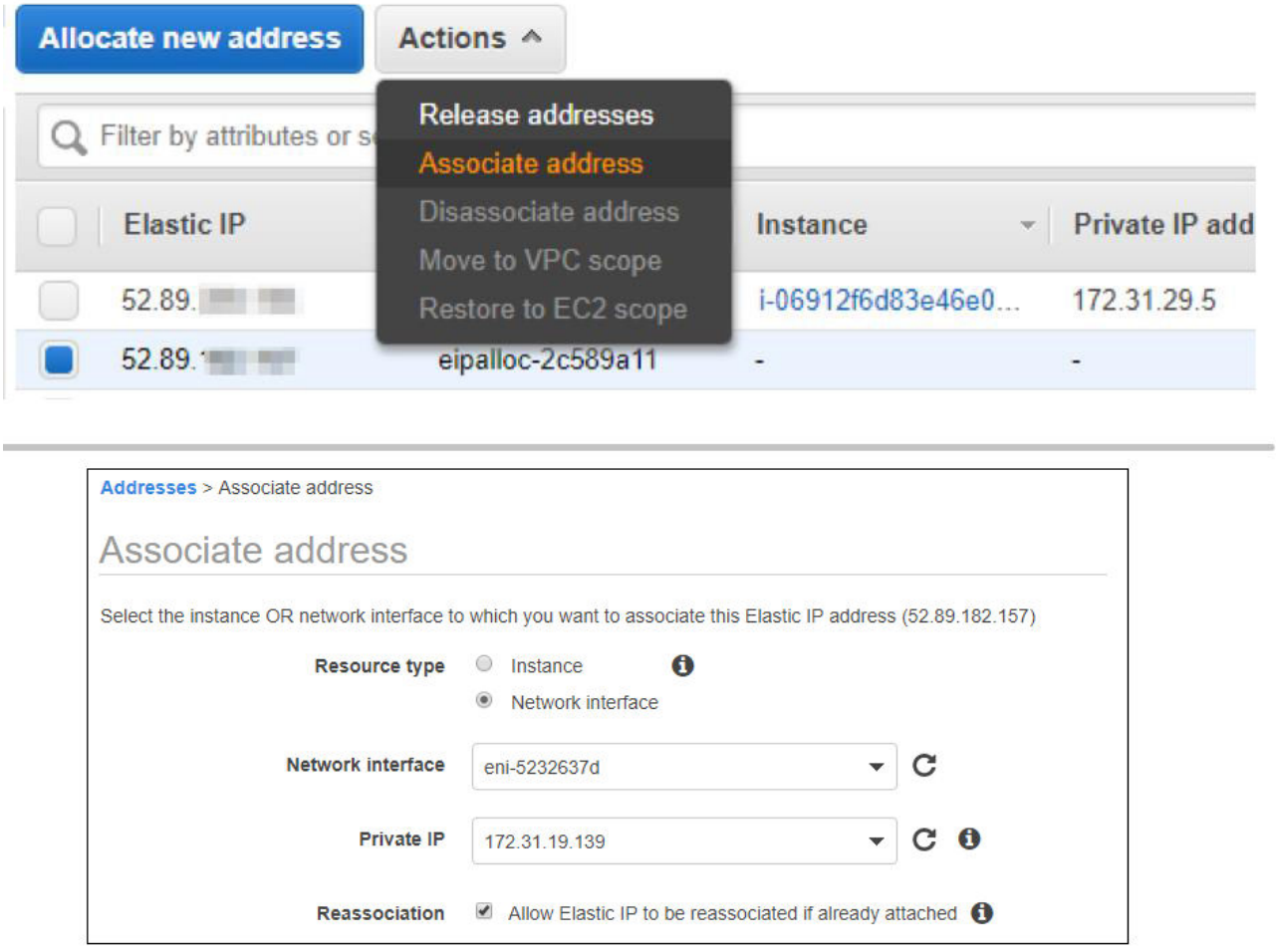
Procedure

1. Log in to AWS and navigate to the **EC2** dashboard
2. Navigate to **Network and Security > Elastic IPs**.
3. Click **Allocate New Address.** The **Allocate New Address** Window opens.
4. Click **Allocate**. The window will display the new Elastic IP address.
5. Click **Close**.

   Next, associate the new elastic IP address to a network interface.

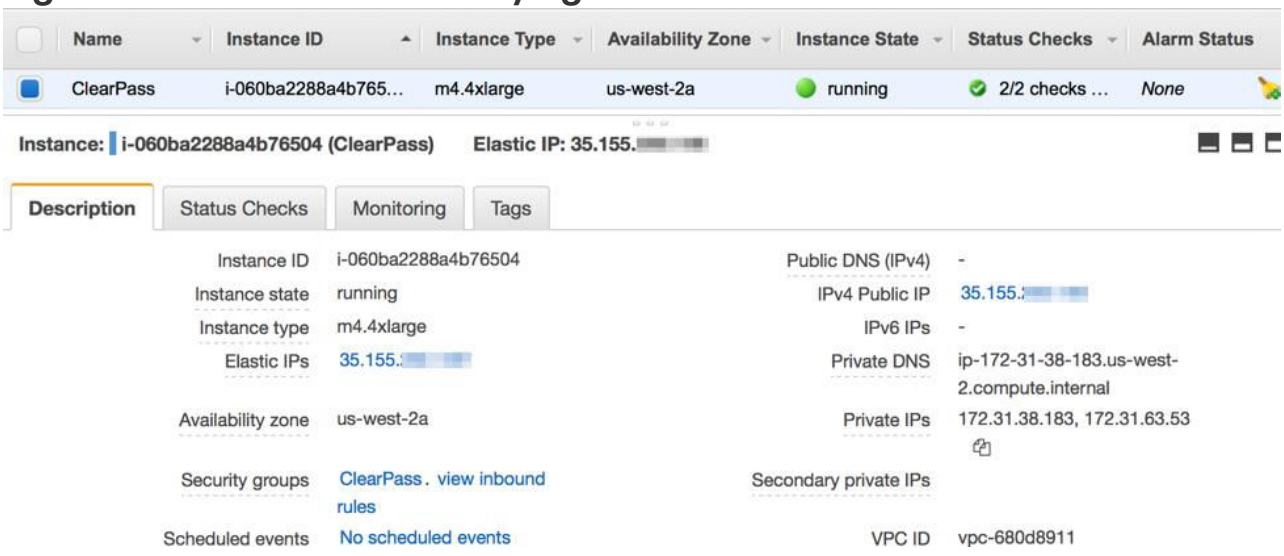
6. Log in to AWS and navigate to the **EC2** dashboard
7. Navigate to **Network and Security > Elastic IPs**.
8. Select the elastic IP you want to attach.
9. Select **Actions** and click **Associate address**. The **Associate Address** window opens.

Figure 1. Steps to Associate an Elastic IP address to an Interface



0. In the **Associate Address** window, configure the following settings
   - Set the **Resource type** option to **Network interface.**
   - Click the **Network Interface** drop down menu and select the Network interface ID of the eth0 network interface of your ClearPass instance.
   - Click the **Private IP** drop-down menu and select the private IP associated to the Network Interface ID.
1. Click **Associate**.
2. (Optional) After assigning an elastic IP, you may want to check the EC2 dashboard to verify that the IP address is reflected in the dashboard.

Figure 2. Instance Status Verifying the Public IP Address



Create and Attach an Elastic Block Store (EBS) Volume

About this task

If you did not attach an EBS volume while you were creating the instance, you can create and attach one using the following procedure.

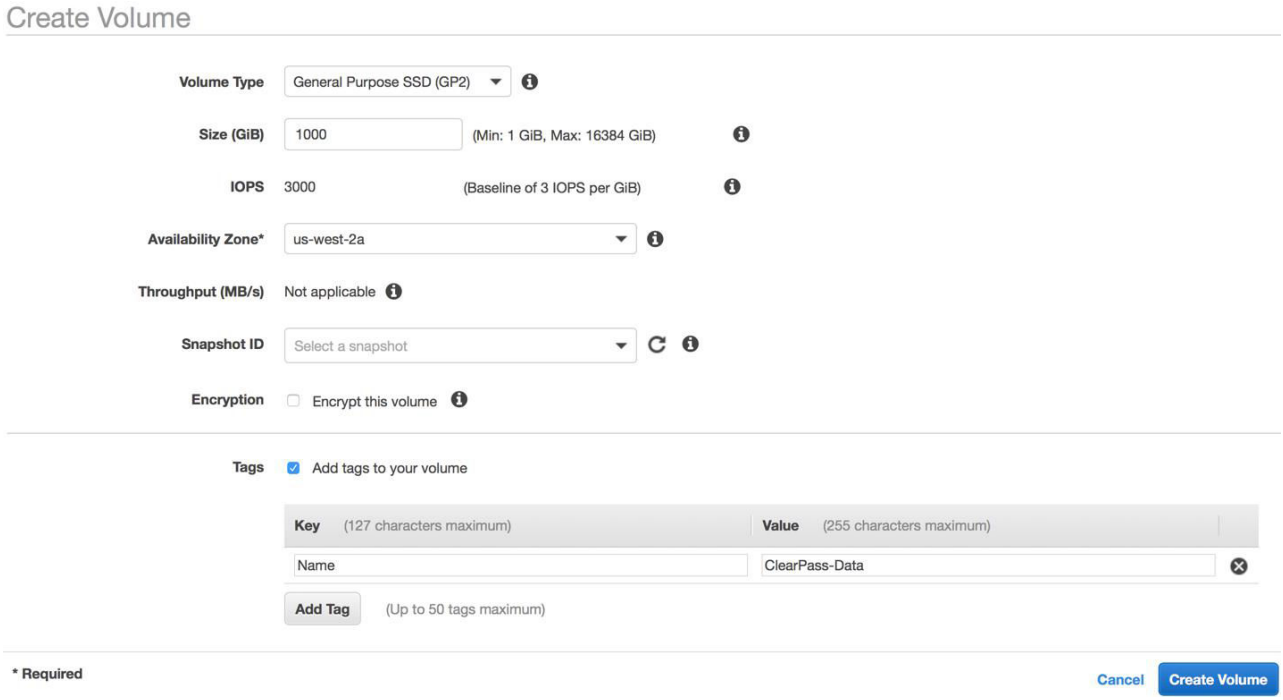
To create an EBS volume:

Procedure

1. Log in to the AWS console and navigate to the **EC2** dashboard.
2. Navigate to **Elastic Block Store > Volumes**.

3. Click **Create Volume**. The **Create Volume** window opens.

Figure 1. EBS Storage Configuration



4. Enter the following values:

   a. In the **Volume** type field, enter **General Purpose SSD**

   b. In the **Size** field, enter a minimum of **1000** GB for CP-VA-SMALL and CP-VA-MEDIUM models, and a minimum of **1800** GB for CP-VA-LARGE models.

   c. Click the **Availability Zone** drop-down list and select the zone of your instance.

5. Click the **Create Additional Tags** option.

6. In the **Key** field, enter **Name**.

7. In the **Value** field, enter the name of your EBS volume.

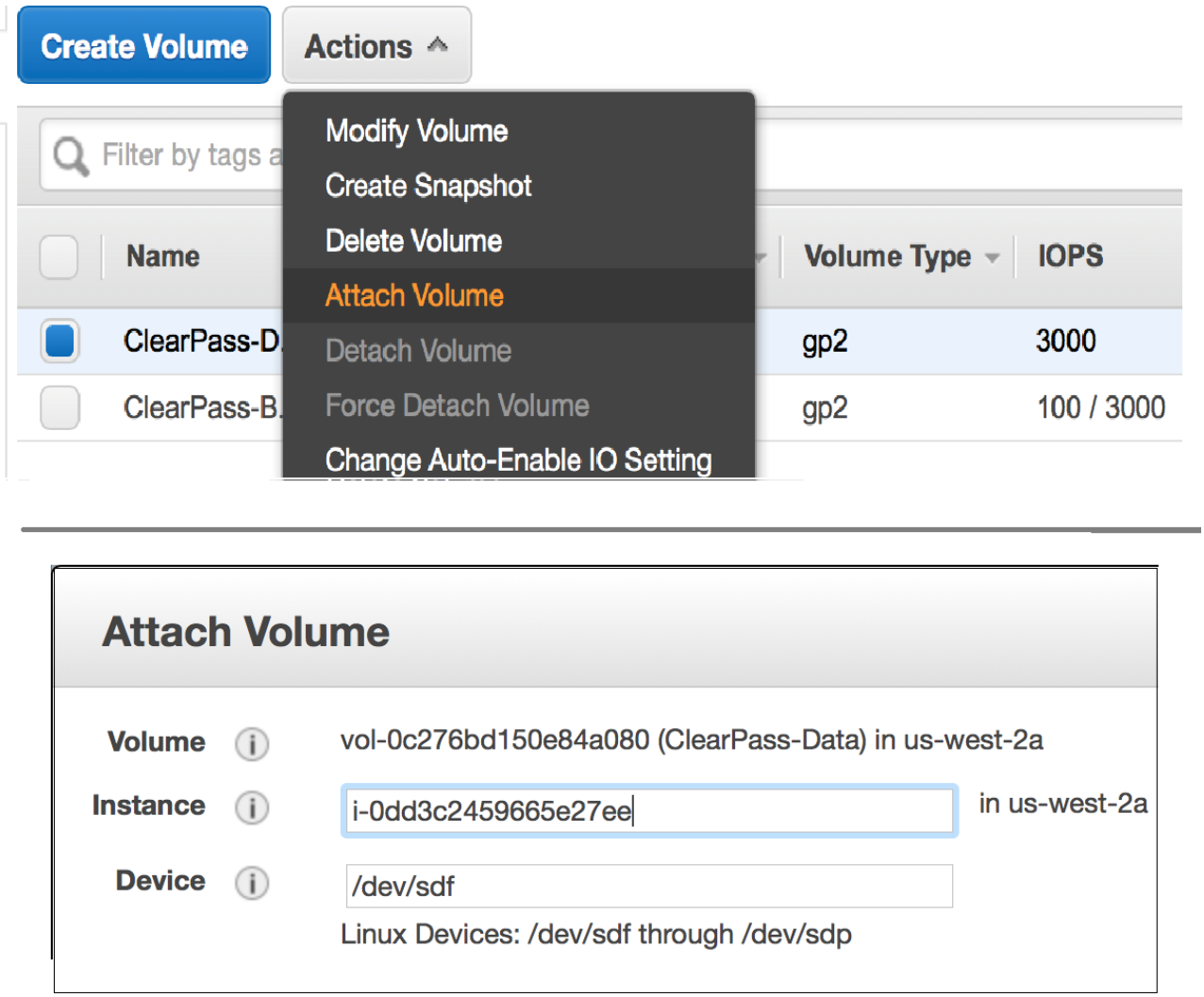
8. Click the **Create Volume** button.

   Next, you must attach the EBS volume to an instance.

9. Select the EBS volume that you wish to attach.

0. Click the **Actions** drop-down menu and select **Attach Volume.** The **Attach Volume** window opens.

Figure 2. Steps to Attach an EBS Volume to an Instance



1. In the **Instance** field, enter the instance ID of the instance to which you want to attach the EBS volume.
2. In the **Device** field, enter **/dev/sdf** . This name is mandatory; the additional volume must be named **/dev/sdf**.
3. Click **Attach.**

Create a Security Group

About this task

You will be prompted to select an existing security group or to create a new security group in the procedure to launch the ClearPass AMI. However, you can also use the procedure described below to create an additional security group outside of this launch workflow.

Procedure

1. Log in to the AWS console and navigate to the **EC2** dashboard.
2. Navigate to **Network and Security > Security Groups**.
3. Click **Create Security Groups**.
4. Enter a security group name and description.
5. Select a valid VPC to which the security group will be associated.
6. Click **Add Rule** to add new inbound and outbound rules to your security group.

   The following images show examples of inbound and outbound rules on a security group.

Figure 1. Security Group Outbound Rules

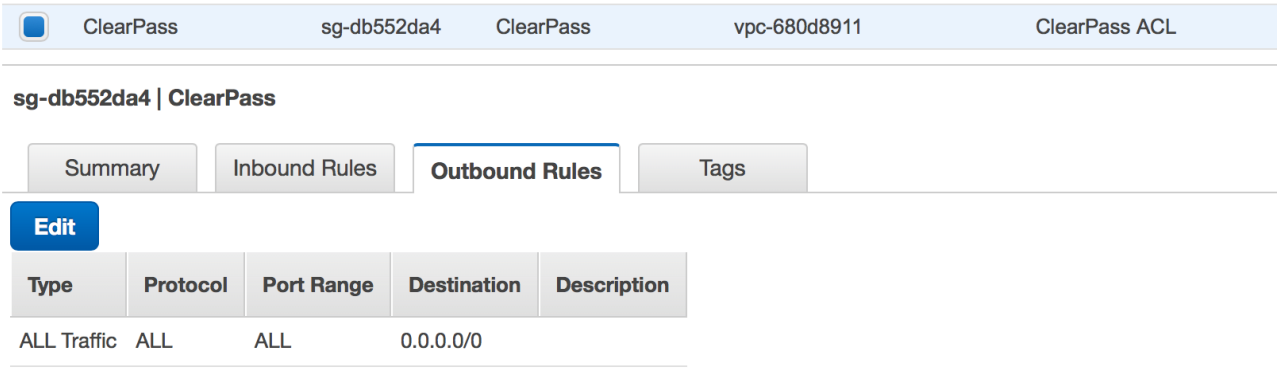
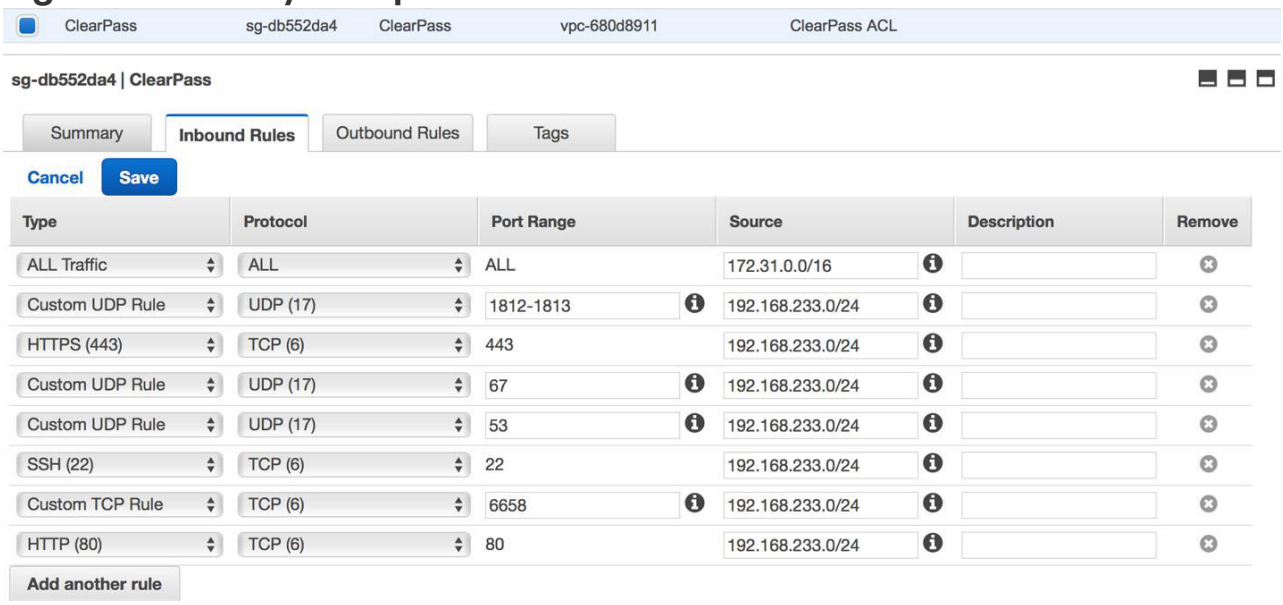


Figure 2. Security Group Inbound Rules



Define and Configure Network Interfaces

About this task

AWS doesn't assign a public IP if you add network interfaces. If you wish to attach a public IP to your interfaces, you can attach an elastic IP after the instance is launched. For more information, refer to Create and Attach an Elastic IP

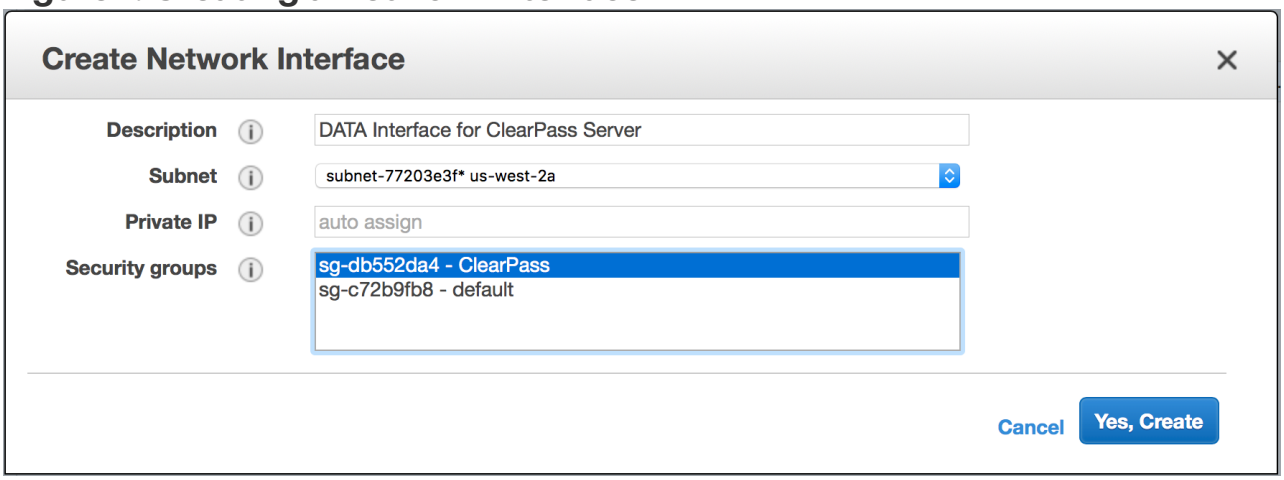
> **CAUTION**
> ClearPass does not support a data port when deployed on AWS. It does not use a configured data port to route any traffic, and all traffic is routed via the management port.

To define a network interface for your region:

Procedure

1. In the AWS EC2 dashboard, navigate to **Network & Security > Network Interfaces**.
2. Click **Create Network Interface**. The **Create Network Interface** window opens.

Figure 1. Creating a Network Interface



3. Enter a description in the **Description** field.
4. Click the **Subnet** drop-down menu and select a subnet.
5. In the **Private IP** field, keep the default **auto assign** setting.
6. Click the **Security Group** drop-down list and select a security group.
7. Click **Yes, Create**.

Attach a Network Interface to an Instance

About this task

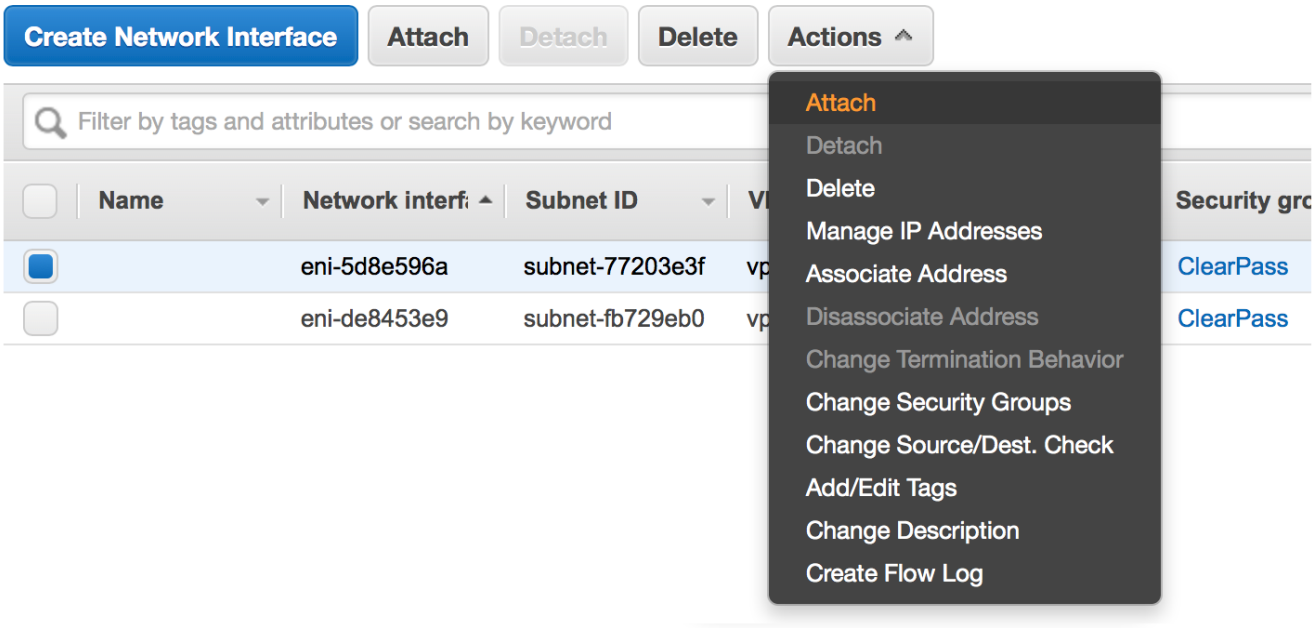
To attach this new network interface to an instance:

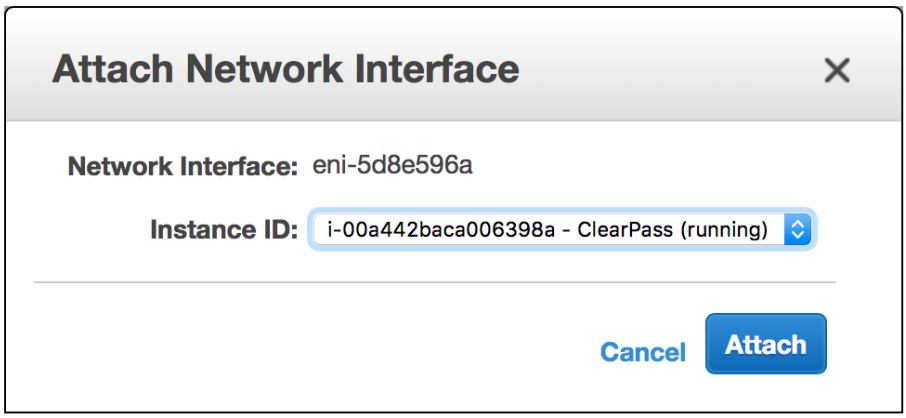
Procedure

Select the network interface that you want to attach to an instance.

2. Click on **Actions** and select **Attach**.

Figure 1. Steps to attach a Network Interface to an Instance





3. Select your instance ID from the dropdown menu.

4. Click **Attach**.

Configuring ClearPass

About this task

After the instance is launched, you must perform some additional configuration steps in ClearPass.

Procedure

1. Log in to ClearPass using the appadmin account but with the eth0 MAC address as the default password.

   > **NOTE**
   >
   > Using the eth0 MAC address as the default password for the admin and appadmin accounts during the initial launch only applies to AWS deployments. All other deployment types continue to use the standard appadmin default account credentials.

2. You are prompted by the command-line interface to enter configuration details. As part of the configuration you will be asked to set a new password.

   > **NOTE**
   >
   > The command-line interface in the AMI version of ClearPass differs slightly from the ClearPass interface on a VA or standalone appliance. For more information on these changes, refer to ClearPass Policy Manager CLI Changes.

3. After the initial configuration is complete, log in to ClearPass using the appadmin account with the new password you configured.
4. Check that all the services are up and only then proceed to the next step.

5. Run the **system morph-vm** command to upgrade your ClearPass instance to either C1000V, C2000V or C3000V model. You will need to have attached an EBS volume with necessary configuration before running **system morph-vm**.

> **CAUTION**
>
> The **system morph-vm** command should be used only once for each instance of a cloud deployment. If this command is used multiple times in the same cloud instance, it can cause disk corruption.

> **NOTE**
>
> The ClearPass CLI may display a warning when running the system morph-vm command indicating that the current active system configuration does not meet the recommended system configuration. Please ignore this warning.

> **NOTE**
>
> Starting with the 6.11.0 release, after the **system morph-vm** command is run during a ClearPass upgrade and the new disk is added, the existing hard disk cannot be removed. This is because the existing disk is used to extend the disk space for the server, and is included in the calculations to determine the required capacity of the additional hard disk.

> **CAUTION**
>
> In an AWS instance with Luks enabled, the **system update-luks-key command** works correctly the first time it is used while morphing the instance; however, if the command is used again after a reboot the instance stops working with the error message "Cannot open access to console, the root account is locked," and the instance cannot be recovered. This issue only occurs in ClearPass 6.11. It is not an issue in earlier ClearPass versions. It will be fixed in a later version. Until then, do not use the system update-luks-key command.

Elastic Block Store (EBS) Requirements

If no EBS volume is attached to your EC2 instance, you get an EBS volume warning message while you are trying connect to ClearPass using SSH, and the ClearPass GUI will display an EC2 EBS warning event on the **Event Viewer** tab in the ClearPass GUI. ClearPass will also display a warning message if an EBS volume is attached but the **system morph-vm** command has not yet been issued.

Figure 1. No EBS Volume Warning Details

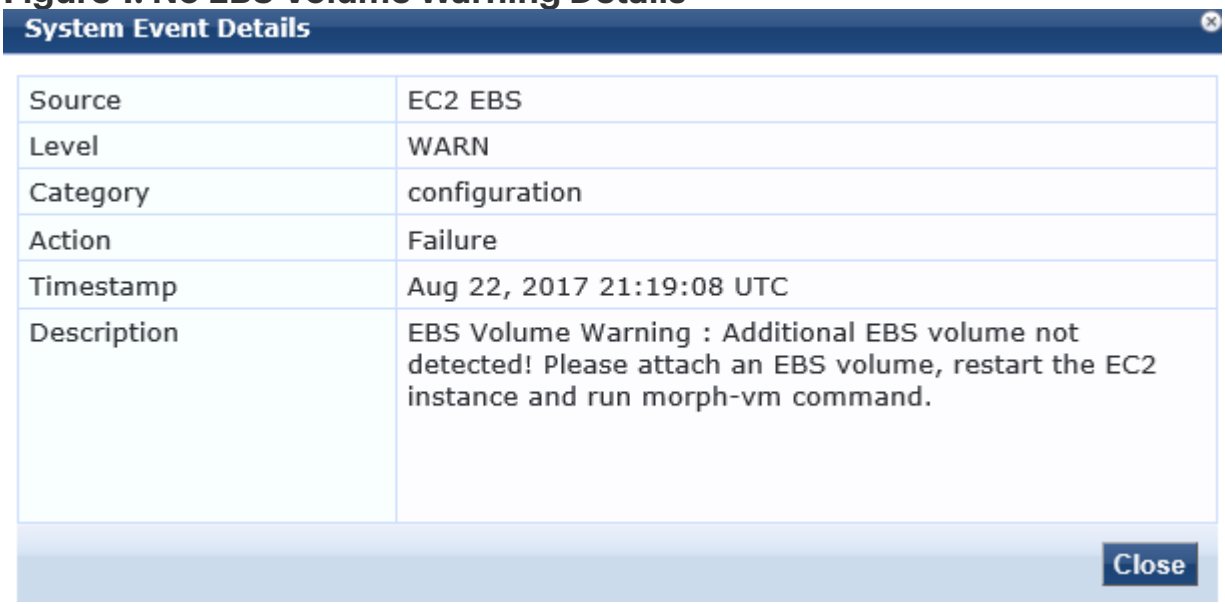
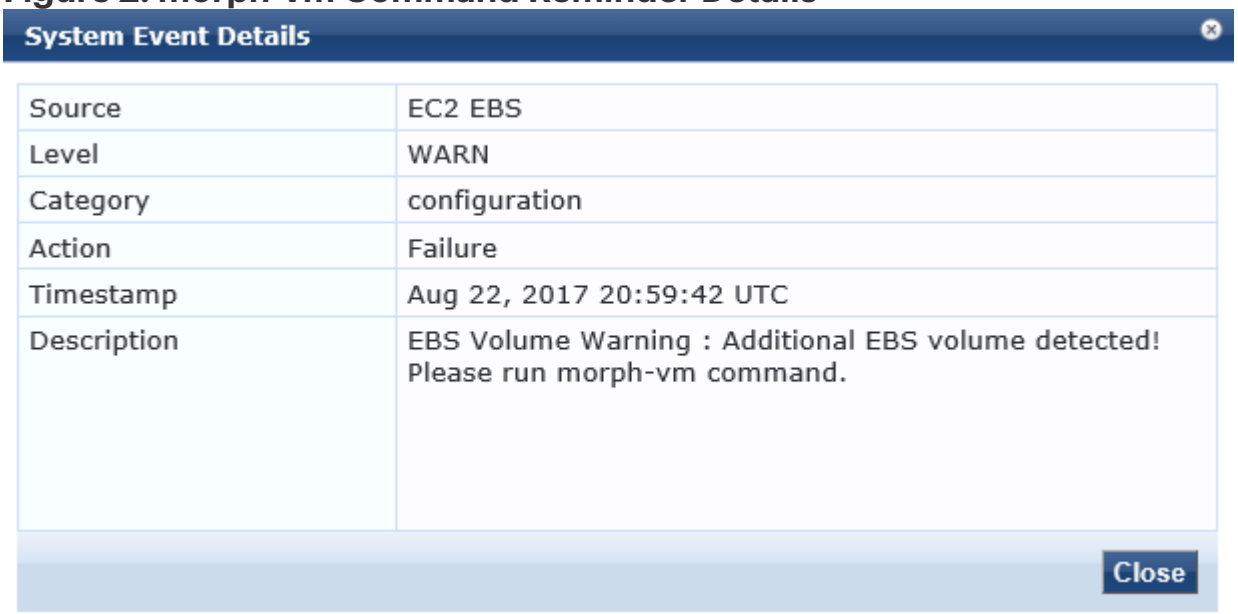


Figure 2. morph-vm Command Reminder Details



ClearPass Policy Manager CLI Changes

AWS assigns its own private IP, gateway, and netmask to all the network interfaces. AWS also creates its own routing table, so there is no need to configure the routing tables. Accordingly, the following CLI changes have been made:

Initial Configuration

The user will not be prompted to enter the IP address, gateway, and netmask for the management and data ports. These options are also disabled in the ClearPass Policy Manager user interface.

FIPS mode is not supported in the initial release. The CLI does not display an option to enable/disable FIPS mode.

Configure Commands

The **ip**, **ipv6** , **and mtu** configuration options are removed. FIPS mode is not supported in the initial relase, so the CLI does not display an option to enable/disable FIPS mode.

System Commands

The **factory reset** option is removed.

Upgrading an M5 Instance Type to an M7 Instance Type

About this task

Users should be aware that an AWS M5 instance cannot be upgraded directly to an M7 instance. To move from an existing M5 instance to an M7 instance:

Procedure

1. First take a backup of the M5 instance.
2. On the appropriate M7 instance type (M7i or M7a), perform a fresh ClearPass 6.12.0 installation.
3. Do the bootstrap configuration steps and update the Platform License.
4. At **Administration > Agents and Software Updates > Software Updates**, download the "ClearPass 6.12.0 Hotfix Patch for system morph-vm on M7i instances" file and install it.
5. Execute the **system morph-vm** appadmin CLI command to deploy the appropriate virtual appliance size (for example, C1000V, C2000V, C3000V) based on the instance type.
6. While the **system morph-vm** command is executing, the following option is displayed:

Do you wish to encrypt all local data? (Y/N)

> **NOTE**
>
> *Note: Yes (Y) is recommended unless virtual system encryption is already enabled.*

This setting cannot be changed after installation.

Press 'Y' or 'N' to proceed

7. To encrypt all local data, enter Y. After the **system morph-vm** command is fully executed, the server will automatically reboot.
8. Restore the backup on the M7 instance.
9. Repeat these steps for every M7 instance.
0. Form the cluster.
1. Update each instance to the latest 6.12.x release.

Cloud Deployments: Google Administration Console MDM

ClearPass supports the Google Administration Console as an MDM External Context Server. To use this feature, a project must first be created in the Google Developer Console, and then the Google Admin Console can be added as a context server in ClearPass Policy Manager.

When all configuration is complete in the Google Developer Console and in ClearPass Policy Manager, subsequent MDM polling cycles will fetch the MDM data for the Google Chrome Devices and add that to the endpoints. This provides profiling data to use with functionality in ClearPass, such as in creating and configuring policies and services.

Subtopics

Steps in the Google Developer Console
Steps in the ClearPass Policy Manager User Interface
Steps in the Google Developer Console
Steps in the ClearPass Policy Manager User Interface

Steps in the Google Developer Console

Procedure

1. Make sure to enable **Admin SDK API**.
2. Create a new **Client ID** and **Client Secret** (select **Web Application** as the **Application Type**).
3. Create a **Consent Screen** with the desired logo and text (This screen is seen by the ClearPass administrator when authorizing ClearPass Policy Manager - more below).
4. Add **Redirect URIs**. This is of the format `https://<clearpass-server>/async_netd/mdm/oauth/google`, where <clearpass-server> should be a fully qualified domain name (FQDN) and not an IP address. This server should be reachable by that FQDN by the device the administrator is using to access the ClearPass Admin UI over a Web browser.
5. Configure the domain's security setting to allow API access.

Steps in the ClearPass Policy Manager User Interface

Procedure

1. Go to **Administration > External Servers > Endpoint Context Servers > Add**. On the **Server** tab, select **Google Admin Console** in the **Select Server Type** drop-down list.
2. Enter the valid **Client ID** and **Client Secret** that were configured in the Google Developer Console.
3. Click **Save** to save the Client ID and Client Secret. This also enables the **Authorize ClearPass** button.
4. Click **Authorize ClearPass**. The Google page for entering the username and password for the Google domain (account) opens in a new tab or window.
5. Enter the credentials. A consent screen (the one that was set up in the Google Console steps) is displayed, where you will be given the choice to authorize ClearPass to communicate with the Google Admin Console to fetch the MDM data for the Google Chrome Devices registered with the domain.
6. After the approval, the status of the operation is displayed — either that a “Refresh Token has been fetched and saved”, or an error message.

7. You can return to the main ClearPass Admin UI window and make additional selections (such as enabling ClearPass to poll for MDM data) before you click **Save** to save the settings.

Steps in the Google Developer Console

Procedure

1. Make sure to enable **Admin SDK API**.
2. Create a new **Client ID** and **Client Secret** (select **Web Application** as the **Application Type**).
3. Create a **Consent Screen** with the desired logo and text (This screen is seen by the ClearPass administrator when authorizing ClearPass Policy Manager - more below).
4. Add **Redirect URIs**. This is of the format `https://<clearpass-server>/async_netd/mdm/oauth/google`, where <clearpass-server> should be a fully qualified domain name (FQDN) and not an IP address. This server should be reachable by that FQDN by the device the administrator is using to access the ClearPass Admin UI over a Web browser.
5. Configure the domain's security setting to allow API access.

Steps in the ClearPass Policy Manager User Interface

Procedure

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2. Enter the valid **Client ID** and **Client Secret** that were configured in the Google Developer Console.
3. Click **Save** to save the Client ID and Client Secret. This also enables the **Authorize ClearPass** button.
4. Click **Authorize ClearPass**. The Google page for entering the username and password for the Google domain (account) opens in a new tab or window.

5. Enter the credentials. A consent screen (the one that was set up in the Google Console steps) is displayed, where you will be given the choice to authorize ClearPass to communicate with the Google Admin Console to fetch the MDM data for the Google Chrome Devices registered with the domain.
6. After the approval, the status of the operation is displayed — either that a "Refresh Token has been fetched and saved", or an error message.
7. You can return to the main ClearPass Admin UI window and make additional selections (such as enabling ClearPass to poll for MDM data) before you click **Save** to save the settings.

Cloud Deployments: Microsoft Azure Cloud Service

This section describes the necessary steps to successfully deploy ClearPass as a virtual appliance (VA) on the Microsoft Azure cloud service.

Subtopics

Behavior Differences in an Azure Deployment
Configuration Workflows
Moving from a Test Environment to a Production Environment
Creating the Azure Custom Storage Account and Virtual Appliance
Upgrading a D3 Instance Type to a D5 Instance Type

Behavior Differences in an Azure Deployment

When ClearPass is hosted in Azure, users should be aware of the following differences:

- In the server configuration settings, editing is disabled for the management port IP address, management port subnet mask, and default gateway. SPAN port settings are also hidden.
- Using more than one cloud service provider in a single ClearPass cluster is not supported. Within each cluster, Azure instances can be combined (hybrid cluster) with virtual appliances and hardware appliances, but not with different cloud providers. To work with more than one cloud service provider, they must be in separate clusters.
- Azure virtual private cloud (VPC) deployments do not support virtual IP (VIP) addresses. The Policy Manager **Cluster-Wide Parameters** settings include the **Virtual**

IP tab and its **Failover Wait Time** field, even though the Virtual IP feature is not supported by ClearPass running on Azure. Although these fields are available and a configuration can be entered, the configuration will not take effect.

- The CLI commands **system factory-reset**, **system install-image**, and **system refresh-network** are not available in ClearPass running on Azure.
- Common Criteria (CC) mode is supported in ClearPass running on Azure.
- Starting with ClearPass 6.10, the **system morph-vm** command can be used to morph an instance of ClearPass on Azure to a larger or production virtual machine.
- Network IP addresses in an Azure instance are managed by Azure, not by ClearPass, and the primary interface is the single default gateway on the management port. The data port is not supported. If a user adds a new data port manually using the network IP routing CLI commands, it will not persist after a reboot. If your organization requires a different default gateway configuration, please contact the Aruba Technical Assistance Center (TAC).
- The footer of the Policy Manager WebUI includes the indicator **[Cloud]** after the version number.

- Starting with ClearPass 6.11, ClearPass on Azure is supported in Content Security Policy (CSP) mode.

> **CAUTION**
>
> While deploying ClearPass on Azure, before you execute the system morph-vm command you must attach a fresh additional disk as an extended volume. Do not attach any volume that was detached from an existing ClearPass instance, and do not re-use any data volume that was previously morphed. Attaching a previously used volume could cause the server to fail to boot after the morph operation.

> **CAUTION**
>
> Do not use the CLI to update an Azure 6.14.x instance to a higher 6.14.x version. Azure instances should only be updated through the Policy Manager user interface on either the **Software Updates** page or the **Cluster Update** page.

> **NOTE**
>
> ClearPass does not support auto-scaling (elastic scaling) in Azure. All appliances in the cluster are expected to be on and connected at all times in order to maintain the cluster. Load balancers may be used instead to balance traffic. For more information, see the "**Deploying CPPM with F5 BIG-IP Local Traffic Manager (LTM)**" Tech Note.

> **CAUTION**
>
> The **system morph-vm** command should be used only once for each instance of a cloud deployment. If this command is used multiple times in the same cloud instance, it can cause disk corruption.

Configuration Workflows

To begin, you must first create a storage account within the Azure portal. Next, you define your VA settings and then add an additional interface to the VA. When you are ready to move your VA from a test environment to a production environment, you will create and add a disk to the VA, back up your ClearPass data, optimize your ClearPass session timeout settings. and then move your data to the new VA.

Moving from a Test Environment to a Production Environment

When you are ready to move your virtual appliance (VA) from a test environment into production, you must create and attach a disk to your VA, back up your ClearPass database, and then use the ClearPass command **system morph-vm** to move your ClearPass data to the virtual appliance.



NOTE

The "ClearPass 6.11.1 Hotfix Patch for Azure Secondary Disk Attach Failures" hotfix patch must be installed on all Azure servers that will be in the cluster (publisher and subscribers) before running the system morph-vm <C1000V|C2000V|C3000V> command during installation, and before forming the cluster. It must be installed on all servers in the cluster (publisher and subscribers).

Subtopics

Adding a Disk to the ClearPass Virtual Appliance
Back Up your ClearPass Database
Change the Backup TImeout Setting
Launch the morph-vm Operation
Create a Backup of your Virtual Appliance

Adding a Disk to the ClearPass Virtual Appliance

About this task

The following procedure adds a data disk to your ClearPass virtual machine deployment. Use this procedure when you are ready to move your ClearPass virtual appliance out of a test environment and into regular production.

Procedure

1. Navigate to the Azure portal and select your ClearPass virtual appliance if it is not already selected.

2. Click the **Overview** menu link, and then select the **Stop** icon to stop the virtual appliance.
3. Click the **Disks** menu link to open the **Disks** page.
4. In the Data disks section, configure the disk settings described in Table 1.

Table 1. Define a Managed Disk

Field	Description
LUN	In the LUN (Logical Unit Number) field, retain the default value of 0.
Disk Name	Enter a unique name for this data disk.
Storage Type	Select Premium SSD as the storage type.
Size	Select the size, in GiB, based on your cost factors and deployment requirements. For v5 instance types: • Enter **1000** GB for a **Standard_D2ds_v5** VA that corresponds to a ClearPass C1000 appliance. • Enter **1000** GB for a **Standard_D4ds_v5** VA that corresponds to a ClearPass C2010 appliance. • Enter **1800** GB for a **Standard_D16ds_v5** VA that corresponds to a ClearPass C3010 appliance. For v3 instance types: • Enter **1000** GB for a **Standard_D2s_v3** VA that corresponds to a ClearPass C1000 appliance.

Field	Description
	• Enter **1000** GB for a **Standard_D4s_v3** VA that corresponds to a ClearPass C2010 appliance. • Enter **1800** GB for a **Standard_D16s_v3** VA that corresponds to a ClearPass C3010 appliance.
Encryption	Select the setting **Platform - managed key**.
Host caching	Select **Read/write**.

5. Click **Save**. The new disk is created and added to your system.
6. Click the **Overview** menu link and select **Start** to restart the VA.

Back Up your ClearPass Database

About this task

To back up the Policy Manager database:

Procedure

1. Access the Policy Manager user interface and navigate to the **Administration > Server Manager > Server Configuration** page.
2. Click the **Back Up** button. Backup files are in the gzipped tar format (tar.gz extension) and are automatically placed in the Shared Local Folder under folder type **Backup Files**.
3. Copy your backup to an external location.

Change the Backup TImeout Setting

About this task

The morph-vm operation may last for hours, depending on factors such as disk type, Input/Output Operations Per Second (IOPS), and the amount of data that needs to be transferred. To prevent an SSH timeout during this operation, best practice is to set the ClearPass session timeout to a higher value than the default setting. A timeout value of three hours is recommended.

Procedure

1. From the Policy Manager user interface, navigate to the **Administration > Server Manager > Server Configuration** page.
2. Select the **Cluster-Wide Parameters** link. The **Cluster-Wide Parameters** page opens to the **General** tab.
3. On the **General** tab, change the **Admin Session Idle Timeout** setting to **180** minutes.
4. Click **Save**.



NOTE

If uninterrupted connectivity to the VA instance can't be guaranteed, best practice is to use the Azure serial console on the **Support + Troubleshooting > Serial Console** page of the Azure portal to access the ClearPass CLI and launch the morph-vm operation.

Launch the morph-vm Operation

Access the ClearPass command-line interface and issue the command **system morph-vm <vm-model>**, where **<vm-model>** is one of the following virtual appliance options:

- **C1000V** for the virtual appliance equivalent of a C1000 appliance
- **C2000V** for the virtual appliance equivalent of a C2010 appliance
- **C3000V** for the virtual appliance equivalent of a C3010 appliance

When the command completes, the system will reboot and use the attached disk. This new data disk will host the ClearPass data.

> **CAUTION**
>
> The **system morph-vm** command should be used only once for each instance of a cloud deployment. If this command is used multiple times in the same cloud instance, it can cause disk corruption.

> **NOTE**
>
> Starting with the 6.11.0 release, after the **system morph-vm** command is run during a ClearPass upgrade and the new disk is added, the existing hard disk cannot be removed. This is because the existing disk is used to extend the disk space for the server, and is included in the calculations to determine the required capacity of the additional hard disk.

Create a Backup of your Virtual Appliance

Best practice is to create a backup of your ClearPass virtual appliance once it is in production mode. You can do this from the **Backup** menu on the Azure portal. For more information on managing virtual appliances in Azure, refer to the **Azure documentation** on the Microsoft documentation portal.

Creating the Azure Custom Storage Account and Virtual Appliance

About this task

Before you create your ClearPass virtual appliance (VA) , you must create a custom storage account within the Azure portal. The ClearPass VA will not function correctly using a default storage account.

Procedure

1. Log in to the **Azure portal**. If you do not have an account, you will be prompted to create one before you can log in.

2. From the Azure portal home page, select **More Services**.
3. in the **All Services** search field, enter **Storage Accounts**. The list of services will filter as you begin to type.
4. Select **Storage Accounts**.
5. Select **Create storage account**. The **Create Storage Account** window opens. Configure the storage account settings as defined in the table below, and then click **Review + Create**.

Table 1. Storage Account Settings

Field	Description
Subscription	Select the Azure subscription you will use to manage this storage account. ClearPass is available in the Microsoft Azure marketplace as a virtual appliance with the Bring Your Own License (BYOL) licensing model only. ClearPass is not supported on the Microsoft Azure Cloud Solution Provider (CSP) program or any other mode.
Resource group	Select an existing resource group to assign the storage account to that group, or click **Create New** to create a new resource group.
Storage account name	Give a unique name to the storage account.
Location	Set the location for the storage account.
Performance	Select **Standard**.
Account Kind	Retain the default **StorageV2 (general - purpose V2)** setting.
Replication	Retain the default **Read - access geo - redundant storage** setting.

Field	Description
Blob access tier	Retain the default **Hot** setting.

Subtopics

Create the ClearPass Virtual Appliance

Create the ClearPass Virtual Appliance

About this task

The following procedure describes the steps to launch the ClearPass VA in Azure.

Procedure

1. Log in to the Azure marketplace (**https://azuremarketplace.microsoft.com/en-us**) and search for **Aruba ClearPass Policy Manager (CPPM)** to locate the ClearPass Policy Manager virtual appliance offer.
2. Select the image.
3. Click **Create**. The **Create a virtual machine** page appears in the Azure portal with the **Basics** tab selected.

> **NOTE**
>
> When creating your virtual appliance, use the **Create** option only. Do not use the **Start with a pre-set configuration** option for creating a VA, as ClearPass does not support pre-set configurations.

Subtopics

Basics
Disks
Networking
Management
Advanced
Tags
Review + Create

Basics

On the **Basics** tab, configure the virtual appliance settings as described in Table 2.

Table 1. Configuring Basic Virtual Appliance Settings

Field	Description
Project details	
Subscription	Select the Azure portal subscription that will manage this VA.
Resource group	Select the resource group used for your storage account for your VA.
Instance Details	
Virtual Machine name	Enter a unique name for this VA.
Region	Choose the region in which you want to deploy the ClearPass VA.
Availability options	Select an availability option, if desired. No infrastructure redundancy is required for the ClearPass VA.
Image	Select the version of the Aruba ClearPass Policy Manager (CPPM) image you wish to install on the VA.
Azure Spot Instance	Select the **No** option. The ClearPass VA should not operate as an Azure spot instance.
Size	Click this drop - down menu and select one of the following recommended sizes. Each of the recommended disk sizes corresponds with a ClearPass hardware appliance model. Using v3 instances as an example: • The **Standard_D2s_v3 - 2 vcpus, 8 Gib memory** option corresponds to a ClearPass C1000 appliance. • The **Standard_D4s_v3 - 4 vcpus, 16 Gib memory** option corresponds to a ClearPass C2010 appliance.

Field	Description
	• The **Standard_D16s_v3 - 16 vcpus, 64 Gib memory** option corresponds to a ClearPass C3010 appliance.
Administrator Account	
Authentication Type	Select the **SSH public key** option. Later in the VA creation process you will be prompted to generate and down an SSH key pair. The Azure VA creation process requires that you create and download SSH keys, but these keys will not be used, and cannot be used to log in to ClearPass.
Username	Enter a username for the VA administrator.
SSH public key source	Select **Generate new key pair**.
Key pair name	Enter a name for the key pair to be used by this VA.
Inbound Port Rules	
Public Inbound Ports	Select **Allow Selected Ports**.
Select inbound ports	Enable the following ports: HTTPS (443) SSH (22)

When you are done configuring the **Basics** settings, click **Next: Disks**.

Disks

On the **Disks** tab, configure the virtual appliance disk settings as described in Table 3.

Table 1. Configuring Disk Settings

Field	Description
Disk Options	

Field	Description
OS disk type	Select an OS disk type. The **Premium SSD** option is supported.
Encryption type	Selected the default option **Encryption at - rest with a platform - managed key**.
Data Disks	
Create and Attach a New Disk	This setting is optional, as you do not need to create and attach a new disk at this time. You may choose to attach a disk when you are ready to move your VA from a test environment into production.

When you are done configuring disk settings, click **Next: Networking**.

Networking

On the **Networking** tab, configure the virtual appliance network interface as described in Table 4. Note that these settings allow you to define only one interface. ClearPass supports only the management interface.

Table 1. Configuring the Networking Interface

Field	Description
Virtual Network	Select an existing virtual network for the VA, or click **Create New** to create a new virtual network.
Subnet	Select an existing subnet, or click **Manage Subnet Configuration** to define a new subnet.
Public IP	Select one of the following options: • Click the drop - down menu to select an existing public IP for the VA • Click the drop - down menu to select **None** if the VA should not use a public IP

Field	Description
	• **Create New** to create a new public IP **NOTE** If you choose a public IP option, you can create rules to restrict access only to VPN - connected devices in a later step.
NIC network security group	Select **Basic**.
Public inbound Ports	Select **Allow selected ports**.
Select inbound ports	Select **HTTPS (443)** and **SSH (22)**.
Accelerated networking	Select **Off**. The ClearPass VA does not support accelerated networking.
Load Balancing	Select **No**. Azure load balancers are not able to load balance RADIUS requests, which would prevent ClearPass from working as expected.

When you are done configuring network interface settings, click **Next: Management**.

Management

On the **Management** tab, configure the virtual appliance management settings as described Table 5.

Table 1. Configuring Virtual Appliance Management Settings

Field	Description
Monitoring	
Boot diagnostics	Select **On**.

Field	Description
Diagnostics storage account	Select the customer storage account you created in Create a Custom Storage Account . **NOTE** The ClearPass VA does not support the default managed storage account, so you must create a custom storage account.
Identity	
System assigned managed identity	Select **Off**.
Microsoft Entra ID	The default **Off** setting cannot be changed as the ClearPass VA does not support Microsoft Entra ID (was Azure Active Directory).
Auto Shutdown	
Enable auto - shutdown	Select **Off**. **WARNING** A ClearPass VA will not operate correctly with this setting enabled. Enabling this setting will cause regular ClearPass outages.

When you are done configuring management settings, click **Next: Advanced**.

Advanced

On the **Advanced** tab, configure the advanced virtual appliance settings as described in Table 6.

Table 1. Configuring Advanced Virtual Appliance Settings

Field	Description
Extensions	
Extensions	Do not select any extensions. The available extensions in this menu are Azure extensions, not ClearPass extensions. The ClearPass VA does not support any Azure extensions.
Custom Data	
Custom data	This field is optional. Custom data is supported if required by your deployment, but it is not required by the ClearPass VA.
Host	
Host group	Do not select a host group.
Proximity placement group	Do not select a proximity placement group.
VM generation	Select **Gen 1**. This is the only option supported by the ClearPass VA.

When you are done configuring advanced VA settings, click **Next: Tags**.

Tags

Tags are not required by the ClearPass VA. However, you can use the **Tags** tab to configure tags for your individual deployment, if desired.

Click **Next: Review + Create** to complete your VA configuration.

Review + Create

About this task

This tab displays the settings you configured in the previous tabs, the product details and pricing for the selected VA size, and Azure terms and conditions. You may be prompted to enter a contact name, email address, and phone number for your VA.

Procedure

1. When you have verified your settings, click **Create**. The **Generate new key pair** window appears.
2. Click **Download private key and create resource**.

> **NOTE**
>
> The Azure VA configuration process requires that you generate these keys, but these keys will not be used by your ClearPass VA, and cannot be used to log in to ClearPass.

3. After you generate the SSH key pair, the **Deployment is in progress** dialog appears and indicates the status of each VA resource. When each resource has been successfully created, the message **Deployment Created** appears.
4. Click **Go to resource** to open your VA in the Azure portal.
5. Initial configuration of ClearPass can be done either through the serial console on Azure or through SSH. When the configuration is done, proceed to the next step.
6. Check that all the required services are up (Ingress services will be down by default; the VIP service will be disabled on Azure ClearPass), either through SSH or the Azure serial console by issuing the service status all command from an appadmin CLI login. After you confirm that all the services are up, allow 20 minutes and *only then* continue to morph ClearPass as defined in the Adding a Disk to the ClearPass Virtual Appliance section.

> **NOTE**
>
> ClearPass supports only the management (MGMT) interface for Azure deployments. Attaching an additional interface to ClearPass must be avoided. Check the following table which shows the list of inbound rules to be defined on different ports.
>
> For additional information on ports for ClearPass, refer to the "Firewall Ports Recommended and Required to be Open" topic in the *ClearPass Policy Manager 6.12 User Guide*.

7. Click the **Networking** menu link.
8. Select **Add Inbound Port** Rule.
9. In the **Protocol** field, select **TCP**, then add ports as a comma-separated list in the **Destination port ranges** field.
0. Select the **UDP** option in the **Protocol** field then add UDP ports as a comma-separated list in the **Destination port ranges** field.
1. Select the **Overview** menu link and click the **Start** icon to restart the VA.

Table 1. Ports Required for ClearPass

Port	Description
ClearPass Policy Manager Cluster (Subscriber - Publisher)	
TCP Port 80	HTTP (Between cluster members)
UDP Port 123	NTP (subscriber to publisher)
TCP Port 443	HTTPS (Bi - directional)
TCP Port 4231	NetWatch (Post - authentication module and the cluster member with Insight enabled)
TCP Port 5432	PostgreSQL for DB replication (subscriber to publisher)
ClearPass Policy Manager used with ClearPass Guest	
TCP Port 80	HTTP (Endpoint - - > Policy Manager)
TCP Port 443	HTTPS (Endpoint - - > Policy Manager)
TCP/UDP Port 49	TACACS+ (NAD < - - > Policy Manager)
UDP 3799	RFC 3576/5176 (CoA) (NAD < - - > Policy Manager)
TCP 25 or 465	SMTP Mail

Port	Description
TCP 22 and 443	SSH and HTTPS and multicast between two servers in a high availability configuration
ClearPass Policy Manager to Active Directory	
TCP/UDP Port 53	DNS from a client to domain controller and between domain controllers
UDP Port 88	Kerberos authentication
TCP/UDP Port 135	Operations between domain controllers and between a client and a domain controller
UDP Port 389	LDAP to handle normal queries from client computers to the domain controllers
TCP/UDP Port 464	Kerberos password changes
TCP Port 3268 and 3269	Global catalog from a client to a domain controller.
ClearPass Policy Manager to OnGuard Client	
TCP Port 6658	OnGuard client to communicate with Policy Manager
Other Ports	
UDP Port 1812, 1645	RADIUS authentication
UDP Port 1813, 1646	RADIUS accounting
TCP/UDP Port 53	DNS
TCP/UDP Port 389	LDAP
TCP/UDP Port 636	LDAP protocol over TLS/SSL

Port	Description
TCP/UDP Port 3269	Microsoft Global Catalog with LDAP/SSL
TCP port 2083	RadSec
UDP Port 5432	ClearPass clustering

Upgrading a D3 Instance Type to a D5 Instance Type

About this task

Users should be aware that an Azure D3 instance cannot be upgraded directly to an D5 instance. To move from an existing D3 instance to a D5 instance:

Procedure

1. First take a backup of the D3 instance.
2. On the appropriate D5 instance type, perform a fresh ClearPass 6.14.0 installation.
3. Do the bootstrap configuration steps and update the Platform License.
4. Execute the **system morph-vm** appadmin CLI command to deploy the appropriate virtual appliance size (for example, C1000V, C2000V, C3000V) based on the instance type.
5. While the **system morph-vm** command is executing, the following option is displayed:

   Do you wish to encrypt all local data? (Y/N)

   > **NOTE**
   >
   > *Note: Yes (Y) is recommended unless virtual system encryption is already enabled.*

   This setting cannot be changed after installation.

   Press 'Y' or 'N' to proceed
6. To encrypt all local data, enter **Y**. After the **system morph-vm** command is fully executed, the server will automatically reboot.

7. Restore the backup on the D5 instance.
8. Repeat these steps for every D5 instance.
9. Form the cluster.
0. Update each instance to the latest 6.14.x release.